Paper 2 – Urology II: All Answers by Section

Compiled from: 2015–2025 actual Paper 2 (12 papers) and Practice Sets 1–17 Paper 2 — 232 answers.

Repeats merged: questions asked more than once are printed once, with every wording and source listed under "Asked as"; their answers are combined so that no fact from any version is lost. 53 merged topics replace 126 original answers; 159 topics in all.

Layout: the 17 revision-book sections; in each, long answer questions (LAQ) first, then short answer questions (SAQ), then a Rapid Revision Sheet of that section's key points.

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Section 1 · Basic Principles — Short answer questions (SAQ)

1. Energy sources in laparoscopic urology
2. Renal isotope scans and DMSA renal scan *(merged: 2 questions)*
3. Microscopic Hematuria

Section 2 · Oncology, DSD, Gender & Ageing — Long answer questions (LAQ)

1. Describe use and principles of immunology and Immunotherapy in Urological Oncology

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

1. Female Circumcision
2. Hereditary urological malignancies.
3. Penile inversion vaginoplasty technique.
4. 46 XX Male

Section 3 · Infections & Inflammation — Long answer questions (LAQ)

1. Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male.

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

1. Interstitial cystitis (Bladder pain syndrome).
2. Malakoplakia

Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

1. Wilms tumour: classification, staging and management *(merged: 2 questions)*

Section 4 · Paediatric Urology 1 — Short answer questions (SAQ)

1. Retractile testis?
2. Retroileal Ureter
3. Medullary sponge kidney
4. Megameatus intact prepuce
5. Undescended testis
6. Techniques of chordee correction
7. Perineo-scrotal hypospadias

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

1. Valve bladder: management *(merged: 2 questions)*
2. Embryogenesis of urinary bladder
3. Anterior urethral valve

Section 6 · Female Urology — Short answer questions (SAQ)

1. Urethral caruncle.
2. Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT *(merged: 3 questions)*

3. VVF repair: early versus delayed timing, and surgery for recurrent VVF *(merged: 2 questions)*
4. ALPP
5. Vaginal prolapse repair.

Section 7 · Voiding Dysfunction — Long answer questions (LAQ)

1. Pathophysiology, classification and management of LUTS.
2. Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

1. Botulinum neurotoxin and its role in urology *(merged: 2 questions)*
2. Uro Dynamic Studies.
3. Uroflowmetry.
4. Management of detrusor underactivity
5. Neuromodulation
6. Bladder dysfunction non genitourinary systemic disease
7. Overactive bladder and its drug therapy *(merged: 2 questions)*
8. Vesicourodynamics

Section 8 · Nocturia & Male Incontinence — Short answer questions (SAQ)

1. Post-prostatectomy incontinence *(merged: 2 questions)*

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

1. Surgical management of erectile dysfunction and penile prosthesis *(merged: 3 questions)*
2. Discuss the role of urologist in reproductive endocrinology and male infertility practice.

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

1. Management of priapism
2. Azoospermia and non-obstructive azoospermia *(merged: 2 questions)*
3. Oligospermia.
4. Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction) *(merged: 3 questions)*
5. CRE & BEP

Section 10 · Testis, Penis & Urethra — Long answer questions (LAQ)

1. Urethral stricture disease: etiology, presentation and management (including recent advances) *(merged: 2 questions)*
2. Importance and management of Inguinal lymph nodes in CA penis.
3. Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis.

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

1. Management of Urethral meatal stenosis.
2. Perineal urethrostomy.
3. Surgical management of bilateral varicocele

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

1. Metabolic evaluation in stone disease (and management of bilateral staghorn calculi) *(merged: 3 questions)*
2. Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management *(merged: 3 questions)*
3. ESWL: mechanism, indications, factors affecting outcome, complications and current status *(merged: 3 questions)*
4. Management of upper ureteric stones

5. Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries *(merged: 3 questions)*
6. Stone formation (etiopathogenesis) and medical management of urolithiasis / renal calculi *(merged: 2 questions)*
7. Uric acid calculi: etiopathogenesis, medical management and allopurinol *(merged: 2 questions)*
8. Urinary crystals

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

1. Renal tubular acidosis
2. Trans Uretero Ureteric anastomoses
3. RIRS / flexible ureteroscopy *(merged: 4 questions)*
4. Retro peritoneal fibrosis medical management
5. Medical expulsive therapy (MET) in urinary calculi *(merged: 2 questions)*
6. Stones (urolithiasis) in pregnancy *(merged: 2 questions)*
7. Uretero-calycostomy.
8. Physiology of UPJ in relation to urine transport
9. Tubeless PCNL *(merged: 2 questions)*
10. Complications of PCNL, including bleeding and persistent haematuria after PCNL *(merged: 2 questions)*
11. Describe various pt positions in PCNL
12. Steinstresse.
13. Treatment of inferior calyceal calculus
14. Primary Hyperoxaluria
15. Fracture penis
16. Ovarian vein syndromes
17. Salvage procedures for Complicated PUJO.

Section 12 · Bladder Cancer & Diversion — Long answer questions (LAQ)

1. CA bladder, etiology, newer investigations and treatment.
2. Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman *(merged: 3 questions)*
3. Use of intestinal segments in urology, and orthotopic neobladder *(merged: 4 questions)*
4. Classification of bladder cancer and bladder-preserving treatment of TCC *(merged: 2 questions)*
5. Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction.
6. Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types *(merged: 2 questions)*

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

1. Bladder diverticulum.
2. Intravesical gene therapy.
3. BCG failure
4. Intravesical therapy for TCC of bladder
5. What is optical coherence tomography. What is its potential role in management of CA bladder?
6. Von Bruns Nest
7. Mitomycin
8. Mullerionosis of UB

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

1. Metastatic RCC (mRCC): assessment and management, including recent concepts *(merged: 3 questions)*
2. Etiopathogenesis and management of Upper tract TCC
3. Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches *(merged: 2 questions)*

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

1. Acquired renal cystic disease
2. ADPKD (Adult polycystic kidney disease)
3. Biomarkers for patients with advanced RCC
4. Tivozanib
5. Immunotherapy in (metastatic) RCC *(merged: 2 questions)*
6. VHL gene and von Hippel–Lindau disease *(merged: 2 questions)*
7. Renal aangiomyolipoma

Section 14 · Adrenal & Retroperitoneum — Long answer questions (LAQ)

1. How will you evaluate and manage 38 yr male with uncontrolled hypertension

Section 14 · Adrenal & Retroperitoneum — Short answer questions (SAQ)

1. Phaeochromocytoma: overview, pre-operative and intra-operative management *(merged: 3 questions)*
2. Cushing syndrome and the dexamethasone suppression test *(merged: 2 questions)*
3. Algorithm for management of an incidentaloma
4. MIBG scan
5. Adrenal Sx.

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

1. Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD) *(merged: 5 questions)*
2. Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval *(merged: 2 questions)*
3. Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment *(merged: 3 questions)*
4. Acute renal failure (AKI) and acute tubular necrosis *(merged: 2 questions)*
5. Principles of vascular anastomosis and vascular access surgery for HD *(merged: 2 questions)*

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

1. Role of vascular grafts in urology
2. Pre-operative evaluation of the living (related and unrelated) renal donor *(merged: 2 questions)*
3. UW solution and organ preservation solutions *(merged: 2 questions)*
4. CAPD (Continuous Ambulatory Peritoneal Dialysis) *(merged: 2 questions)*
5. Pre transplant nephrectomy
6. GFR estimation: clinical assessment and in the prospective transplant donor *(merged: 2 questions)*
7. Post transplant malignancy
8. Pre transplant Immunosupression treatment options
9. Brainstem death: biology and clinical implications *(merged: 2 questions)*
10. Anomalies of renal artery.
11. Mycophenolate Mofetil.
12. Metabolic abnormalities in CKD

13. Renal transplant in neurogenic bladder
14. Pre-emptive renal transplant
15. THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules *(merged: 2 questions)*
16. Marginal kidney donor

Section 16 · BPH — Long answer questions (LAQ)

1. BPH: medical management and surgical modalities *(merged: 2 questions)*

Section 16 · BPH — Short answer questions (SAQ)

1. Millin's prostatectomy
2. Catheter free trial
3. Irrigation fluids for TURP.
4. LASERS in BPH.
5. Silodosin.
6. Dutasteride.
7. Measures to decrease complications of TURP
8. Tamsulosin
9. Prostatic stents
10. Role of UDS before prostate surgery
11. Monopolar v/s Bipolar TURP.

Section 17 · Prostate Cancer — Long answer questions (LAQ)

1. Castration-resistant prostate cancer (CRPC): definition and treatment *(merged: 2 questions)*
2. Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid) *(merged: 3 questions)*
3. PSA screening for prostate cancer: evidence, problems and solutions *(merged: 2 questions)*
4. Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy *(merged: 3 questions)*
5. 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis
6. 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.
7. Discuss management of organ confined CA prostate

Section 17 · Prostate Cancer — Short answer questions (SAQ)

1. Transrectal ultrasonography (TRUS) *(merged: 2 questions)*
2. Radiotherapy and brachytherapy for CA prostate *(merged: 2 questions)*
3. TRUS / prostate biopsy *(merged: 2 questions)*
4. PIN
5. Maximum (combined) androgen blockade and bicalutamide *(merged: 2 questions)*
6. Gleason's score and its significance
7. HIFU
8. Saturation biopsy in CA prostate
9. Biological effects of cryosurgery
10. Imaging measures in CA prostate
11. LHRH analogue
12. Expectant treatment of CA prostate
13. Molecular staging of CA prostate

Section 1 · Basic Principles — Short answer questions (SAQ)

Energy sources in laparoscopic urology

Type: SAQ (10 marks)

Source: 2015-Summer Paper 2, Q3d

Principles

- Devices seal by denaturing protein. Electrosurgery uses ~500 kHz current, too fast to stimulate nerve or muscle.
- **Cut**: continuous low-voltage waveform boils cell water. **Coagulation**: interrupted high-voltage waveform desiccates, spreading deeper. Also blend and spray (fulguration).

Types

Device	Principle	Lateral spread	Notes
Monopolar	Tip → patient → return pad	**5–7 mm**	Versatile; stray current
Bipolar	Current only between jaws	1–6 mm	No pad or capacitive coupling
Vessel sealer (LigaSure)	Compression plus impedance-controlled energy	1–3 mm	Seals vessels **≤7 mm**; single application
Ultrasonic (Harmonic)	Blade vibrating at 25–55 kHz; no current	**≈1 mm**	Safe with pacemakers; blade >200 °C, cools slowly
Argon beam	Monopolar current through argon, non-contact	Superficial	Raw surfaces; **gas embolism**

- **Also know:** lasers (CO2, KTP (potassium titanyl phosphate), holmium), now rarely used; laparoscopic RFA (radiofrequency ablation, target 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles) for small renal masses.

Electrosurgical injury

- Unseen injuries present late as perforation.

1. **Insulation failure**.
2. **Direct coupling**: active electrode touches another metal instrument.
3. **Capacitive coupling**: current induced through intact insulation; avoid hybrid trocars (metal cannula, plastic anchor).
4. **Thermal spread** to ureter, nerves or bowel.
5. **Return pad burns**; 6. **surgical fire**.

Safety

- **AEM (active electrode monitoring)** detects insulation failure and capacitive coupling; **REM (return electrode monitoring)** detects poor pad contact.
- Lowest effective power (cut 50–80 W, coagulation 30–50 W), tip in view, inspect insulation; bipolar near vital structures and in pacemaker patients.

Diagram to draw: Monopolar vs bipolar circuit.

Key points to remember:

- Spread: monopolar 5–7 mm > bipolar 1–6 > LigaSure 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz.
- Stray current: insulation failure, direct and capacitive coupling; use AEM and REM.

Renal isotope scans and DMSA renal scan

Asked as:

- Renal isotope scans — *Winter 2017 Paper II Q3f (SAQ)*
- DMSA Renal Scan. — *Summer 2018 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)

Principle

- A radiotracer follows a physiological process; a **gamma camera** (sodium iodide crystal, photomultipliers, collimator) images the photons. SPECT (single-photon emission computed tomography) gives 3D images.
- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon, so it is the workhorse.

Renal tracers

Tracer	Handling	Study	Main uses
DTPA (diethylenetriaminepentaacetic acid)	Glomerular filtration only	Dynamic	GFR (glomerular filtration rate); captopril renography; poor images in renal impairment
MAG3 (mercaptoacetyltriglycine)	Tubular secretion (protein-bound), high extraction	Dynamic	**Split function and obstruction (first choice)**; transplant
DMSA (dimercaptosuccinic acid)	Filtered, then retained by proximal tubules	Static at 2 h	**Pyelonephritis and scars**

Diuretic renography

- **Preparation:** good hydration; catheter if outlet obstruction or neurogenic bladder; cap a nephrostomy.
- **Dose:** 5–10 mCi (185–370 MBq) intravenously.
- **Phases:** perfusion (first minute); function: **split function** from the area under each curve at 1–3 min (normal 45:55 to 55:45); drainage.
- **Furosemide** 0.5 mg/kg (adult maximum 40 mg). Timing: F+20 (classic), F+15, F0 or F−15. Always note when it was given.

Post-furosemide T½ (half-clearance time)	Meaning
<10 min	Not obstructed
10–20 min	Indeterminate
>20 min	Obstructed

- **Curve types (O'Reilly):** I normal; II obstructed (rising); IIIa dilated, not obstructed (falls after furosemide); IIIb equivocal.
- **Pitfalls:** poor function, dehydration, full bladder, capacious pelvis.

DMSA renal scan

Principle

- **^{99m}Tc-DMSA** (technetium-99m dimercaptosuccinic acid) is filtered and then taken up by **proximal tubular cells** (megalin and cubilin receptors). It is retained in the cortex and excreted very slowly.
- So it gives a **static map of functioning cortex**, not drainage.

Technique

- Intravenous dose: **50 µCi/kg in children, 5 mCi (185 MBq) in adults**.
- Images at **2–4 hours** (later in renal failure): posterior and oblique views, **pinhole collimator** or **SPECT** (single-photon emission computed tomography), which is slightly more sensitive.
- Radiation about 1 mSv.

What it measures

- **DRF (differential renal function):** from geometric-mean counts; normal 45–55%. Better than MAG3 (mercaptoacetyltriglycine) for ectopic, horseshoe or duplex kidneys and very poor function.
- **Cortical defects:** photopenic areas.

Indications

- **Acute pyelonephritis** (gold standard), especially in infants with febrile UTI (urinary tract infection).
- **Renal scarring** in VUR (vesicoureteric reflux) and recurrent UTI; top-down approach (DMSA first, VCUG [voiding cystourethrogram] only if abnormal).
- Function of a duplex upper moiety, ectopic or horseshoe kidney; MCDK (multicystic dysplastic kidney: no uptake).
- Reconstruct or remove a poorly functioning kidney.

Acute pyelonephritis versus scar

	Acute pyelonephritis	Scar
Timing	Within 1 week of fever (sensitivity falls by 1 month)	**≥6 months** after infection (scars persist at about 6 months)
Defect	Ill-defined, may be multiple	Sharp, with cortical thinning
Kidney size	Normal or swollen	Small or contracted
Repeat scan	Resolves (about 85%) (acute pyelonephritis defects resolve)	Persists

Goldraich scar grading

Type	Finding
1	No more than 2 scarred areas
2	More than 2 scars, normal tissue between
3	Generalised damage, small kidney
4	End-stage, shrunken kidney, little or no uptake

Limitations

- Cannot tell **congenital reflux dysplasia** (boys, grade IV–V VUR, no UTI) from acquired scar.
- Gross hydronephrosis gives false defects; no information on drainage; poor uptake in renal failure and proximal tubular disorders.
- **Also know:** MAG3 is best for obstruction; NICE (National Institute for Health and Care Excellence): late DMSA 4–6 months after atypical or recurrent UTI.

Other uses

- **Captopril renography (DTPA):** captopril removes angiotensin II support of GFR in the stenotic kidney. Improvement of ≥10% off ACE (angiotensin-converting enzyme) inhibitors

means renovascular hypertension is likely.
- **Transplant:** acute rejection (perfusion falls, worsens) versus delayed graft function (perfusion kept, improves); urine leak.
- **DMSA:** acute pyelonephritis defects resolve; scars persist at about 6 months.
- **Nuclear cystography** for follow-up of reflux (more sensitive, lower radiation than VCUG [voiding cystourethrogram]).
- **Also know:** bone scan (MDP, methylene diphosphonate); labelled leucocyte scans; PSMA (prostate-specific membrane antigen) PET (positron emission tomography). Radiation dose is low.

Diagram to draw: renogram curves: normal, obstructed (rising) and dilated non-obstructed (falls after furosemide).

Diagram to draw: posterior DMSA image: normal kidney and a scarred kidney with polar defects.

Key points to remember:

- DTPA = filtration; MAG3 = secretion (best for obstruction); DMSA = cortex.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h.
- Gold standard for acute pyelonephritis and scars.
- T½ <10 min not obstructed; >20 min obstructed.
- Normal split (DRF) 45–55%; furosemide 0.5 mg/kg.
- Scar only if the defect persists ≥6 months.

Microscopic Hematuria

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 2, Q3e (also answered in 2016-Summer Paper 3, Q3d)

Definition

- **MH (microscopic haematuria): ≥3 RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020).
- **Dipstick alone is insufficient**: myoglobin, semen and bacterial peroxidases give false positives.
- Prevalence about 6.5%; malignancy in about 3% (over 85% bladder cancer).

Causes

- Malignant: bladder, UTUC (upper tract urothelial carcinoma), RCC (renal cell carcinoma), prostate.
- Benign: UTI (urinary tract infection), stones, BPE (benign prostatic enlargement), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal:** IgA nephropathy (dysmorphic RBCs, casts, proteinuria).
- Anticoagulants do not explain haematuria: evaluate as usual.

Initial evaluation

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch syndrome), examination, **blood pressure, creatinine**.
- Suspected UTI → culture, treat, **repeat urinalysis**; persistent MH → full evaluation.
- Renal features → nephrology **and** urological evaluation.

AUA/SUFU 2020 risk stratification

	Low (all)	Intermediate (any)	High (any)
Age	Women <50; men <40	Women 50–59; men 40–59	≥60
Smoking (pack-years)	Never or <10	10–30	>30
RBCs/HPF	3–10	11–25	>25
Gross haematuria history	No	No	**Yes**
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Shared decision: repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT (computed tomography) urography**

- Additional risk factors (irritative symptoms, radiation, cyclophosphamide, occupational exposure) move low to intermediate.
- CT contraindicated → MR (magnetic resonance) urography, or retrograde pyelograms.
- **No cytology or urine markers initially**; white-light cystoscopy.
- Persistent MH after a negative ultrasound → CT or retrograde imaging. Hereditary RCC risk (VHL (von Hippel–Lindau)) → upper tract imaging in any stratum.

Follow-up

- Negative evaluation → consider repeat urinalysis within 12 months; release if negative.
- Re-evaluate for gross haematuria, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up.

> **Key points to remember:**
> - ≥3 RBCs/HPF on microscopy; dipstick not enough.
> - Intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.

Rapid Revision Sheet – Section 1 · Basic Principles

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Energy sources in laparoscopic urology (SAQ)

- Spread: monopolar 5–7 mm > bipolar 1–6 > LigaSure 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz.
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Renal isotope scans and DMSA renal scan (SAQ)

- DTPA = filtration; MAG3 = secretion (best for obstruction); DMSA = cortex.
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Section 2 · Oncology, DSD, Gender & Ageing — Long answer questions (LAQ)

Describe use and principles of immunology and Immunotherapy in Urological Oncology

Type: LAQ (25 marks)

Source: 2025-Winter Paper 2, Q2 (immunomodulation in RCC (renal cell carcinoma): see 2025-Winter Paper 4, Q3a)

Introduction

- Immunotherapy works by either **supplying a danger signal** that starts a Th1 response (BCG (bacille Calmette-Guérin), vaccine adjuvant) or **removing a brake** on an existing response (PD-1 (programmed death-1), CTLA-4 (cytotoxic T-lymphocyte antigen-4)).

Principles of tumour immunology

Arm	Cells and role
Innate (rapid, pattern-based)	PAMPs (pathogen-associated molecular patterns) bind **TLRs (Toll-like receptors)**; macrophages, neutrophils; urothelium as sentinel (IFN (interferon), IL-8 (interleukin-8))
Bridge	**Dendritic cell (DC)**: takes up antigen, matures on danger signal, migrates to lymph node
Adaptive (specific, memory)	**CD8 killer** (sees class I MHC (major histocompatibility complex), ~9 amino acids); **CD4 helper** (class II); B cells
Brake	**Treg** (regulatory T cell, FoxP3+ CD4); MDSCs (myeloid-derived suppressor cells); M2 macrophages

- **Two signals**: TCR (T-cell receptor) meets MHC–peptide (signal 1) plus **CD28–B7** costimulation (signal 2).
- **Th1 vs Th2**: IL-12 gives Th1 (IL-2, IFN-γ, TNF-α (tumour necrosis factor-α)), which is antitumour. IL-4 gives Th2 (antibody, chronic inflammation).
- **Checkpoints**: CTLA-4 (on Tregs) competes for B7; PD-1 on CD8 cells binds **PD-L1** (programmed death ligand-1); LAG-3 binds class II.
- **Adaptive immune resistance**: CD8 cells secrete IFN-γ, which induces PD-L1 on the tumour, which switches the same T cells off. So PD-L1 marks a suppressed ongoing response.

Immunoediting (three Es)

- **Elimination** (small tumours destroyed) → **Equilibrium** (growth held) → **Escape** (clinical cancer).
- Escape: loss of antigen or MHC, Treg expansion, checkpoint ligands.
- **GU (genitourinary) landscape**: bladder: inflammation-driven (schistosomiasis), CD8 infiltration good. Kidney: **CD8 paradox** (more CD8, worse outcome). Prostate: Treg-dominated, PD-L1 low, so checkpoint inhibitors fail.

Immunotherapy in practice

1. BCG in NMIBC (non-muscle-invasive bladder cancer)

- Bacilli attach via **fibronectin**; TLR2/4 activation → IL-8 and TNF-α → neutrophil wave → DC priming → **Th1 CD4** response → memory.

- Th1 predicts success, Th2 failure. About half stay in remission for years. Used for intermediate/high-risk NMIBC and CIS (carcinoma in situ) with induction plus maintenance.

2. Cytokines and gene therapy

- **IFN-α** and **high-dose IL-2** (durable complete responses in a few mRCC (metastatic renal cell carcinoma) patients; capillary leak). Intravesical IFN-α: mixed.
- **Nadofaragene firadenovec** (adenovirus with IFN-α2b gene + Syn3): BCG-unresponsive CIS, CR (complete response) 53.4%; FDA (Food and Drug Administration) 2022.
- **N-803** (IL-15 superagonist) with BCG: FDA approved 2024 for BCG-unresponsive CIS.

3. Cancer vaccines

- Antigen plus adjuvant (often GM-CSF (granulocyte-macrophage colony-stimulating factor)) to activate DCs.
- **Sipuleucel-T**: autologous blood cells pulsed with **PAP (prostatic acid phosphatase)–GM-CSF**, three infusions. Asymptomatic mCRPC (metastatic castration-resistant prostate cancer): **OS (overall survival) 25.8 vs 21.7 months**, **no delay in progression**. First FDA-approved solid-tumour vaccine.
- Failed: PROSTVAC (prostate), IMA901 and AGS-003 (RCC).

4. Checkpoint inhibitors

Site	Use (trial)
BCG-unresponsive CIS	Pembrolizumab if cystectomy declined/unfit (KEYNOTE-057: CR 41%)
Adjuvant MIBC (muscle-invasive bladder cancer) and UTUC (upper tract urothelial carcinoma)	Nivolumab after radical surgery (**CheckMate 274**: DFS (disease-free survival) 20.8 vs 10.8 months; included UTUC, benefit uncertain in that small subgroup). Atezolizumab (IMvigor010) negative
Metastatic UC (urothelial carcinoma)	Avelumab **maintenance** after platinum (JAVELIN Bladder 100, OS gain); pembrolizumab second line (**KEYNOTE-045**: OS 10.3 vs 7.4 months); **enfortumab vedotin (EV) + pembrolizumab**
RCC	Nivolumab + ipilimumab (CheckMate 214); IO (immuno-oncology) + TKI (tyrosine kinase inhibitor) combinations; adjuvant pembrolizumab (KEYNOTE-564, DFS HR (hazard ratio) 0.68)
Prostate	Monotherapy poor; pembrolizumab only for **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient) tumours

- VEGF (vascular endothelial growth factor) suppresses DCs and T-cell entry, so TKIs synergise with IO.
- **Biomarkers**: PD-L1 CPS (combined positive score) only for first-line single-agent pembrolizumab in cisplatin-ineligible UC; TMB (tumour mutational burden), MSI.

5. Adoptive cell therapy (brief)

- **CAR-T (chimeric antigen receptor T cells)**: T cells engineered against a surface antigen (e.g. PSMA (prostate-specific membrane antigen)). Investigational in GU; limited by poor trafficking, hostile microenvironment and cytokine release syndrome.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg

Grade	Action
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies need hormone replacement. BCG toxicity: cystitis, BCG sepsis.

Recent advances

- **EV-302**: enfortumab vedotin + pembrolizumab first line in metastatic UC, OS 31.5 vs 16.1 months versus platinum; now the preferred first line.
- **NIAGARA**: perioperative durvalumab with neoadjuvant cisplatin chemotherapy improved event-free and overall survival in MIBC.
- **Also know:** CheckMate 901 (nivolumab + gemcitabine–cisplatin); ADT (androgen deprivation) + IO trials; anti-CD38 and LAG-3 targets.

Diagram to draw: T cell–tumour synapse with TCR–MHC, CD28–B7, CTLA-4 and PD-1/PD-L1 blocked by antibodies.

Key points to remember:

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG: TLR2/4 → neutrophils → Th1 memory.
- Sipuleucel-T: OS +4.1 months, no delay in progression.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

Section 2 · Oncology, DSD, Gender & Ageing — Short answer questions (SAQ)

Female Circumcision

Type: SAQ (10 marks)

Source: 2019-Summer Paper 2, Q3f

Assumed to mean female genital mutilation/cutting. Not covered in the reference chapters; answer follows WHO (World Health Organization) definitions.

Definition

- **FGM (female genital mutilation):** all procedures that partly or totally remove the external female genitalia, or otherwise injure them, for non-medical reasons (WHO).
- Over 200 million living women are affected, mainly in Africa, the Middle East and Asia; usually in childhood. Practised in some Indian communities ("khatna").

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoral glans and/or prepuce
II (excision)	Removal of the clitoral glans and labia minora, with or without the labia majora
III (infibulation)	Narrowing of the vaginal opening by cutting and joining the labia into a seal, with or without clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterisation

Complications

- **Immediate:** pain, haemorrhage, shock, infection (tetanus), **acute urinary retention**, urethral injury, death.
- **Urological:** slow stream, dysuria, **recurrent UTI (urinary tract infection)**, meatal obstruction under the scar, urethral stenosis, incontinence, **vesicovaginal fistula** after obstructed labour.
- **Gynaecological and sexual:** inclusion cysts, keloid, dyspareunia, infertility.
- **Obstetric:** obstructed labour, perineal tears, postpartum haemorrhage, neonatal death.
- **Psychological:** anxiety, depression, post-traumatic stress disorder.

Evaluation

- Sensitive history; examination to classify type; flow rate, residual urine; cystoscopy if fistula suspected.

Management

- **Deinfibulation (type III):** anterior midline incision of the scar under local or regional anaesthesia to expose the urethral meatus and introitus; raw edges oversewn.
  - **Timing:** ideally before pregnancy, or antenatally, or in labour. Relieves retention, UTI and dyspareunia.
  - **Never reinfibulate** (illegal in the UK).
- Excise cysts; treat stenosis; repair fistula; counselling. Clitoral reconstruction has limited evidence.

Legal and ethical aspects

- WHO and the UN (United Nations) condemn FGM as a human rights violation; WHO opposes its performance by health workers.
- Illegal in many countries (e.g. UK FGM Act 2003, with mandatory reporting in girls under 18). India has no specific law.

Diagram to draw: vulva showing types I–III and the deinfibulation incision line.

> **Key points to remember:**
> - WHO types: I clitoridectomy, II excision, III infibulation, IV others.
> - Urological: retention, recurrent UTI, stenosis, fistula.
> - Deinfibulation for type III; never reinfibulate.

Hereditary urological malignancies.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 2, Q3f

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: **early, bilateral, multifocal** tumours.
- **4–6% of RCC (renal cell carcinoma)** is familial; all familial RCC syndromes are **autosomal dominant**; only hereditary papillary RCC (HPRC) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma, pancreatic cysts
HPRC	MET (7q31) oncogene	Multiple bilateral papillary (type 1-like)	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive**, often solitary papillary	Skin and uterine leiomyomas
BHD (Birt–Hogg–Dubé)	FLCN (folliculin, 17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13)	Multiple AMLs (angiomyolipomas); RCC 1–3%	Angiofibromas, cortical tubers

- **VHL type 1** (deletions, truncations): no phaeochromocytoma; **type 2** (missense): phaeochromocytoma.
- **Also know:** SDH (succinate dehydrogenase)-deficient RCC (paraganglioma); BAP1 (uveal melanoma, mesothelioma); Cowden (PTEN; breast, thyroid).

Management principles for kidney

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat operations. **Exception: HLRCC**: excise promptly, wide margins.
- TSC AML: everolimus; embolisation or partial nephrectomy if ≥5 cm. **Belzutifan** (HIF-2α [hypoxia-inducible factor 2α] inhibitor, FDA [Food and Drug Administration]-approved 2021) for VHL.

Other urological cancers

- **Prostate: BRCA2**, BRCA1, ATM, CHEK2, PALB2, HOXB13, mismatch repair genes; father plus brother: 9-fold risk. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): 14-fold risk, about 75-fold with MSH2. Suspect if age <60 or Lynch cancer history; MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.

- **Also know:** Wilms tumour (WT1 gene: WAGR [Wilms, aniridia, genitourinary anomalies, retardation] and Denys–Drash; Beckwith–Wiedemann).
- **Approach:** genetic counselling, germline testing, screening relatives.

Diagram to draw: VHL loss → HIF accumulation → VEGF (vascular endothelial growth factor) → angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Penile inversion vaginoplasty technique.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 2, Q3b

Introduction

- Commonest gender-affirming genital operation for transgender women: a functional, aesthetic, sensate vulva and vagina.

Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care, version 8):** one assessment, **6 months of hormones** before gonadectomy.
- **Permanent hair removal** of the scrotal graft area (prevents intravaginal hair).
- BMI (body mass index) usually ≤35 kg/m²; stop smoking; sperm banking; continue oestrogen.

Principle: homologous tissues

Male structure	Becomes
Glans on dorsal neurovascular bundle	**Clitoris**
Preputial mucosal collar	Clitoral hood, inner labia minora
Dorsal bulbar urethral plate	Vestibule
Penile shaft skin (inverted)	**Neovaginal lining**
Scrotal skin (full-thickness graft)	Extra canal lining
Lateral scrotal skin + cord fat	Labia majora

Steps

```
Lithotomy; tranexamic acid 1 g; catheter
→ harvest, defat and tubularise scrotal skin graft
→ bilateral orchidectomy (leave cord fat for labial bulk)
→ deglove penis between dartos and Buck's fascia
   (keep dartos with skin = blood supply)
→ separate dorsal neurovascular bundle + glans;
   excise ALL corporal erectile tissue
→ clitoroplasty; fix at level of adductor longus tendons
→ shorten and spatulate urethra; excise bulbospongiosus
→ canal dissection: central tendon, behind prostate,
   preserve Denonvilliers fascia, repeated rectal checks
→ invert penile skin + graft into canal; labiaplasty
→ pack canal; negative-pressure dressing
```

- **Why excise all erectile tissue:** residual tissue engorges and obstructs the canal.
- Depth depends on canal dissection, not penile length. **Robotic peritoneal flaps** add depth with less rectal injury; **intestinal vaginoplasty** if skin is inadequate or for revision.

Postoperative care

- Pack and catheter out on day 5.
- **Lifelong dilation**, e.g. 4 times daily for 6 weeks, tapering to daily after 1 year; pelvic floor physiotherapy.

Complications

- **Wound dehiscence** (commonest, 12–33%); granulation tissue.
- **Rectovaginal fistula** (0.8% primary, 6.3% revision); urethrovaginal fistula; meatal stenosis, spraying.
- **Vaginal stenosis** (poor dilation); prolapse 4–10%.
- Most achieve orgasm (median about 6 months).

Diagram to draw: sagittal view of the lined canal between prostate and rectum.

> **Key points to remember:**
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; keep Denonvilliers fascia intact.
> - Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

46 XX Male

Type: SAQ (10 marks)

Source: 2024-Winter Paper 2, Q3c (also answered in 2022-Summer Paper 1, Q3e)

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition

- **Testes and male genitalia in a 46,XX individual.** Old term "XX sex reversal"; renamed **46,XX testicular DSD** (Chicago consensus 2006). About **1 in 20,000** males.
- In the DSD classification it is a 46,XX DSD of gonadal development, with ovotesticular DSD; **CAH (congenital adrenal hyperplasia)** is the androgen-excess type.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) crosses from Yp to X during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY switches on SOX9 → Sertoli cells → **AMH (anti-Müllerian hormone)**: no uterus. Leydig testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis always fails.

Clinical features

- Male identity; found at **infertility work-up**, or prenatally (cell-free DNA 46,XX, male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia; progressive testicular regression.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	**Never (no AZF)**	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis; FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY**; gene panel if SRY-negative; scrotal ultrasound.
- In an infant with atypical genitalia, exclude CAH (17-hydroxyprogesterone) and ovotesticular DSD.

Management

- **Testosterone replacement** when hypogonadal (libido, bone, muscle).
- **Infertility:** sperm retrieval is futile; counsel **donor sperm or adoption**.

- Repair hypospadias or undescended testis; reduction mammoplasty for persistent gynaecomastia.
- Genetic counselling, psychological support, bone density follow-up.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

Key points to remember:

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Rapid Revision Sheet – Section 2 · Oncology, DSD, Gender & Ageing

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe use and principles of immunology and Immunotherapy in Urological Oncology (LAQ)

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG: TLR2/4 → neutrophils → Th1 memory.
- Sipuleucel-T: OS +4.1 months, no delay in progression.
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- WHO types: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, fistula.
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- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

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- Shorter than Klinefelter; FISH for SRY confirms.

Section 3 · Infections & Inflammation — Long answer questions (LAQ)

Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male.

Type: LAQ (25 marks)

Source: 2018-Summer Paper 2, Q1

Introduction

- GU (genitourinary) TB (tuberculosis) is caused by the *Mycobacterium tuberculosis* complex (MTBC). It is the second commonest extrapulmonary site after lymph nodes. Untreated, it destroys the kidney and contracts the bladder.

Pathophysiology

- **Primary infection:** inhaled bacilli spread by lymph and blood, before immunity develops, and seed the kidney and epididymis.
- **Granuloma:** macrophages, epithelioid and giant cells inside a lymphocyte cuff, kept alive by IFN-γ (interferon-γ). In 90–95% it contains the infection; viable bacilli stay dormant. Reactivation follows old age, diabetes, renal failure, HIV (human immunodeficiency virus) or steroids, sometimes decades later. Then **caseation** releases bacilli into urine.

Route	Lesion
Haematogenous (principal)	Parenchymal disease: kidney, epididymis, prostate
Descending urinary	Luminal disease: ureteric stricture, cystitis, urethra
Ascending (intravesical BCG [bacille Calmette–Guérin])	Cystitis, prostatitis, epididymo-orchitis

Site	Frequency	Pathology
Kidney	80%	Periglomerular granulomas, then caseating cavities, papillary necrosis, calcification; **autonephrectomy** ("putty kidney")
Ureter	—	Stricture, commonest at the **VUJ** (vesicoureteric junction); "beaded corkscrew" or "pipestem" ureter
Bladder	—	Starts at the ureteric orifices ("golf-hole" orifice); ulcers; after about a year of fibrosis it contracts; **thimble bladder <20 mL**
Epididymis	Commonest genital site	Starts in globus minor; beaded vas, posterior scrotal sinus
Prostate, seminal vesicles	22–49%	Calcification, abscess; infertility

- **Key concept:** drugs cure by fibrosis, and fibrosis obstructs. Strictures and bladder contraction can **worsen during treatment**.

Diagnosis (brief)

- **Sterile pyuria** with or without haematuria (>90%).
- **Urine AFB (acid-fast bacilli) culture** (gold standard): 3–5 first-void morning samples on consecutive days. LJ (Löwenstein–Jensen) solid medium takes 4–6 weeks; use a liquid

system as well. Culture gives drug susceptibility.
- **GeneXpert MTB/RIF:** rapid DNA test that also detects rifampicin resistance (marker of MDR [multidrug-resistant] TB).
- **Imaging:** chest radiograph in all; CT (computed tomography) urography: calcification, phantom calyx, Kerr's kink, ureteric strictures, small bladder. Biopsy if culture is negative.

Medical treatment (first step in all)

- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months, with pyridoxine. Some extend to 9–12 months in complicated disease.
- Repeat ultrasound during treatment for worsening obstruction; stent or PCN (percutaneous nephrostomy) early. Steroids: anecdotal only.
- **Surgery after at least 4–6 weeks of ATT (antituberculous therapy)**, when inflammation and bacillary load have settled.

Assessment of a thimble bladder (30-year-old man)

Assess	How	Why
Disease activity	Repeat urine AFB culture; bladder biopsy	Do not reconstruct with active TB
Capacity and compliance	Cystogram, cystoscopy under anaesthesia, urodynamics	Size and type of reconstruction
Upper tracts	Ultrasound, CT urography, MCU (micturating cystourethrogram)	VUJ stricture or reflux needs reimplantation
Renal function	Creatinine, GFR (glomerular filtration rate), split function	Bowel reservoir needs creatinine <2 mg/dL, GFR >35–40 mL/min
Urethra, bladder neck, sphincter	Urethrogram, endoscopy	Orthotopic reservoir needs a patent, continent outlet
Patient	Dexterity, motivation, fertility wishes	May need CIC (clean intermittent catheterisation)

Surgical options

Bladder state	Operation
Half the bladder diseased, capacity <100 mL	**Augmentation ileocystoplasty** ("clam" or onlay of detubularised ileum)
Severely contracted	Augmentation with **ileocaecum** or sigmoid
Thimble bladder (<20 mL)	**Substitution cystoplasty:** supratrigonal excision of the diseased bladder, keeping trigone and bladder neck, and an orthotopic detubularised ileal or ileocaecal reservoir

- **Principles:** detubularise and fold the bowel into a spherical, low-pressure reservoir (Laplace's law; interrupts peristalsis). Anastomose widely to the trigone.
- **Ureters:** if the VUJ is strictured or refluxing, excise to healthy ureter and **reimplant into the reservoir**: an antireflux tunnel under a colonic taenia, or an isoperistaltic afferent ileal limb. A Boari flap or psoas hitch is not possible in a contracted bladder.
- **Avoid orthotopic reconstruction** with: poor renal function; strictured or incompetent urethra; inability to perform CIC; active TB; short bowel or inflammatory bowel disease. Then use a continent catheterisable pouch (Mitrofanoff) or an **ileal conduit**.
- **Also know:** laparoscopic and robotic augmentation; transurethral incision of bladder neck contracture.

Complications

- **Mucus** and catheter blockage; **stones**; bacteriuria and UTI (urinary tract infection); **hyperchloraemic metabolic acidosis**; vitamin B12 deficiency if ileum is lost; **perforation** (needs regular emptying); incomplete emptying needing CIC; nocturnal incontinence; late tumour in the reservoir.

Follow-up

- Urine AFB culture, creatinine, bicarbonate, B12 and ultrasound. GU TB relapses in 6–22% (mean about 5 years), so follow up for up to 10 years.

Diagram to draw: supratrigonal excision of a thimble bladder with an ileocaecal substitute and ureteric reimplantation.

> **Key points to remember:**
> - Kidney 80%; bladder disease descends from the kidney; thimble bladder <20 mL.
> - Urine AFB culture: 3–5 early-morning samples; GeneXpert detects rifampicin resistance.
> - 2HRZE/4HR; operate after 4–6 weeks of ATT.
> - Augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.

Section 3 · Infections & Inflammation — Short answer questions (SAQ)

Interstitial cystitis (Bladder pain syndrome).

Type: SAQ (10 marks)

Source: 2017-Winter Paper 2, Q3b

Definitions

- **AUA (American Urological Association):** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or another identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (>6 months) bladder-related pain with at least one other urinary symptom (persistent urge or frequency), after confusable diseases are excluded.
- NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987) research criteria missed about 60% of patients.

Phenotypes

- **Hunner lesion (HL) disease, "IC":** about 20%, mainly over 50; true inflammation (lymphoplasmacytic infiltrate, mast cells).
- **Non-Hunner BPS (bladder pain syndrome):** bladder normal or shows glomerulations.
- **ESSIC type:** cystoscopy 1 normal, 2 glomerulations, 3 Hunner lesion; biopsy A normal, B inconclusive, C positive (e.g. 3C).

Pathogenesis (theories)

- Leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation and central sensitisation; stress-axis dysfunction; small-fibre neuropathy. No infection proven. 75–80% have pain beyond the pelvis (IBS [irritable bowel syndrome], fibromyalgia).

Evaluation (diagnosis of exclusion)

- History (**ask about ketamine**), pelvic-floor palpation, PVR (post-void residual), urinalysis, culture, voiding chart, symptom scores; cytology if risk factors.
- **Office cystoscopy if over 50 or HL suspected;** biopsy to exclude CIS (carcinoma in situ).
- **Local anaesthetic challenge** localises pain to the bladder. Potassium sensitivity test is not recommended.
- **Also know:** urodynamics optional; hydrodistension glomerulations are non-specific.

Treatment (AUA 2022: phenotype-based, no fixed tiers)

Group	Options
Behavioural	Education, dietary triggers, stress; **pelvic-floor myofascial therapy** if tender
Oral	**Amitriptyline** 10 mg rising to about 50 mg; hydroxyzine; cimetidine; **PPS** (pentosan polysulfate) 100 mg three times daily (**pigmentary maculopathy**: eye checks)
Intravesical	**DMSO** (dimethyl sulfoxide) 50%, 50 mL, 15–20 min dwell; heparin; alkalinised lidocaine
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short low-pressure hydrodistension; onabotulinumtoxinA; sacral neuromodulation
Last resort	Ciclosporin A for refractory HL; diversion ± cystectomy (<10%)

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term systemic steroids.

Diagram to draw: stellate Hunner lesion with radiating vessels.

> **Key points to remember:**
> - AUA definition: bladder pain with LUTS >6 weeks, no other cause.
> - Hunner lesion: cystoscopy over 50; fulguration or triamcinolone.
> - Amitriptyline, PPS, DMSO; potassium test obsolete.

Malakoplakia

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 2, Q3e

Definition

- "Soft plaque": a rare chronic granulomatous inflammation caused by **defective macrophage killing of bacteria**, mostly *E. coli*. Described by Michaelis and Gutmann (1902) and von Hansemann (1903).
- Commonest in the **bladder**; also ureter, kidney, prostate, testis, GI (gastrointestinal) tract, skin, lung.

Pathogenesis

- Macrophages ingest bacteria but cannot digest them (lysosomal defect, low intracellular cGMP [cyclic guanosine monophosphate]).
- Bacterial remnants become mineralised with calcium phosphate, forming laminated **Michaelis–Gutmann bodies**.
- Hosts are often immunocompromised, diabetic, chronically ill or **renal transplant recipients**.

Pathology

- Gross: soft yellow-brown plaques or nodules; renal tumour-like masses.
- Micro: sheets of large foamy histiocytes (**von Hansemann cells**) with **Michaelis–Gutmann bodies** (pathognomonic; "owl's eye" targetoid; stain with von Kossa, PAS [periodic acid–Schiff] and Prussian blue).
- MG bodies may be absent early; **α1-antitrypsin** immunostaining helps.

Clinical features

- Mostly >50 years; urinary tract **female:male 4:1**.
- Bladder: irritative LUTS (lower urinary tract symptoms), haematuria, recurrent *E. coli* UTI (urinary tract infection); plaques or masses mimicking cancer; distal ureteric stricture.
- Kidney: masses, fever, flank pain; renal vein or IVC thrombosis. Testis: epididymo-orchitis. Prostate: mimics carcinoma.

Investigations

- Urine culture (*E. coli*). Cystoscopy and **biopsy is diagnostic**.
- Imaging: enlarged kidneys with filling defects, **no stones, calcification or hydronephrosis** (unlike XGP).

	XGP (xanthogranulomatous pyelonephritis)	Malakoplakia
Cell	Lipid-laden foamy macrophages	Von Hansemann cells with MG bodies
Stone/obstruction	Usually (staghorn)	Absent
Laterality	Unilateral	Uni- or multifocal, bilateral
Treatment	Nephrectomy	Antibiotics; nephrectomy if unilateral, symptomatic

Management

- **Long-term antibiotics active inside macrophages:** fluoroquinolones (concentrate in macrophages), trimethoprim–sulfamethoxazole, rifampicin, doxycycline.

- **Bethanechol + ascorbic acid** raise intracellular cGMP and correct the degradative defect.
- Reduce immunosuppression where possible.
- Surgery if progression despite medical therapy: TUR (transurethral resection) of bladder lesions; **nephrectomy** for symptomatic unilateral renal disease.

Prognosis

- Bilateral or transplant-kidney disease: death often within 6 months; overall mortality can exceed 50%. Unilateral: long survival after nephrectomy.

Diagram to draw: histiocyte with Michaelis–Gutmann bodies.

> **Key points to remember:**
> - Defective macrophage digestion of *E. coli*.
> - Michaelis–Gutmann bodies are pathognomonic.
> - Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

Rapid Revision Sheet – Section 3 · Infections & Inflammation

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male. (LAQ)

- Kidney 80%; bladder disease descends from the kidney; thimble bladder <20 mL.
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Malakoplakia (SAQ)

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Section 4 · Paediatric Urology 1 — Long answer questions (LAQ)

Wilms tumour: classification, staging and management

Asked as:

- Wilm's tumor — *Practice Set 10 Paper II Q3b (SAQ)*
- Describe classification and management of Wilm's bladder — *Practice Set 15 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(Practice Set 15 Paper II Q1: bilateral Wilms tumour also answered in 2020-Summer Paper 3, Q3e and 2016-Summer Paper 3, Q3c; Wilms tumour in Practice Set 10 Paper 2, Q3b)

Read as: "Wilms tumour" (nephroblastoma); "bladder" is taken as a typo.

Introduction

- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.
- **Genetics:** mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in anaplasia (in most anaplastic tumours)**.

Syndromes:

Syndrome	WT risk
WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability; WT1 deletion)	50%
Denys–Drash (46,XY DSD [disorder of sex development], mesangial sclerosis; WT1 point mutation)	50%
BWS (Beckwith–Wiedemann syndrome), hemihypertrophy (11p15)	5–10%

- **Screening:** screen by ultrasound every 3–4 months (3–4-monthly) until (to age) 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

Classification

A. Histological (pathology)

Group	Features	Significance
Favourable histology	**Triphasic:** blastema, epithelium (tubules), stroma; pseudocapsule	Survival >90%
Anaplasia (unfavourable)	Nuclei ≥3× neighbours (larger), hyperchromasia, abnormal mitoses; **focal** or **diffuse** (diffuse worse than focal)	Chemoresistant; poor even at stage I

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

Risk	Histology
Low	Completely necrotic
Intermediate	All others (regressive, mixed, epithelial, stromal, focal anaplasia)

Risk	Histology
High	**Blastemal-predominant**; diffuse anaplasia

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

B. Staging: COG (Children's Oncology Group), surgical-pathological, before chemotherapy

Stage	Description
I	Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion
II	Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely (fully) resected
III	Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants
IV	Haematogenous metastases (lung commonest, liver, bone, brain)
V	Bilateral at diagnosis (each side staged separately)

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.
- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

Clinical features and evaluation

- Abdominal mass; haematuria about (~) 20%; hypertension about (~) 25% (renin). Look for aniridia, hemihypertrophy, varicocele (IVC thrombus). Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus. Acquired von Willebrand disease in up to 8%.
- Investigations: ultrasound with Doppler (IVC) → contrast CT (computed tomography) or MRI abdomen → **chest CT** (lung is the commonest metastatic site; before anaesthesia). Blood count, renal and liver function, coagulation.

Management: two philosophies

	COG (North America)	SIOP (Europe)
Sequence	**Upfront (transperitoneal) radical nephrectomy**, then chemotherapy	**Preoperative VCR + AMD** (vincristine + dactinomycin; 4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy (surgery)
Advantage	Untreated histology and biology for risk assignment	Tumour shrinks; spill almost zero
Risk basis	Stage + histology + LOH	Post-chemotherapy histology + stage

(VCR = vincristine; AMD = actinomycin D; DOX = doxorubicin.)

Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).
- **Lymph node sampling of at least 5 nodes** (**sample ≥5 lymph nodes**: hilar, paracaval, para-aortic, interaortocaval) is essential; formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins → preoperative chemotherapy to avoid bypass.

Postoperative therapy (COG, favourable histology)

Group	Treatment
Very low risk (<2 years, <550 g, stage I)	Nephrectomy alone (AREN0532)
Stage I–II	VCR + AMD (vincristine + dactinomycin)
Stage III	VCR + AMD + DOX (add doxorubicin) + flank RT (radiotherapy)
Stage IV	Three drugs + whole-lung RT; lung RT omitted if (lung) nodules clear by week 6 (AREN0533)
Anaplasia or 1p/16q LOH (loss of heterozygosity)	Intensified regimens (regimen M adds cyclophosphamide + etoposide)

Bilateral WT (stage V) and nephron sparing

- Bilateral WT occurs in 5–7%. **No upfront nephrectomy or biopsy.** The aim is cure while preserving nephrons.
- Three-drug chemotherapy, then **NSS (nephron-sparing surgery) by week 12**:

```
VCR + AMD + DOX × 6 weeks → reimaging
  ├─ resectable → bilateral NSS (nephron-sparing surgery)
  └─ not → 6 more weeks → surgery by week 12
       poor response → bilateral open biopsy
```

- **Why by week 12:** in NWTS-4 (National Wilms Tumor Study-4) prolonged chemotherapy delayed surgery and some tumours became anaplastic.
- **AREN0534:** 4-year EFS (event-free survival) 82.1% and OS (overall survival) 94.9%.
- Bilateral nephrectomy and dialysis are rare; transplant after 1–2 years.
- **Unilateral NSS:** for syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

Prognosis, relapse and late effects

- Favourable histology: survival >90%.
- Salvage survival: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.
- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).

Diagram to draw: COG staging diagram showing capsule, renal sinus, nodes and spill.

Key points to remember:

- WT1 at 11p13 (WAGR, Denys-Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: three drugs, surgery by week 12, NSS.

Section 4 · Paediatric Urology 1 — Short answer questions (SAQ)

Retractile testis?

Type: SAQ (10 marks)

Source: 2020-Summer Paper 2, Q3b

Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

Testis	Finding	Action
Retractile	Reaches scrotal bottom and stays	Observe yearly
Gliding (unstable high scrotal)	Pulled down but springs back	Treat as undescended
Undescended (palpable)	Cannot reach mid-scrotum	Orchidopexy by 18 months
Ascending (acquired)	Previously scrotal, now not	Orchidopexy

Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

Diagram to draw: inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

Key points to remember:

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

Retroileal Ureter

Type: SAQ (10 marks)

Source: 2025-Winter Paper 2, Q3b

*"Retroileal" is taken as **retroiliac ureter**, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).*

Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

Comparison with retrocaval ureter

	Retroiliac	Retrocaval (circumcaval)
Vessel	Common iliac artery	IVC (inferior vena cava)
Embryology	Persistent ventral umbilical artery root	Persistent subcardinal vein
Side	Either, may be bilateral	Right
Level	L5–S1	L3–L4, "fishhook"
Associations	Vasal/mesonephric ectopia	Horseshoe kidney, left renal anomalies

- **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels.

Diagram to draw: ureter behind the common iliac artery at L5–S1, dilated above.

Key points to remember:

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Medullary sponge kidney

Type: SAQ (10 marks)

Source: Practice Set 9 Paper 2, Q3e

Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
  - **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
  - **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
  - Impaired urinary concentration.
  - Mutations in H^+-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.
- Found in **2.6–21% of stone formers**, more often women.

Diagnosis

- **Urographic triad:**
  1. Enlarged kidneys, with or without papillary calcification.
  2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**.
  3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

Management

Agent	Target	Note
Potassium citrate	Hypocitraturia, hypercalciuria	**First choice**; reduces calciuria and recurrence
Thiazide	Hypercalciuria	Useful even without hypercalciuria
Inorganic phosphate	Hypercalciuria	If thiazide unsuitable; avoid with urease infection
Fluids, low-sodium diet	Volume, calciuria	All patients

- Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable.
- **Prognosis:** usually benign; renal failure is rare.

Diagram to draw: pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

Key points to remember:

- Ectatic collecting ducts; sporadic; "bristles on a brush".
- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

Megameatus intact prepuce

Type: SAQ (10 marks)

Source: Practice Set 15 Paper 2, Q3f

Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.
- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):**
  1. Glans traction suture; retract prepuce.
  2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft.
  3. Dissect the wide distal urethra free of the glans wings.
  4. **Excise a ventral wedge** of redundant urethra to taper it.
  5. Tubularise over a catheter; cover with dartos.
  6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

Diagram to draw: fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

Key points to remember:

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

Undescended testis

Type: SAQ (10 marks)

Source: Practice Set 16 Paper 2, Q3d (also answered in 2018-Summer Paper 3, Q3a and 2024-Winter Paper 1, Q2)

Definition and classification

- Testis not in the normal scrotal position: **palpable** (canalicular, high scrotal, **ectopic**, commonest at the superficial inguinal pouch) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).
- **Retractile:** brought down and stays; review yearly. **Acquired:** previously scrotal, ascends later.

Epidemiology and aetiology

- 2–4% of term boys at birth, 1–2% at 3 months; commoner in preterm infants.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgens** (inguinoscrotal phase), with genitofemoral nerve/CGRP (calcitonin gene-related peptide) guiding the gubernaculum.

Evaluation

- Warm room, frog-leg position; "milk" from the anterior superior iliac spine. Contralateral hypertrophy suggests vanished testis.
- **No ultrasound or MRI** (magnetic resonance imaging): they do not change management.
- **Bilateral non-palpable, or with hypospadias:** urgent karyotype and hormones to exclude **CAH** (congenital adrenal hyperplasia) or other DSD (differences of sex development).

Management

- **Observe to 6 months corrected age; operate after 6 months, preferably by 12, and by 18 months at the latest** (AUA [American Urological Association] 2014). Early surgery restores testicular growth.
- **Hormones (hCG [human chorionic gonadotrophin], LHRH [luteinising hormone-releasing hormone]): not recommended**; descent in about 20% only.

Situation	Operation
Palpable	Inguinal orchidopexy (mobilise cord, ligate processus, subdartos pouch) or scrotal (Bianchi)
Non-palpable	Examination under anaesthesia, then laparoscopy
Testis seen, vessels long	Laparoscopic orchidopexy (primary most successful)
Short vessels	**Fowler–Stephens** one- or two-stage (testicular vessels divided; survives on vasal collaterals) or Shehata traction
Vessels and vas enter ring	Inguinal or scrotal exploration; excise nubbin
Blind-ending vessels	Stop: vanished testis
Postpubertal, normal contralateral testis	Orchidectomy

- **Complications:** atrophy (higher after Fowler–Stephens), re-ascent, vas injury.

Prognosis

- **Fertility:** paternity about 80% unilateral, one-third bilateral.

- **Cancer:** 2–5-fold risk; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh). Teach self-examination.

Diagram to draw: laparoscopic view of the internal ring: testis, vessels entering ring, blind-ending vessels.

Key points to remember:

- Operate from 6 months, preferably by 12, latest 18 months.
- No imaging, no hormones; laparoscopy for non-palpable testis.
- Bilateral non-palpable: exclude CAH/DSD.

Techniques of chordee correction

Type: SAQ (10 marks)

Source: Practice Set 17 Paper 2, Q3b (related: Practice Set 10 Paper 1, Q3d)

Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test
  ├─ Straight        → urethroplasty
  ├─ Residual <30°   → dorsal plication
  └─ >30°            → divide urethral plate → still >30° → ventral lengthening (stage)
```

	Dorsal shortening (plication)	Ventral lengthening (corporotomy ± graft)
Methods	**Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication	1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect
Use	≤15–30°	Moderate–severe after plate division
Recurrence	28% (one series)	9%
Risk	Shortening; dorsal nerve injury off midline	Longer, staged; theoretical erectile dysfunction

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

Diagram to draw: penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

Perineo-scrotal hypospadias

Type: SAQ (10 marks)

Source: Practice Set 17 Paper 2, Q3e (related: 2017-Winter Paper 3, Q1)

Definition

- A severe **proximal hypospadias**: meatus at the penoscrotal junction, scrotum or perineum, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

Evaluation

- Examine stretched penile length, glans width, transposition and **both testes**.
- **DSD (disorder of sex development) work-up** if there is an undescended testis (especially non-palpable), a perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone). DSD risk is 64% in proximal vs 7% in distal cases.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

Preparation

- **Timing:** 6–18 months.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery; evidence mixed.

Surgery

1. Curvature correction: deglove, artificial erection, plication if <30°, plate division and ventral corporotomy ± graft if >30° (see Practice Set 17 Paper 2, Q3b).

2. Urethroplasty: staged repair is now preferred

Repair	Technique	Comment
Two-stage Bracka / STAG (staged tubularised autograft)	Stage 1: correct curvature, divide plate, quilt inner-preputial or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer	Preferred for most
Two-stage Byars	Dorsal preputial flaps moved ventrally, tubularised later	Supple flap bed
One-stage TPIF (transverse preputial island flap, Duckett tube) or onlay island flap	Flap urethroplasty	Higher complications
Proximal TIP (tubularised incised plate)	Plate kept	Stricture, recurrent curvature: abandoned for severe cases

- **Why stage?** A urethroplasty over a fresh corporal graft fails more often.
- Principles: magnification, tension-free closure, **vascularised barrier layer** (dartos, tunica vaginalis); **scrotoplasty** for transposition.

Complications

- **23–68%**; 25–75% need further surgery. **Fistula** commonest (3–45%), glans dehiscence, meatal stenosis, stricture, **diverticulum** (long rigid neourethra raises resistance: Poiseuille's law), recurrent curvature, hair-bearing urethra.

Diagram to draw: two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

Key points to remember:

- Perineoscrotal hypospadias with undescended testis → DSD work-up.
- Curvature >30°: plate division ± graft and staging.
- STAG/Bracka staged repair preferred; fistula commonest complication.

Rapid Revision Sheet – Section 4 · Paediatric Urology 1

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Wilms tumour: classification, staging and management (LAQ)

- WT1 at 11p13 (WAGR, Denys-Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: three drugs, surgery by week 12, NSS.

Retractile testis? (SAQ)

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

Retroileal Ureter (SAQ)

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Medullary sponge kidney (SAQ)

- Ectatic collecting ducts; sporadic; "bristles on a brush".
- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

Megameatus intact prepuce (SAQ)

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

Undescended testis (SAQ)

- Operate from 6 months, preferably by 12, latest 18 months.
- No imaging, no hormones; laparoscopy for non-palpable testis.
- Bilateral non-palpable: exclude CAH/DSD.

Techniques of chordee correction (SAQ)

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

Perineo-scrotal hypospadias (SAQ)

- Perineoscrotal hypospadias with undescended testis → DSD work-up.
- Curvature >30°: plate division ± graft and staging.
- STAG/Bracka staged repair preferred; fistula commonest complication.

Section 5 · Paediatric Urology 2 — Short answer questions (SAQ)

Valve bladder: management

Asked as:

- Management of valve bladder. — *Summer 2020 Paper II Q3e (SAQ)*
- Valve bladder. — *Practice Set 12 Paper II Q3e (SAQ)*

Type: SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q3e)

Definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV (posterior urethral valve) ablation, without residual obstruction, because of bladder dysfunction.

Pathophysiology (vicious cycle)

```
Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual
+ polyuria (nephrogenic diabetes insipidus) → bladder never empty
→ pressure transmitted to dilated upper tracts → they refill the bladder → renal damage
```

- Three processes: **polyuria, poor sensation, residual urine.**
- Age pattern: hyperreflexia in infancy → better compliance in childhood → large atonic bladder (myogenic failure) after puberty.

Clinical features

- Late continence (about 5.5 years), incontinence, UTI (urinary tract infection), worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**

Evaluation

- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

Management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones).

- Reflux is treated through the bladder; reimplantation is rarely needed. Manage CKD (chronic kidney disease).

Transplantation

- ESRD (end-stage renal disease) in 20–50%. Optimise the bladder first (videourodynamics, CIC or overnight drainage); graft survival is then similar.

Diagram to draw: vicious-cycle diagram of the valve bladder.

> **Key points to remember:**
> - Polyuria, poor sensation, residual urine.
> - Overnight drainage breaks the cycle.
> - Ureteric augmentation rather than bowel; optimise before transplant.

Embryogenesis of urinary bladder

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 2, Q3c (also answered in 2021-Summer Paper 3, Q1)

1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**.
- **Mesonephric (Wolffian) duct entry** is the landmark:
  - **Cranial vesicourethral canal** (continuous with the allantois) → **bladder and pelvic urethra**.
  - **Caudal (definitive) UGS** → phallic urethra (male); distal vagina and vestibule (female).

2. Layers of the wall

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, serosa:** surrounding splanchnic mesenchyme; epithelium and mesenchyme induce each other.

3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS.
- **Apoptosis** separates ureters from mesonephric ducts. Ureteric orifices move **cranially and laterally**; mesonephric ducts end caudally as ejaculatory ducts.
- **Classic view:** trigone muscle is ureteric. **Current view** (mouse lineage tracing): mostly bladder smooth muscle.
- **Weigert–Meyer rule:** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** → median umbilical ligament.

5. Function

- Urine production starts at **week 10**. Fetal bladder seen in about half at 10 weeks, almost all by 13 weeks; empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

Anomalies explained by embryology

Anomaly	Basis
Exstrophy	Cloacal membrane not reinforced by infraumbilical mesoderm
Urachal (patent urachus, sinus, cyst, diverticulum)	Failed obliteration at different levels
Agenesis, duplication	UGS or trigone failure; septation
Congenital (Hutch) diverticulum	Detrusor deficiency at the ureteric hiatus
VUR, ectopic ureter	Abnormal bud position

Diagram to draw: cloaca divided by the urorectal septum, labelled vesicourethral canal, UGS, allantois.

Key points to remember:

- Urorectal septum weeks 4–6; mesonephric duct entry is the landmark.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.

Anterior urethral valve

Type: SAQ (10 marks)

Source: Practice Set 17 Paper 2, Q3c

Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

AUV versus PUV

Feature	AUV	PUV
Frequency	25–30 times rarer	Commonest infant outlet obstruction
Site	Bulbar, penoscrotal, penile	From the verumontanum
Sign	Penile bulge	Weak stream, palpable bladder
Bladder dysfunction	Up to 80%	Near universal
Renal outcome	78% normal function	ESRD (end-stage renal disease) 20–50%

Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

Diagram to draw: VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

Key points to remember:

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function.

Rapid Revision Sheet – Section 5 · Paediatric Urology 2

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Valve bladder: management (SAQ)

- Polyuria, poor sensation, residual urine.
- Overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Embryogenesis of urinary bladder (SAQ)

- Urorectal septum weeks 4–6; mesonephric duct entry is the landmark.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.

Anterior urethral valve (SAQ)

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function.

Section 6 · Female Urology — Short answer questions (SAQ)

Urethral caruncle.

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3f

Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

Differential diagnosis

Lesion	Distinguishing feature
Urethral mucosal prolapse	**Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women
Urethral carcinoma	Firm, ulcerated, indurated, fixed; bleeding mass; biopsy
Urethral diverticulum	Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging)
Skene gland cyst or abscess	Lateral to meatus; distorts it; does not communicate
Others	Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus

- Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied.

Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions:
  1. Grasp the tip and apply traction to expose the base.
  2. Excise at the base.
  3. Reapproximate mucosa with absorbable sutures if the defect is large.
  4. Foley catheter for a few days.
- Complications: bleeding, recurrence, meatal stenosis.

Diagram to draw: meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT

Asked as:

- TOT. — *Summer 2020 Paper II Q3f (SAQ)*
- TOT versus TVT in SUI. — *Summer 2021 Paper II Q3e (SAQ)*
- TVT — *Practice Set 16 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)
(TOT answered in 2020-Summer Paper 2, Q3f.) (TVT versus TOT also answered in 2021-Summer Paper 2, Q3e)

Assumed meaning: TOT = transobturator tape, a mid-urethral sling for female stress urinary incontinence.

Introduction (definition and history)

- Both are polypropylene **MUS (mid-urethral slings)** for SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** retropubic polypropylene MUS; Ulmsten, 1995–1998.
- **TOT (transobturator tape):** a synthetic MUS of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) described it to avoid bladder, bowel and vessel injury.

Mechanism

- **Integral theory** (Petros and Ulmsten, 1990): continence needs pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus (pubourethral ligaments, suburethral hammock and pubococcygeus maintain continence). Hypermobility is a sign of their damage, not the cause.
- The loose tape at mid-urethra secures (supports) the **posterior urethral wall** during stress, which compresses against it on straining (compression, dynamic kinking); it is not tight at rest.
- TOT anchors in the obturator muscles.

Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 µm: allows macrophage entry and fibrous ingrowth, so it resists infection.

Indications and evaluation

- SUI or stress-predominant mixed incontinence after failed pelvic floor training; index SUI with urethral hypermobility.
- TVT preferred over TOT in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh, mesh risks and alternatives (colposuspension, autologous sling, bulking) (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

Technique differences

	TVT (retropubic)	TOT (transobturator)
Path	Vagina → retropubic space → suprapubic skin, hugging the pubis; cystoscopy essential	Labiocrural fold → obturator membrane → vagina (outside-in), or reverse (inside-out)
Structures at risk	Bladder, bowel, iliac and epigastric vessels	Obturator nerves and vessels, vaginal fornix

TVT technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

TOT technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.
2. Stab incision in the labiocrural fold, at the notch between inferior pubic ramus and adductor longus tendon.
3. Trocar perforates the obturator membrane, turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornix.
5. Tension-free: a clamp passes easily between tape and urethra.

- **Inside-out** (vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

Outcomes

- TVT: Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. Long-term cure varies 48–97% by definition.
- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not different); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane review (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.
- TOT poorer with **ISD** or a **fixed urethra**: retropubic sling preferred.

Complications (Cochrane, 2017)

Complication	TVT (retropubic)	TOT
Bladder perforation	**4.5%**	**0.6%**
Groin pain	1.3%	**6.4%**
Voiding dysfunction, blood loss (bleeding), vascular injury	More	Less
Mesh exposure	About 2%	About 2%

TVT complications

Complication	Notes
Bladder perforation	4.5% (TOT 0.6%); re-pass and continue

Complication	Notes
Urethral injury	Abandon mesh; staged repair
Bleeding/haematoma	2–3%; rarely bowel or vascular injury
Voiding dysfunction	Surgery needed 2.7%; incise tape in midline within 3 months
Mesh exposure	About 2%
Others	De novo urgency, suprapubic pain (groin pain less than TOT)

TOT complications

- Groin pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- Urethral injury during either: **do not place mesh** (AUA [American Urological Association] guideline).

Choosing

- **TVT preferred: ISD (intrinsic sphincter deficiency)** (ALPP [abdominal leak point pressure] <60 cm H_2O or MUCP [maximum urethral closure pressure] <20 cm H_2O), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** single-incision mini-slings (non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (colposuspension, autologous fascial sling, bulking agents).

Diagram to draw: coronal pelvis: U-shaped retropubic TVT at mid-urethra versus flat TOT, tape under mid-urethra through both obturator foramina.

Key points to remember:

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% vs 77.7%).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

VVF repair: early versus delayed timing, and surgery for recurrent VVF

Asked as:

- Early versus delayed VVF repair. — *Summer 2021 Paper II Q3b (SAQ)*
- Surgery For Recurrant Vesico Vaginal Fistulae — *Winter 2025 Paper II Q3f (SAQ)*

Type: SAQ (10 marks)
(timing of VVF repair: see 2021-Summer Paper 2, Q3b)

Introduction

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina communication causing continuous leakage.
- Causes: **obstetric** (over 90% in low-resource settings; obstructed labour) and **iatrogenic** (over 75% in high-resource settings, mainly hysterectomy); radiation; malignancy.
- Common mechanism: **ischaemic necrosis**, then a tract that epithelialises with time.
- Presents: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.
- Recurrent VVF: persists after a failed repair. **The first repair has the best chance**; redo tissue is scarred and poorly vascularised.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.

The debate: early versus delayed repair

	Early repair	Delayed repair
Definition (varies)	Days, <6 weeks or <3 months	Traditionally ≥3 months
Rationale	Clean iatrogenic tissue heals well; ends wetness and distress early	Inflammation settles, necrotic tissue sloughs, margins become clear
Evidence	**Comparable closure** in most series; Waaldijk: high first-attempt closure with immediate obstetric repair	One small series: more recurrence when repaired <6 weeks
Drawback	Friable tissue if inflamed	Prolonged distress, no better outcome

- Some authors **avoid 4–12 weeks** (active healing phase).
- **EAU (European Association of Urology) consensus:** no minimum delay if **no active infection and no recent irradiation**.

Recommended timing by cause

- **Clean post-hysterectomy:** early repair. **Obstetric with necrosis:** catheter trial, wait about 3 months.
- **Radiation fistula:** wait **≥12 months** after radiotherapy; biopsy for recurrence; interposition flap; diversion if the bladder is poor.
- **Recurrent (redo) fistula:** wait until inflammation settles (commonly about 3 months). Radiation: **≥12 months** after radiotherapy; failed radiation repairs were redone after **≥6 months**.

Conservative trial (during the wait)

- Catheter drainage cures about 15%: best if small (<1 cm, ideally <3 mm), started **within 3 weeks**, fresh and non-irradiated. Once epithelialised (≥6 weeks), unlikely. EAU: if tried, at least 12 weeks.
- **Also know:** fulguration of tiny tracts; fibrin glue.

Evaluation before redo

- **Biopsy the tract** if any history of cancer (fistula may signal recurrence).
- **Cystoscopy and vaginoscopy**: size, number, relation to ureters and bladder neck.
- **CT urography (computed tomography urography)** to exclude a **ureterovaginal fistula**; dye test.
- Urodynamics for capacity and compliance; treat UTI (urinary tract infection) / infection, optimise nutrition.

Choice of route

- Vaginal or abdominal (open, laparoscopic, robotic): about 90% first-attempt closure.

Vaginal	Abdominal (open, laparoscopic, robotic)
Distal or trigonal, accessible	**Recurrent ≥2 cm**, supratrigonal, narrow vagina
Failed **Latzko** can be cured by a second Latzko	Ureter involved, needs reimplantation
Flap-splitting, offset flaps	Augmentation or **Tanagho bladder flap** needed
Less pain, shorter stay	**O'Conor** bivalving finds the scarred plane

- Robotic repair of recurrences gives high closure.

Principles at repair (operative principles)

- Treat infection, optimise nutrition; ureteric stents if within 1 cm of a ureter.
- Excise scar; wide mobilisation; tension-free, multilayer, **non-overlapping** closure.
- **Interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent. **Interposition flap is widely used in recurrent fistula** (accepted though not proven superior; an RCT showed no benefit in simple VVF):

Flap	Route	Use
Martius (labial fat pad, pudendal vessels)	Vaginal	Mid–distal, bladder neck; not apex
Peritoneal	Vaginal or abdominal	Apical post-hysterectomy
Omentum (J-flap)	Abdominal	Radiation, recurrent, large
Gracilis, Singapore flaps	Vaginal/perineal	Refractory, large tissue loss

- Fill test 200–300 mL; catheter often ≥2 weeks (recurrent: **≥10–14 days**, then cystogram).

When repair is unwise

- Multiple failures, severe radiation damage, massive tissue loss: **conduit or continent diversion**; if unfit, ureteric occlusion with nephrostomies.
- **Also know:** rectus abdominis flap (last resort); more than one repair predicts residual incontinence.

Diagram to draw: timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation); Martius flap between bladder and vaginal closures.

Key points to remember:

- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months obstetric, ≥12 months radiation.
- Catheter trial works only if small, fresh and started within 3 weeks.
- First repair has the highest success.
- Biopsy; exclude ureteric fistula before redo.
- Interposition flap (Martius, peritoneum, omentum) in recurrent repair.
- Diversion after multiple failures or severe radiation.

ALPP

Type: SAQ (10 marks)

Source: 2024-Winter Paper 2, Q3a (also answered in 2020-Summer Paper 1, Q3e and 2019-Summer Paper 1, Q3b)

Definition

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**.
- It measures sphincter strength: **lower ALPP = weaker sphincter**. Seen only in SUI (stress urinary incontinence); there is **no normal value**.
- Provoked by Valsalva (**VLPP**, Valsalva leak point pressure) or cough (**CLPP**, cough leak point pressure). VLPP is lower than CLPP, so state the method.

Technique

1. Urethral and rectal catheters; standing; fill to **150 mL**.
2. Graded Valsalva or cough.
3. Record **total Pves (vesical pressure) at visible leak**, including baseline.
4. No leak: repeat every 50 mL to capacity; then remove the urethral catheter and read pressure from the rectal line.
5. Report volume, catheter size and provocation.

Interpretation (McGuire 1993)

ALPP (cmH_2O)	Inference
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal
≥90	**Urethral hypermobility**, little ISD

Clinical uses

- Separates ISD from hypermobility; low ALPP traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

Limitations

- Falls with larger catheters and volumes; invalid with DO (detrusor overactivity); prolapse masks leak.
- ISD and hypermobility coexist; ALPP alone predicts surgical success poorly.
- **VALUE trial (2012):** office evaluation was non-inferior to urodynamics before surgery for uncomplicated SUI.

ALPP versus DLPP

ALPP	DLPP (detrusor leak point pressure)
Sphincter against abdominal pressure	Bladder pressure against outlet resistance
Low = bad	**High = bad**: >40 cmH_2O → hydronephrosis or reflux in 85% (McGuire 1981)
SUI	Neurogenic, low-compliance bladder

- **Also know:** newer term "urodynamic stress test" (ICS, International Continence Society); EAU (European Association of Urology): do not use leak point pressure to grade severity.

Diagram to draw: Pves, Pabd and Pdet traces with leak on Valsalva, flat Pdet.

Key points to remember:

- Pves at leak with strain, no detrusor contraction; start at 150 mL.
- <60 ISD; 60–90 equivocal; ≥90 hypermobility.
- VLPP < CLPP; catheter size and volume alter it.

Vaginal prolapse repair.

Type: SAQ (10 marks)

Source: Practice Set 17 Paper 2, Q3d (related: Practice Set 12 Paper 1, Q3d)

Read as: surgical repair of vaginal wall and vault prolapse (pelvic organ prolapse).

Principles

- POP (pelvic organ prolapse) is a quality-of-life condition: operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q (Pelvic Organ Prolapse Quantification). **The apex is the cornerstone**: failing to support it greatly raises recurrence.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI (stress urinary incontinence)** with prolapse reduced; consider a concomitant sling. Finish with **cystoscopy** to confirm ureteric efflux.

Operations by compartment

Compartment	Operations	Key points
Anterior (cystocele)	**Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP (arcus tendineus fasciae pelvis))	Central vs lateral defect; does not support the apex
Apical (uterus, vault)	**Uterosacral suspension** (ureter injury 1–11%); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; hysteropexy; Manchester	OPTIMAL trial: uterosacral = sacrospinous at 2 years
Apical, abdominal	**Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to sacral promontory	Most durable (>90%); mesh exposure ~10% long term (CARE)
Posterior (rectocele)	**Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy	Levator plication causes dyspareunia; constipation may persist
Obliterative	**Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis	Elderly, not sexually active; success >90%; coital regret ≤13%

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- Transvaginal synthetic mesh for POP was withdrawn in the USA (FDA (Food and Drug Administration), 2019).

Complications

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia; recurrence (anterior after sacrospinous fixation); mesh exposure.

Diagram to draw: DeLancey levels of support with sacrospinous and sacrocolpopexy fixation points.

Key points to remember:

- Always restore apical support.
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.

Rapid Revision Sheet – Section 6 · Female Urology

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Urethral caruncle. (SAQ)

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.

Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT (SAQ)

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% vs 77.7%).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

VVF repair: early versus delayed timing, and surgery for recurrent VVF (SAQ)

- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months obstetric, ≥12 months radiation.
- Catheter trial works only if small, fresh and started within 3 weeks.
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ALPP (SAQ)

- Pves at leak with strain, no detrusor contraction; start at 150 mL.
- <60 ISD; 60–90 equivocal; ≥90 hypermobility.
- VLPP < CLPP; catheter size and volume alter it.

Vaginal prolapse repair. (SAQ)

- Always restore apical support.
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.

Section 7 · Voiding Dysfunction — Long answer questions (LAQ)

Pathophysiology, classification and management of LUTS.

Type: LAQ (25 marks)

Source: 2024-Winter Paper 2, Q1 (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

Classification

1. By phase (ICS, International Continence Society)

Storage	Voiding	Post-micturition
Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence	Weak or split stream, hesitancy, intermittency, straining, terminal dribble	Feeling of incomplete emptying, post-micturition dribble

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

2. Functional (Wein)

	Because of the bladder	Because of the outlet
Failure to store	DO (detrusor overactivity), low compliance, hypersensitivity	Sphincter weakness (SUI, stress urinary incontinence)
Failure to empty	DUA (detrusor underactivity)	BPO, stricture, bladder neck contracture, sphincter dyssynergia

3. Severity: IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

4. By cause: prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT (dihydrotestosterone)**; androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.
- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.

- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.
- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

Management: EAU algorithm

```
Bothersome LUTS?  No  -> watchful waiting, lifestyle advice
Absolute surgical indication? Yes -> surgery
Otherwise:
  Nocturnal polyuria      -> lifestyle +/- desmopressin
  Storage predominant     -> antimuscarinic or beta3-agonist
  Prostate >40 mL         -> 5ARI (5-alpha-reductase inhibitor)
                             +/- alpha1-blocker
  Others                  -> alpha1-blocker or PDE5 inhibitor
  Residual storage        -> add antimuscarinic or beta3-agonist
Failure or complication   -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

Drug	Example	Mechanism and notes
α1-blocker	Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg	Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome)
5ARI	Finasteride 5 mg, dutasteride 0.5 mg	Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery
Antimuscarinic	Solifenacin, tolterodine	Block detrusor M3 receptors; for storage symptoms if PVR <200 mL
β3-agonist	Mirabegron 50 mg, vibegron	Relax detrusor during storage
PDE5I (phosphodiesterase-5 inhibitor)	Tadalafil 5 mg daily	Smooth-muscle relaxation; useful with ED (erectile dysfunction)

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

Prostate volume	Options
<30 mL	TUIP, TURP (transurethral resection of the prostate)
30–80 mL	**TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation
>80 mL	**HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.
- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

Diagram to draw: BPH → BPE → BPO → BOO → bladder changes → LUTS.

Key points to remember:

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

Type: LAQ (25 marks)

Source: Practice Set 1 Paper 2, Q2

Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.

Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

Phase	Detrusor	Outlet	Clinical
Spinal shock (usually 6–12 weeks; up to 1–2 years)	Areflexic, smooth bladder	Bladder neck closed; reduced closure pressure	**Retention**; overflow only
Recovery (suprasacral pattern)	**NDO**, no sensation below lesion	**Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia**	Reflex leakage, incomplete emptying, high pressure

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm H_2O) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or seizures.
- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with

terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm H_2O** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

Management

Goals: protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

Aim	Options
Emptying	**CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended
Storage (lower pressure)	**Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off)
Refractory small bladder	**Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff)
Men unable to do CIC	**External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm H_2O afterwards = failure. Urethral stents abandoned
Neuromodulation	**SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand
Last resort	Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

```
Cervical SCI → spinal shock (retention): CIC
   → VUDS at 3 months: NDO + DSD
   → antimuscarinic + CIC → Botox → augmentation
   → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy
```

Diagram to draw: neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

Key points to remember:

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm H_2O threatens the kidneys.

Section 7 · Voiding Dysfunction — Short answer questions (SAQ)

Botulinum neurotoxin and its role in urology

Asked as:

- Botulinim neurotoxin. — *Summer 2016 Paper II Q3d (SAQ)*
- Botulinum Toxin and its role in Urology — *Winter 2025 Paper II Q3d (SAQ)*

Type: SAQ (10 marks)

(also answered in 2017-Winter Paper 3, Q3b)

Assumed meaning: botulinum neurotoxin (spelling error in the question).

Introduction

- Toxin of *Clostridium botulinum* (neurotoxin); serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**; type B is short-acting.
- 150 kDa: heavy chain (100 kDa) plus zinc-dependent light chain (50 kDa).

Mechanism

1. Heavy chain binds the cholinergic nerve terminal; the toxin is internalised.
2. Light chain enters the cytosol.
3. It **cleaves SNAP-25 (synaptosomal-associated protein 25)**, a SNARE (vesicle-docking) protein, so vesicles cannot fuse and **acetylcholine release stops**.

- **Afferent action**: less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) receptors on C fibres, so urgency falls too.
- Effect wears off (wanes) as nerve terminals sprout (about 6–9 months).

Urological uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK); incontinence down about 59%, about 25% dry. **ABC** trial: equal to anticholinergic (antimuscarinic), more complete dryness. **ROSETTA**: 200 U equal (comparable) to SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY trials): less incontinence, lower detrusor pressure, larger capacity
Children with NDO (**paediatric NDO**, spina bifida)	About **10 U/kg (maximum 360 U)** in Campbell series; FDA (Food and Drug Administration) label (age ≥5) 6 U/kg, maximum 200 U
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection in men unable to do CISC (clean intermittent self-catheterisation)
Others: **IC/BPS (interstitial cystitis/bladder pain syndrome)**	100 U, often trigonal; AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	Intraprostatic (prostate injection for benign enlargement): failed versus sham; placebo-controlled RCTs (randomised controlled trials) showed **no benefit**; not recommended

- **Also know:** pelvic floor spasm.

Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor (cystoscopic detrusor injection) at **about 20 sites (100 U) or 30 sites (200 U)**, sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

Side effects (complications)

- **Retention needing CISC**: 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; higher (more) in NDO. Each 10 mL rise in baseline PVR raises risk (about) 27%. **Patients must accept CISC beforehand** (patient must accept CISC beforehand).
- **UTI** about 20%; haematuria; rarely generalised weakness.
- Contraindications: active UTI, myasthenia gravis, pregnancy, unwillingness to self-catheterise.
- Repeat injections when the effect fades; response is maintained.

Diagram to draw: cholinergic synapse showing SNAP-25 cleavage by the light chain.

Key points to remember:

- Cleaves SNAP-25; acts on efferent and afferent nerves (efferent and afferent action).
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

Uro Dynamic Studies.

Type: SAQ (10 marks)

Source: 2017-Summer Paper 2, Q3d

Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first**, **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

Components

Test	Method	Main values
Uroflowmetry	Voided volume ≥150 mL	Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture
PVR	Bladder scan or catheter	Emptying
Filling cystometry	Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis	Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cmH_2O impaired)
Leak point pressures	ALPP (abdominal LPP): cough or Valsalva at 150 mL	Low ALPP = weak sphincter (ISD (intrinsic sphincter deficiency))
	DLPP (detrusor LPP)	**>40 cmH_2O** risks upper tracts (McGuire)
Pressure–flow study	Void with lines in	High pressure–low flow = obstruction; low–low = underactivity
EMG (electromyography)	Perineal surface electrodes	DESD (detrusor–external sphincter dyssynergia)
UPP (urethral pressure profile)	Catheter withdrawn	Limited value
Videourodynamics	Above with contrast and fluoroscopy	VUR (vesicoureteric reflux), bladder neck, site of obstruction

Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

Indications

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP (transurethral resection of the prostate)); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS (multiple sclerosis), spina bifida), radiation cystitis, before augmentation.

- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

Urodynamic risk factors

- Impaired compliance, DESD, DISD (detrusor–internal sphincter dyssynergia), high-pressure DO throughout filling.

Also know: ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

Diagram to draw: normal trace: Pves, Pabd, Pdet, flow, EMG.

> **Key points to remember:**
> - Pdet = Pves − Pabd.
> - BOOI >40 obstructed; BCI <100 weak.
> - DLPP >40 cmH_2O: upper-tract risk.
> - Ask a question first; reproduce the symptom.

Uroflowmetry.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 2, Q3f

Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical.
- Measure **PVR** (postvoid residual) by ultrasound afterwards.

Parameters

Parameter	Meaning
Qmax (maximum flow rate)	Most useful value; normal men >15 mL/s, women >20 mL/s
Qave (average flow rate)	Volume ÷ flow time; less useful
Voided volume (VV)	Must be ≥150 mL
Flow time / voiding time	Voiding time includes interruptions
Time to Qmax	Normally in the first third of the void
Voiding efficiency	VV ÷ (VV + PVR) × 100

Curve patterns

Pattern	Suggests
Bell-shaped	Normal
Plateau (flattened, box)	Fixed obstruction: **urethral stricture**
Prolonged, low, skewed	BPO (benign prostatic obstruction) or detrusor underactivity
Intermittent / straining	Detrusor underactivity, abdominal straining or obstruction
Staccato	Dysfunctional voiding (sphincter overactivity)
Tower (high, short)	Overactive bladder

Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.
- **Nomograms** correct for volume: **Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.

Limitations

- **Cannot separate BOO (bladder outlet obstruction) from detrusor underactivity**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

Uses

- BPH (benign prostatic hyperplasia) evaluation before treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy.
- Children: dysfunctional voiding, after posterior urethral valve ablation.
- Women: voiding dysfunction before stress incontinence surgery.
- **EAU** (European Association of Urology): before surgery, pressure–flow study if Qmax >10 mL/s.

Diagram to draw: flow curves: normal bell, plateau (stricture), prolonged low (BPO), intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = stricture; intermittent = underactivity or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.

Management of detrusor underactivity

Type: SAQ (10 marks)

Source: 2019-Summer Paper 2, Q3d

Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span." **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** symptoms of slow stream, hesitancy and straining, with or without incomplete emptying. Found in 40–48% of men over 70 having urodynamics.

Causes

- **Neurogenic:** cauda equina, disc prolapse, spinal injury, diabetes, multiple system atrophy, early stroke, radical pelvic surgery.
- **Myogenic:** long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing.
- **Drugs:** anticholinergics, opioids. **Functional:** Fowler syndrome, dysfunctional voiding.

Evaluation

- Bladder diary, flow rate, PVR (post-void residual), sacral neurological examination, spinal MRI if indicated, drug review.
- **Pressure-flow study** is the only accepted test. **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Low Pdet with low Qmax.
- Ambulatory urodynamics if no contraction on standard study (the "bashful bladder").

Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms.

Option	Details
Conservative	Timed and double voiding; treat constipation; stop causative drugs
CISC (clean intermittent self-catheterisation)	**Mainstay** for high PVR; fewer infections than indwelling catheter. Suprapubic catheter if CISC impossible
α-blockers	Lower outlet resistance; modest benefit
Parasympathomimetics (bethanechol, distigmine)	**Weak evidence**, systemic side effects (bronchospasm, rarely cardiac arrest); not routine
SNM (sacral neuromodulation)	For non-obstructive retention, especially Fowler syndrome; needs some detrusor function
Sacral anterior root stimulator (Brindley)	Complete spinal cord injury, with posterior rhizotomy
Intrasphincteric botulinum toxin	Experimental
Outlet surgery (TURP, transurethral resection of the prostate)	Little help in low-PVR men; may free catheter-dependent fit men
Diversion	Mitrofanoff when urethral CISC fails; rarely ileal conduit

- Avoid Credé or Valsalva voiding (reflux risk).

Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

Diagram to draw: pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug.
> - SNM helps non-obstructive retention; intact sensation predicts response.

Neuromodulation

Type: SAQ (10 marks)

Source: 2023-Winter Paper 2, Q3b (also answered in 2019-Summer Paper 4, Q1)

Taken as neuromodulation for lower urinary tract dysfunction.

Definition and mechanism

- Electrical stimulation of a peripheral nerve that changes central reflex pathways controlling the bladder. It acts on **afferents**, not directly on muscle.
- **OAB (overactive bladder):** afferent input damps PMC (pontine micturition centre) firing, so fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling and relaxes the external sphincter. Hence one therapy treats opposite disorders.
- Exclude anatomical causes first (obstruction, stones, cancer).

Modalities

Modality	Method	Main use
SNM (sacral neuromodulation)	Tined lead in **S3** foramen + implanted generator	Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence
PTNS (percutaneous tibial nerve stimulation)	34-gauge needle above medial malleolus	Refractory OAB
Implantable tibial devices	Small device over tibial nerve	OAB without clinic visits
Pudendal stimulation	Lead at Alcock's canal	SNM failure (off-label)
SARS (sacral anterior root stimulator)	Anterior S2–S4 roots + dorsal rhizotomy	Complete spinal cord injury

Sacral neuromodulation

- **Third-line** for OAB (AUA, American Urological Association; EAU, European Association of Urology).
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or staged tined lead (up to 4 weeks). **Implant if ≥50% improvement** in diary symptoms.
- **S3 response:** perineal "bellows" plus great-toe flexion.
- **Evidence:** InSite: success 61% vs 42% with medical therapy. ROSETTA: onabotulinumtoxinA 200 U slightly better for UUI (urgency urinary incontinence) but more UTI (urinary tract infection) and self-catheterisation.
- **Problems:** pain, lead migration, loss of effect; revision >30% at 5 years; cost. Newer rechargeable, MRI-conditional devices.

Tibial nerve stimulation

- Tibial nerve carries **S2–S3** fibres. Needle 3–4 cm above medial malleolus; toe flexion confirms placement; **30 min weekly × 12**, then maintenance.

- **SUmiT:** response 55% vs 21% with sham.
- **Also know:** IVES (intravesical electrical stimulation) in children; dorsal genital nerve stimulation.

Diagram to draw: S3 lead with buttock generator, and PTNS needle above the medial malleolus.

> **Key points to remember:**
> - Afferent modulation explains benefit in both OAB and retention.
> - S3: bellows + great-toe flexion; implant if ≥50% improvement.
> - PTNS: 12 weekly sessions; SUmiT 55% vs 21%.

Bladder dysfunction non genitourinary systemic disease

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 2, Q3a

Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.

Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
  - Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
  - Spinal cord: DO **with DSD**.
  - Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.
- **Urodynamics** before treatment, because symptoms do not predict findings.

Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

Other systemic diseases

Group	Disease	Usual finding
Neurological	Stroke	Acute retention, then DO; no true DSD
	Parkinson disease	DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy)
	Multiple sclerosis	DO about 38%, DO with DSD 29%, areflexia 26%; changes over time
Infective	HIV/AIDS	DO, DSD; areflexia with CMV (cytomegalovirus)
	Herpes zoster/simplex, Guillain–Barré	Retention, areflexia; reversible
	Spinal tuberculosis, Lyme, HTLV-1	DO with DSD or areflexia
Autoimmune and connective tissue	SLE (systemic lupus erythematosus)	DO; lupus cystitis
	Systemic sclerosis	Areflexia, low compliance
Infiltrative	Amyloidosis (transthyretin)	Low sensation, underactivity, SUI (stress urinary incontinence)
Nutritional	B12 deficiency, alcohol (thiamine)	Mixed; reversible with replacement
Endocrine	Hyperthyroidism	Poor emptying; resolves with treatment

Also know: myasthenia gravis; schizophrenia; gastroparesis.

Management principles

- Treat the disease; reversible therapy.
- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

Diagram to draw: neuraxis with lesion levels and the bladder pattern at each.

> **Key points to remember:**
> - Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
> - Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
> - Urodynamics before treating; reversible treatment first.

Overactive bladder and its drug therapy

Asked as:

- Overactive bladder — *Practice Set 15 Paper II Q3b (SAQ)*
- Drug therapy in OAB — *Practice Set 17 Paper II Q3a (SAQ)*

Type: SAQ (10 marks)
(OAB also answered in Practice Set 4 Paper 1, Q2; detrusor relaxants (drug therapy) also answered in Practice Set 1 Paper 1, Q3e)

Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** vs **wet** (with UUI). OAB is a symptom diagnosis; DO (detrusor overactivity) is urodynamic, found in only 44–69% with urgency.

Pathophysiology

- **Afferent:** silent C fibres become active and hypersensitive.
- **Urothelial:** urothelium releases ATP (adenosine triphosphate) and acetylcholine onto suburothelial afferents.
- **Myogenic:** patchy denervation → spontaneous micromotions that spread.
- **Central:** loss of forebrain inhibition of the pontine micturition centre.
- **Associations:** ageing, BOO (bladder outlet obstruction), metabolic syndrome.

Evaluation

- **Minimum:** history, examination, **urinalysis**. Bladder diary (3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases.

Management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

Line	Treatment
Behavioural	Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks
Drugs	**Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine); **β3-agonists** (mirabegron, vibegron); combinations
Third line	**PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor (about 25% dry; UTI about 20%; retention); **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years)
Last resort	Augmentation cystoplasty or urinary diversion

- **Trials:** ROSETTA (Botox beat SNM at 6 months, more UTI); SUmiT (PTNS 55% vs sham 21%).

Drug therapy: place and mechanisms

- In OAB (overactive bladder), drugs are second line, after behavioural therapy. Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.

- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent signalling.
- **β3 agonists:** activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage.

Agents

Drug	Feature	Usual dose
Oxybutynin	Non-selective; most dry mouth; patch, gel	5 mg two to three times daily; ER (extended-release)
Tolterodine	Bladder-selective; QT prolongation at high dose	ER 4 mg daily
Fesoterodine	Prodrug of tolterodine's active metabolite	4–8 mg daily
Solifenacin	M3-selective; long half-life	5–10 mg daily
Darifenacin	About 30-fold M3-selective	7.5–15 mg daily
Trospium	Quaternary; little brain entry	20 mg twice daily
Mirabegron (β3)	Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise	25–50 mg daily
Vibegron (β3)	No CYP2D6 interaction	75 mg daily

Comparison

Feature	Antimuscarinics	β3 agonists
Efficacy	Similar	Similar
Side effects	Dry mouth, constipation, blurred vision	Near placebo
Cognition	**Dementia risk 10–65% higher with use >3 months** (antimuscarinic use >3 months raises dementia risk)	No anticholinergic burden
Contraindications	Retention, uncontrolled narrow-angle glaucoma, gastric retention	Uncontrolled severe hypertension

- **Choice:** elderly → prefer β3 agonist (β3-agonists in the elderly), or ER trospium, darifenacin or fesoterodine. ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.

Combination and special groups

- **Solifenacin + mirabegron** beats monotherapy (SYNERGY; BESIDE in non-responders).
- **Men:** α-blocker first; add an antimuscarinic (if PVR is low) or β3 agonist to an α-blocker; retention up to 3%, so check PVR (post-void residual).
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; third line onabotulinumtoxinA 100 U and neuromodulation.

Diagram to draw: bladder showing urothelial afferents and C fibres with brainstem and cortical inhibition; detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.

Key points to remember:

- OAB is defined by urgency; DO is urodynamic.
- Behavioural therapy first; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk.
- Long-term antimuscarinics → dementia risk.
- Solifenacin + mirabegron combination works (SYNERGY).
- Botox 100 U, PTNS, SNM for refractory OAB.

Vesicourodynamics

Type: SAQ (10 marks)

Source: Practice Set 16 Paper 2, Q3a (also answered in 2016-Summer Paper 4, Q3d)

Assumed meaning: videourodynamics (VUDS), i.e. urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.

Definition

- **VUDS:** multichannel urodynamics (pressure–flow ± EMG [electromyography]) recorded with fluoroscopy, using **contrast as the filling fluid**.

Set-up

- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm or tilting table.
- SUI (stress urinary incontinence) views standing; void in the usual position. **Screen only at key moments** (provocation, pressure rise, voiding).

What fluoroscopy adds

Situation	VUDS finding
NLUTD (neurogenic lower urinary tract dysfunction)	**Gold standard**; DESD (detrusor–external sphincter dyssynergia) and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula
VUR (vesicoureteric reflux)	Pressure and volume at onset, grade; reflux "pops off" pressure, so compliance looks falsely safe
Bladder neck	Only way to assess it; **PBNO** (primary bladder neck obstruction)
Female obstruction	**Nitti criteria:** radiographic narrowing between bladder neck and meatus with a sustained detrusor contraction
Dysfunctional voiding	"Spinning-top" urethra at the external sphincter
SUI	Leakage with ALPP (abdominal leak point pressure); open bladder neck at rest suggests ISD (intrinsic sphincter deficiency)
Male obstruction	Localises prostatic obstruction, bladder neck contracture or stricture

PBNO in young men: type I high pressure, low flow; type II normal pressure, low flow, narrow bladder neck; type III delayed bladder-neck opening.

Indications

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS (lower urinary tract symptoms); complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant; children with PUV (posterior urethral valves) or dysraphism.
- Shows risk factors: poor compliance, DLPP (detrusor leak point pressure) >40 cm H_2O, DESD.

Limitations

- Radiation, cost, space, contrast allergy. EAU (European Association of Urology): optimum test in NLUTD; AUA (American Urological Association): lowest dose that answers the question.

Diagram to draw: VUDS trace (Pves, Pabd, Pdet, flow, EMG) beside a voiding cystourethrogram image.

Key points to remember:

- VUDS = urodynamics + fluoroscopy with contrast.
- Only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

Rapid Revision Sheet – Section 7 · Voiding Dysfunction

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Pathophysiology, classification and management of LUTS. (LAQ)

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm H_2O threatens the kidneys.

Botulinum neurotoxin and its role in urology (SAQ)

- Cleaves SNAP-25; acts on efferent and afferent nerves (efferent and afferent action).
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

Uro Dynamic Studies. (SAQ)

- Pdet = Pves − Pabd.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 cmH_2O: upper-tract risk.
- Ask a question first; reproduce the symptom.

Uroflowmetry. (SAQ)

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = stricture; intermittent = underactivity or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.

Management of detrusor underactivity (SAQ)

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; no effective drug.
- SNM helps non-obstructive retention; intact sensation predicts response.

Neuromodulation (SAQ)

- Afferent modulation explains benefit in both OAB and retention.
- S3: bellows + great-toe flexion; implant if ≥50% improvement.
- PTNS: 12 weekly sessions; SUmiT 55% vs 21%.

Bladder dysfunction non genitourinary systemic disease (SAQ)

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

Overactive bladder and its drug therapy (SAQ)

- OAB is defined by urgency; DO is urodynamic.

- Behavioural therapy first; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk.
- Long-term antimuscarinics → dementia risk.
- Solifenacin + mirabegron combination works (SYNERGY).
- Botox 100 U, PTNS, SNM for refractory OAB.

Vesicourodynamics (SAQ)

- VUDS = urodynamics + fluoroscopy with contrast.
- Only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

Section 8 · Nocturia & Male Incontinence — Short answer questions (SAQ)

Post-prostatectomy incontinence

Asked as:

- Post prostatectomy incontinence. — *Summer 2021 Paper II Q3a (SAQ)*
- Prostatectomy incontinence — *Practice Set 11 Paper II Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 11 Paper II Q3f: also answered in 2021-Summer Paper 2, Q3a)

Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad") and reporting; expert no-pad continence is in the mid-90s % at 1 year; continence improves for up to 2 years.

Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends (entirely) on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches (sphincter or nerve injury at the apex), short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g (another source: mild <150 g, moderate <400 g, severe more; thresholds vary).
- **Standing cough test**, grade (grades) 0 to 4 (0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, creatinine, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges (shows) residual sphincter coaptation; urodynamics when needed (if mixed symptoms).

Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative (started before surgery); pads, condom catheter or (penile) clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

Situation	Choice
Mild–moderate, no radiotherapy	**Male sling**: AdVance / AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system])

Situation	Choice
Severe, or any radiotherapy	**AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm H_2O
ProACT balloons	Expert centres; avoid after radiotherapy

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts (lifespan) 8–10 years. Erosion risks: radiotherapy (risk higher after radiotherapy), prior erosion, urethroplasty (use a **transcorporal cuff** for fragile urethra).
- **Also know:** climacturia; treat VUAS first.

Diagram to draw: AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

> **Key points to remember:**
> - Wait 6–12 months with PFMT before surgery.
> - Sling for mild–moderate; AUS for severe or irradiated.
> - MASTER: sling non-inferior to AUS at 12 months.

Rapid Revision Sheet – Section 8 · Nocturia & Male Incontinence

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Post-prostatectomy incontinence (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months.

Section 9 · Andrology & Sexual Medicine — Long answer questions (LAQ)

Surgical management of erectile dysfunction and penile prosthesis

Asked as:

- Discuss surgical management of erectile dysfunction. — *Summer 2020 Paper II Q2 (LAQ)*
- Penile Prosthesis. — *Summer 2018 Paper II Q3d (SAQ)*
- Describe various penile prosthesis in ED — *Practice Set 15 Paper II Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3c and 2018-Summer Paper 2, Q3d; surgery for ED in 2020-Summer Paper 2, Q2)

Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Prosthesis: most effective, but last-line, treatment of ED; **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- **Indications:** failed or unacceptable medical therapy (ED not responding to, contraindicated for (contraindication to), or unacceptable with (refusal of) oral PDE5Is, injections or vacuum devices; common after prostatectomy, pelvic radiotherapy, diabetes); cavernosal (corporal) fibrosis (after ischaemic priapism, infection or implant removal; early implant preserves length); Peyronie disease (Peyronie's disease) with ED; penile or pelvic trauma.
- Detailed vascular testing is needed only before vascular surgery.

A. Penile vascular surgery

	Arterial revascularisation	Venous surgery
Defect	Arteriogenic (poor inflow)	Veno-occlusive (venous leak)
Work-up	Duplex; selective pudendal angiography (also checks the inferior epigastric donor)	DICC (dynamic infusion cavernosometry and cavernosography) locates the leak
Technique	Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation)	Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication
Selection	Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma	Not defined
Status	Strictly selected men	Investigational
Complications	Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia	Glans numbness, shortening, curvature, skin necrosis

B. Penile prosthesis

Types

Type	Devices / Example	Features	Best for
Malleable (semirigid) rods	Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel)	Two bendable rods; simple, cheap; almost never fail (mechanical failure exceedingly rare); always (permanently) firm; erode more in insensate men	Poor manual dexterity (suited to poor dexterity); cost
Mechanical rods	Spectra (discontinued)	Articulating segments	–
2-piece (two-piece) inflatable	Ambicor (discontinued)	Cylinders + scrotal pump; no reservoir; poor flaccidity	–
3-piece (three-piece) inflatable	AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10	Two cylinders, scrotal pump, abdominal reservoir; **best rigidity with (and) true flaccidity**; most popular	Most men

Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery; lockout valve prevents autoinflation.

Contraindications: active skin, urinary (UTI [urinary tract infection]) or systemic infection; uncontrolled diabetes; unrealistic expectations; reversible ED. In insensate men (spinal cord injury) use an inflatable device, because rods erode.

Consent: the erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

Preoperative preparation

- Treat phimosis and skin or fungal lesions (skin and urine infection); sterile urine; stop anticoagulants and aspirin.
- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

Operative steps (3-piece, penoscrotal approach)

```
Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures → corporotomy
→ dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13 mm)
→ goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation
→ close corporotomies → pump in dependent subdartos (dartos) pouch → reservoir in space of Retzius
→ connect tubing → test inflation → irrigate → close in layers
```

- Dilate with Hegar or Brooks dilators, instruments pointing dorsolaterally; check for crossover or perforation; cylinders sized to corporal length (equal to or 1 cm shorter than measured length).
- **Approaches:** penoscrotal (most used, commonest; secures the pump / pump secured; urethra at risk); infrapubic (reservoir placed under vision (direct reservoir view); dorsal nerve at risk); subcoronal or ventral penile for rods.
- Malleable rods (rods are placed slightly shorter): 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Ectopic submuscular reservoir** after pelvic surgery or cystectomy; a lockout valve prevents autoinflation.
- **Surgical principles:** antibiotics, glucose control, no-touch technique.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation / urethral injury (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling; modelling for Peyronie's disease); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

Complications

- **Mechanical failure** (commonest; inflatables): repair in about 15% at 5 years, 30% at 10 years.
- **Infection (1–3%)**: biofilm on the device. Prevent with intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h), chlorhexidine–alcohol preparation, **antibiotic-coated devices** (halve infection; antibiotic-coated devices halve it) and **no-touch technique** (gives <0.5%). Treatment (treat by removing): remove **all** parts (components). **Mulcahy salvage** (antiseptic washout and immediate new device / replacement) succeeds in 84%; delayed reimplantation loses length.
- Others: erosion or extrusion (catheterisation, rods in insensate men), reservoir erosion into bladder or bowel, autoinflation, penile shortening, **glans necrosis** (diabetes, smoking, radiation, previous implant), cylinder aneurysm.

Special situations (also know): ischaemic priapism >36 hours (early implantation preserves length); transplant recipients (no extra infection); after phalloplasty (single cylinder, high erosion).

Outcome (outcomes): patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

Diagram to draw: 3-piece (three-piece) inflatable prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.

Key points to remember:

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece (three-piece) inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Discuss the role of urologist in reproductive endocrinology and male infertility practice.

Type: LAQ (25 marks)

Source: 2022-Summer Paper 2, Q1

Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.
- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

Test	What it tells
LH (luteinising hormone)	Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus
FSH (follicle-stimulating hormone) / inhibin B	Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction
Oestradiol (E2)	**T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity)
Prolactin	Only with low T, low libido/ED (erectile dysfunction), headache or field defect
TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone	Only if signs; CAH (congenital adrenal hyperplasia)

3. Medical (endocrine) treatment

Condition	Mechanism	Treatment
Hypogonadotropic hypogonadism (Kallmann, pituitary)	Testis normal, signal missing: most treatable cause	**hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years
Anabolic steroids / exogenous T	Negative feedback suppresses LH → intratesticular T falls → azoospermia	Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine)
Hyperprolactinaemia	Prolactin inhibits GnRH	Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine)
Partial hypogonadism, obesity	Aromatisation of T to E2 suppresses LH	Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10
CAH	Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours	Glucocorticoids
Primary testicular failure (Klinefelter 47,XXY)	Germ cells absent or few	No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction)

4. Surgical role

Problem	Operation
Clinical varicocele + abnormal semen	**Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele
Vasal or epididymal obstruction	**Vasovasostomy / vasoepididymostomy** (microsurgical intussusception)
EDO	**TURED** (transurethral resection of ejaculatory ducts)
Non-obstructive azoospermia	**Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found)
Unreconstructable obstruction	PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration)
Retrograde ejaculation; anejaculation	Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation

5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

Recent advances: sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

Also know: antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

Diagram to draw: hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

Key points to remember:

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Section 9 · Andrology & Sexual Medicine — Short answer questions (SAQ)

Management of priapism

Type: SAQ (10 marks)

Source: 2015-Summer Paper 2, Q3a

Definition and classification

- Full or partial erection **≥4 hours** beyond, or unrelated to, sexual stimulation (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America)).
- **Ischaemic** (95%; veno-occlusive, rigid, painful, **emergency**); **non-ischaemic** (arterial fistula after trauma, partly rigid, painless); **stuttering** (recurrent ischaemic episodes, typically SCD (sickle cell disease)).
- **Causes**: ICI (intracavernosal injection), SCD (29–42% of males), α-blockers, trazodone, cocaine, leukaemia, metastases.

Diagnosis

- **Corporal blood gas**:

	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	**<30**	**>60**	**<7.25**
Mixed venous (flaccid)	40	50	7.35
Non-ischaemic (arterial)	>90	<40	7.40

- Colour Doppler if equivocal (absent flow or fistula); blood count, haemoglobin electrophoresis.

Ischaemic priapism

```
Penile block → aspiration ± irrigation
→ Phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h
→ Distal shunt ± tunnelling → proximal shunt
→ >36 h or refractory → immediate prosthesis
```

- Aspiration alone succeeds in under a quarter. Monitor blood pressure and pulse.

Shunt	Method
Winter (distal)	Biopsy needle through glans
Ebbehoj / T-shunt (distal)	Scalpel through glans; T-shunt turns blade 90°
Al-Ghorab (distal, open)	Excise tunica tips via dorsal glans incision
Snake tunnelling	Dilator passed proximally through distal shunt
Quackels (proximal)	Cavernosum to spongiosum at the bulb
Grayhack (proximal)	Cavernosum to saphenous vein
Penoscrotal decompression	Proximal corporotomy, clot evacuation

- **Duration predicts ED (erectile dysfunction)**: 0% ≤12 h, 22% at 12–24 h, 56% at 24–36 h, 100% >36 h.
- **Early prosthesis** gives less shortening than delayed insertion.

- Oral agents and hydration are adjuncts only; in SCD, treat intracavernosally with concurrent haematology care.

Non-ischaemic priapism

- Observe (about 60% resolve); reassess at 4 weeks. Then **selective embolisation**; repeat it, rather than surgery, for recurrence.

Stuttering priapism

- Prophylaxis: daily PDE5i (phosphodiesterase-5 inhibitor, which resets PDE5 regulation), pseudoephedrine, self-injected phenylephrine; hormonal agents (GnRH (gonadotropin-releasing hormone) agonists, antiandrogens) at the cost of libido.

Diagram to draw: Distal and proximal shunt sites.

> **Key points to remember:**
> - Ischaemic gas: pO2 <30, pCO2 >60, pH <7.25.
> - Distal shunt ± tunnelling before proximal; prosthesis if >36 h.
> - High-flow: observe (~60% resolve), then embolise.

Azoospermia and non-obstructive azoospermia

Asked as:

- Azospermia — *Summer 2017 Paper II Q3e (SAQ)*
- Non obstructive azoospermia — *Winter 2025 Paper II Q3a (SAQ)*

Type: SAQ (10 marks)
(also answered in 2019-Summer Paper 1, Q3d)

Assumed meaning: azoospermia.

Definition

- **No sperm in the ejaculate** after **centrifugation** of the sample, on **at least two** semen analyses; 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** no sperm in the **centrifuged** ejaculate on two samples, due to **failure of spermatogenesis**. About 60% of azoospermic men.

Classification

Type	Cause	Clues
Pre-testicular (hypogonadotropic)	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone), testosterone
Testicular (NOA (non-obstructive azoospermia), spermatogenic failure)	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic	Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm
Post-testicular (OA (obstructive azoospermia))	CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes, **FSH ≤7.6**, dilated epididymis
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, commonest; **Y microdeletion** (AZF (azoospermia factor) a, b, c).
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic**: commonest overall.
- *Note:* hypogonadotropic hypogonadism is pre-testicular and treated medically (hCG (human chorionic gonadotropin) ± FSH (follicle-stimulating hormone)).

Evaluation

- History (mumps, surgery, drugs, testosterone); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones**: FSH (reflects germ-cell mass), LH, testosterone, prolactin; oestradiol.
- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.

```
Azoospermia → vasa absent? → CBAVD → CFTR testing (both partners), renal US
            → volume <1 mL? → sperm in post-ejaculate urine → retrograde
                                → none → TRUS for EDO
            → normal volume: FSH ≤7.6 + testis >4.6 cm → obstructive
                             FSH >7.6 + small testes → spermatogenic failure
```

- **Genetics** (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine) guideline): **karyotype** and **Y microdeletion** in NOA (**in all**, with counselling); Klinefelter (47,XXY) commonest. **AZF (azoospermia factor) a or b deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** in CBAVD.
- **TRUS (transrectal ultrasound)**: seminal vesicle >1.5 cm, midline cyst → EDO.
- **Testis biopsy**: combined with sperm extraction and freezing.

Histology	Meaning	Sperm retrieval
Hypospermatogenesis	All stages, fewer cells	Best
Maturation arrest	Halts at one germ-cell stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Management

- **Hypogonadotropic**: stop testosterone; **hCG (human chorionic gonadotropin) ± FSH**; dopamine agonist for prolactinoma.
- **Obstructive**: **vasovasostomy** or **vasoepididymostomy** (microsurgical); **TURED (transurethral resection of ejaculatory ducts)** for EDO; or sperm retrieval (MESA (microsurgical epididymal sperm aspiration), PESA, TESA) + ICSI (intracytoplasmic sperm injection).
- Retrograde ejaculation: sympathomimetics, or sperm from alkalinised urine.
- **Non-obstructive**: **micro-TESE (microdissection testicular sperm extraction)** + ICSI; sperm in about half. Repair clinical varicocele; hormonal optimisation (off-label). In detail:

1. Optimisation: stop exogenous testosterone; repair clinical varicocele (evidence weak); off-label clomiphene, anastrozole or hCG for low testosterone.

2. Micro-TESE (microdissection testicular sperm extraction)

- **Principle:** spermatogenesis in NOA is **focal**. At ×20–25 magnification, **larger, opaque tubules** are taken; little tissue removed, blood supply spared.
- Sperm found in **about 50%**; better than conventional TESE.

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

- FSH and testis size do not reliably predict retrieval.

3. ICSI (intracytoplasmic sperm injection) with fresh or frozen sperm. Sons of AZF deletion carriers inherit the deletion.

4. No sperm: donor sperm or adoption; genetic counselling.

Also know: in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagram to draw:

- Hypothalamic–pituitary–testis axis with the three levels of failure.

- Opened testis under microscope: plump tubules versus thin Sertoli cell–only tubules.

Key points to remember:

- Confirm on two centrifuged samples.
- FSH ≤7.6 and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in NOA (in all); AZFa/b (complete deletion): do not retrieve; no retrieval.
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.

Oligospermia.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 2, Q3a

Definition

- **Oligozoospermia:** sperm concentration below the lower reference limit. **WHO (World Health Organization) 2021 (6th edition): 16 million/mL** (total count 39 million per ejaculate). Earlier thresholds were 20 million (1999) and 15 million (2010).
- These are 5th centile values of fertile men, not fertility cut-offs. A single value must be read with testis size and FSH (follicle-stimulating hormone).
- **Severity:** mild to moderate 5–16 million/mL; **severe <5 million/mL**; cryptozoospermia (sperm only in a centrifuged pellet).

Other WHO 2021 lower limits	Value
Volume	1.4 mL
Progressive motility	30%
Normal forms (strict)	4%
Vitality	54%

Causes

- **Pre-testicular:** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, hyperprolactinaemia, obesity (raised oestradiol).
- **Testicular: varicocele** (commonest correctable; heat and oxidative stress), cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, Klinefelter mosaic, **AZFc (azoospermia factor c) deletion**; **idiopathic** (largest group).
- **Post-testicular:** partial ejaculatory duct obstruction (low volume), infection, antisperm antibodies.
- Lifestyle: smoking, alcohol, heat, drugs.

Evaluation

- **Two semen analyses** after 2–7 days of abstinence.
- History and examination: testis volume, varicocele, vasa, secondary sex characters.
- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **<5 million/mL: karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] guideline).
- Selected: scrotal Doppler; TRUS (transrectal ultrasonography) if volume is low; semen culture; DNA fragmentation.

Management

- **General:** stop testosterone and gonadotoxins; weight loss; stop smoking.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen. It improves semen and pregnancy rates. Do not treat a subclinical one.
- **Medical:** hypogonadotropic: hCG (human chorionic gonadotropin) with or without FSH. Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone to oestradiol ratio is low.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank

sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days; judge response after 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: 16 million/mL; severe <5 million/mL.
> - Karyotype and Y-microdeletion if <5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction)

Asked as:

- Sperm extraction in Obstructive azoospermia. — *Winter 2024 Paper II Q3f (SAQ)*
- Testicular aspiration of sperms and assisted reproduction — *Summer 2019 Paper II Q3c (SAQ)*
- Sperm Retrival techniques — *Practice Set 16 Paper II Q3e (SAQ)*

Type: SAQ (10 marks)
(also answered in 2016-Summer Paper 3, Q3b and 2021-Summer Paper 4, Q2)

Assumed meaning: sperm retrieval techniques.

Introduction

- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- Causes: **CBAVD (congenital bilateral absence of the vas deferens)**, vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury.
- **Clues:** normal testes, **FSH (follicle-stimulating hormone) ≤7.6 IU/L and testis long axis >4.6 cm**; CBAVD → CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners.
- TESA (testicular sperm aspiration) retrieves sperm from the testis by needle for ICSI (intracytoplasmic sperm injection), which needs only one sperm per oocyte.

Indications

- **OA (obstructive azoospermia)** with normal spermatogenesis: CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction (backup at vasoepididymostomy, frozen for IVF [in vitro fertilisation]), or when epididymal aspiration fails.
- **NOA (non-obstructive azoospermia):** only testicular retrieval works; TESA gives poor yield; micro-TESE preferred.
- Anejaculation (spinal cord injury): vibratory stimulation or electroejaculation first; retrieval after failed vibratory stimulation or electroejaculation.
- Retrieved sperm need **ICSI (intracytoplasmic sperm injection)**, because sperm from obstructed systems have poor motility; cryopreserve spares.

Techniques (comparison of retrieval methods)

Technique	Method / Site	Best for	Merits and limits; drawback; notes
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput (caput epididymis), local anaesthesia	OA	Simple; few sperm (fewer sperm); may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened under microscope at 16–25×; aspirated; micropipette by **capillary action**, never suction	**OA (best)**	**Best in OA**: many motile sperm, multiple vials (multiple frozen vials); needs microsurgery (microsurgical skill)
TESA (testicular sperm aspiration)	Testis, needle: 23 G needle with suction, many passes	OA	Least invasive; immature, immotile sperm, few sperm, small yield; poor in NOA

Technique	Method / Site	Best for	Merits and limits; drawback; notes
Percutaneous core biopsy	14 G biopsy gun	OA	Fresh retrieval in OA; blind; vessel injury
TESE (testicular sperm extraction)	Open biopsy, one or more sites	OA, NOA	When epididymis is absent or empty; gold standard for tissue (gold standard for diagnosis); blind sampling
Micro-TESE	Testis bivalved; larger opaque tubules picked (chosen) under the microscope	**NOA (best)**	Sperm in about half; long operation
Vasal aspiration	Vasotomy, testicular end flushed	At reconstruction	

- **MESA:** move proximally until motile sperm appear (most motile in the efferent ductules); 4–12 pipettes, frozen in aliquots.
- **Micro-TESE (Schlegel 1999):** spermatogenesis in NOA is **focal**. Plump tubules hold sperm; thin white ones are Sertoli cell–only. Map positive sites; if negative, explore the other testis.

MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally**, up to the efferent ductules (proximal sperm are more motile).
4. Collect by **capillary action** into micropipettes, never suction.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

TESA technique

1. Local anaesthesia with spermatic cord block.
2. 23 G needle on a high-suction syringe; many passes draw tubules into the needle.
3. Wet examination at once; repeat until sperm are found. Cryopreserve spare sperm.

Choice and outcome

- Epididymal sperm first; testicular if that fails.

Assisted reproduction

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF (in vitro fertilisation):** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved sperm (poor motility).
- Frozen retrieved sperm work as well as fresh (frozen sperm match / equal fresh), except (possibly) in spinal cord injury.

Results, outcomes and concerns

- OA: retrieval almost always succeeds (succeeds in nearly all OA); good pregnancy rates. NOA micro-TESE: sperm in **about half**.
- NOA: retrieval depends on the cause:

Condition	Retrieval
Post-orchidopexy	74%
AZFc (azoospermia factor c) deletion	70%

Condition	Retrieval
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
Sertoli cell–only (Sertoli-cell-only)	30%
AZFa/AZFb deletion	**0%: do not operate**

- NOA pregnancies: more miscarriage (more miscarriages occur) (aneuploidy, DNA damage).
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR (cystic fibrosis transmembrane conductance regulator) testing.
- Female burden: ovarian stimulation, multiple pregnancy.

Complications

- Haematoma (commonest), infection, pain, vascular injury (TESA), epididymal obstruction (PESA), testicular atrophy or fall in testosterone (TESE).
- **Also know:** micro-TESE is for NOA (non-obstructive azoospermia); post-mortem retrieval needs prior consent; microfluidic sperm sorting.

Diagram to draw: testis and epididymis showing PESA, MESA and TESA sites; needle in testis with syringe suction; flow to ICSI; micro-TESE: bivalved testis with plump tubules beside thin Sertoli cell–only tubules.

Key points to remember:

- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; micro-TESE finds sperm in about half of NOA.
- AZFa/AZFb deletion: 0% (means no sperm), do not operate.
- All surgically retrieved sperm need ICSI; cryopreserve extras.

CRE & BEP

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 2, Q3f

"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.

Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

Mechanism	Example
Class A carbapenemase	KPC (Klebsiella pneumoniae carbapenemase)
Class B metallo-β-lactamase	**NDM (New Delhi metallo-β-lactamase)**: common in India
Class D	OXA-48
Non-enzymatic	ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
  - KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
  - **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
  - Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).
- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

IGCCCG (International Germ Cell Cancer Collaborative Group) risk	Regimen
Good	BEP ×3 or EP ×4
Intermediate/poor	BEP ×4 (VIP ×4 if lungs poor)
Stage I NSGCT (non-seminomatous GCT), high risk	BEP ×1 adjuvant

- **Toxicity:** bleomycin lung fibrosis (keep FiO_2 low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

Diagram to draw: BEP 21-day cycle timeline.

> **Key points to remember:**
> - CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
> - BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
> - Bleomycin lung: low FiO_2 and restricted fluids at surgery.

Rapid Revision Sheet – Section 9 · Andrology & Sexual Medicine

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Surgical management of erectile dysfunction and penile prosthesis (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece (three-piece) inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Management of priapism (SAQ)

- Ischaemic gas: pO2 <30, pCO2 >60, pH <7.25.
- Distal shunt ± tunnelling before proximal; prosthesis if >36 h.
- High-flow: observe (~60% resolve), then embolise.

Azoospermia and non-obstructive azoospermia (SAQ)

- Confirm on two centrifuged samples.
- FSH ≤7.6 and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in NOA (in all); AZFa/b (complete deletion): do not retrieve; no retrieval.
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.

Oligospermia. (SAQ)

- WHO 2021: 16 million/mL; severe <5 million/mL.
- Karyotype and Y-microdeletion if <5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction) (SAQ)

- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; micro-TESE finds sperm in about half of NOA.
- AZFa/AZFb deletion: 0% (means no sperm), do not operate.
- All surgically retrieved sperm need ICSI; cryopreserve extras.

CRE & BEP (SAQ)

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low FiO_2 and restricted fluids at surgery.

Section 10 · Testis, Penis & Urethra — Long answer questions (LAQ)

Urethral stricture disease: etiology, presentation and management (including recent advances)

Asked as:

- Describe etiology, clinical presentation and management of urethral strictures. — *Summer 2016 Paper II Q2 (LAQ)*
- Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease — *Winter 2025 Paper II Q1 (LAQ)* (also answered in 2016-Summer Paper 2, Q2)

Type: LAQ (25 marks)

"Structure" is taken to mean urethral stricture disease.

Definition

- A fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar with (often with) **spongiofibrosis**.
- Posterior narrowing is named differently (terminology: WHO (World Health Organization) 2004, ICUD (International Consultation on Urological Diseases) 2010): after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery or radiation, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses (about) three-quarters of the lumen area, so symptoms appear late.

Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries and/plus retrograde flow from the glans) allows (lets) the bulb to be divided and mobilised.

Aetiology

Category	Examples and mechanism
Idiopathic (commonest, with iatrogenic; about equal to iatrogenic)	No cause found
Iatrogenic	Catheters (common / very common in LMICs (low- and middle-income countries)), TURP (transurethral resection of prostate), bladder tumour resection (TURBT (transurethral resection of bladder tumour)), cystoscopy, hypospadias repair, radiotherapy. Mechanism: mucosal injury, pressure ischaemia, infection
Traumatic	**Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI) gives a posterior obliterative defect
Inflammatory	Gonorrhoea (historically commonest, now rare); **lichen sclerosus (LS)**. Chlamydia: no proven link
Congenital	Rarest; infants only (short anterior stricture in infants)

- **LS**: chronic skin disease starting at (LS starts at) the glans and meatus. High-pressure voiding forces urine (urine forced) into Littre glands, causing (causes) inflammation and deep spongiofibrosis. It **recurs in any (genital) skin** used to repair it.
- **Jordan's spectrum**: mucosal fold → iris (iris constriction) → full-thickness → deep spongiofibrosis → periurethral fibrosis (fibrosis beyond spongiosum) → fistula or abscess.

Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- **Retention** after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

Evaluation (define 4 things: location, length, depth, density)

Test	Role
Uroflowmetry and PVR (post-void residual)	Low, plateau-shaped flow (flat plateau curve); screen and follow-up
Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)	Site and length; steep oblique views; done by the surgeon
Flexible urethroscopy	Calibre, appearance; paediatric scope sees beyond
Sonourethrography	Length and depth of spongiofibrosis
MRI (magnetic resonance imaging)	PFUI defect alignment; not routine (for PFUI only)

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation. In retention, place a **suprapubic catheter** rather than force a dilation.

Management (standard management, brief in the 2025 answer)

First agree the aim: **palliation or cure**. The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned (archaic), because repeated endoscopy makes later urethroplasty harder.

1. Dilation: soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Curative only for (cures only) epithelial strictures without spongiofibrosis.

2. DVIU (direct vision internal urethrotomy)

- Heals by **secondary intention**: a race between epithelialisation and wound contraction.
- Best (only) for **bulbar strictures <1.5 cm**, without dense spongiofibrosis, at first attempt. A second urethrotomy (DVIU) adds almost nothing. Laser gives no advantage over cold knife (laser equals cold knife).
- Complications: recurrence (commonest), bleeding, extravasation, ED (erectile dysfunction) from deep cuts.
- Adjuncts: **intermittent self-dilation**; mitomycin C injection. **Optilume** paclitaxel-coated balloon (**ROBUST III**: 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy).

3. Urethroplasty

Technique	Indication and key points
Excision and primary anastomosis (EPA)	**Gold standard**, about 90–95% success; short bulbar strictures (1–2 cm, up to / sometimes 3–5 cm proximally). Excise all scar, spatulate widely, tension-free. **Non-transecting** and **vessel-sparing** variants preserve bulbar arteries
BMG (buccal mucosal graft) onlay	Longer bulbar strictures: **dorsal onlay (Barbagli)**, ventral onlay, **dorsal inlay (Asopa)**. Graft equals flap. Kulkarni one-sided repair for panurethral strictures

Technique	Indication and key points
Augmented anastomosis	Excise the narrowest part, anastomose the floor, graft the roof
Penile skin flaps (Orandi, Quartey, circular)	Penile strictures (not LS); skin island carried on dartos
Staged repair (Johanson)	Stage 1 open and graft; tubularise after 6 months or more. For **LS**, failed hypospadias, infection
Perineal urethrostomy	Complex or panurethral disease, elderly or after (multiple) failures

- **Onlay is better than (beats) tube** repairs; **avoid genital skin in LS** (use oral mucosa). Topical clobetasol for early LS meatal disease.
- Meatal stenosis: meatotomy; fossa navicularis: flap or graft.
- **PFUI**: suprapubic catheter, then delayed **perineal anastomotic urethroplasty (repair)** after 3 months or more. Progressive steps: bulbar mobilisation, intracrural (corporal / crural) separation, inferior pubectomy, rerouting of the urethra.
- **Urethral stents** (UroLume) are withdrawn; AUA (American Urological Association): no stent if urethroplasty is possible.

Recent advances in management

1. **Ladder abandoned**: the "reconstructive ladder" is archaic. Repeated endoscopy lowers later urethroplasty success, so **early urethroplasty** is offered after one failed DVIU (AUA (American Urological Association) and EAU (European Association of Urology) guidelines).
2. **Optilume paclitaxel-coated balloon** (FDA (Food and Drug Administration) 2023): paclitaxel inhibits fibroblasts. **ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after failed endoscopy; IPSS (International Prostate Symptom Score) 9 vs 19.9.
3. **Anti-scarring adjuncts after DVIU**: mitomycin C injection reduces recurrence; intermittent self-dilation; removable stents. Permanent UroLume stents withdrawn.
4. **Vessel-sparing and non-transecting EPA**: preserve bulbar arteries, theoretically less ED (erectile dysfunction).
5. **Augmented anastomosis**: excise the narrowest segment, graft the rest.
6. **Kulkarni one-sided dorsal BMG** through a perineal incision for panurethral strictures.
7. **Staged oral mucosa repair for LS**; avoid skin. **BMG perineal urethrostomy** reduces neomeatal stenosis.
8. **Robotic reconstruction** of bladder neck contracture and posterior stenosis (Y-V plasty, BMG).
9. **Tissue-engineered oral mucosa** (investigational). **Prevention**: silicone and hydrophilic catheters, smaller endoscopes.
10. **PROMs (patient-reported outcome measures)**: USS-PROM (Urethral Stricture Surgery PROM) is increasingly used to judge success.

- **Also know:** dartos-island fossa repair; lingual graft; clobetasol for early LS.

Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

Follow-up

- Uroflow, PVR and symptom score; RUG or cystoscopy if flow falls. Most EPA recurrences occur in the first 6 months (EPA: mostly in first 6 months); graft results decline with time (grafts decline with time).

Diagram to draw: RUG of a bulbar stricture, and cross-section of dorsal onlay versus ventral onlay.

Key points to remember:

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Assess (define) location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU only (once) for bulbar <1.5 cm at first attempt; do not repeat.
- EPA is gold standard; BMG onlay for longer; staged repair for LS.
- Optilume (ROBUST III 74.6% vs 26.8%); early urethroplasty, ladder abandoned.

Importance and management of Inguinal lymph nodes in CA penis.

Type: LAQ (25 marks)

Source: 2017-Summer Paper 2, Q1

Importance

- Presence and extent of inguinal metastasis is the **strongest prognostic factor** in penile SCC (squamous cell carcinoma).
- A long **locoregional phase** precedes distant spread, so the groin is a **curable station**; lymphadenectomy alone can cure.
- **Kroon (2005)**: early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about 84% vs about 35% when delayed until nodes were palpable (ENE (extranodal extension) about 20% vs 95%).
- Manage both groins **within 3 months** of diagnosis; aim for **at least 8 nodes per side**.
- About 80% 5-year survival with 2 or fewer unilateral nodes, no ENE, no pelvic nodes.

Lymphatic drainage

- **Superficial system** (prepuce, shaft skin) → superficial inguinal nodes. **Deep system** (glans, frenulum) → corona → within Buck's fascia → presymphyseal plexus → femoral triangle nodes.
- Presymphyseal crossover makes drainage **bilateral** and unpredictable in side; with one palpable groin, the other has micrometastasis in about 30%.
- **Stepwise, no skip**: superficial → deep inguinal → external iliac; so superficial status gates deep dissection and inguinal status gates pelvic dissection.
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**:

Zone	Related vessel
Central	Saphenofemoral junction
Superolateral	Superficial circumflex iliac vein
Inferolateral	Lateral femoral cutaneous and superficial circumflex veins
Superomedial (sentinel nodes mostly here)	Superficial external pudendal and superficial epigastric veins
Inferomedial	Great saphenous vein

- **Deep nodes** (1–3) medial to the femoral vein; **Cloquet's node** is the highest.

Staging (TNM (tumour–node–metastasis) 8th edition)

N	Clinical	Pathological
1	One mobile unilateral node	2 or fewer unilateral, no ENE
2	Mobile, 2 or more unilateral or bilateral	3 or more (unilateral or bilateral)
3	Fixed mass or pelvic nodes	ENE or pelvic nodes

- **T1a**: no LVI (lymphovascular invasion), no PNI (perineural invasion), not high grade. **T1b**: LVI or PNI or high grade.

Evaluation

- **Careful bilateral palpation**: cN0 or cN+ (number, side, mobile or fixed). Imaging (US (ultrasound), CT (computed tomography), MRI) only if examination is unreliable (obesity, previous groin surgery).

- **cN+ or high risk**: FDG-PET/CT (fluorodeoxyglucose positron emission tomography–CT) or CT chest–abdomen–pelvis.
- About half of palpable nodes are reactive, yet an antibiotic trial is **no longer recommended** (delays treatment). US-guided **FNAC (fine-needle aspiration cytology)** is specific; a negative result never ends the pathway.

Management of the impalpable groin (cN0): risk-based

Risk (EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023)	Primary	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance (compliant) or surgical staging
High	**pT1b or more**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if unavailable

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk, staged surgically (ILND first).
- **DSNB**: US ± FNAC first (a tumour-replaced node diverts flow → false negative); then 99mTc (technetium-99m) nanocolloid, blue dye and gamma probe; serial sectioning. False-negative about 7% at expert centres.
- **Modified ILND (Catalona)**: smaller field (nothing lateral to the femoral artery or below the fossa ovalis), **saphenous vein preserved**, no sartorius transposition; positive frozen section → complete ILND.
- **Also know:** superficial ILND; fascia-sparing ILND (FS-ILND); **VEIL (video-endoscopic inguinal lymphadenectomy)**/robotic: same yield, fewer wound complications, non-bulky nodes only; Cabanas anatomical sentinel biopsy abandoned.

Management of palpable nodes (cN+)

```
Palpable node → US-guided FNAC
   positive → radical ILND same side + stage other side
              (superficial ILND, FS-ILND or DSNB)
   negative → excisional biopsy → negative: observe
Bulky (≥4 cm), fixed, bilateral or pelvic nodes
   → neoadjuvant TIP × 4 → responders: ilioinguinal resection
   → cisplatin-unfit: chemoradiation
```

- **Radical ILND** (cN2 or more; cN1 may have FS-ILND): incision 2 cm below and parallel to the inguinal ligament; flaps deep to **Camper's fascia**; boundaries inguinal ligament, sartorius, adductor longus; fascia lata opened; stay medial to the femoral artery (femoral nerve); **sartorius transposition** over the vessels; suction drains.
- **PLND (pelvic lymph node dissection)**: if **more than 2 inguinal nodes or ENE** (pN2–3); ipsilateral; external iliac commonest, then obturator.
- **Neoadjuvant TIP** (paclitaxel, ifosfamide, cisplatin), **Pagliaro (2010)**: objective response 50%; survival mainly in responders.
- **Adjuvant**: chemotherapy for pN2–3 (weak evidence); radiotherapy for multiple nodes or ENE (at least 50 Gy; 54 Gy for ENE).

Complications

- Wound infection, **flap necrosis**, dehiscence, seroma, **lymphocele**, **lymphoedema**, DVT/PE (deep vein thrombosis/pulmonary embolism).

- Reduced by transverse incision, saphenous sparing, drain removal by volume (<30 mL/day), thromboprophylaxis, minimally invasive access; sartorius transposition increases them.

Follow-up

- Examination 3-monthly for 2 years, then 6-monthly to 5 years (recurrence mostly within 2 years); then self-examination.

Diagram to draw: femoral triangle with Daseler's zones and Cloquet's node.

> **Key points to remember:**
> - Nodal status is the strongest prognostic factor; ILND can cure.
> - pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
> - Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
> - PLND if more than 2 nodes or ENE; bulky → neoadjuvant TIP (InPACT (International Penile Advanced Cancer Trial) testing the sequence).

Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis.

Type: LAQ (25 marks)

Source: 2018-Summer Paper 2, Q2

Lymphatic drainage of the testis

- **Why retroperitoneal:** the testis develops beside the kidney (L1–L2) and its lymph returns to where it began.
- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run up the cord with the testicular vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum.
- From there lymph passes to the **cisterna chyli** (in front of L1–L2), the **thoracic duct** and the left innominate vein. This explains **left supraclavicular** node metastases.

Retroperitoneal nodal regions

Region	Extent
Left para-aortic	Aortic midline to the left ureter
Interaortocaval	Between IVC (inferior vena cava) and aorta
Right paracaval	IVC midline to the right ureter

Primary landing zones

	Right testis	Left testis
Primary	**Interaortocaval**	**Left para-aortic** (and left hilar)
Secondary	Right paracaval, precaval	Interaortocaval, pre-aortic
Crossover	Small but real to the left side	Rare to the right, unless bulky

- **Crossover is right to left** only, because lymph flows from right to left.
- **Cauda of epididymis** drains with the vas to the **external iliac** nodes.
- **Pelvic and inguinal nodes** are involved only if normal drainage is altered: previous scrotal or inguinal surgery (orchidopexy, hernia repair), scrotal violation, or tumour invading the tunica vaginalis or scrotum.

Lymphatic drainage of the scrotum

- Scrotal skin drains to the **ipsilateral superficial inguinal nodes**. Lymphatics, like vessels and nerves, do not cross the median raphe.

Clinical relevance

- **Trans-scrotal orchidectomy or biopsy is contraindicated.** It leaves cord behind and opens scrotal lymphatic routes, causing local and inguinal or pelvic recurrence.
- Explains RPLND (retroperitoneal lymph node dissection) templates and the para-aortic or "dog-leg" radiotherapy fields.

Stage I seminoma: definition and staging

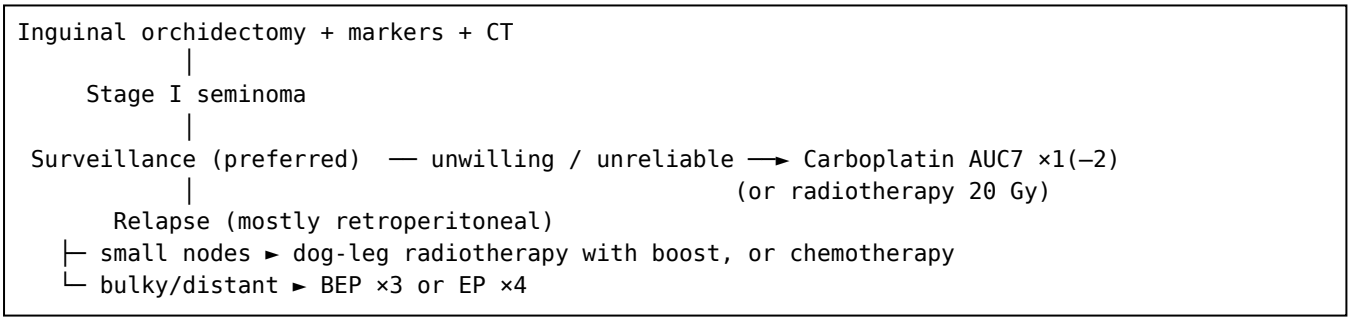
- Disease confined to the testis: any pT, N0, M0 on imaging. Stage IA = pT1; IB = pT2–4; IS = markers still raised after orchidectomy.
- **Work-up:** AFP (alpha-fetoprotein), β-hCG (human chorionic gonadotropin) and LDH (lactate dehydrogenase) before and after orchidectomy. CT abdomen and pelvis; chest CT or radiograph. Baseline testosterone; **offer sperm banking**. A raised AFP means non-seminoma.

- **Radical inguinal orchidectomy:** early cord control at the internal ring. It cures 80–85% of stage I seminoma.
- **Risk factors for relapse: tumour >4 cm** and **rete testis invasion**. Five-year relapse is 12%, 16% and 32% with 0, 1 and 2 factors. LVI (lymphovascular invasion) predicts relapse in non-seminoma, not seminoma.

Options after orchidectomy (cancer-specific survival about 100% with each)

	Surveillance	Carboplatin	Radiotherapy
Method	Clinical review, markers, CT	**One cycle, AUC (area under the curve) 7**	**20 Gy** in 10 fractions to the para-aortic strip
Relapse	14–20%	About 5%	3–5%
Relapse site	Retroperitoneum (84–100%)	Retroperitoneum	Pelvis, chest, left supraclavicular
Drawbacks	Late relapse, many scans, compliance	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity, oligospermia
Status	**Preferred**	If unwilling or unsuitable for surveillance; risk factors	If unwilling or unsuitable for surveillance

- **Carboplatin (Calvert):** 7 × (GFR [glomerular filtration rate] + 25) mg, using isotope GFR.
- **Trials: MRC TE18** (Medical Research Council): 20 Gy equals 30 Gy, with less toxicity. **MRC TE19/EORTC (European Organisation for Research and Treatment of Cancer) 30982:** one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE10: para-aortic field equals dog-leg, with 2% pelvic relapse.
- **Risk-adapted (EAU [European Association of Urology]):** surveillance for low risk; with risk factors discuss surveillance or carboplatin (treating all would overtreat two-thirds).



- BEP: bleomycin, etoposide, cisplatin; EP: etoposide, cisplatin.

Relapse and IGCCCG

- Relapses are almost always cured; median 14 months, but 10–20% occur after 4 years.
- **IGCCCG** (International Germ Cell Cancer Collaborative Group): seminoma is good risk unless there are non-pulmonary visceral metastases (intermediate). **There is no poor-risk seminoma**, and markers do not change the risk group.

Follow-up (surveillance)

- Examination and markers every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly; CT every 4–6 months (years 1–3), then 6-monthly to year 7, then yearly. CT finds >85% of relapses.

Recent advances

- **miR-371a-3p** serum microRNA detects relapse better than standard markers.

- MRI (magnetic resonance imaging) instead of CT for surveillance (TRISST trial) to reduce radiation.

Diagram to draw: retroperitoneal nodes with right (interaortocaval) and left (para-aortic) landing zones and the right-to-left crossover arrow.

Key points to remember:

- Right testis to interaortocaval; left to para-aortic; crossover only right to left.
- Scrotum to ipsilateral superficial inguinal nodes; never operate trans-scrotally.
- Risk factors: >4 cm and rete testis invasion (not LVI).
- Surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

Section 10 · Testis, Penis & Urethra — Short answer questions (SAQ)

Management of Urethral meatal stenosis.

Type: SAQ (10 marks)

Source: 2019-Summer Paper 2, Q3a

Definition and causes

- Narrowing of the external urethral meatus below the normal calibre for age.

Age	Normal calibre
Under 4 months	6–10 Fr
1–12 months	10–12 Fr
1–10 years	12–14 Fr
Over 10 years	14–18 Fr

- **Children:** almost always after **neonatal circumcision**: wet nappies irritate the exposed meatus (ammoniacal meatitis) and the ventral lips fuse into a thin membrane. Rarely congenital with hypospadias.
- **Adults:** LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]); catheter or instrument trauma; failed hypospadias repair. Usually the tip of a **fossa navicularis stricture**.

Clinical features

- Thin, forceful stream deflected **upwards**; straining, frequency, incomplete emptying; pain at the start of voiding.

Evaluation

- Watch the child void; calibrate with a bougie à boule. Uroflowmetry. Urethrography only for UTI (urinary tract infection) or suspected proximal lesion.
- Adults: **retrograde urethrogram (RUG) before meatal surgery**, as a proximal stricture often coexists, especially in LS.

Management

Situation	Treatment
Meatitis, early narrowing (child)	Topical 0.05% clobetasol; gentle dilation; nappy hygiene
Symptomatic stenosis (child)	**Ventral meatotomy** (EMLA [eutectic mixture of local anaesthetics] cream in clinic, or general anaesthesia); parents spread the edges with a dilator tip 3 times daily for 7–10 days
Simple adult stenosis	Ventral meatotomy with 3 sutures (apex and each side), mucosa to glans; dilator afterwards
Scarred or recurrent	Dorsal meatotomy (Y-V plasty after excising the scar ridge); effective but poorer cosmesis
Fossa navicularis involved	Urethroplasty: penile skin island flap (Jordan, De Sy), not in LS

Situation	Treatment
LS	Topical steroid; extended meatotomy; **buccal mucosa graft** (never genital skin); perineal urethrostomy for pananterior disease
Unfit or elderly	Self-dilation with clobetasol as lubricant

Complications

- Bleeding, restenosis (commoner in LS), spraying.

Also know: biopsy atypical LS (squamous carcinoma risk up to 8%).

Diagram to draw: ventral meatotomy: crush, incise ventrally, suture mucosa to glans at apex and both sides.

> **Key points to remember:**
> - Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
> - Adult: RUG first; often a fossa navicularis stricture.
> - LS: steroid, buccal mucosa, never genital skin.

Perineal urethrostomy.

Type: SAQ (10 marks)

Source: 2021-Summer Paper 2, Q3c

Definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses** rather than reconstructs a diseased anterior urethra.

Indications

- **Complex anterior strictures:** LS (lichen sclerosus) panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk patients** who do not want or cannot withstand staged reconstruction.
- **First stage** of a staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for penile or urethral cancer.
- Temporary: stricture after phalloplasty; resectoscope access in morbid obesity; Fournier gangrene.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU is easy in LS.

Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS increase restricture.

Techniques

1. Classical (Blandy) PU: inverted-U perineal skin flap advanced into a ventral bulbar urethrotomy (a random-pattern advancement flap).

2. BMG (buccal mucosal graft) PU, steps:

1. High lithotomy; inverted-U incision; divide bulbospongiosus in the midline.
2. Free the spongiosum; **transect** it; **dorsal urethrotomy** until healthy mucosa.
3. Bring the healthy ventral urethra to perineal skin; calibrate proximally to 30 Fr; close the distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin.

- The graft prevents the circumferential scar that narrows a deep PU.

Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or periodic dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Most succeed at mid-term with high satisfaction in selected patients.

Diagram to draw: sagittal perineum showing inverted-U flap sutured into the opened bulbar urethra.

> **Key points to remember:**
> - PU bypasses, rather than reconstructs, the diseased anterior urethra.
> - Ideal for LS, failed repairs and elderly patients; patient voids sitting.
> - BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Surgical management of bilateral varicocele

Type: SAQ (10 marks)

Source: Practice Set 17 Paper 2, Q3f (related: Practice Set 6 Paper 3, Q3e)

Introduction

- Dilated pampiniform veins: ~15% of men, 35% of primary and 75–81% of secondary infertility. Mostly left (left vein enters the renal vein 8–10 cm higher). With sensitive tests most are bilateral; clinical significance of subclinical ones is uncertain.
- Harm: heat (lost countercurrent exchange), oxidative stress, DNA fragmentation, Leydig dysfunction; progressive.
- Grades: subclinical (Doppler only), I (palpable on Valsalva), II (palpable), III (visible).

Indications for repair

- Infertile couple, **palpable** varicocele, abnormal semen, normal female partner.
- Pain (dull ache on standing), hypogonadism, high DNA fragmentation; adolescent testicular asymmetry >20%.
- **Do not repair subclinical** sides. Repair each palpable side.

Techniques

Technique	Recurrence	Hydrocele	Bilateral use
Microsurgical subinguinal / inguinal	**0.5–1%**	**~0**	**Preferred**: two small incisions, local or regional anaesthesia
Laparoscopic	3–15%	12%	Both sides through same ports; general anaesthesia, serious complications possible
Retroperitoneal (Palomo)	15–25%	7%	Children; artery ligation acceptable
Embolisation / sclerotherapy	15–25%	0	Femoral (left) or jugular (right) access; technical failure

Microsurgical steps (each side)

- Subinguinal incision over the external ring; deliver the cord.
- Microscope and **micro-Doppler** to map the testicular artery; papaverine for spasm.
- Ligate all internal and external spermatic veins; preserve **artery, lymphatics, vas and one set of vasal veins**.
- Deliver the testis to ligate gubernacular veins (lower recurrence).
- **Impulse test** confirms no missed vein.

Bilateral-specific points

- **Never ligate both testicular arteries**: in adults this has caused azoospermia and atrophy. So artery-sparing microsurgery is safest bilaterally.
- An isolated right varicocele, or one that does not decompress lying down: image for a renal or retroperitoneal mass.
- Repair any coexisting hydrocele.

Complications

- Hydrocele (lymphatic ligation), testicular artery injury (atrophy), recurrence (upright colour Doppler shows reflux).

Diagram to draw: spermatic cord cross-section: artery, lymphatics and vas preserved, veins ligated.

Key points to remember:

- Repair only palpable sides with abnormal semen.
- Microsurgical subinguinal: recurrence <1%, hydrocele nil.
- Never ligate both testicular arteries.

Rapid Revision Sheet – Section 10 · Testis, Penis & Urethra

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Urethral stricture disease: etiology, presentation and management (including recent advances) (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Assess (define) location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU only (once) for bulbar <1.5 cm at first attempt; do not repeat.
- EPA is gold standard; BMG onlay for longer; staged repair for LS.
- Optilume (ROBUST III 74.6% vs 26.8%); early urethroplasty, ladder abandoned.

Importance and management of Inguinal lymph nodes in CA penis. (LAQ)

- Nodal status is the strongest prognostic factor; ILND can cure.
- pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
- Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
- PLND if more than 2 nodes or ENE; bulky → neoadjuvant TIP (InPACT (International Penile Advanced Cancer Trial) testing the sequence).

Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis. (LAQ)

- Right testis to interaortocaval; left to para-aortic; crossover only right to left.
- Scrotum to ipsilateral superficial inguinal nodes; never operate trans-scrotally.
- Risk factors: >4 cm and rete testis invasion (not LVI).
- Surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

Management of Urethral meatal stenosis. (SAQ)

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; often a fossa navicularis stricture.
- LS: steroid, buccal mucosa, never genital skin.

Perineal urethrostomy. (SAQ)

- PU bypasses, rather than reconstructs, the diseased anterior urethra.
- Ideal for LS, failed repairs and elderly patients; patient voids sitting.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Surgical management of bilateral varicocele (SAQ)

- Repair only palpable sides with abnormal semen.
- Microsurgical subinguinal: recurrence <1%, hydrocele nil.
- Never ligate both testicular arteries.

Section 11 · Stones, Obstruction & Trauma — Long answer questions (LAQ)

Metabolic evaluation in stone disease (and management of bilateral staghorn calculi)

Asked as:

- Discuss metabolic evaluation in urinary calculus disease. — *Summer 2015 Paper II Q2 (LAQ)*
- Describe metabolic evaluation in stone disease — *Practice Set 5 Paper II Q1 (LAQ)*
- Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi — *Winter 2017 Paper II Q2 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2015-Summer Paper 2, Q2 and 2025-Winter Paper 1, Q3b)

Introduction and rationale

- Stone disease is a chronic metabolic disorder (disease): first-time formers have about a **50% risk of recurrence within 10 years** (another source: up to 50% recur within 5–10 years).
- Evaluation finds the treatable abnormality; therapy matched to 24-hour urine results reduces stone events. The **ROKS (Recurrence of Kidney Stone) nomogram** estimates individual risk.

Physicochemical basis (why we measure these)

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention, often on **Randall plaque**.
- In the **metastable zone** (between solubility and formation products) there is no spontaneous nucleation, but crystals grow on existing nuclei but do not form spontaneously; above it nucleation is spontaneous.
- **Promoters** (calcium, oxalate, uric acid, low volume) raise supersaturation; **inhibitors** (citrate, magnesium, Tamm-Horsfall protein) oppose crystallisation / raise the supersaturation needed to crystallise. The 24-hour urine measures both.

Evaluation for urolithiasis (Winter 2017)

- **History:** colic, UTI (urinary tract infection), previous stones, fluid and diet, drugs (calcium, vitamins C and D, acetazolamide, steroids), GI (gastrointestinal) disease, gout, family history.
- **Laboratory:** urinalysis (pH, crystals, pyuria) and culture; serum creatinine, electrolytes, calcium, uric acid; PTH (parathyroid hormone) if calcium is high; bicarbonate if RTA (renal tubular acidosis) is suspected.

Imaging	Role
NCCT (non-contrast CT)	Gold standard: sensitivity 97%, specificity 95%. Size, site, density (uric acid 200–400 HU [Hounsfield units]; calcium oxalate 600–1200 HU), skin-to-stone distance
Low-dose NCCT (≤3 mSv)	Equal accuracy for stones ≥3 mm
Ultrasound	First line in pregnancy and children
KUB (plain X-ray)	Follow-up of radio-opaque stones only
CT urography	Anatomy before PCNL (percutaneous nephrolithotomy)

Imaging	Role
MAG3 (mercaptoacetyltriglycine) renogram or DMSA (dimercaptosuccinic acid) scan	Split function

- **Stone analysis:** infrared spectroscopy or X-ray diffraction.
- **Metabolic evaluation:** screening (history, serum, urine, stone analysis) for every first-time former. **High-risk formers** (recurrent, children, GI disease, obesity, diabetes, hyperparathyroidism, RTA, gout, family history, solitary kidney, large bilateral stones) need one or two **24-h urines** on a usual diet: volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine (cystine if suspected). Bilateral staghorns are high risk by definition.

Step 1: Screening evaluation (every first-time former; AUA [American Urological Association] and EAU [European Association of Urology])

Component	What is assessed
History	GI (gastrointestinal) disease or bariatric surgery / bowel disease, gout, sarcoidosis, recurrent UTI (urinary tract infection), family history; diet; lithogenic drugs: calcium, vitamins C and D, acetazolamide, topiramate, steroids, triamterene, indinavir
Blood	Creatinine, sodium, potassium (electrolytes), calcium, uric acid, bicarbonate; **PTH (parathyroid hormone)** if calcium is high or high-normal
Urine	pH, crystals (hexagonal = cystine), culture; cyanide–nitroprusside cystine screen, confirmed by quantitative 24-h cystine
Stone analysis	Infrared spectroscopy or X-ray diffraction: uric acid points to metabolic syndrome, cystine to a genetic cause, struvite to infection
Imaging	NCCT (non-contrast computed tomography): burden predicts future events; uric acid usually <500 HU (Hounsfield units)

- Screening is a **triage**: low risk gets general advice; high risk goes on to 24-hour urine.

Who is high risk (who needs evaluation)

- **Screening evaluation** for every first-time former (AUA, EAU); **24-h urine** for high-risk and recurrent formers: recurrent formers, **children**, solitary kidney, bilateral or large burden, GI disease or bypass, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, family history; uric acid, cystine or infection stones.

Step 2: Extended metabolic evaluation (24-h urine)

- **One or two 24-hour urines** (collections) on the usual diet, at least (≥) 4 weeks after the episode, stone-free and infection-free. Creatinine checks completeness / adequacy (men about 20–25, women about 15–20 mg/kg/day).

Parameter	Desirable per day	Abnormality and meaning / causes
Volume	**>2.5 L**	Low volume: commonest, easiest to fix
Calcium	<250 mg women, <300 mg men (or <4 mg/kg)	**Hypercalciuria**: absorptive (PTH suppressed), renal leak (PTH high, serum calcium normal), resorptive (primary hyperparathyroidism)
Oxalate	**<40 mg**	Dietary; **enteric** (fat malabsorption, gastric bypass: fat binds calcium, freeing oxalate; often 60–70 mg); **primary** (hepatic, often >100 mg, childhood; type 1 (AGXT gene) ~70%, type 2 (GRHPR), type 3 (HOGA1))

Parameter	Desirable per day	Abnormality and meaning / causes
Citrate	**>320 mg**	Hypocitraturia: distal RTA, diarrhoea, hypokalaemia, high animal protein
Uric acid	<600 mg (labs: <750 women, <800 men)	Hyperuricosuria promotes calcium oxalate stones
pH	5.8–6.2	<5.5 uric acid (metabolic syndrome); >6.5–7 calcium phosphate, RTA, infection
Sodium	<100 mmol	High sodium raises calcium excretion
Cystine	Negative screen	Cystinuria; keep <250 mg/L

- **Uric acid stones** depend mainly on pH: below the pKa (5.5) uric acid is undissociated and insoluble.
- **Infection stones**: urease-splitting organisms, pH >7, struvite.
- **Also know:** magnesium, potassium, phosphate; hypomagnesiuria; RSS (relative supersaturation) is calculated from these and guides therapy; medullary sponge kidney and calyceal diverticulum (stasis plus metabolic defects); 2,8-dihydroxyadenine stones (APRT (adenine phosphoribosyltransferase) deficiency).

Interpreting hypercalciuria (Pak classification)

Type	Defect	Serum calcium	PTH	Fasting urine calcium
Absorptive (20–40%)	Increased gut absorption	Normal	Normal or suppressed	Normal
Renal leak (5–8%)	Tubular calcium loss	Normal	High (secondary)	High
Resorptive (3–5%)	Primary hyperparathyroidism	High	High	High

- Fast and calcium-load subtyping is no longer advised, because it does not change treatment.

Special tests

- **Distal RTA:** urine pH >6, low citrate, calcium phosphate stones. Incomplete form (forms): oral **ammonium chloride load** (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria:** autosomal recessive transport defect (cystine, ornithine, lysine, arginine); quantitative 24-hour cystine; solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
- **Primary hyperoxaluria:** genetic testing (type 1, AGXT gene, about 70%).

```
Stone episode → screening evaluation
  Low risk → general advice
  High risk → 24-h urine ×1–2 → targeted therapy → repeat ≤6 months (within 6 months) → yearly
```

Linking results to treatment (treatment linked to abnormality)

- **General**: urine >2.5 L, sodium ≤2.3 g/day, normal calcium, less animal protein, more fruit and vegetables.

Abnormality	Treatment
Low volume	Fluid for urine >2.5 L/day

Abnormality	Treatment
Hypercalciuria	**Thiazide** (hydrochlorothiazide 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate; plus sodium restriction
Primary hyperparathyroidism	**Parathyroidectomy**
Hypocitraturia, distal RTA	**Potassium citrate** 30–60 mEq/day
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; magnesium
Primary hyperoxaluria type 1	Pyridoxine, **lumasiran** (RNAi (ribonucleic acid interference)); liver ± kidney transplant
Hyperuricosuria	**Allopurinol** 300 mg/day
Uric acid stones	**Potassium citrate** to (urine) pH 6.0–6.5
Cystinuria	Urine ≥3 L/day; pH **7.0–7.5**; **tiopronin**
Struvite	Complete clearance; AHA (acetohydroxamic acid) if surgery impossible

- **Borghi 1996**: urine >2 L/day cut 5-year recurrence (12% vs 27%). **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low calcium. **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence.

Special groups and follow-up

- **Children:** all evaluated (spot ratios if not toilet-trained; consider Dent disease, monogenic causes). **Pregnancy:** defer 24-hour urine (physiological hypercalciuria).
- **Follow-up (AUA):** repeat 24-hour urine **within 6 months** of starting therapy, then yearly (as in the flowchart); blood tests for drug effects (potassium on thiazide or citrate); analyse / re-analyse every new stone. Adherence is poor; citrus juices are only adjuncts.

Recent advances

- **Lumasiran** (RNA interference) for primary hyperoxaluria type 1; **NOSTONE (2023):** hydrochlorothiazide did not reduce recurrence.

Staghorn calculus

- **Definition:** a branched stone filling the renal pelvis and extending into the calyces. **Partial:** pelvis and some calyces; **complete:** pelvis and all major calyces.
- **Composition:** classically struvite (magnesium ammonium phosphate) with carbonate apatite. Bacterial **urease** (Proteus, Klebsiella, Ureaplasma) splits urea into ammonia, so urine pH rises above 7.2 and struvite crystallises inside a bacterial biofilm. Fewer than half of modern staghorns are infection stones; calcium phosphate, cystine and uric acid also occur.
- **Why treat:** untreated staghorns destroy the kidney and cause sepsis, XGP (xanthogranulomatous pyelonephritis) and death.

Work-up before surgery

- NCCT with coronal and 3D views: burden, calyceal anatomy, colon and spleen.
- **Renal function:** eGFR (estimated glomerular filtration rate) and split function, because both kidneys are at risk.
- **Infection:** culture-specific antibiotics before surgery; drain an obstructed or infected kidney (nephrostomy through a calyx usable later for PCNL). Culture the stone.
- **Guy's stone score (2011)** predicts stone-free rate:

Grade	Stone
I	Single stone in mid or lower pole or pelvis, simple anatomy

Grade	Stone
II	Single upper-pole stone; multiple stones in simple anatomy; or single stone in abnormal anatomy
III	Multiple stones in abnormal anatomy; calyceal diverticulum; partial staghorn
IV	Staghorn; or any stone in spina bifida or spinal injury

- **Also know:** STONE score, CROES nomogram, S-ReSC score.

Treatment options (AUA [American Urological Association], EAU [European Association of Urology])

- **Goal: complete clearance**, because each fragment is an infected nidus; also clear infection and preserve function.

Option	Role
PCNL	**First line.** Multiple tracts if needed; upper-pole access reaches pelvis, lower pole and ureteropelvic junction. Flexible nephroscopy raises clearance; second-look for residue
ECIRS (endoscopic combined intrarenal surgery)	PCNL (usually supine) plus retrograde flexible ureteroscopy: fewer tracts, less bleeding, higher clearance
Staged RIRS (retrograde intrarenal surgery)	Selected partial staghorns; patients who cannot stop anticoagulants
SWL (shock wave lithotripsy)	Not as monotherapy; adjunct for residual fragments
Anatrophic nephrolithotomy (Boyce–Smith)	Incision along Brödel's avascular line, artery clamped, cold ischaemia, infundibula reconstructed. Now rare; when endoscopy cannot clear the stone in a reasonable number of sessions
Nephrectomy	Poorly functioning kidney (split below about 10–20%) that is infected or painful; XGP

Bilateral staghorns: sequence

```
Sepsis or obstruction on either side?
  Yes -> drain + antibiotics -> treat that side once sterile
  No  -> treat the better-functioning (or symptomatic) kidney first
              |
  NCCT: stone-free? -> second look if residue
              |
  Other kidney after recovery
  (selected: same-session bilateral PCNL, or PCNL + contralateral RIRS)
```

- **Why the better kidney first:** it secures renal function early; some start with the simpler side. No trial defines the order.
- **Same-session bilateral surgery:** feasible at experienced centres if the first side was short and uncomplicated; saves an anaesthetic but risks bilateral bleeding or obstruction.
- Check creatinine after each side. Remove a non-functioning kidney only after the other side is cleared.

Complications and follow-up (staghorn)

- PCNL risks: bleeding (transfusion about 1%; angioembolisation), sepsis, pleural, colon, spleen or liver injury.
- NCCT for stone-free status; treat residual fragments >4 mm. Surveillance cultures.
- 24-h urine even in struvite formers, then targeted therapy; urine ≥2.5 L/day.
- **AHA (acetohydroxamic acid)** 250 mg three times daily, a urease inhibitor, only when surgery is exhausted. Avoid if creatinine >2.5 mg/dL or in pregnancy; causes DVT (deep

vein thrombosis), haemolytic anaemia.

Diagram to draw: supersaturation zones (undersaturated, metastable, unstable) and the evaluation algorithm; staghorn with an upper-pole PCNL tract and a retrograde flexible scope (ECIRS).

Key points to remember:

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below (<) 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months.
- NCCT gold standard (97%/95%); 24-h urine for high-risk formers.
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management

Asked as:

- Traumatic renal vascular parenchymal injuries : diagnosis and management. — *Summer 2017 Paper II Q2 (LAQ)*
- Diagnosis and Treatment of renal trauma — *Practice Set 9 Paper II Q1 (LAQ)*
- Grade 4 Renal trauma. — *Practice Set 3 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 9 Paper II Q1: also answered in 2017-Summer Paper 2, Q2; grade IV in Practice Set 3 Paper 2, Q3c) (Practice Set 3 Paper II Q3c: renal trauma also answered in 2017-Summer Paper 2, Q2)

Assumed meaning: diagnosis and management of traumatic renal parenchymal and renal vascular injuries.

Introduction

- Kidney is the **commonest urological organ injured** (up to 5% of all trauma); **blunt injury causes 80–90%**; firearms cause most penetrating injuries.
- **Rapid deceleration** stretches the kidney at its fixed points (hilum and UPJ, ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle or UPJ avulsion.
- Main renal artery injury is rare (about 0.05% of blunt trauma).

Classification: AAST (American Association for the Surgery of Trauma) 2018

Grade	Injury
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation
III	Laceration >1 cm without collecting-system injury; **any vascular injury or active bleeding contained within Gerota's fascia**
IV	Laceration into collecting system with extravasation; renal pelvis laceration and/or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction (thrombosis)
V	**Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for multiple injuries, up to grade III. Change from 1989: main vessel injury moved from IV to V; pseudoaneurysm and AVF (arteriovenous fistula) now grade IV.

Definition: AAST 2018, grade IV in detail

- Laceration into the **collecting system with urinary extravasation**.
- **Renal pelvis laceration and/or complete UPJ (ureteropelvic junction) disruption.**
- **Segmental artery or vein injury** (pseudoaneurysm, AVF [arteriovenous fistula]).
- **Active bleeding beyond Gerota's fascia.**
- **Segmental or complete infarction** from thrombosis without active bleeding.
- Change from 1989: main-vessel injury moved to grade V; segmental vascular and UPJ injuries added to grade IV.

Clinical diagnosis

History and examination

- Mechanism is the most important (key) history; mechanism matters. Signs: flank bruise or tenderness, **fractured ribs 9–12**, flank or low chest wound.

Haematuria

- Test the **first urine sample**. Significant: gross haematuria, or microscopic (microhaematuria) (>5 RBC [red blood cells] per high-power field, hpf) with **shock (SBP [systolic blood pressure] <90 mmHg)**.
- Degree does **not** match severity; it may be **absent in pedicle injury** and UPJ avulsion (in about half of UPJ injuries), and in grade IV.

Indications for imaging (AUA (American Urological Association) and EAU (European Association of Urology))

1. Blunt trauma with gross haematuria.
2. Blunt trauma with microhaematuria and **SBP (systolic blood pressure) <90 mmHg** at any time.
3. Rapid deceleration (high-speed collision, fall), even without haematuria.
4. Penetrating trauma with likely renal injury, any haematuria.
5. Children: image liberally (hypotension appears late).

- Microhaematuria without shock or deceleration (microhaematuria alone without shock): observe without imaging.

Imaging

- **Contrast-enhanced CT (computed tomography)** is the gold standard. Arterial/nephrographic phase (~80 s) shows bleeding; **delayed excretory phase** (5–10 min / minutes) shows urinary extravasation (urine leak).
- **CT red flags:**

Finding	Suggests
Medial haematoma	Pedicle (vascular) injury
Medial extravasation, no contrast in distal ureter (without distal ureteric contrast)	Renal pelvis or UPJ avulsion
Absent enhancement, cortical rim sign	Main renal artery occlusion
Active contrast blush (vascular contrast blush)	Arterial bleeding needing AE (angioembolisation)
Haematoma rim >4 cm (another source: >3.5–4 cm), medial laceration	High risk of intervention; bleeding likely to need intervention (≥2 of rim, blush, medial laceration predict intervention)

- **One-shot IVP (intravenous pyelography)** on table (2 mL/kg, one film at 10 min) in the unstable patient: confirms a **functioning contralateral kidney**. FAST (focused assessment with sonography for trauma) is non-specific. Renal vein injury is poorly seen on CT.

Management

```
Unstable despite resuscitation → laparotomy (± damage control)
                                  (or angioembolisation / AE if team ready, else laparotomy)
Stable → stage with CT
   Grade I–III → non-operative
   Grade IV–V  → non-operative in selected; AE for active bleeding
   Active blush / pseudoaneurysm / AVF → superselective AE
   Extravasation → observe (90% heal); stent ± nephrostomy if urinoma enlarges,
                   fever, obstruction, or UPJ/pelvis injury
```

- **Non-operative**: bed rest until gross haematuria clears, serial haemoglobin (**every 4–6 h**); success >90% in blunt injury and selected penetrating ones. **Non-operative success** about 85% for grade IV–V even in penetrating series; most failures need only AE, stent or nephrostomy.
- Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- Urinary extravasation heals spontaneously in **76–90%** (another source: up to 90%).
- **Angioembolisation**: superselective coils for active bleeding, pseudoaneurysm, AVF or delayed bleeding; repeat AE often succeeds.
- **Reimaging** (reimage / repeat CT at ~48 hours) for grade IV–V with deep laceration (deep lacerations), or new fever, pain or falling haematocrit (routine reimaging debated).

Indications for exploration

Absolute	Relative
Haemodynamic instability, no or transient response	Extravasation with devitalised parenchyma (>25%; lowers the threshold)
Expanding or pulsatile retroperitoneal haematoma	Arterial thrombosis
Suspected pedicle avulsion	Associated colon or pancreatic injury
UPJ avulsion	Incomplete staging

Operative technique (operative steps)

- Midline transabdominal (laparotomy); life-threatening injuries first.
- **Vascular control before opening Gerota's fascia**: lift small bowel; incise retroperitoneum over the aorta above the inferior mesenteric artery → find the left renal vein crossing the aorta → loop vein and both arteries; then reflect colon, open Gerota's fascia.
- **Renorrhaphy**: debride → suture-ligate vessels → watertight collecting-system closure → close defect over bolster → omental cover → drain. Polar injury: partial nephrectomy.
- **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, resuscitate in ICU (intensive care unit), relook (return) at 24–48 h.

Renovascular injury

- **Main artery thrombosis** (intimal tear): unilateral with normal other kidney → observe (revascularisation rarely succeeds). **Solitary or bilateral → revascularise** (warm ischaemia ideally <120 min); endovascular stent for (or stent an) intimal flap.
- **Vein**: left renal vein can be ligated near the IVC (inferior vena cava) (gonadal, adrenal collaterals); right cannot. Repair with 5-0 vascular suture; segmental veins may be ligated.

Complications

- **Delayed bleeding** (within 3 weeks; usually within 21 days): AE (embolise).

- Urinoma or abscess: stent, nephrostomy, drain.
- **Hypertension**: Goldblatt (arterial stenosis), **Page kidney** (haematoma compression), segmental ischaemia, AVF.
- Hydronephrosis, loss of function; **also know:** AKI (acute kidney injury).

Follow-up

- Blood pressure, urinalysis, creatinine; EAU: DMSA (dimercaptosuccinic acid) scan for salvaged grade III–V kidneys; imaging if new symptoms.

Diagram to draw: left renal vein crossing the aorta, with vessel loops on the vein and artery. For grade IV: kidney with laceration into a calyx, urinoma and segmental pseudoaneurysm.

Key points to remember:

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Delayed CT phase is essential; blush means AE.
- Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
- Revascularise only a solitary kidney or bilateral injuries.

ESWL: mechanism, indications, factors affecting outcome, complications and current status

Asked as:

- Describe indication, mechanism of stone fragmentation and current status of ESWL? — *Summer 2019 Paper II Q2 (LAQ)*
- Describe factors affecting outcome of ESWL. Discuss complications of ESWL and strategies to avoid them. — *Summer 2021 Paper II Q1 (LAQ)* (Mechanism, indications and current status are answered in 2019-Summer Paper 2, Q2.)
- Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. — *Practice Set 13 Paper II Q2 (LAQ)* (also answered in 2021-Summer Paper 2, Q1; mechanism and indications in 2019-Summer Paper 2, Q2)

Type: LAQ (25 marks)

Introduction

- ESWL (extracorporeal shock wave lithotripsy), now SWL, breaks stones with focused acoustic shock waves generated outside the body (generated at F1 and focused on the stone at F2). First human use 1980 (Dornier HM1).
- A shock wave is a short pulse: peak positive pressure 30–100 MPa, then a negative tail of about −10 MPa.
- Stones break by spallation, shear, squeezing and **cavitation**. Cavitation also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

Components of a lithotripter

- Energy source at F1 (first focus); focusing system; coupling (water or gel); fluoroscopy and/or ultrasound to place the stone at F2. A focal zone wider than the stone improves breakage.

Energy source	Principle	Advantages	Disadvantages
Electrohydraulic (HM3)	Underwater spark; ellipsoid reflector	Most effective	Electrode wear; variable shocks
Piezoelectric	Ceramic elements on a spherical dish	Long life; little pain	Low energy; tiny focal zone
Electromagnetic (modern machines)	Coil moves a metal membrane; lens focuses	Consistent; high energy at F2	—

Mechanisms of stone fragmentation

- Each shock creates stress inside the stone. Above a critical value, cracks form; repeated shocks make them grow and join (**fatigue**).

Mechanism	How it works
Spallation	Wave reflects and inverts at the back surface of the stone; tension breaks off the distal face
Shear and tear (superfocusing)	Waves inside the stone reflect and focus; cracks form where shear peaks
Quasistatic squeezing	Wave travels faster in stone than fluid; a ring of pressure compresses the stone
Dynamic squeezing	Shear waves at the stone edges combine; strongest force in initial breakage

Mechanism	How it works
Cavitation	Negative phase forms bubbles; collapse erodes the surface. Excess bubbles shield the stone and injure tissue

Indications

- **Renal (AUA/EAU):** <2 cm outside the lower pole and lower pole <1 cm (SWL or URS [ureteroscopy]); EAU: 10–20 mm equal to endourology; >20 mm only after PCNL (percutaneous nephrolithotomy).
- **Ureteric:** any level, best <10 mm; emergency SWL within 48 hours works well. Patients avoiding anaesthesia and stents; children.
- **Poor choices:** >2 cm or staghorn; lower pole >1 cm (**Lower Pole I**: stone-free 21% SWL vs 91% PCNL); cystine, brushite, calcium oxalate monohydrate; infection stones.

Contraindications

- **Absolute:** pregnancy; uncorrected coagulopathy or (ongoing) antithrombotics; untreated UTI (urinary tract infection); obstruction distal to the stone; nearby arterial aneurysm (aneurysm near the stone; aortic >4 cm); severe uncontrolled (untreated) hypertension.
- **Relative:** obesity or skeletal deformity preventing targeting; pacemaker.

Factors affecting (influencing) outcome / predictors of success

1. Stone factors

Factor	Favourable	Unfavourable
Size (most important)	<10 mm	Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy)
Density on NCCT (non-contrast computed tomography / non-contrast CT [computed tomography])	<900 HU (Hounsfield units) clears best	>1000 HU predicts failure
Composition	Calcium oxalate dihydrate, uric acid	Cystine, brushite, calcium oxalate monohydrate (dense, homogeneous)
Impaction (ureter)	Recent, mobile	Chronic impaction (soft-tissue ring of thick ureteral wall around the stone on CT, unchanged position): may be the strongest predictor
Burden and site	Single, renal pelvis or upper calyx	Multiple sites; lower pole >1 cm

- Infection stones and stones in several sites: prefer a more efficient modality.

2. Renal anatomical factors

- **Lower pole (lower pole anatomy):** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR **21% SWL vs 91% PCNL**. A steep infundibulopelvic angle and a long (>10 mm), narrow (<5 mm) infundibulum reduce clearance.
- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

3. Patient factors

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.

- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

4. Machine and technique factors (technique to optimise results)

1. **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
2. Treat infection; stop antithrombotics; control blood pressure. No routine stent: it does not improve clearance.
3. **Analgesia:** good analgesia, GA (general anaesthesia) or sedation keeps the stone in focus, because stone movement reduces breakage.
4. **Coupling:** gel as a mound, free of air; air pockets in the gel cut fragmentation by up to two-fifths.
5. **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks (bubbles clear between shocks), so less shielding and renal injury.
6. **Ramping:** start at low energy and step up (ramping energy up also helps); this causes protective vasoconstriction.
7. Positioning (prone for mid-ureteric stones over the sacrum), continuous imaging (monitor with imaging), experienced team. Little gain beyond two sessions (more than two sessions add little).
8. **Adjuncts:** α-blockers (faster fragment clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

Complications

Group	Complication	Comment
Fragment-related	Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments	Steinstrasse rises several-fold with burden >2 cm^2
Infective	UTI (urinary tract infection), pyelonephritis, urosepsis (sepsis)	Infection stones, stents
Renal tissue injury	Haematuria (transient, common); **subcapsular or perirenal haematoma** (minority; rarely embolisation)	Risk: age, hypertension, obesity, diabetes, coronary disease, thrombocytopenia, anticoagulation, high energy, many shocks, narrow-focus machines
Other organs	Liver, spleen, pancreas, colon / bowel, lung (children), skeletal muscle	Very rare
Cardiac	Arrhythmia during treatment	Pause or ECG (electrocardiogram) gating
Skin	Petechiae, bruising	Minor

- **Refuted long-term effects** (**not** proven): new hypertension, diabetes, decline in (loss of) renal function and infertility.

Strategies to avoid complications

1. **Select.** Absolute contraindications as above. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI, bowel reconstruction). Stop antithrombotics before and for the day and (about) 48 hours after. Control blood pressure; reprogramme / check pacemakers.

3. **Protect the kidney.** Slow rate; **slow ramping** (ramp over at least 4 minutes) (induces protective renal vasoconstriction and lowers haematoma risk); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

Current status

- **Declining use worldwide.** URS has grown with small digital scopes and lasers, giving higher single-session stone-free rates. Newer lithotripters are also less effective than the HM3.
- **SWL vs URS:** SWL has fewer complications, no stent or general anaesthesia, lower cost, but more retreatment. Guidelines still list it first-line for small renal and ureteric stones.

Recent advances

- **Burst wave lithotripsy:** bursts of 5–100 ultrasound cycles at 4–8 MPa; less cavitation; done awake (no anaesthesia); uniform fragments.
- **Ultrasonic propulsion** moves fragments to help them pass.

Diagram to draw: ellipsoid reflector with F1 (spark) and F2 (stone), stone and gel coupling, with spallation and squeezing arrows on the stone; box of failure predictors.

Key points to remember:

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

Management of upper ureteric stones

Type: LAQ (25 marks)

Source: Practice Set 2 Paper 2, Q2 (SWL also answered in 2019-Summer Paper 2, Q2 and 2021-Summer Paper 1, Q1; flexible URS in 2019-Summer Paper 2, Q3b)

Introduction

- **Upper (proximal) ureter:** UPJ (ureteropelvic junction) to the upper border of the sacroiliac joint.
- Proximal stones pass least often, so many need intervention. Dangers: renal damage and sepsis.

Evaluation

- **NCCT (non-contrast CT)**: gold standard; gives size, density (HU, Hounsfield units), SSD (skin-to-stone distance) and impaction. KUB (kidney–ureter–bladder radiograph) for SWL follow-up; creatinine, urine culture, coagulation.

1. Emergency: drain first

- **AUA (American Urological Association) indications for urgent intervention:** infection with obstruction, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Obstructed infected kidney: drain only** (double-J stent or PCN, percutaneous nephrostomy; equal for sepsis); definitive treatment later. PCN suits large stones (stent fails if transverse diameter ≥9.5 mm).

2. Observation and MET (medical expulsive therapy): limits

Spontaneous passage	Rate
By size	1–4 mm 78%; 5–7 mm 60%; 8–10 mm 39%
By site	UVJ (ureterovesical junction) 92%; distal 75%; mid 60%; **proximal 48%**

- **Observation** suits small (<5–6 mm) uncomplicated stones: NSAIDs (non-steroidal anti-inflammatory drugs) first.
- **MET (alpha-blockers, tamsulosin or silodosin):** reduce spasm so a larger urine bolus pushes the stone. Benefit is clear only for **distal stones 5–10 mm**; meta-analysis shows **no benefit for proximal stones**. SUSPEND (2015) showed no overall benefit.
- Trial of passage is limited to **4–6 weeks**, with imaging.

3. Choice of active treatment

- Factors: stone (size, density, impaction), patient (habitus, anticoagulation, preference), surgeon and equipment.

EAU (European Association of Urology) recommendation for proximal ureteric stones

Size	First choice	Alternative
<10 mm	SWL or URS (ureteroscopy)	–
>10 mm	URS (retrograde or antegrade)	SWL
Large (>15–20 mm), impacted	PCNL with antegrade URS	Laparoscopic ureterolithotomy

SWL versus URS

	SWL (shock wave lithotripsy)	URS
Stone-free rate (AUA/EAU meta-analysis, approximate)	<10 mm about 90%; >10 mm about 70%	<10 mm about 80%; >10 mm about 80%
Sessions	More retreatment	Single session; higher SFR at 2 weeks, equal at 3 months
Anaesthesia	Sedation or none	Usually general
Complications	Fewer	Perforation, stent pain
Anticoagulation	Must stop	Laser URS can continue

- **SWL fails** with size >10–15 mm, impaction (CT "ring" sign), HU >1000, SSD >10 cm, cystine or COM (calcium oxalate monohydrate) stones.
- **SWL technique:** supine; 60–90 shocks/min; ramp energy; no routine pre-stent. Contraindicated in pregnancy, bleeding disorder, untreated UTI (urinary tract infection), distal obstruction, uncontrolled hypertension.

4. Ureteroscopy for proximal stones

- **Flexible URS** preferred: it follows a stone that migrates into the kidney.
- Safety wire; access sheath (12/14 Fr); tight ureter: pre-stent and return after 2–4 weeks.
- **Laser:** holmium:YAG (2120 nm) or thulium fibre laser; dusting or fragmenting; keep **≤10 W in the ureter** to avoid thermal injury.
- **Retropulsion** into the kidney: reduced by low-energy settings and antiretropulsion devices (Stone Cone, NTrap); pneumatic lithotripsy works poorly above the brim.
- **Push-back:** stone deliberately pushed into the renal pelvis with a catheter, then treated by SWL, RIRS (retrograde intrarenal surgery) or PCNL; now largely replaced by in-situ flexible URS.
- Stent only for perforation, impaction, infection or a planned second stage.

5. Antegrade approach (PCNL with antegrade URS)

- Indications: large (>15–20 mm), hard or **impacted proximal stones**, failed retrograde access, urinary diversion, transplant, concomitant renal stones.
- **Upper or mid-pole posterior calyx** gives a straight line to the UPJ; flexible nephroscope or antegrade URS down the dilated ureter; high SFR.

6. Laparoscopic (or robotic) ureterolithotomy

- Indications: large impacted stones, failed SWL/URS, where endourology is unavailable, or with concomitant UPJ obstruction (pyeloplasty).
- Steps: retroperitoneal or transperitoneal access; identify the bulge; **clamp the ureter above the stone** to prevent migration; longitudinal ureterotomy; remove; double-J stent; close with 4-0 absorbable sutures; drain.
- High single-stage SFR; risk of urine leak. **Open ureterolithotomy** is now rare.

Complications of treatment

Complication	Points
Perforation	0–4%; stop, stent about 4 weeks
Avulsion	0.06–0.5%; **proximal third most at risk**; nephrostomy and delayed reconstruction
Stricture	3–6%; image all after URS
Sepsis	About 5% after URS

Complication	Points
SWL	Steinstrasse, haematoma, colic

Also know: pregnancy: conservative, then stent or URS (no SWL).

Recent advances

- Single-use digital scopes; pressure sensors; thulium fibre laser; robotic URS; **BWL (burst wave lithotripsy)**.

Diagram to draw: algorithm for proximal stone: drain if infected; <10 mm SWL or URS; >10 mm URS; large impacted antegrade or laparoscopic.

Key points to remember:

- Proximal stones pass in only 48%; MET does not help proximal stones.
- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries

Asked as:

- Classification and treatment of lower track trauma in # pelvis — *Practice Set 6 Paper II Q2 (LAQ)*
- Pelvic # assessment and management — *Practice Set 8 Paper II Q3d (SAQ)*
- Early surgical intervention in urethral injuries — *Practice Set 16 Paper II Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(lower tract trauma in pelvic fracture also answered in Practice Set 6 Paper 2, Q2) (posterior urethral injury also answered in Practice Set 5 Paper 3, Q1)

Assumed meaning: lower urinary tract trauma (bladder and urethra) in pelvic fracture; urological assessment and management of lower urinary tract injury in pelvic fracture.

Introduction

- Pelvic fracture accompanies **83–95% of bladder injuries (ruptures)**, yet only **5–10% of pelvic fractures** injure the bladder. Urethral injury occurs in about **10% of male** and up to 6% of female pelvic fractures.
- 80–94% have other major injuries (associated injuries), which cause the deaths; long-term disability comes from wrong management.

Mechanism

- **EPR (extraperitoneal rupture):** pelvic disruption (shearing) tears the bladder at its fascial attachments, or bone spicules lacerate it.
- **IPR (intraperitoneal rupture):** a blow on a **full bladder** bursts the dome, the weakest part covered by peritoneum.
- **PFUI (pelvic fracture urethral injury):** the posterior urethra is fixed by the puboprostatic ligaments, so displacement tears it at the **bulbomembranous junction**. The sphincter is usually avulsed but stays functional. Highest risk: straddle fracture (all four rami), vertically unstable fractures.

Classification

Bladder: AAST (American Association for the Surgery of Trauma) grades I–V

Grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal wall laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Practical: contusion, EPR (simple or complex), IPR, combined.

Urethra: Goldman I–V

Type	Injury
I	Posterior urethra stretched, intact
II	Disruption above the urogenital diaphragm
III	Disruption of membranous urethra through the urogenital diaphragm (commonest)

Type	Injury
IV	Bladder neck injury extending into urethra (IVa: bladder-base injury mimicking IV)
V	Anterior urethral injury

- Chapter's practical grading: **partial injury, minor distraction, major distraction** (a cystoscope cannot pass into the bladder); practically partial vs complete disruption. Stricture risk rises with each.
- Grades: posterior partial, minor or major distraction; anterior contusion, incomplete or complete.

Assessment

- **ATLS (Advanced Trauma Life Support)** resuscitation first; pelvic binder and stabilisation.

Clinical features

- Bladder: **gross haematuria (nearly all)**, suprapubic pain, inability to void, free fluid.
- Urethra: **triad: blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate and butterfly perineal haematoma are unreliable. **Vaginal examination** in every female; **vaginal and rectal examination.**

Investigations

- **RUG (retrograde urethrogram) first (before any catheter / before catheterisation) if blood at the meatus** (AUA [American Urological Association]); blood at the meatus mandates RUG. Method: 16 Fr catheter (catheter tip) in fossa navicularis, 3 mL balloon, penile traction, oblique position, 20 mL contrast, film during injection. Females: urethroscopy.
- **Cystography:** absolute indication: **gross haematuria + pelvic fracture** (rupture in ~29%). Relative: gross haematuria alone; microscopic haematuria with fracture.
  - **CT cystography:** retrograde fill with **350 mL** of 2–4% contrast; no drainage film needed.
  - Plain: scout, full-bladder and **drainage films** (posterior leaks otherwise missed); plain cystogram needs a drainage film.
  - Clamping the catheter after IV contrast is inadequate.
- **EPR:** flame-shaped pelvic contrast. **IPR:** contrast outlines (outlining) bowel loops.

Treatment

Resuscitation first (ATLS [Advanced Trauma Life Support]); pelvic stabilisation.

Principles (urethra)

- Key decision: **immediate repair versus deliberate delay**. Clean, localised injuries are repaired at once; PFUI (pelvic fracture urethral injury) is drained and reconstructed later.

Bladder

- **Uncomplicated EPR:** 22 Fr urethral catheter 2–3 weeks; cystogram before removal; repair if leak persists beyond 4 weeks.
- **Immediate repair:** IPR; penetrating injury; bladder neck injury; bone fragments (bone fragment) in the bladder; rectal or vaginal injury; poor drainage or clots; open pelvic fixation or laparotomy for other reasons.
- IPR: open or laparoscopic two-layer closure; inspect ureteric orifices; cystogram at 7–10 days. At laparotomy, repair EPR from inside the bladder, leaving the pelvic haematoma undisturbed.
- Suprapubic tube adds nothing over urethral catheter (AUA). Antibiotics only around IPR repair.

- Rectal or vaginal injury: separate the walls, no overlapping suture lines, interpose tissue.

Urethra: acute (posterior urethra, PFUI)

```
Blood at meatus → RUG
  Partial → one gentle catheter attempt (confirm position)
  Complete → SPC (suprapubic catheter): percutaneous 16 Fr
             or open (if bladder repair needed)
             ± brief primary realignment if stable
             (flexible cystoscopy, antegrade + retrograde)
  Bladder neck/rectal injury → immediate repair
Delayed (2–3 months) → up-and-down-o-gram → perineal excision
  and primary anastomosis (corporal split → pubectomy → rerouting)
```

- **SPC (suprapubic cystostomy) is the standard first step:** percutaneous 16 Fr Foley via peel-away sheath, or open. Avoid small "punch" tubes. SPC is safe with pelvic plating (AUA); place it high.
- **Primary realignment** (AUA [American Urological Association] Urotrauma option): brief attempt in a stable patient, acutely or within days, by combined antegrade and retrograde flexible cystoscopy (avoid prolonged attempts).
  - Catheter out at 4–6 weeks; keep the SPC; remove it 7–14 days later if voiding well.
  - It does **not** lower urethroplasty rates (**no difference** in urethroplasty rates vs SPC); most still form a **1–2 cm stenosis**. Realignment does not add ED (erectile dysfunction) or incontinence.
- Partial tear: one gentle catheter attempt.
- **Immediate open repair** only for **bladder neck or rectal injury**, or when laparotomy is needed anyway.
- **Also know:** female urethral injuries are usually repaired early; in boys tears may extend into the bladder neck.

Urethra: delayed (2–3 months)

- **Up-and-down-o-gram** (cystogram + RUG) for gap length; MRI for complex cases; flexible cystoscopy if bladder neck does not open.
- **Perineal excision and primary anastomosis (EPA)** (delayed anastomotic urethroplasty) is the treatment of choice: lithotomy → mobilise bulbar urethra → excise all scar → identify prostatic apex → tension-free spatulated anastomosis.
- Escalating steps for long gaps: **corporal splitting → inferior pubectomy → supracrural rerouting**. Abdominoperineal approach for fistula, bladder neck injury, failed repair, children.
- **Not** "cut-to-the-light" or repeated DVIU (direct visual internal urethrotomy) for gaps ≥1 cm.

Anterior urethra: early management

Injury	Early management
Contusion, incomplete straddle tear	Urethral catheter alone
Major straddle injury	SPC (standard); primary realignment for lesser injuries; delayed anastomotic urethroplasty
Low-velocity gunshot or stab	**Immediate primary repair:** debride minimally, spatulate, tension-free anastomosis over a catheter
High-velocity or close-range shotgun	SPC, staged repair
Penile fracture with urethral tear	**Immediate repair:** partial tear oversewn; complete tear debrided and anastomosed; catheter 2–3 weeks

- Penile fracture: urethral injury in about 20% of Western and 3% of Asian series; **intraoperative flexible cystoscopy** before catheterisation. Limit spongiosal debridement.

Complications

- ED (erectile dysfunction) up to 82%, mainly from the injury (nerve, arterial, venous).
- Stricture: restenosis 5–15% (DVIU usually works); incontinence <4% (uncommon).
- Missed IPR: ascites, acidosis, azotaemia (peritoneal reabsorption).
- Fistula, abscess, neurogenic voiding dysfunction; follow up ≥1 year.

Diagram to draw: pelvis with EPR (flame-shaped) vs IPR on cystogram, and bulbomembranous distraction defect; RUG showing partial vs complete posterior urethral disruption.

Key points to remember:

- Blood at meatus → RUG before catheter.
- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC (standard), then delayed perineal anastomotic urethroplasty at 2–3 months.
- Primary realignment: brief, endoscopic; equal urethroplasty rates.
- Immediate repair: bladder neck or rectal injury (repair at once), penile fracture urethra, low-velocity penetrating injury.

Stone formation (etiopathogenesis) and medical management of urolithiasis / renal calculi

Asked as:

- Describe etiopathogenesis of stone formation and medical management of urolithiasis — *Practice Set 7 Paper II Q2 (LAQ)*
- Describe medical management of renal calculi — *Practice Set 12 Paper II Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 4, Q2)

Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to 50% within 5 years. Removal is surgical; recurrence is metabolic.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20%, uric acid and struvite about 7% each, cystine 1–3%.

Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Anatomical**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	< Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors work. **Zone of therapy**
Unstable	> Kf	Spontaneous nucleation, growth, aggregation

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

Steps of stone formation

1. **Supersaturation** (low volume, high calcium, oxalate, uric acid; low citrate).
2. **Nucleation**, then **growth** and **aggregation** (aggregation can occlude a tubule within the 5–7 minute nephron transit).
3. **Retention**: fixed-particle theory. **Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) suggest a secondary cause (enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria).

Inhibitors, promoters, matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm-Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

Pathogenesis by abnormality

Abnormality	Mechanism
Hypercalciuria (commonest)	Absorptive (gut), renal leak (high PTH (parathyroid hormone)), resorptive (primary hyperparathyroidism)
Hyperoxaluria	Primary (hepatic enzyme defect, PH1 AGXT), **enteric** (fat binds calcium, freeing oxalate), dietary
Hyperuricosuria	Urate promotes calcium oxalate nucleation
Hypocitraturia	Distal RTA, diarrhoea, hypokalaemia, animal protein (intracellular acidosis)
Uric acid	**Low urine pH** (<5.5; pKa 5.5) more than excess uric acid
Cystine	Autosomal recessive defect in renal cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate + carbonate apatite; biofilm

Medical management

Aims: Relieve pain, dissolve suitable stones and **prevent recurrence** (up to 50% at 5 years).

1. Acute colic (acute care) and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory. Urgent drainage if infection with obstruction.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm** after migration from the kidney: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

2. Dissolution (chemolysis)

- **Uric acid** stones (radiolucent) dissolve with oral potassium citrate to alkalinise (EAU (European Association of Urology): pH 7.0–7.2 for dissolution); allopurinol if hyperuricosuric.
- **Cystine:** alkalinise to pH 7.0–7.5 with thiol drugs. **Struvite:** acetohydroxamic acid occasionally dissolves.

3. Prevention: general measures (every former / all formers)

- **Fluid** for urine **≥2.5 L/day** (Borghi 1996: 12.1% vs 27% recurrence at 5 years; recurrence halved).
- **Normal calcium**, low sodium (low salt), less animal protein, more fruit and vegetables (Borghi 2002: halved recurrence versus low-calcium diet).

4. Metabolic evaluation (high-risk and recurrent formers): stone analysis, serum chemistry (serum calcium, uric acid, creatinine, PTH (parathyroid hormone) if needed), one or two **24-h (24-hour) urines**; repeat 24-hour urine within 6 months of therapy, then yearly.

5. Selective drug therapy

Abnormality	Drug and dose	Rationale
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption; add potassium citrate (hypokalaemia → hypocitraturia; thiazide lowers potassium and citrate)

Abnormality	Drug and dose	Rationale
Hypocitraturia, distal RTA (renal tubular acidosis)	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular citrate reabsorption
Hyperuricosuria	Allopurinol 300 mg/day	Watch hypersensitivity
Uric acid stones	Potassium citrate to **pH 6.0–6.5**; avoid >7.0 (calcium phosphate)	pH disease
Enteric hyperoxaluria	**Calcium with meals**, low oxalate and fat, magnesium	Binds gut oxalate
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA (ribonucleic acid) interference)	ILLUMINATE-A
Cystinuria	Urine ≥3 L (another source: >3 L), pH **7.0–7.5**; **tiopronin** or D-penicillamine	Cystine <250 mg/L; mixed disulfide
Struvite	Complete clearance; AHA (acetohydroxamic acid) 250 mg three times daily (another source: two to three times daily)	Urease inhibitor; thromboembolism

- **Also know:** recurrent calcium stones with normal work-up: thiazide or citrate (AUA (American Urological Association): thiazide or citrate for recurrent calcium formers even without abnormality); NOSTONE (2023): hydrochlorothiazide did not reduce recurrence; citrus juices are adjuncts.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a papilla; flowchart from stone analysis to selective therapy.

Key points to remember:

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt; MET for distal 5–10 mm stones.
- Thiazide for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

Uric acid calculi: etiopathogenesis, medical management and allopurinol

Asked as:

- Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, — *Practice Set 9 Paper II Q2 (LAQ)*
- Allopurinol — *Practice Set 10 Paper II Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Uric acid stones form **8–10%** of stones in the USA and are increasing with obesity and diabetes. They are **radiolucent** and can be **dissolved medically**.

Etiopathogenesis

1. Why humans form uric acid stones

- Humans and apes lack hepatic **uricase**, which converts uric acid to allantoin (10–100 times more soluble). Blood and urine uric acid are therefore high.

2. Physical chemistry: pH is the key

- Uric acid is a weak acid, **pKa about 5.35–5.5**. At pH 5.5 half is free uric acid and half is urate.
- Free uric acid is poorly soluble (about 96–100 mg/L); sodium urate is about 20 times more soluble. At **pH 6.5, over 90% is ionised**.
- **No natural inhibitor** of uric acid crystallisation is known, so stone formation depends almost entirely on supersaturation.

3. The three determinants

Determinant	Causes	Importance
Low urine pH (<5.5)	Insulin resistance, obesity, type 2 diabetes, gout, high animal-protein diet, chronic diarrhoea, ileostomy	**Most important**; most formers have normal uric acid excretion
Low urine volume	Hot climate or occupation, diarrhoea, dehydration	Concentrates uric acid
Hyperuricosuria (>600 mg/day)	Purine-rich diet, gout, myeloproliferative disorders, tumour lysis, uricosuric drugs, inherited defects	Least often the driver in idiopathic cases

4. Why urine is too acid: insulin resistance

- Insulin normally stimulates **ammonia production** from glutamine in the proximal tubule and the Na^+/H^+ exchanger NHE3.
- Insulin resistance reduces ammonium excretion. Acid is then buffered as titratable acid, so urine pH falls.
- **Renal lipotoxicity:** fat accumulates in the proximal tubule, which then burns fatty acids instead of glutamine, so ammoniagenesis falls further.
- Endogenous acid production is also higher. Uric acid stones are thus a **renal manifestation of the metabolic syndrome**; about a third of stones in diabetics are uric acid.

5. Aetiological categories

- **Congenital:** HGPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency (Lesch–Nyhan), PRPP (phosphoribosylpyrophosphate) synthetase overactivity, hereditary renal

hypouricaemia (URAT1, the urate transporter).
- **Idiopathic:** low urine pH without gout ("gouty diathesis").

6. Uric acid and calcium stones

- In **hyperuricosuric calcium oxalate stone disease**, urate promotes calcium oxalate by removing inhibitors, by "salting out" and by **epitaxy** (urate crystals act as a nidus).
- **Also know:** ammonium acid urate stones (laxative abuse, diarrhoea, ileostomy); theobromine (cocoa) inhibits uric acid crystallisation in vitro.

Diagnosis in brief

- Radiolucent on X-ray; CT **200–400 HU (Hounsfield units)**; **dual-energy CT** identifies uric acid.
- Urine pH persistently <5.5; serum uric acid; 24-hour urine; screen for diabetes.

Rationale of medical management

Principle: uric acid stones are a **"pH disease" more than a uric acid disease**. Raising pH turns insoluble uric acid into soluble urate. Diluting urine lowers supersaturation. Lowering uric acid production helps only when excretion is high.

1. Fluids

- Urine output **at least 2.5 L/day** lowers uric acid concentration.

2. Diet and lifestyle

- Limit non-dairy **animal protein** (it gives purines and an acid load; dietary purines provide about 30% of urinary uric acid).
- More fruit and vegetables (alkali load). Weight loss, exercise and glucose control correct the underlying insulin resistance.

3. Urinary alkalinisation (mainstay)

- **Potassium citrate 30–60 mEq/day** in two or three doses. Citrate is metabolised to bicarbonate, raising urine pH.
- **Target pH 6.0–6.5** for prevention (EAU [European Association of Urology]: 6.2–6.8); **avoid pH >7.0**, which risks calcium phosphate stones.
- **Why potassium, not sodium:** sodium alkali raises urinary calcium and can cause calcium stones. Sodium bicarbonate or sodium citrate is used if potassium is unsafe (renal impairment, hyperkalaemia).
- Patients **monitor urine pH with dipsticks**.

4. Dissolution (oral chemolysis)

- Pure uric acid stones dissolve over weeks to months. **EAU: pH 7.0–7.2**; stent or nephrostomy if obstructed or infected; monitor by imaging.

5. Xanthine oxidase inhibitors

- **Allopurinol 300 mg/day** lowers uric acid production. It is **not first-line**, because low pH, not hyperuricosuria, is the usual cause.
- Indications: hyperuricosuria with hyperuricaemia (e.g. gout), recurrence despite alkalinisation (AUA, American Urological Association), hyperuricosuric calcium oxalate stones.
- Reduce the dose in renal impairment. Risks: rash; **allopurinol hypersensitivity syndrome** (2–8% have reactions; may be fatal); Stevens–Johnson syndrome.
- **Febuxostat** is an alternative. **Rasburicase** (recombinant uricase) prevents uric acid nephropathy in tumour lysis.

Landmark trial: Ettinger (1986), randomised placebo-controlled trial: allopurinol more than halved recurrence in hyperuricosuric, normocalciuric calcium oxalate stone formers.

Allopurinol (in detail)

Introduction and mechanism

- A **purine (hypoxanthine) analogue** that inhibits **xanthine oxidase (xanthine dehydrogenase)**. This blocks hypoxanthine → xanthine → uric acid, lowering serum and urinary uric acid.
- Metabolised in the liver to **oxypurinol**, the active metabolite. It has a long half-life and is cleared by the kidney, so the dose is cut in renal impairment.

Uses in urology

Indication	Rationale
Recurrent calcium oxalate stones with hyperuricosuria (>600 mg/day) and normal urinary calcium	Urate "seeds" calcium oxalate (heterogeneous nucleation, salting out, inhibitor adsorption). **Ettinger (1986) RCT (randomised controlled trial): recurrence more than halved**; AUA (American Urological Association) recommends it
Uric acid stones	**Not first line:** low urine pH is the main cause, so alkalinise with potassium citrate to pH 6.0–6.5. Allopurinol is added for hyperuricosuria, gout or failure of alkalinisation
2,8-DHA (dihydroxyadenine) stones (APRT (adenine phosphoribosyltransferase) deficiency)	Blocks adenine → DHA; febuxostat more effective
Tumour lysis prophylaxis; gout; HPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency, PRPS (phosphoribosylpyrophosphate synthetase) overactivity	Prevents urate nephropathy

Dose

- Adult **300 mg/day** (start 100 mg, titrate); child 4–10 mg/kg/day (maximum 300 mg). Reduce for low creatinine clearance.

Adverse effects

- Rash (commonest), myalgia, gastrointestinal upset; gout flare on starting; interstitial nephritis.
- **Hypersensitivity in 2–8%**, from skin reactions to **allopurinol hypersensitivity syndrome** (fatal in up to 27%) and Stevens-Johnson syndrome. Risk is higher with high dose, renal failure, diuretics and statins. **HLA-B*58:01** carriers (some Asian populations) have higher risk.
- **Xanthine stones** (radiolucent): rare, mainly with high doses in Lesch-Nyhan syndrome.

Drug interactions

- **Azathioprine and 6-mercaptopurine:** xanthine oxidase metabolises them, so toxicity (marrow suppression) rises. Reduce their dose sharply (important after transplant).

Alternatives

- **Febuxostat:** non-purine xanthine oxidase inhibitor, no dose change in mild to moderate renal impairment. Carries a cardiovascular-death warning.
- **Also know:** rasburicase (urate oxidase) for tumour lysis.

Diagram to draw:

- Sigmoid curve of uric acid solubility against urine pH, marking pKa 5.5 and target 6.0–6.5.
- Purine pathway (hypoxanthine → xanthine → uric acid) showing the xanthine oxidase block.

Key points to remember:

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.
- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Urinary crystals

Type: LAQ (25 marks)

Source: Practice Set 11 Paper 2, Q2

Introduction

- Crystals are precipitated urinary solutes seen on microscopy of the sediment. They occur in both normal and diseased states.
- Their value lies in suggesting stone type, metabolic disease, infection or drug toxicity. **Cystine is the one crystal that is never normal.**

How crystals form (physicochemistry)

- Urine is normally **supersaturated** with calcium oxalate and phosphate. Crystals form only when the concentration product passes a critical level.

Saturation zone	What happens
Undersaturated (below Ksp, solubility product)	No crystals; stones may dissolve
Metastable (Ksp to Kf, formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors effective
Unstable (above Kf)	Spontaneous nucleation, growth and aggregation

- **Nucleation** in vivo is mostly heterogeneous (on cell debris or other crystals). Crystals then grow, aggregate and are retained, for example on **Randall plaque**.
- **Inhibitors:** citrate, magnesium, pyrophosphate, Tamm–Horsfall protein, nephrocalcin, osteopontin. No inhibitor exists for uric acid or cystine, so these depend on pH and concentration.
- **Determinants of crystalluria:** urine volume, solute excretion, pH, temperature and inhibitors.

Examination of urine for crystals

- Fresh midstream specimen, examined within 2 hours at body or room temperature. **Refrigeration precipitates crystals** that may not redissolve.
- Dipstick pH (physiological range 4.5–8.0); centrifuged sediment; polarised light helps identify crystals.
- Phosphaturia (cloudy alkaline urine) **clears with acetic acid**; pyuria does not.
- Candida mimics calcium oxalate and red cells; budding and hyphae separate it.

Classification by urine pH and shape

Urine pH	Crystal	Shape	Clinical significance
Acidic	**Uric acid**	Rhomboid or lemon-shaped plates, rosettes; yellow-brown	Uric acid stones (pH <5.5; pKa 5.35), gout, tumour lysis
Acidic	Amorphous urates	Pink "brick-dust" granules	Usually normal (concentrated, cooled urine)
Acidic	**Cystine**	**Hexagonal** colourless plates	Always abnormal: cystinuria
Acidic or alkaline	**Calcium oxalate dihydrate**	**Envelope** (octahedral)	Common, often normal; hypercalciuria

Urine pH	Crystal	Shape	Clinical significance
Acidic or alkaline	Calcium oxalate monohydrate	Dumbbell or oval	Hyperoxaluria; ethylene glycol poisoning
Alkaline	**Triple phosphate (struvite)**	**Coffin-lid** prisms	Urease-producing infection (Proteus); staghorn stones
Alkaline	Amorphous phosphate	White granules	Benign phosphaturia
Alkaline	Calcium phosphate (brushite, apatite)	Needles or rosettes; amorphous	RTA (renal tubular acidosis), hyperparathyroidism
Alkaline	Ammonium biurate	"Thorny apple" spheres	Old specimens; ammonium urate stones (diarrhoea, laxative abuse, ileostomy)
Any	Cholesterol	Notched flat plates	Nephrotic syndrome, lipiduria

- **Rare metabolic crystals:** 2,8-dihydroxyadenine, brown spiculated spheres (APRT [adenine phosphoribosyltransferase] deficiency, radiolucent stones); xanthine (hereditary xanthinuria, high-dose allopurinol); tyrosine needles and leucine spheres (severe liver disease).

Drug crystals

- **Two routes:** the drug itself precipitates (direct), or it changes urine chemistry (indirect).

Drug	Crystal and features	Prevention
Indinavir (protease inhibitor)	Needles or fan-shaped plates; crystalluria in about two-thirds of new users; stones 4–13%; radiolucent, often invisible on CT	Hydration; more soluble below pH 5.5
Atazanavir	Highest stone risk of newer antiretrovirals	Hydration
Sulfonamides (sulfadiazine)	"Sheaves of wheat"	Hydration, alkalinise
Acyclovir (high-dose intravenous)	Birefringent needles; acute kidney injury	Slow infusion, hydration
Amoxicillin, ciprofloxacin (high dose)	Needles; worse in alkaline urine	Hydration
Methotrexate (high dose)	Crystal nephropathy	Alkalinise, hydration
Triamterene, ephedrine, guaifenesin, silicate antacids	Stones of drug or metabolite	Stop the drug

- **Indirect:** loop diuretics and steroids (hypercalciuria, calcium stones); acetazolamide and topiramate (alkaline urine, hypocitraturia, calcium phosphate); cytotoxics (uric acid); laxative abuse (ammonium urate).

Clinical significance

1. **Diagnosis of stone type** before stone analysis is available; crystals in a stone former help direct metabolic work-up.
2. **Recurrence risk:** persistent crystalluria in repeated early-morning samples, and large or aggregated crystals, suggest active stone disease.
3. **Infection:** struvite crystals with urine pH >7.2 signal a urease-producing organism.
4. **Acute kidney injury:** uric acid (tumour lysis), calcium oxalate monohydrate (ethylene glycol) and drug crystals cause intratubular obstruction.
5. **Monitoring therapy:** absence of cystine or uric acid crystals confirms adequate alkalinisation.

- Most crystalluria in healthy people is transient and harmless.

Management principles

- Repeat on a fresh, warm specimen; confirm pH; 24-hour urine metabolic evaluation and stone analysis.
- **Fluids** for urine output above 2–2.5 L/day.
- **pH manipulation:** alkalinise (potassium citrate) for uric acid, cystine, methotrexate and sulfonamides; avoid alkali in calcium phosphate and struvite.
- Treat the cause: antibiotics and stone clearance for struvite; stop or change the drug; specific therapy (thiols for cystine, allopurinol for uric acid).

Diagram to draw: sketches of hexagon (cystine), envelope (calcium oxalate), coffin lid (struvite), rhomboid (uric acid), sheaf of wheat (sulfonamide).

Key points to remember:

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

Section 11 · Stones, Obstruction & Trauma — Short answer questions (SAQ)

Renal tubular acidosis

Type: SAQ (10 marks)

Source: 2015-Summer Paper 2, Q3b

Definition

- **RTA (renal tubular acidosis)**: hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular H+ secretion or HCO3 reabsorption.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	H+ secretion (α-intercalated cell)	Proximal HCO3 reabsorption	Aldosterone deficiency or resistance
Urine pH	**Always >6**	<5.5 at steady state	Low
Potassium	Low	Low	**High**
Serum HCO3	Low, may be very low	15–18 mEq/L	Mildly low
Urine calcium	High	Normal or mildly high	Low
Urine citrate	**Profoundly low**	Near normal	–
Stones	**Up to 70%; calcium phosphate, nephrocalcinosis**	Uncommon	Uncommon
Causes	Sjögren, lupus, inherited	Fanconi	Diabetes, obstruction
Treatment	**Potassium citrate** 1–2 mEq/kg/day	Alkali 10–15 mEq/kg/day plus potassium	Fludrocortisone, low-potassium diet, stop offending drugs

- **Why distal RTA forms stones**: fixed alkaline urine, profound hypocitraturia (acidosis increases citrate reabsorption) and hypercalciuria from bone buffering.
- **Incomplete distal RTA**: normal bicarbonate, but urine cannot be acidified below 5.5.

Diagnosis

- **UAG (urine anion gap) = (Na + K) – Cl** reflects urinary ammonium: negative in GI (gastrointestinal) loss, positive in renal causes.
- **Ammonium chloride load** (0.1 g/kg orally, hourly urine pH for about 6 h): pH fails to fall below 5.5 in distal RTA. **Furosemide–fludrocortisone test** is a better-tolerated alternative.
- **Fractional bicarbonate excretion** >15% during bicarbonate loading indicates proximal RTA.

Urological relevance

- Obstruction causes hyperkalaemic distal RTA, reversible by relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).

- Potassium citrate beats sodium alkali, which raises calcium excretion.
- **Also know:** CKD (chronic kidney disease) acidosis; amiloride and trimethoprim cause hyperkalaemic distal RTA; mixed type (carbonic anhydrase II deficiency); medullary sponge kidney; laxative abuse (ammonium urate stones); inherited forms (recessive with deafness); growth failure in children.

Diagram to draw: Nephron with defect sites.

Key points to remember:

- Type 1: urine pH >6, hypokalaemia, hypocitraturia, calcium phosphate stones (up to 70%).
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause; NH4Cl load fails to lower pH <5.5 in distal RTA.

Trans Uretero Ureteric anastomoses

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3a

Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

Contraindications

Absolute	Relative (disease that may involve both ureters)
Donor ureter too short to reach without tension	Stone disease, **urothelial carcinoma**
Diseased recipient ureter	Retroperitoneal fibrosis, pelvic radiation
	Chronic pyelonephritis, genitourinary tuberculosis

- Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk).

Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

Diagram to draw: donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

Key points to remember:

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
- Risk: unilateral injury becomes bilateral.

RIRS / flexible ureteroscopy

Asked as:

- RIRS (Retrograde intrarenal surgery) — *Winter 2017 Paper II Q3e (SAQ)*
- RIRS — *Practice Set 5 Paper II Q3f (SAQ)*
- Flexible Ureteroscopy. — *Summer 2019 Paper II Q3b (SAQ)*
- Flexible URS — *Practice Set 1 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)

(also answered in 2017-Winter Paper 2, Q3e) *Covered in part in 2017-Summer Paper 1, Q3b; written in full here.* (also answered in 2019-Summer Paper 2, Q3b; digital scopes in 2019-Summer Paper 4, Q3c)

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy (fURS; flexible ureteroscopy, flexible URS (ureteroscopy)) inside the kidney through the natural route; it examines and treats the whole upper tract by the natural route.
- Marshall first reported it; a working channel and active deflection made it therapeutic. Active deflection, miniaturised digital scopes and the holmium laser made it practical.

Instrument

Feature	Details
Tip	4.5–7.5 Fr, tapered (shaft larger) (digital tips up to about 8.5 Fr)
Working channel	**3.6 Fr** (laser fibre, basket)
Deflection	**≥270° both ways** (reaches lower pole); lower pole needs about 140° (lower pole angle averages 140°; up to / maximum 175°)
Types	Fibreoptic; digital; single-use (rows below)
Fibreoptic	Smallest tip; image degrades with fibre breakage (fibres break → black spots); repair about every 15 cases
Digital ("chip-on-tip", CCD [charge-coupled device] or CMOS (complementary metal-oxide semiconductor) chip on tip)	Better image; larger tip (about 8.5 Fr); not more durable
Single-use (LithoVue, 2016)	7.7 Fr; no repair, reprocessing or sterilisation cost; cheaper below about 99 cases a year

Indications

- **Stones:** renal stones **<2 cm**; lower pole <1 cm (or >1 cm / 1–2 cm / larger as an alternative to PCNL [percutaneous nephrolithotomy]); patients who cannot stop (**must continue / must stay on**) anticoagulants; failed SWL (shock wave lithotripsy); obesity; bleeding disorders; staged treatment of larger stones.
- **Others:** diagnosis and (laser) ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; strictures and endopyelotomy; calyceal diverticulum; infundibular stenosis; essential haematuria; foreign bodies.

Technique

1. Sterile urine; single-dose antibiotic; α-blocker or **pre-stenting** if the ureter is tight.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.

3. **Ureteral access sheath (UAS)** (usually / often 12/14 Fr) over a stiff wire: lowers intrarenal pressure and temperature and allows (eases) repeated passes, but causes ureteric injury in nearly half (severe ureteric injury occurs in about 1 in 8).
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) or thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket (fragmenting and basketing). Keep power ≤10 W in the ureter and ≤25 W in the kidney (tissue damage above 43 °C; heat above 43 °C damages tissue).
5. Lower pole stones: relocate (move them) the stone to an upper calyx with a basket.
6. Inspect the ureter on withdrawal; stent unless uncomplicated. **Omit stent** (stent may be omitted if anatomy and function are normal) if the case was uncomplicated with normal anatomy, normal renal function (and function), no (ureteric) injury and no second stage (AUA [American Urological Association]).

- **Difficult (unfavourable) lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum.

Outcomes

- Stone-free rate is higher than SWL for renal stones <2 cm but falls with size and lower pole location. Larger stones may need staged procedures.
- Reported rates depend on the imaging used (CT strictest).

Complications

- Stent symptoms (commonest); **sepsis about 5%**; stricture about 3%; perforation (a few per cent); bleeding or subcapsular haematoma <1%; **avulsion <1%** (nephrostomy, delayed reconstruction / repair); scope damage.

Recent advances

- Single-use digital scopes; **thulium fibre laser** and pulse-modulated (Moses) holmium (less retropulsion, finer dust); **suction and flexible navigable sheaths** (FANS; suction access sheaths); intrarenal pressure sensing (sensors; pressure-sensing scopes); robotic flexible ureteroscopy (robotic fURS / URS); AI (artificial intelligence) navigation.

Diagram to draw: flexible scope (through an) access sheath, deflected into the lower pole calyx.

Key points to remember:

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but injures (may injure) the ureter in about 1 in 8 (ureteric injury).
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm and patients on anticoagulants; first choice (modality of choice) when anticoagulants cannot be stopped (cannot stop).
- Sepsis about 5%, stricture about 3%.

Retro peritoneal fibrosis medical management

Type: SAQ (10 marks)

Source: 2019-Summer Paper 2, Q3e

Introduction

- RPF (retroperitoneal fibrosis): inflammatory fibrous tissue around the distal aorta (L4–L5) that encases the ureters. Age 40–60; male:female 2–3:1.
- **Idiopathic (about 70%):** chronic periaortitis, often autoimmune. Many cases are **IgG4-related disease** (raised serum IgG4, IgG4-positive plasma cells, storiform fibrosis).
- **Secondary (about 30%):** drugs (methysergide, ergots, β-blockers), malignancy (lymphoma commonest), radiotherapy, asbestos, infection (TB, tuberculosis).

Evaluation before treatment

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein), anaemia, renal failure; serum IgG4.
- CT/MRI: mass around aorta and IVC (inferior vena cava), medially deviated ureters; high T2 signal means activity. **PET-CT** is the most sensitive test of activity.
- Biopsy if malignancy is suspected.

Medical management

1. **Relieve obstruction first:** stents or nephrostomy (nephrostomy if uraemic and anuric); manage post-obstructive diuresis.
2. **Stop the culprit drug;** treat any secondary cause.
3. **Steroids are first line:**
    - Prednisone about 60 mg daily (about 1 mg/kg), tapered to 5 mg, total course up to 2 years.
    - About 80% respond: pain and symptoms settle in days, ESR falls, the mass shrinks.
    - Relapse on tapering is 25–50%. Active inflammation predicts response; burnt-out fibrosis responds poorly.
4. **Tamoxifen:** thought to inhibit fibroblasts. **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen, so steroids remain first line.
5. **Steroid-sparing or second line:** mycophenolate mofetil (with steroids), azathioprine, cyclophosphamide, ciclosporin, colchicine; medroxyprogesterone.
6. **Rituximab:** for refractory or IgG4-related disease.

Monitoring

- Symptoms, ESR/CRP, creatinine, ultrasound, periodic CT/MRI or PET-CT. Remove stents when ureters drain.
- Watch steroid side effects (diabetes, osteoporosis).

When medical therapy fails

- About two-thirds clear obstruction with drainage plus drugs; about 15% need **ureterolysis**: bilateral, with biopsies, and the ureters protected by intraperitonealisation or omental wrap (open, laparoscopic or robotic). Autotransplantation is rarely needed.

Diagram to draw: mass around aorta at L4–L5 pulling both ureters medially.

Key points to remember:

- Drain first, then steroids (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease; ureterolysis for about 15%.

Medical expulsive therapy (MET) in urinary calculi

Asked as:

- Medical expulsive therapy in urinary calculi. — *Summer 2021 Paper II Q3d (SAQ)*
- Medical expulsive Rx for urinary calculi. — *Practice Set 11 Paper II Q3e (SAQ)*

Type: SAQ (10 marks)
(also answered in 2021-Summer Paper 2, Q3d)

Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

Rationale: natural history

- Spontaneous passage depends on **width**, size and site:

Width	Spontaneous passage	Location	Spontaneous passage
0–3 mm	98%	Vesicoureteric junction (small stones)	92%
5 mm	65%	Distal ureter	75%
>6.5 mm	9%	Proximal ureter	48%

Mechanism

- **α1 receptors** (mainly **α1A and α1D**) are abundant in (dominate in) the distal ureter and VUJ (vesicoureteric junction). Blockade reduces spasm and the force and frequency of peristalsis (peristaltic force and frequency), so a larger urine bolus pushes the stone / builds higher pressure above the stone.
- Others: calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) relieve pain but do not aid passage.**

Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses but twice the retrograde ejaculation); alfuzosin, doxazosin.

Evidence

Study	Result
SUSPEND (Pickard, 2015): 1136 patients; tamsulosin, nifedipine or placebo	No difference in need for intervention at 4 weeks (19% vs 20%)
STONE (Meltzer, 2018): stones <9 mm (up to 9 mm)	Tamsulosin no better than placebo
Meta-analyses and a large distal-stone RCT (randomised controlled trial)	Benefit for **distal stones >5 mm**

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm** (smaller ones pass anyway; larger rarely pass).

Indications and guideline position (AUA [American Urological Association] and EAU [European Association of Urology])

- AUA and EAU: offer MET for **ureteric stones <10 mm**, especially distal, when immediate intervention is not needed (EAU: distal >5 mm).
- **After SWL (shock wave lithotripsy)** for faster fragment clearance (to clear fragments). Children: off-label but effective.

Contraindications (need urgent drainage instead; drain urgently)

- Infection with obstruction or sepsis, solitary or bilateral obstruction, renal deterioration (impairment), refractory pain or vomiting.

Protocol

- NSAID analgesia; counsel on off-label use, dizziness, hypotension, retrograde ejaculation.
- Periodic KUB (kidney–ureter–bladder radiograph) or ultrasound follow-up; **intervene after 4–6 weeks** if not passed.

Diagram to draw: ureter with α1 receptors concentrated distally (at the distal ureter and VUJ [vesicoureteric junction]) and a stone above it (above the vesicoureteric junction).

> **Key points to remember:**
>
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

Stones (urolithiasis) in pregnancy

Asked as:

- Management of Stones in pregnancy. — *Summer 2022 Paper II Q3d (SAQ)*
- Urolithiasis in pregnancy — *Practice Set 4 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)
(Summer 2022 Paper II Q3d: lower ureteric stone in pregnancy also answered in 2020-Summer Paper 3, Q3a) (Practice Set 4 Paper II Q3b: also answered in 2022-Summer Paper 2, Q3d)

Background

- 1 in 200 to 1 in 1500 pregnancies; **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Why calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Rising (raised) citrate, magnesium and urine flow protect, so stone rates barely change.
- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; transvaginal US for distal ureter; a rise in Doppler **RI (resistive index)** suggests acute obstruction.
- Second line: **MRI (magnetic resonance imaging)** (stone as a filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy**.

Management

```
Symptomatic stone in pregnancy
 ├─ sepsis, solitary kidney, renal failure, uncontrolled pain
 │     → urgent stent or nephrostomy (+ antibiotics)
 ├─ pain controlled, no infection → observe (50–80% pass)
 └─ failed observation → ureteroscopy + holmium laser
       or temporary stent/nephrostomy until delivery
```

- **Conservative:** hydration, paracetamol, small doses of morphine. **NSAIDs (non-steroidal anti-inflammatory drugs) are contraindicated** (third-trimester ductus closure). **MET (medical expulsive therapy)** is off-label.
- **URS (ureteroscopy):** first-line intervention for a non-complex stone (non-complex stones) in any trimester; second trimester is the safest window. Neuraxial anaesthesia, minimal fluoroscopy (lead shield beneath the patient, US guidance), fetal monitoring. Third-trimester URS may raise preterm labour.
- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid (rapid encrustation): change stents about **every 4 weeks** (4–6 weeks in series), nephrostomy about every 3 weeks; definitive treatment after delivery.
- **Asymptomatic renal stones:** observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** until after delivery except in expert centres.

After delivery

- Clear residual stones; **metabolic evaluation** postpartum.

Diagram to draw: the management flowchart above.

Key points to remember:

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks.

Uretero-calycostomy.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 2, Q3f (also answered in 2020-Summer Paper 3, Q3d)

Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

Procedure (open, laparoscopic or robotic)

```
Flank position (45–60°); pre-placed double-J stent
→ mobilise ureter with generous periureteral tissue
→ mobilise kidney; partial lower-pole nephrectomy
   to expose the calyx widely (not simple nephrotomy)
→ ligate pelvic outlet; spatulate ureter laterally
→ apex stitch to lateral calyx wall; second at 180°
→ interrupted mucosa-to-mucosa sutures, placed then tied
→ capsule closed over cut surface, away from anastomosis
→ cover with perinephric fat, peritoneum or omentum; drain
```

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.
- Stent out at 4–6 weeks; urogram about 1 month later, then renogram. Later endoscopy needs **antegrade upper-pole access**.

Complications

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, loss of the renal unit.
- Success about **70%** in the largest (mostly redo) series; better in small paediatric robotic series. The access route does not affect outcome.

Also know: alternatives are endopyelotomy, redo pyeloplasty, flap repair, ileal ureter, buccal graft, autotransplantation, nephrectomy (poor kidney); calicovesicostomy for a low-lying kidney.

Diagram to draw: kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
>
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
> - Stricture is the main failure; success about 70% in redo series.

Physiology of UPJ in relation to urine transport

Type: SAQ (10 marks)

Source: 2023-Winter Paper 2, Q3f

UPJ = ureteropelvic (pelviureteric) junction.

Anatomy relevant to function

- The UPJ is the apex of a funnel-shaped pelvis; its spiral smooth muscle is continuous with calyces and ureter.

Pacemaker and electrical activity

- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain and only modulate.
- **Primary pacemaker: atypical smooth muscle cells** at the pelvicalyceal border; they fire spontaneously (HCN, hyperpolarisation-activated cyclic nucleotide-gated, channels).
- **ICC-like (interstitial cells of Cajal-like) cells** (c-Kit positive) in the pelvis and UPJ relay the signal to typical smooth muscle and act as back-up pacemakers.
- Signal spreads through **gap junctions**. Calcium entry gives a plateau action potential; calcium–calmodulin activates myosin light-chain kinase, causing contraction.

Bolus formation and transport

1. Urine fills the pelvis at low pressure; the UPJ is closed.
2. Pelvic contraction pushes urine through the opening UPJ into the upper ureter.
3. The UPJ closes behind it, forming a **bolus**.
4. Ureteric walls **coapt** behind the bolus; the wave (2–6 per minute, 20–80 cm H_2O) carries it to the bladder.

- At **low flow**, the UPJ acts as a filter: several pelvic contractions occur for each ureteric contraction.
- As **flow rises**, ureteric frequency reaches 1:1, then bolus volume rises.
- At **very high flow**, boluses merge into a continuous column and the ureter dilates **without obstruction**.

Effect of obstruction

- **Acute:** stronger, more frequent contractions.
- **Chronic:** dilatation, walls fail to coapt, peristalsis becomes ineffective; raised pelvic pressure lowers GFR (glomerular filtration rate).
- **Intrinsic UPJO (UPJ obstruction):** abnormal muscle, excess collagen, fewer ICC-like cells, so an **aperistaltic segment** cannot transmit the wave.
- **Also know:** endotoxin and progesterone reduce peristalsis. Diuretic renogram and Whitaker test (chapter: pressure difference >12 cm H_2O suggests obstruction; classic teaching: <15 unobstructed, >22 obstructed) assess transport.

Diagram to draw: pelvis, UPJ and ureter showing bolus formation and coaptation.

Key points to remember:

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

Tubeless PCNL

Asked as:

- Tubeless PCNL — *Winter 2025 Paper II Q3e (SAQ)*
- Tubeless PCNL — *Practice Set 9 Paper II Q3d (SAQ)*

Type: SAQ (10 marks)

(also answered in 2025-Winter Paper 2, Q3e and 2016-Summer Paper 4, Q3e)

Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

Selection

Suitable	Keep a nephrostomy
Complete clearance, single tract	Pyonephrosis or infected stone
Minimal bleeding	Significant bleeding
Intact collecting system	Significant collecting-system perforation
Supracostal access, bilateral cases (supracostal and bilateral cases shown safe)	Residual stone needing a second look

Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent or string.
3. Withdraw the sheath under vision; **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

Evidence

- Meta-analyses of randomised trials: shorter stay and operative time, and less pain.
- Less pain than large tubes (**>16 Fr**) (pain benefit is over large tubes), but **no benefit over small 8–9 Fr tubes**; small tubes (≤10 Fr) give similar comfort without more bleeding.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.

- Bleeding without tamponade: re-insert (reinsert) a balloon or Kaye catheter, then **angioembolisation**.

Current position

- Choose tubeless, small tube or stent case by case (AUA (American Urological Association)). Pairs well with mini-PCNL.

Diagram to draw: kidney with sealed tract and double-J stent.

> **Key points to remember:**
> - Tubeless = stent only; totally tubeless = neither.
> - Benefit is mainly over large (>16 Fr) tubes.
> - Keep a tube for bleeding, perforation, infection or second look.

Complications of PCNL, including bleeding and persistent haematuria after PCNL

Asked as:

- Complications of PCNL — *Practice Set 2 Paper II Q3e (SAQ)*
- What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!? — *Practice Set 6 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)
(mini-PCNL complications in 2022-Summer Paper 1, Q1) (PCNL complications also in Practice Set 2 Paper 2, Q3e)

Introduction

- PCNL (percutaneous nephrolithotomy) is the standard for renal stones >2 cm. Most complications arise from **access and tract dilatation**.
- Graded by the **Clavien–Dindo** system (I: no intervention; II: drugs or transfusion; III: intervention; IV: ICU [intensive care unit]; V: death).
- Bleeding is the commonest serious complication of PCNL. Transfusion was historically ~7%, now nearer 1% with smaller tracts. Angioembolisation is needed in <1%.

Incidence (systematic review, Seitz 2012, quoted by EAU [European Association of Urology])

Complication	Rate
Fever	10.8%
Transfusion	7% (nearer 1% with modern and mini tracts)
Thoracic	1.5%
Sepsis	0.5%
Organ injury	0.4%
Angioembolisation	0.4%
Death	0.05%

Main complications

Complication	Cause	Prevention and management
Acute bleeding	Infundibular or segmental vessel injury	Fornix puncture; place sheath, tamponade with balloon or **Kaye catheter**, clamp nephrostomy; transfuse
Delayed bleeding	Pseudoaneurysm, AV (arteriovenous) fistula	**Selective angioembolisation**
Infection, SIRS (systemic inflammatory response syndrome), sepsis	Infected stone, high intrarenal pressure	Sterile urine, culture-specific antibiotics, low-pressure irrigation; stop and drain if pus
Pleural (pneumothorax, hydrothorax)	Supracostal access, highest above 11th rib	Subcostal where possible, puncture in expiration; chest drain
Colon	Retrorenal colon, left lower pole	CT review; extraperitoneal: double-J stent and withdraw nephrostomy into colon (colostomy tube)
Spleen, liver	Upper-pole access, organomegaly	Ultrasound guidance; conservative or surgery

Complication	Cause	Prevention and management
Collecting system	Perforation, extravasation	Stop, drain; later infundibular stenosis
Fluid absorption	Long operation, perforation	**Normal saline**; hyponatraemia, overload

Also know: venous air embolism, DVT (deep vein thrombosis) and pulmonary embolism, neuropraxia and eye injury (prone), residual fragments, loss of renal function (multiple tracts), urinoma, nephrocutaneous fistula, retained hardware.

Risk factors

- Staghorn, multiple tracts, supracostal access, large tract, long operation, CKD (chronic kidney disease), diabetes, positive urine culture, inexperience.
- **Bleeding risk factors:** infundibular or upper-pole puncture, multiple or large tracts, staghorn stone, long operation, antithrombotic drugs, coagulopathy, diabetes.

Reducing risk

- **Tranexamic acid** reduces transfusion (Cochrane).
- Ultrasound-guided access, mini-PCNL, ECIRS (endoscopic combined intrarenal surgery).
- **Bleeding prevention:** puncture through the **calyceal fornix** along a posterior calyx (Sampaio: no arterial injury); correct clotting; mini-tracts.

Bleeding in PCNL

Causes

Timing	Cause
Intraoperative	Parenchymal or venous tract bleeding; infundibular artery injury; through-and-through puncture; posterior segmental artery injury (upper-pole access); torn infundibulum or pelvis
Early postoperative	Tract bleeding after sheath removal; clot retention
Delayed (about 1 week)	**Pseudoaneurysm**, **AV (arteriovenous) fistula**

Mechanism of delayed bleeding

- **Pseudoaneurysm:** clot seals an arterial tear, then ruptures intermittently.
- **AV fistula:** artery and vein heal together; the thin vein cannot withstand arterial pressure.

Management at day 6 (persistent haematuria)

- At day 6, assume an **arterial lesion** until proved otherwise.

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Resuscitate: IV access, crossmatch, transfuse; Hb, clotting
→ Stop anticoagulants; catheter, irrigate clots
→ Stable, mild: bed rest, clamp nephrostomy, serial Hb
→ Persistent or recurrent bleeding or Hb fall
   → CT angiography (arterial blush) or direct angiography
→ SUPERSELECTIVE ANGIOEMBOLISATION (coils)
→ Fails or unstable after embolisation
   → open exploration, partial or total nephrectomy
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- **Angiography** diagnoses >90% and treats in the same sitting. It may miss intermittent bleeds; non-selective embolisation loses renal tissue.
- Clot colic or obstruction: double-J stent or nephrostomy.
- Follow-up: Hb and creatinine; check for infection and abscess.

Intraoperative ladder (for completeness)

- Pressure on tract → nephrostomy clamped (tamponade) → balloon reinflation or **Kaye tamponade catheter** → angioembolisation.

Diagram to draw: kidney cross-section with posterior calyx, Brödel line, pleura and retrorenal colon; angiogram sketch: pseudoaneurysm before and after coiling.

Key points to remember:

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.
- Supracostal access increases pleural injury; colon injury at left lower pole.

Describe various pt positions in PCNL

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 2, Q3b (also answered in 2024-Winter Paper 1, Q3f)

Introduction

- PCNL (percutaneous nephrolithotomy) position decides calyceal access, working space, airway safety and simultaneous retrograde access.

Positions

Position	Set-up	Advantages	Disadvantages
Prone (Goodwin, 1955)	Lithotomy for ureteric catheter, then turned prone; horizontal bolsters at nipple line and ASIS (anterior superior iliac spine) bring the kidney down more than vertical ones	**Posterior calyx and upper pole** access; large working area; familiar	Turning; airway hard to reach; pressure injuries; retrorenal colon commoner prone
Prone split-leg / prone-flexed	Legs apart for flexible cystoscopy; hip flexion opens the costovertebral space	ECIRS (endoscopic combined intrarenal surgery) prone	Needs split-leg table; prone cystoscopy learning curve
Supine (Valdivia-Uría, 1987)	Ipsilateral flank at table edge on a lumbar bolster; arm across chest	Airway retained; no turning; **gravity drains fragments**, low pelvic pressure	Less working area; anterior calyces; mobile kidney; harder upper pole
Galdakao-modified supine Valdivia (Scoffone)	Supine flank tilt with ipsilateral leg extended and other leg in lithotomy	**Simultaneous retrograde access (ECIRS)** in one position	Longer, more horizontal tract
Bart's flank-free modified supine (Bach)	Bolster under the thorax, flank free	More flank space	Fewer data
Lateral decubitus (flank)	Kidney-up flank position	Obese, kyphoscoliosis, poor respiratory reserve	Unfamiliar fluoroscopic orientation

Positioning safety (prone)

- Bolster shins so toes float; pad knees; arms ≤90° to the torso (brachial plexus); axillary and pectoral rolls; padded face and eyes.
- **Prone-specific complications:** neuropraxia, ocular injury, endotracheal tube displacement.

Choosing a position

- **Prone:** standard anatomy, upper pole or multiple posterior tracts.
- **Supine variants:** obesity, cardiopulmonary disease, ECIRS, ileal conduit or continent channel, combined ureteric stone.
- **Spinal deformity:** individualise; pad every pressure point.
- **Evidence:** similar stone-free and complication rates; choice is mainly surgeon preference.

Diagram to draw: axial CT sketch showing prone (vertical) and supine (horizontal) tracts.

Key points to remember:

- Prone: upper pole and posterior calyx access, large working area.
- Galdakao-modified supine: one position for ECIRS, airway safe, gravity drainage.
- Outcomes equal; horizontal bolsters drop the kidney more.

Steinstresse.

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 2, Q3e

Assumed meaning: steinstrasse.

Definition

- **Steinstrasse** (German, "stone street"): a column of stone fragments in the ureter that fails to pass and may obstruct. Typically after **SWL (shock wave lithotripsy)**; occasionally after PCNL (percutaneous nephrolithotomy) or URS (ureteroscopy).
- Incidence is low overall (about 4–7% after SWL, EAU [European Association of Urology]) but rises several-fold with stone burden **>2 cm²**.

Risk factors

- Large burden (>2 cm), renal pelvic stones, hard stones that break into large fragments, high energy, impaired ureteric drainage.

Classification (Coptcoat)

Type	Composition
I	Small fragments only
II	Larger lead fragment (4–5 mm) followed by small fragments
III	Large fragments throughout

Clinical features

- Asymptomatic (silent obstruction), colic, nausea; fever and sepsis if infected; anuria and AKI (acute kidney injury) in a solitary kidney. Most lie in the distal ureter.

Evaluation

- KUB (kidney, ureter, bladder) X-ray, ultrasound, or non-contrast CT (computed tomography); creatinine, urine culture, blood counts.

Management

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Steinstrasse
 ├─ infection / sepsis  → urgent PCN (percutaneous nephrostomy) + antibiotics
 ├─ asymptomatic, no obstruction → MET + analgesia, imaging every 1–2 weeks
 └─ failure, pain, obstruction:
      ├─ lead fragment (type II/III) → SWL to lead fragment
      └─ URS with laser lithotripsy (most definitive)
```

- **MET (medical expulsive therapy):** α-blockers (tamsulosin) improve fragment clearance and reduce pain.
- **PCN** is preferred in infected obstruction: drains without manipulating the ureter, and fragments often pass once the kidney is decompressed. A stent is an alternative.
- Retained fragments with poor progress: URS clears the column.

Prevention

- Choose PCNL or RIRS (retrograde intrarenal surgery) for large burdens.
- Energy ramping, slower shock rate; MET after SWL.
- **Stent before SWL:** may reduce steinstrasse with large stones but does not improve stone-free rate and adds stent symptoms. The chapters disagree: Chapter 97 says it may

reduce steinstrasse; Chapter 98 advises against stenting at SWL.

Diagram to draw: ureter with lead fragment and trailing column (Coptcoat types I–III).

Key points to remember:

- Risk rises several-fold with burden >2 cm².
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

Treatment of inferior calyceal calculus

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 2, Q3b

Introduction

- Lower pole is the **commonest site** of renal stones. Gravity-dependent position makes fragments hard to clear, so **SWL (shock wave lithotripsy) works poorly** here.

Factors in decision

- **Stone:** size, density (<900 HU [Hounsfield units] favours SWL), composition.
- **Anatomy unfavourable for SWL clearance:** acute **infundibulopelvic angle**, long and narrow infundibulum, skin-to-stone distance >10 cm.
- **Patient:** obesity, anticoagulants (favour URS [ureteroscopy]), pregnancy, preference.
- Evaluation: NCCT (non-contrast CT), urine culture, renal function.

Treatment by size

Size	Options
Asymptomatic, small	Observation acceptable; about half have an event in 5 years. Treat if growing, symptomatic, infected or occupational
<10 mm	SWL or **RIRS** (retrograde intrarenal surgery). AUA (American Urological Association): either
10–20 mm	EAU (European Association of Urology): SWL only if favourable anatomy; otherwise **RIRS or PCNL** (percutaneous nephrolithotomy)
>10 mm (AUA)	**PCNL** (highest SFR [stone-free rate]) or URS; SWL not first-line
>20 mm	PCNL first; staged RIRS an alternative

Evidence

- **Lower Pole I (Albala 2001):** stones >1 cm: SFR 91% PCNL vs 21% SWL.
- Lower Pole II: stones ≤1 cm, URS vs SWL: no significant SFR difference.

Technique points

- **RIRS:** flexible ureteroscope (deflection up to 270°); relocate the stone to an upper calyx with a basket before laser (holmium or thulium fibre) lithotripsy; access sheath.
- **PCNL:** direct lower pole posterior calyx puncture; mini-PCNL reduces bleeding (transfusion near 1%).
- **SWL adjuncts:** percussion, diuresis, inversion therapy and alpha-blockers improve fragment clearance.
- **Also know:** lower pole calyceal diverticulum or infundibular stenosis favours PCNL; residual fragments >4 mm after URS need treatment.

Follow-up

- Imaging for residual fragments; metabolic evaluation if recurrent.

Diagram to draw: infundibulopelvic angle and infundibular length/width on an IVU (intravenous urogram) tracing.

Key points to remember:

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

Primary Hyperoxaluria

Type: SAQ (10 marks)

Source: Practice Set 10 Paper 2, Q3f

Definition

- Rare **autosomal recessive** errors of **hepatic glyoxylate metabolism**: glyoxylate is not converted to glycine and is oxidised to oxalate, an end product cleared only by the kidney.
- Hyperoxaluria = urinary oxalate >40 mg/day; in PH (primary hyperoxaluria) often **>100 mg/day**.

Types

	PH1 (~70%)	PH2 (~10%)	PH3 (~10%)
Enzyme (gene)	**AGT** (alanine-glyoxylate aminotransferase; **AGXT**), liver peroxisome, **pyridoxine (B6) cofactor**	GRHPR (glyoxylate reductase/hydroxypyruvate reductase)	HOGA1 (4-hydroxy-2-oxoglutarate aldolase), mitochondria
Urine marker	**Glycolate**	**L-glycerate**	4-hydroxy-2-oxoglutarate
Course	**Most severe**; ESKD (end-stage kidney disease) in 2nd–3rd decade	Milder, can reach ESKD	Childhood stones; rarely ESKD

Pathophysiology

- High urinary oxalate → calcium oxalate crystals in tubules → **nephrocalcinosis** and recurrent stones.
- **Vicious circle:** falling GFR (glomerular filtration rate) reduces oxalate clearance → plasma oxalate rises → **systemic oxalosis** (bone, retina, heart, vessels, skin).

Clinical features

- Spectrum from **infantile oxalosis with renal failure** to occasional adult stones; some diagnosed only after a transplant fails.
- Clues: childhood stones, nephrocalcinosis, family history, **calcium oxalate monohydrate** stones.

Diagnosis

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24-h urinary oxalate ↑ (child: spot oxalate:creatinine ratio)
   → urinary glycolate / L-glycerate / 4-hydroxy-2-oxoglutarate
   → genetic testing (AGXT, GRHPR, HOGA1)
   all normal → secondary (enteric, dietary) hyperoxaluria
```

- Plasma oxalate when GFR is low; screen siblings and related living donors.

Management

- **Hyperhydration** (about 3 L/m²/day), **potassium citrate**, normal calcium intake.
- **Pyridoxine** trial in all PH1: responsive mainly with the Gly170Arg mutation; it restores AGT activity and peroxisomal targeting.
- **Lumasiran** (RNA interference against glycolate oxidase, which cuts glyoxylate supply): **ILLUMINATE-A**: 84% vs 0% reached urinary oxalate ≤1.5 × upper limit of normal. **Also know:** nedosiran.

- Stones: ureteroscopy or PCNL (percutaneous nephrolithotomy); limit SWL (shock wave lithotripsy).
- **ESKD:** intensive dialysis, then **combined liver–kidney transplant for PH1** (only the liver replaces the enzyme; a lone kidney would be re-damaged). Kidney alone usually suffices for PH2.

Diagram to draw: hepatic glyoxylate pathway showing the AGT block (PH1) and the lumasiran target (glycolate oxidase).

> **Key points to remember:**
> - PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
> - Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
> - Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

Fracture penis

Type: SAQ (10 marks)

Source: Practice Set 15 Paper 2, Q3c

Definition and mechanism

- **Penile fracture** is a tear of the **tunica albuginea** with rupture of the corpus cavernosum, in the **erect** penis.
- Abnormal bending of the erect penis causes an abrupt rise in intracavernosal pressure beyond tunical strength → **transverse tear**.
- The tunica (inner circular, outer longitudinal layers) is **thinnest ventrolaterally**, so most tears are ventral or lateral, proximal to mid shaft.
- **Causes:** intercourse (commonest; rear-entry), masturbation, rolling over; self-inflicted "taqaandan" (Middle East); collagenase injection for Peyronie disease.
- **Bilateral tears in about 10%**; urethral injury about 20% in the West vs about 3% in Asia and the Middle East.

Clinical features

- **Cracking or popping sound → pain → rapid detumescence → swelling and discoloration.**
- **"Eggplant deformity"** when Buck's fascia is intact; if torn, haematoma spreads to scrotum, perineum and suprapubic area.
- Penis deviates **away from the tear**; tunical defect may be palpable.
- **Urethral injury:** blood at meatus, haematuria, retention.

Investigations

- **Diagnosis is clinical** (AUA [American Urological Association] Urotrauma Guideline).
- **Ultrasound** for equivocal cases (localises the tear); **MRI** (magnetic resonance imaging) if still equivocal (100% negative predictive value). Cavernosography is discouraged.
- Urethra: **intraoperative flexible cystoscopy** (AUA and EAU [European Association of Urology] preference) or RUG (retrograde urethrogram).
- **"False fracture":** dorsal vein or artery rupture.

Management

- **Immediate surgical repair** (less curvature, ED [erectile dysfunction] and nodules than conservative care).

1. Incision: ventral penoscrotal or lateral over the defect; subcoronal degloving if site unclear (preputial ischaemia risk).
2. Evacuate haematoma; find the tear (artificial erection helps).
3. Close the tunica with **interrupted 2-0/3-0 absorbable sutures**; avoid debriding erectile tissue.
4. Urethra: partial → close over catheter; complete → tension-free repair; catheter 2–3 weeks.

- Abstinence about 1 month. Delay up to 7 days does not worsen outcome. After collagenase injection without urethral injury, conservative care is acceptable.

Complications

- About 20%: nodule, curvature, ED, painful erection, infection, stricture or fistula.

Diagram to draw: cross-section of penis showing tunical tear, Buck's fascia and haematoma spread.

Key points to remember:

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

Ovarian vein syndromes

Type: SAQ (10 marks)

Source: Practice Set 15 Paper 2, Q3e

Definition

- **Ovarian vein syndrome** (Clark, 1964) is extrinsic compression of the **right ureter** by a dilated or thrombosed **right ovarian vein**, causing intermittent hydroureteronephrosis. It is a **controversial** entity; many cases are now attributed to other causes.
- The plural also covers ovarian vein thrombosis and pelvic congestion.

Anatomy (basis)

- Ovarian veins run in the infundibulopelvic ligaments. The **right** enters the IVC (inferior vena cava) obliquely; the **left** enters the left renal vein at a right angle.
- Gonadal vessels **cross anterior to the ureter**; ovarian veins dilate greatly in pregnancy.

Types

Syndrome	Mechanism	Features
Ovarian vein (ureteric) syndrome	Dilated right ovarian vein compresses the ureter at the crossing	Multiparous women; right flank pain (premenstrual or in pregnancy); recurrent UTI (urinary tract infection)
Ovarian vein thrombosis	Postpartum hypercoagulability and venous stasis; mostly right-sided	Fever and right lower-quadrant or flank pain after delivery; may obstruct ureter; risk of pulmonary embolism
Pelvic congestion syndrome	Incompetent, refluxing ovarian veins with pelvic varices; sometimes secondary to nutcracker (left renal vein compression)	Chronic pelvic pain worse on standing, dyspareunia, post-coital ache

Controversy

- Right-sided dilatation in pregnancy is mostly **physiological hydronephrosis** (progesterone plus uterine compression; stops at the pelvic brim). Venous compression is rarely proven, so diagnosis is by **exclusion**.

Investigations

- Ultrasound; **CT urography** or IVU (intravenous urography): right hydroureter to a smooth oblique extrinsic indentation, normal ureter below.
- **Diuretic renogram** to prove obstruction; retrograde pyelography.
- CT or MR (magnetic resonance) venography, Doppler; **gonadal venography** (gold standard for reflux).
- **Exclude:** stone, retrocaval ureter, iliac vessel crossing, retroperitoneal fibrosis, endometriosis.

Management

- **Asymptomatic:** observe. **Obstruction:** stent or nephrostomy if infected or pregnant; definitive **ureterolysis with ligation and division of the ovarian vein** (open or laparoscopic).
- **Thrombosis:** anticoagulation and antibiotics.
- **Pelvic congestion:** progestogens or GnRH (gonadotropin-releasing hormone) analogues; **coil or sclerotherapy embolisation**; ligation.

Diagram to draw: right ovarian vein crossing anterior to the right ureter, with proximal hydroureteronephrosis.

Key points to remember:

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

Salvage procedures for Complicated PUJO.

Type: SAQ (10 marks)

Source: Practice Set 16 Paper 2, Q3f

Assumed meaning: salvage options for recurrent PUJO (pelviureteric junction obstruction) after a failed pyeloplasty or endopyelotomy.

Definition and causes

- **Failure:** persistent pain or infection, unchanged or worse hydronephrosis, obstructed renogram or falling function. Most failures occur within 2 years, but up to 30% appear later.
- **Causes:** missed crossing vessel, ischaemic or tense anastomosis, urine leak with fibrosis, kinking, non-dependent anastomosis, infection.

Evaluation

- Diuretic renogram (MAG3, mercaptoacetyltriglycine): split function and drainage.
- CT (computed tomography) angiography for a crossing vessel; retrograde or antegrade pyelogram for stricture length.
- Drain first (stent or nephrostomy) if infected or function is uncertain, then reassess.

Salvage options

Option	Indication	Results/notes
Endopyelotomy (antegrade or retrograde ureteroscopic, laser)	Short stricture, no significant crossing vessel, moderate hydronephrosis	About 70% after failed pyeloplasty; lateral incision; stent 4–6 weeks
Redo pyeloplasty (open, laparoscopic, robotic)	Most failures; failed endopyelotomy	About 83–89%; wide mobilisation of kidney and ureter; transperitoneal (scarred retroperitoneum)
Ureterocalicostomy	Scarred or small intrarenal pelvis, dilated lower-pole calyx	About 70% in redo series; **resect** lower-pole parenchyma (not nephrotomy)
Buccal mucosa onlay graft	Long scarred PUJ or proximal ureter, dense scar	Robotic with omental wrap; failure 15–25% (CORRUS)
Appendiceal onlay flap	Right-sided proximal stricture	Pedicled on appendiceal artery
Flaps (spiral, vertical), downward nephropexy	Gain length	Combine with redo repair
Ileal ureter	Long defect	Contraindicated if creatinine >2 mg/dL, bladder dysfunction, IBD (inflammatory bowel disease), radiation
Autotransplantation	Solitary or poor contralateral kidney; other repairs impossible	About 90% long term
Nephrectomy	Function <15% with normal contralateral kidney	Laparoscopic

- Rule of thumb: failed pyeloplasty → try endourology first if anatomy suits; failed endopyelotomy → redo pyeloplasty.
- **Also know:** robotic near-infrared fluorescence (indocyanine green) to identify ureter and perfusion; Davis intubated ureterotomy (historical).

Diagram to draw: ureterocalicostomy: lower-pole parenchyma excised, spatulated ureter sewn to calyx.

Key points to remember:

- Redo pyeloplasty is most successful (about 83–89%); endopyelotomy about 70%.
- Ureterocalicostomy: resect lower-pole parenchyma; about 70% in redo series.
- Function <15% with normal other kidney: nephrectomy.

Rapid Revision Sheet – Section 11 · Stones, Obstruction & Trauma

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Metabolic evaluation in stone disease (and management of bilateral staghorn calculi) (LAQ)

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below (<) 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months.
- NCCT gold standard (97%/95%); 24-h urine for high-risk formers.
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management (LAQ)

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Delayed CT phase is essential; blush means AE.
- Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
- Revascularise only a solitary kidney or bilateral injuries.

ESWL: mechanism, indications, factors affecting outcome, complications and current status (LAQ)

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

Management of upper ureteric stones (LAQ)

- Proximal stones pass in only 48%; MET does not help proximal stones.
- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries (LAQ)

- Blood at meatus → RUG before catheter.
- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC (standard), then delayed perineal anastomotic urethroplasty at 2–3 months.

- Primary realignment: brief, endoscopic; equal urethroplasty rates.
- Immediate repair: bladder neck or rectal injury (repair at once), penile fracture urethra, low-velocity penetrating injury.

Stone formation (etiopathogenesis) and medical management of urolithiasis / renal calculi (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor; ductal plugs suggest secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt; MET for distal 5–10 mm stones.
- Thiazide for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

Uric acid calculi: etiopathogenesis, medical management and allopurinol (LAQ)

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.
- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Urinary crystals (LAQ)

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

Renal tubular acidosis (SAQ)

- Type 1: urine pH >6, hypokalaemia, hypocitraturia, calcium phosphate stones (up to 70%).
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause; NH4Cl load fails to lower pH <5.5 in distal RTA.

Trans Uretero Ureteric anastomoses (SAQ)

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
- Risk: unilateral injury becomes bilateral.

RIRS / flexible ureteroscopy (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but injures (may injure) the ureter in about 1 in 8 (ureteric injury).
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm and patients on anticoagulants; first choice (modality of choice) when anticoagulants cannot be stopped (cannot stop).
- Sepsis about 5%, stricture about 3%.

Retro peritoneal fibrosis medical management (SAQ)

- Drain first, then steroids (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease; ureterolysis for about 15%.

Medical expulsive therapy (MET) in urinary calculi (SAQ)

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

Stones (urolithiasis) in pregnancy (SAQ)

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks.

Uretero-calycostomy. (SAQ)

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
- Stricture is the main failure; success about 70% in redo series.

Physiology of UPJ in relation to urine transport (SAQ)

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

Tubeless PCNL (SAQ)

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

Complications of PCNL, including bleeding and persistent haematuria after PCNL (SAQ)

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.
- Supracostal access increases pleural injury; colon injury at left lower pole.

Describe various pt positions in PCNL (SAQ)

- Prone: upper pole and posterior calyx access, large working area.
- Galdakao-modified supine: one position for ECIRS, airway safe, gravity drainage.
- Outcomes equal; horizontal bolsters drop the kidney more.

Steinstresse. (SAQ)

- Risk rises several-fold with burden >2 cm^2.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

Treatment of inferior calyceal calculus (SAQ)

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

Primary Hyperoxaluria (SAQ)

- PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.

- Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
- Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

Fracture penis (SAQ)

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

Ovarian vein syndromes (SAQ)

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

Salvage procedures for Complicated PUJO. (SAQ)

- Redo pyeloplasty is most successful (about 83–89%); endopyelotomy about 70%.
- Ureterocalicostomy: resect lower-pole parenchyma; about 70% in redo series.
- Function <15% with normal other kidney: nephrectomy.

Section 12 · Bladder Cancer & Diversion — Long answer questions (LAQ)

CA bladder, etiology, newer investigations and treatment.

Type: LAQ (25 marks)

Source: 2016-Summer Paper 2, Q1

Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

Aetiology (risk factors)

Factor	Key facts
Smoking	30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting
Occupation (5–10%)	Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps)
Environment	**Arsenic** in water (West Bengal, Bangladesh, Taiwan)
Radiation	Pelvic radiotherapy 1.5–4 times; latency 15–30 years
Drugs	**Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting)
Chronic inflammation	*Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma
Genetic	First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion

- **Also know:** obesity raises risk; diet has no proven role.

Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

TNM (tumour, node, metastasis), 2017	Extent
Ta / Tis	Papillary non-invasive / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	One pelvic node / several / common iliac

Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.
- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris**

System reporting).

Newer investigations

Test	Principle	Value
Blue-light cystoscopy (PDD, photodynamic diagnosis) with HAL (hexaminolevulinate)	Porphyrins in tumour fluoresce red under blue light	About one-third more CIS; recurrence benefit inconsistent
NBI (narrow-band imaging)	415 and 540 nm light absorbed by haemoglobin highlights vessels	More tumours; many false positives
Urine biomarkers	NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder	More sensitive, less specific than cytology; **none replaces cystoscopy**
mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)	Muscle-line integrity scored 1–5	**≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT
En bloc TURBT (transurethral resection of bladder tumour)	Tumour removed in one piece with detrusor	Better staging and muscle sampling, fewer perforations; recurrence similar

- **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases.

Treatment

1. TURBT: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

2. NMIBC by risk (AUA (American Urological Association))

Risk	Definition	Treatment
Low	Solitary low-grade Ta ≤3 cm; PUNLMP	Single instillation; surveillance
Intermediate	Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm	Chemotherapy or BCG; maintenance 1 year
High	High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion	**BCG induction + 3-year maintenance**; early cystectomy if very high risk

- **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression.
- **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200.

3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.

- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

4. Metastatic: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

Diagram to draw: bladder wall layers with Ta, Tis, T1–T4 marked.

Key points to remember:

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman

Asked as:

- Describe in detail the management of T1G3 TCC of urinary bladder? — *Summer 2019 Paper II Q1 (LAQ)*
- Discuss management of T1G3 bladder tumor — *Practice Set 7 Paper II Q3a (SAQ)*
- Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13 — *Practice Set 4 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 7 Paper II Q3a: also answered in 2019-Summer Paper 2, Q1) (Practice Set 4 Paper II Q1: T1G3 management also answered in 2019-Summer Paper 2, Q1)

Practice Set 4 Paper II Q1: "C13" is read as CIS (carcinoma in situ): T1 N0 M0 urothelial carcinoma with concurrent CIS in a 25-year-old woman.

Introduction and risk category

- T1G3 is high-grade (HG) urothelial carcinoma (the current name for TCC, transitional cell carcinoma) that invades (invading) the lamina propria but not the muscularis propria (not muscle). TCC is now urothelial carcinoma. CIS is flat and always high grade.
- T1 is 20–30% of NMIBC (non-muscle-invasive bladder cancer) and its most dangerous form: recurrence >80%, progression to muscle invasion 30–50%.
- Grade, not stage, drives progression. So every HG T1 is **high risk** (AUA [American Urological Association]/SUO [Society of Urologic Oncology] and EAU [European Association of Urology]).
- **T1 with CIS is high-risk NMIBC** (AUA and EAU). CIS is the second strongest progression factor after grade. Untreated CIS progresses to muscle invasion in 40–83%.
- **EAU 2021 very high risk:** T1 high grade (HG) with CIS plus at least one of age >70, multiple papillary tumours, or diameter ≥3 cm. In this woman (25-year-old), age does not count, so number and size decide. LVI (lymphovascular invasion) and variant histology also make it very high risk.
- **Young woman:** long life expectancy, fertility, pregnancy and sexual function shape every choice.

Risk factors for progression

- **Adverse factors:** concurrent CIS (carcinoma in situ), LVI (lymphovascular invasion), variant histology, size >3 cm, multifocality, prostatic urethral involvement, residual T1 at re-resection.

Factor	Significance
Concurrent CIS (carcinoma in situ)	Second most important factor after grade
Depth of invasion (substaging)	Invasion deep to the muscularis mucosae is worse; T1a/T1b proposed, no agreed method
LVI (lymphovascular invasion)	Suggests occult spread
Variant histology	Micropapillary (any amount), plasmacytoid, sarcomatoid, nested
Size >3 cm, multifocal, recurrent	More residual tumour and progression
Prostatic urethra, diverticulum	Poor BCG contact; diverticulum lacks muscle
Residual T1 at re-TURBT	Progression up to 80%

Factor	Significance
HG T1 at 3 months after BCG	Progression about 82%

- **Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; hydronephrosis suggests muscle invasion.

Evaluation

- Cystoscopy, urine cytology (usually positive in CIS), CT (computed tomography) urography (upper tract tumour <5%, more with CIS), bimanual examination under anaesthesia (including vagina).
- Smoking and occupational history.
- **Pregnancy test** before CT and BCG (bacille Calmette-Guérin); expert pathology review.

Step 1: Complete, good-quality TURBT (transurethral resection of bladder tumour)

- Resect all visible tumour **with detrusor muscle** in the specimen; send the base separately.
- Blue-light cystoscopy or NBI (narrow-band imaging) finds more CIS (**blue-light cystoscopy** finds about one-third more CIS). **En bloc resection** (where possible) keeps orientation for substaging.
- Biopsy the prostatic urethra if tumour is at the bladder neck or trigone, or with CIS.
- Biopsy red velvety areas and the **bladder neck** (relevant to a later neobladder).

Step 2: Re-TURBT (restaging) in all T1

- Within **2–6 weeks** (within 6 weeks), before BCG (bacille Calmette-Guérin).
- **Why:** residual tumour in ≥40%; upstaging to muscle invasion in 25–30% (about half if no muscle in the first specimen). **Residual T1: progression up to 80%.** Re-TURBT improves recurrence- and progression-free survival: Divrik's RCT (randomised controlled trial) showed better recurrence-free and progression-free survival.

```
HG T1 on TURBT → re-TURBT at 2–6 weeks
   ├─ ≥T2                          → neoadjuvant chemotherapy + cystectomy
   ├─ residual T1 / very high risk → offer early radical cystectomy
   └─ T0, Ta or CIS                → BCG induction + 3-year maintenance
```

- Practice Set 7 version: HG T1 → re-TURBT; T0, Ta, CIS, T1 → BCG (bacille Calmette-Guérin); very high risk → offer early radical cystectomy.
- Practice Set 4 version: T1 + CIS → re-TURBT within 6 weeks; T0/Ta/CIS → BCG induction + 3-year maintenance.

Step 3a: BCG – standard for most patients (bladder preservation)

- **Mechanism:** BCG attaches to (binds) **fibronectin**, triggers cytokines (IFN-γ [interferon-gamma], IL [interleukin]-2) and a **Th1 (T-helper 1)** immune (cytokine) response that kills tumour. It is the **only intravesical agent that reduces progression**.
- **Schedule:** start 2–4 weeks after TURBT, never immediately (intravasation and sepsis); induction weekly ×6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** 3 weekly doses at 3 and 6 months, then every 6 months (6-monthly) to **3 years**. Median recurrence-free survival 76.8 vs 35.7 months. **EORTC 30962:** full dose for 3 years in high risk.
- **CIS:** tumour-free up to 84%, about half durable; persistent CIS at 3 months: second course gives a further 30–50% complete response.
- BCG is live: avoid pregnancy and breastfeeding during treatment.

- **Contraindications (avoid with):** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis common; BCG sepsis needs isoniazid, rifampicin, ethambutol and steroids (antituberculous drugs and steroids).

Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Patients who progress under surveillance do worse than those presenting with muscle invasion (progressors do worse than primary muscle-invasive disease). Immediate cystectomy gives about **92% 10-year disease-free survival**.
- **AUA:** offer to fit patients with persistent HG T1 on re-resection, or T1 with CIS, LVI or variant histology.
- **EAU very-high-risk group (2021 model; risk factors: age >70, multiple papillary tumours, diameter ≥3 cm):** T1 HG with CIS plus ≥1 risk factor; T1 HG without CIS but with all three; also LVI, variant histology, CIS of the prostatic urethra (prostatic urethral CIS), persistent T1 at re-TURBT. Offer immediate cystectomy; BCG if declined or unfit.
- **Offer for (Practice Set 4):** very high risk, residual HG T1, LVI, variants, diffuse CIS, or patient preference.
- Also: diverticular tumour, tumour not controllable endoscopically.

Step 3c: Cystectomy tailored to a young woman

- Standard female operation is anterior exenteration (bladder, urethra, anterior vagina, uterus, cervix, ovaries).
- **Organ-sparing** (≤cT2, bladder neck uninvolved): keep the **uterus and ovaries** (fertility, hormones, avoids surgical menopause; ovaries also make testosterone). Keep the **vagina and neurovascular tissue** (vaginal dissection no more dorsal than 2 and 10 o'clock) for sexual function and continence. Oncological safety of sparing is not fully known.
- **Orthotopic neobladder** suits her: needs a negative bladder-neck biopsy and **urethral frozen section** (negative bladder neck means negative urethra; CIS does not predict urethral involvement). Day continence 80–90%; **retention 20–30% in women**, so she must accept CIC (clean intermittent self-catheterisation). See Practice Set 4 Paper 2, Q3e.
- **Fertility:** if uterus or ovaries must go, offer oocyte or embryo cryopreservation first. Pregnancy after a neobladder is possible; caesarean uses a high midline incision to avoid the pouch.

Step 4: BCG failure (BCG-unresponsive disease)

- **Adequate BCG for T1:** ≥5 of 6 induction doses.
- **BCG-unresponsive:** HG T1 at the 3-month check after induction; persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Standard:** timely radical cystectomy (EAU, AUA).
- **Unfit or refusing (declined):** pembrolizumab (KEYNOTE-057: CIS complete response 41%); nadofaragene firadenovec (53%); N-803 with BCG; gemcitabine–docetaxel; TAR-200; trials.
- **Also know:** BCG-intolerant; delayed relapse may respond to BCG rechallenge.

Follow-up (AUA/SUO high risk)

- Bladder preserved: cystoscopy and cytology every 3–4 months (3-monthly) for 2 years, 6-monthly for 2 years, then yearly for life.
- Upper-tract imaging yearly for 2 years (regular upper-tract imaging).
- Positive cytology with normal cystoscopy: check prostatic urethra and upper tracts.

- After neobladder: B12, metabolic panel, upper tracts.

Recent advances

- Checkpoint inhibitors with BCG in BCG-naive disease (trials); urine markers predicting BCG response; TAR-200.

Diagram to draw: bladder wall layers showing CIS, Ta, T1a/T1b (above and below the muscularis mucosae) and T2.

Key points to remember:

- Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

Use of intestinal segments in urology, and orthotopic neobladder

Asked as:

- Use of intestinal segments in urology — *Winter 2023 Paper II Q2 (LAQ)*
- Various uses of intestines in Uro — *Practice Set 17 Paper II Q1 (LAQ)*
- Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder. — *Practice Set 14 Paper II Q2 (LAQ)*
- Neobladder — *Practice Set 4 Paper II Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(also answered as an SAQ in 2022-Summer Paper 1, Q3a) (also answered in 2023-Winter Paper 2, Q2) (related: intestinal segments in 2023-Winter Paper 2, Q2; continent diversion and neobladder in Practice Set 3 Paper 4, Q2) (neobladder also answered in 2020-Summer Paper 4, Q3a)

Read as: the various uses of intestinal segments in urology.

Introduction

- Bowel is used to divert urine (conduit, continent pouch, neobladder), store urine, carry it to the skin, enlarge the bladder (augmentation), replace the ureter, make catheterisable channels and line a neovagina.
- Uses of bowel: conduit (ileal, colon), continent cutaneous pouch (Indiana, Mainz), **orthotopic neobladder**, augmentation cystoplasty, ileal ureter, catheterisable channels (Mitrofanoff, Yang–Monti).
- Choice depends on renal function, previous surgery or radiation, bowel disease, mesenteric length and the reconstruction needed.

Uses at a glance

Use	Segment and operation
Incontinent diversion	**Ileal conduit** (10–15 cm ileum, 10–15 cm from the ileocaecal valve); colon conduit (transverse) after pelvic radiation; jejunal conduit only if nothing else (avoid)
Continent cutaneous pouch	Indiana (right colon, plicated ileum as channel), Mainz I, Kock (ileal nipple valves); Mitrofanoff channel; with appendix or tapered ileum channel
Orthotopic neobladder	Studer (40–44 cm ileum plus afferent limb), Hautmann (~70 cm ileum, W shape), joined to the urethra; about 50–70 cm detubularised ileum
Rectal reservoir	Ureterosigmoidostomy, Mainz II: now rare (acidosis, cancer)
Augmentation cystoplasty	"Clam" ileocystoplasty (20–40 cm ileum; ~25 cm ileum), sigmoid, caecum; gastrocystoplasty in renal failure. For refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation)
Ureteric replacement	**Ileal ureter**, isoperistaltic ileum, for long defects; Yang–Monti tubes
Catheterisable channels	**Mitrofanoff** appendicovesicostomy; **Yang–Monti** ileal tube (1–2 cm ileum retubularised transversely); MACE (Malone antegrade continence enema) for bowel
Neovagina	**Sigmoid** preferred (wide lumen, mobile mesentery, less stenosis); ileum if pelvis irradiated; MRKH (Mayer–Rokitansky–Küster–Hauser), cloaca, gender-affirming vaginoplasty

- Ileal ureter contraindications: creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.

Choice of segment

Segment	Electrolyte disturbance (metabolic effect)	Advantages / use	Drawbacks
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes H^+ and Cl^-; less mucus; usable for renal failure, short gut, dense adhesions	**Haematuria–dysuria syndrome**, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis**	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Default: mobile, compliant, constant (reliable) blood supply; default conduit, ureteric replacement	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis, hypokalaemia	Easy antireflux tunnel; transverse colon outside pelvic radiation field	Diarrhoea if ileocaecal valve used; obstruction ~5%
Appendix	—	Mitrofanoff catheterisable channel	—

Technical (surgical) principles of using bowel

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Assess renal function. Mark the stoma site before surgery when a stoma is possible, through the rectus.
- **Mesentery:** transilluminate, preserve (keep) a major mesenteric arcade, wide base; avoid watershed areas (Griffiths point at splenic flexure, Sudeck point at rectosigmoid).
- **Bowel anastomosis:** good blood supply (well-vascularised), no tension, no spillage, close the mesenteric window.
- **Detubularisation** (Kock): opening the bowel on its antimesenteric border and refolding it into a sphere gives a spherical reservoir. This **abolishes coordinated (peristaltic) contractions** and increases volume (gives more volume) for the same surface area; a **sphere** gives the largest volume for its surface area; by Laplace's law (pressure ∝ wall tension/radius) a larger radius means lower pressure, so pressure stays low.

Ureterointestinal anastomoses

Technique	Type	Stricture
Bricker	Refluxing, end to side	4–22% (mean 6%)
Wallace	Refluxing, ureters joined end to end	~3% (lowest)
Leadbetter–Clarke, Goodwin	Nonrefluxing (antireflux) tunnel in colon	8–14%

- Principles: spatulate (spatulated), preserve periureteric tissue, tension-free, mucosa to mucosa, soft stents (stented; leak 3–5% without). Bring the bowel to the ureter, not the ureter to the bowel.
- Refluxing (Bricker, Wallace; stricture about 3–6%) versus antireflux tunnels (more strictures).
- Fix and retroperitonealise the anastomosis. The left ureter passes behind the sigmoid mesentery; avoid angulation under the IMA (inferior mesenteric artery), a common stricture site.
- Reflux with normal unobstructed ureters has not been shown to harm the adult kidney; antireflux tunnels stricture more. Obstruction, not low-pressure reflux, damages kidneys.

Applications

- **Incontinent diversion:** ileal conduit (10–15 cm ileum, 10–15 cm from ileocaecal valve); colon conduit after pelvic radiation.
- **Continent cutaneous pouch** (Indiana, Mainz, Kock) with appendix or tapered ileum channel.
- **Orthotopic neobladder** (Studer, Hautmann): about 50–70 cm detubularised ileum joined to the urethra.
- **Ureterosigmoidostomy:** now rare (acidosis, cancer).
- **Augmentation cystoplasty** ("clam" ileocystoplasty, ~25 cm ileum) for refractory neurogenic or small contracted bladder; needs CIC.
- **Ileal ureter:** isoperistaltic ileum for long defects. Contraindicated if creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.
- **Catheterisable channels:** Mitrofanoff appendicovesicostomy, Yang–Monti ileal tube.

Stomas

- **Protruding (nipple, "rosebud") stoma** for appliance wearers: less stenosis, better appliance fit. **Flush stoma** for continent pouches that are catheterised (protruding "rosebud" for appliances; flush for catheterised pouches).
- **Loop-end (Turnbull) stoma** for obese patients with a short mesentery. Site is marked before surgery, through the rectus muscle.

Selecting conduit versus continent diversion (patient selection)

- **Renal function needed for continent diversion:** creatinine <2 mg/dL or GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria; normal liver function; ability to self-catheterise.
- **Contraindications to continent diversion:** poor renal or hepatic function, inability or unwillingness to self-catheterise, inflammatory bowel disease or short gut. A neobladder also needs a tumour-free urethra and a competent sphincter.
- A conduit is simplest and fastest; continent options improve body image but carry more reoperations and metabolic problems.

Complications (to plan for)

Metabolic (worse with longer segment, longer dwell (contact) time, poorer renal function)

- **Hyperchloraemic acidosis (ileum, colon):** NH_4^+ (ammonium) replaces Na^+ (sodium) on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate; chlorpromazine or nicotinic acid limit chloride transport.
- **Hypokalaemia:** worse with colon; replace potassium before correcting acidosis (paralysis risk).
- **Jejunal syndrome:** treat with sodium chloride and thiazide. **Gastric alkalosis:** H2 blockers or PPI (proton-pump inhibitor).
- **Altered sensorium:** hyperammonaemia (drain the pouch, lactulose), hypomagnesaemia.
- **Drug reabsorption:** phenytoin, methotrexate; drain pouches during chemotherapy.
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation (delay) in children.
- **Nutrition:** B12 deficiency (give parenteral B12 after 5–7 years of ileal diversion), bile-salt diarrhoea.

Other

- Infection (bacteriuria in about three-quarters of conduits; common; treat pure Proteus or Pseudomonas; treat only symptoms or urea-splitting organisms).
- Mucus.
- **Stones:** colon conduit 3–4%, ileal conduit 10–12%, continent reservoir (reservoirs) ~20% (mucus, staples).
- **Augment perforation (pouch rupture):** surgical emergency; regular CIC prevents it.
- Ureterointestinal stricture (lifelong risk), renal deterioration, bowel obstruction, stomal stenosis and parastomal hernia.
- **Cancer (secondary malignancy):** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years). Neovagina: stenosis, mucus discharge, adenocarcinoma reported.

Orthotopic neobladder

Definition / indication: an **orthotopic** detubularised bowel (usually ileal) reservoir joined to the native urethra after radical cystectomy, when the urethra is tumour-free, to allow **volitional voiding** per urethra without a stoma. 80–90% of men and about 75% of women are medically eligible.

Principles:

1. **Reliable outlet:** intact rhabdosphincter (pudendal nerve), healthy, tumour-free urethra.
2. **Low pressure:** complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area.
3. **Adequate volume (capacity):** ≥500 mL when mature (44 cm double-folded ileum: <200 mL initially (early), **500–600 mL** by 1 year).
4. Unobstructed upper tracts.

Selection / contraindications:

- **Absolute contraindications:** tumour at the **urethral margin** (positive urethral margin, frozen section), urethral stricture, poor renal function (eGFR (estimated glomerular filtration rate) <35–40 mL/min), poor or severe liver disease, inability to self-catheterise (catheterise).
- **Relative:** prostatic stromal invasion (men); bladder-neck or anterior vaginal tumour (women); poor sphincter; prior radiation (irradiated pelvis); frailty. Age, obesity and CIS (carcinoma in situ) alone are not bars.

Neobladder techniques compared (types)

Design	Segment and shape	Ureters	Advantage	Drawback
Studer	40–44 cm ileum, double-folded U + afferent limb	Refluxing, isoperistaltic limb	Simple, no staples, short ureters; stricture <3%	Reflux on cystogram (benign)
Hautmann W	About 70 cm ileum, W	Refluxing (since 1997); largest capacity	Largest early capacity, early continence	More late retention, more absorption
T pouch	44 cm + 15 cm T limb	Flap-valve antireflux (antireflux flap valve)	Accepts dilated or short ureters	More reoperations than Studer
Abol-Enein	40 cm ileum, W	Serous-lined tunnel	Durable antireflux	Needs long, non-dilated ureters
Camey II	65 cm ileum, U	Le Duc tunnel (Le Duc antireflux)	Early detubularised design	Le Duc strictures 7–29%

Design	Segment and shape	Ureters	Advantage	Drawback
Hemi-Kock	61 cm ileum	Intussuscepted nipple	Good early results	Nipple stenosis, staple stones
Padua (VIP)	About 42 cm ileum, funnel	Le Duc (Le Duc antireflux)	Anatomical funnel; robotic	—
Mainz III	10–15 cm caecum + 20–30 cm ileum (ileocaecal)	Submucosal tunnel (tunnelled)	Good compliance	Loses ileocaecal valve
Sigmoid (Reddy)	35 cm sigmoid (colon), U	Tunnelled	Spares ileum (short bowel)	Poor compliance, urgency

- **Hautmann vs Studer:** larger initial capacity gives earlier continence but more late retention and absorption.
- **Studer steps (technique, brief):** isolate ileum (ileum taken) 15–20 cm proximal to the ileocaecal valve; open distal 40–44 cm antimesenterically; sew into a U and fold into a sphere; stented ureters end-to-side (ureteric anastomoses) to the afferent limb; dependent point to urethra.
- **USC (University of Southern California) RCT (randomised controlled trial):** Studer vs T pouch, no difference in renal function at 3 years; T pouch needed more secondary procedures.

Neobladder outcomes

Outcome	Result
Day continence	80–90% (3–6 months)
Night incontinence	20–50% (no sphincter reflex, mucosal water secretion)
Retention needing CIC (clean intermittent catheterisation)	Men 4–10%; women 20–30%: men leak, women fail to empty
Urethral recurrence (men)	About 7% at 5 years

Neobladder complications

- **Early:** urine leak, ileus, infection, mucus plugging.
- **Late: hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, incisional hernia, urethral recurrence.

Care and follow-up

- Neobladder: mucus irrigation, bicarbonate; voiding training (sit, relax, strain), timed voiding, night alarm. Annual metabolic panel, B12, upper tracts; urethral cytology if CIS.
- Lifelong: electrolytes, bicarbonate, renal function, B12, upper-tract imaging (ultrasound or CT) for stricture and stones; endoscopy of colonic or augmented segments as above.

Recent advances

- Robotic intracorporeal diversion (complications similar to open in experienced hands); robotic intracorporeal neobladder comparable to open; tissue-engineered bladder (experimental) and alternatives to bowel (ureterocystoplasty, autoaugmentation) remain limited.
- **Also know:** neobladder uptake is low despite 80–90% of men being eligible; lifelong follow-up of electrolytes, B12, upper tracts.

Diagram to draw: ileal conduit with mesenteric arcade and Wallace anastomosis.

Diagram to draw: Studer (U-)pouch with afferent limb, ureters and urethral anastomosis.

Key points to remember:

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Classification of bladder cancer and bladder-preserving treatment of TCC

Asked as:

- Classify CA bladder. Discuss in details organ preserving surgeries for TCC of bladder. — *Practice Set 10 Paper II Q2 (LAQ)*
- Bladder preserving approach in invasive TCC of bladder — *Practice Set 5 Paper II Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2019-Summer Paper 4, Q2)

Read as: classification of bladder cancer, and bladder-preserving treatment of TCC (transitional cell carcinoma, urothelial carcinoma).

A. Classification

1. Histological

Type	Share	Note
Urothelial carcinoma	~90%	Variants: micropapillary, plasmacytoid, sarcomatoid, nested; squamous or glandular differentiation
Squamous cell carcinoma	~5%	Schistosomiasis, catheter, stones
Adenocarcinoma	~1–2%	Urachal (dome), exstrophy
Small cell / others	<1%	Neuroendocrine; sarcoma

2. Grade (WHO 2004/2016, World Health Organization)

- PUNLMP (papillary urothelial neoplasm of low malignant potential), low-grade and high-grade carcinoma. WHO 1973 (G1–G3) is still reported by some.
- Low-grade (FGFR3 pathway) recurs but rarely progresses. High-grade and CIS (carcinoma in situ; TP53/RB pathway) progress.

3. Stage: TNM 2017

T	Definition
Ta / Tis	Non-invasive papillary / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall

- N1 one true-pelvic node; N2 ≥2 pelvic nodes; N3 common iliac; M1a distant nodes; M1b organs.

4. Clinical groups

- **NMIBC (non-muscle-invasive, ~70%)**: Ta, Tis, T1. Risk groups: low (solitary primary low-grade Ta <3 cm), intermediate, high (high-grade, T1 or CIS).
- **MIBC (muscle-invasive)**: T2–T4a. Metastatic.
- **Molecular:** luminal papillary versus basal/squamous subtypes.

B. Organ-preserving surgery

Why (introduction): Standard for MIBC (muscle-invasive bladder cancer) is cisplatin NAC (neoadjuvant chemotherapy) plus RC (radical cystectomy). RC is morbid and patients are

elderly. Bladder preservation suits **unfit** patients and **fit, motivated patients with limited disease**. Single modalities fail because they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%, occult nodes about 25%).

Options:

Option	Use
Radical TURBT (transurethral resection of bladder tumour) alone	Not a primary treatment
Partial cystectomy + PLND (pelvic lymph node dissection)	Solitary tumour **<3 cm** in mobile bladder (dome), **no CIS (carcinoma in situ)**, 1–2 cm margin, frozen sections; **urachal adenocarcinoma**
RT (radiotherapy) or chemotherapy alone	Inferior
TMT (trimodality therapy)	Best studied, guideline-accepted

1. TURBT (transurethral resection of bladder tumour) for NMIBC: the standard

- Complete resection with **detrusor muscle in the specimen**; separate base biopsies; cold-cup biopsy of suspicious areas.
- **En bloc resection** (loop or laser) keeps architecture, gives more detrusor and fewer perforations.
- **Repeat TURBT within 6 weeks** for all T1, incomplete resection or no muscle. Residual tumour in ≥40%.
- Then intravesical chemotherapy or BCG (bacillus Calmette–Guérin) by risk group.

2. Radical TURBT for MIBC

- Only for tumour in superficial muscle with a negative re-resection; about 10% are pT0 at cystectomy.
- About a third later develop MIBC. Not a primary treatment.

3. Partial cystectomy

Ideal candidate	Contraindication
Solitary tumour **<3 cm** in a mobile part (dome)	**CIS**, multifocal tumour
No hydronephrosis	Trigone or bladder neck tumour
1–2 cm margin possible; adequate capacity left	Small bladder; prostatic urethral involvement
Urachal adenocarcinoma (treatment of choice)	

Steps: NAC (neoadjuvant cisplatin chemotherapy) if eligible → PLND (pelvic lymph node dissection) → cystotomy away from tumour → full-thickness excision with perivesical fat → **frozen-section margins** → ureteric reimplantation if needed → watertight closure → drain. For urachal cancer, en bloc dome, urachus and umbilicus.

Outcomes: 5-year OS (overall survival) 67–70%; intact bladder 65–74%; salvage RC up to a quarter. Late recurrence, so lifelong follow-up.

4. TMT (trimodality therapy): the best-studied option

- **Components:** (1) maximal **visibly complete TURBT**; (2) RT (radiotherapy) to bladder ± nodes; (3) concurrent **radiosensitising chemotherapy**.
- **Selection:** cT2, solitary (small solitary tumour), **no hydronephrosis, no extensive CIS (no CIS)**, complete TURBT, good bladder function, compliant patient. Only 6–19% of MIBC qualify.

Radiosensitisers	Evidence
Mitomycin C + 5-FU (fluorouracil)	**BC2001:** 2-year locoregional DFS (disease-free survival) 67% vs 48% for RT alone
Cisplatin-based	MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols
Gemcitabine; carbogen + nicotinamide	Cisplatin-ineligible; hypoxia modification

- **Dose:** 64–66 Gy in 32 fractions or **55 Gy in 20** (non-inferior).

```
TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr → cystoscopy + biopsy at 3–4 months
     ├─ Complete response → surveillance (NMIBC recurrence: TURBT + BCG)
     └─ Persistent / recurrent MIBC (invasive) → salvage cystectomy (10–30%)
```

- **Salvage cystectomy** in **10–30%**: the designed safety net. Non-muscle-invasive (NMIBC) recurrence: TURBT and intravesical BCG (bacillus Calmette–Guérin).
- **Outcomes:** 5-year CSS (cancer-specific survival) 50–84%. MGH/RTOG: complete response 69%; 80% of 5-year survivors keep the bladder. Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and similar CSS (81% vs 84%). No completed randomised trial (SPARE closed).

Comparison

	Radical cystectomy	Bladder preservation
Staging	Pathological	Clinical only
Function	Diversion, sexual dysfunction	Native bladder; RT toxicity
Follow-up	Upper tract, urethra	**Cystoscopy 3-monthly for 2 years, to 10 years**, plus CT

- **Follow-up (bladder preservation):** Cystoscopy every 3–4 months for 2 years, then less often, for 10 years; CT (computed tomography) urogram and chest imaging; cytology.

Recent advances

- Checkpoint inhibitors with chemoradiation (trials ongoing; under trial: S1806, KEYNOTE-992); bladder sparing after complete response to NAC (RETAIN: about half recurred); robotic partial cystectomy.
- **Also know:** RT alone is quasi-palliative; chemotherapy alone not recommended.

Diagram to draw: TNM T-stage diagram of the bladder wall with a TMT flowchart (as above).

Key points to remember:

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; T2, no hydronephrosis, no CIS.
- TMT: BC2001 MMC/5-FU (mitomycin C/5-FU) + RT, DFS 67% vs 48%; salvage cystectomy 10–30%.

Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction.

Type: LAQ (25 marks)

Source: Practice Set 8 Paper 2, Q2 (principles also answered in Practice Set 3 Paper 4, Q2)

Assumed meaning: GI = gastrointestinal (bowel) reconstruction, i.e. antegrade continence enema channels.

Introduction

- A continent diversion empties at will: by voiding through the urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous**).
- **Continence is a pressure balance:** reservoir pressure must stay below outlet resistance. Failure of either side causes leakage.

Principles of continence

A. Low-pressure, adequate-capacity reservoir

- **Detubularisation** (opening bowel on its antimesenteric border) and cross-folding break up coordinated peristaltic contractions.
- **Spherical shape:** greatest volume for a given surface area. By **Laplace's law** ($P = 2T/R$), a larger radius gives lower pressure for the same wall tension.
- **Capacity ≥500 mL** when mature. Bowel stretches with time (ileum more than colon); 44 cm ileum holds <200 mL at first and 500–600 mL within a year.
- Ileum is most compliant; colon has pressure waves.

B. Reliable outlet

- **Orthotopic:** intact **rhabdosphincter** (pudendal nerve), healthy urethra, careful apical dissection, nerve sparing. Voiding is by Valsalva and pelvic floor relaxation. Night leakage (20–50%) reflects loss of the guarding reflex.
- **Continent cutaneous:** an efferent channel that is easy to catheterise and has a durable continence mechanism.

Mechanism	Example	How it works	Weakness
Nipple valve (intussuscepted)	Kock, Mainz I	Reservoir pressure compresses the intussuscepted segment	Same pressure effaces it; failure ≥15%, reoperation about one-third
Flap valve	**Mitrofanoff**, Yang–Monti, Kropp	Narrow tube in a submucosal tunnel is coapted as pressure rises	No pop-off; stomal stenosis
Imbricated ileocaecal valve + tapered ileum	Indiana pouch	Continence zone lies at the plicated valve	Bowel frequency, B12 loss

- **Also know** (native outlet): Kropp, Pippi Salle, slings, AUS (artificial urinary sphincter), bulking, bladder neck closure.

C. Protect the upper tracts

- Unobstructed ureterointestinal anastomosis; eGFR (estimated glomerular filtration rate) about ≥35–40 mL/min for the acid load; ability to self-catheterise.

The Mitrofanoff principle

- **Mitrofanoff (1980):** any supple, narrow tube implanted in a **submucosal tunnel** of a low-pressure reservoir, with its other end at skin, gives a **continent catheterisable channel**.
- As the reservoir fills, pressure transmitted through the mucosa flattens the tunnelled segment against the muscle backing: a **flap valve**. Continence lies in the tunnel, not in the conduit.
- Appendix: only a **2 cm tunnel** suffices because of its small calibre.

Conduits

Conduit	Points
Appendix (first choice)	Natural tube, minimal harvest morbidity; needs length ≥5–6 cm and ≥12 Fr; caecal cuff or **Mitchell caecal extension** adds length
Yang–Monti	2–3 cm ileum opened on the antimesenteric border and retubularised transversely over 12 Fr; double or spiral Monti for obese; twice the revisions
Tapered ileum	Long enough; kinks if redundant
Ureter, fallopian tube	More stenosis
Casale bladder flap	No bowel; consumes bladder capacity

Technique (appendicovesicostomy)

1. Mobilise caecum; detach appendix with a caecal cuff, preserving mesoappendix.
2. Spatulate tip; tunnel submucosally (posterolateral or anterior bladder wall), intra- or extravesically.
3. Bring the base to the umbilicus or right iliac fossa with skin flaps, without tension or twist; keep it short and straight; fix the bladder to the abdominal wall.
4. Catheterise after each step. Channel catheter 2–4 weeks; then CIC every 4 hours by day.

Applications in urinary reconstruction

- **Neurogenic bladder** (spina bifida, spinal injury), often with augmentation, when urethral CIC is difficult (wheelchair-bound, sensate urethra).
- After bladder neck reconstruction or closure; exstrophy; valve bladder.
- **Continent cutaneous diversion** after cystectomy: Penn and right colon pouch (appendix in taenia; up to 100% continence); Yang–Monti if no appendix.
- Unusable urethra: complex stricture, urethrectomy.

Applications in GI reconstruction: MACE

- **MACE (Malone antegrade continence enema; Malone, 1990):** appendix brought to skin as an appendicocaecostomy with an antireflux tunnel (flap valve) so stool does not leak.
- Antegrade washouts empty the colon, giving **faecal continence** in neurogenic bowel (spina bifida), anorectal malformations, cloaca and refractory constipation.
- Options: Monti-MACE; **split appendix** for both MACE and urinary channel; caecostomy button.

Results and complications

- **Continence >95%** for flap valves; nipple valves about 85%.
- **Stomal stenosis is the commonest problem**; also false passage, channel polyps or granulation, pouch stones, revision (suprafascial or subfascial).
- Missed CIC: no pop-off, so **reservoir rupture** or upper tract damage.

Diagram to draw: flap valve (tunnelled appendix collapsing as pressure rises) beside a nipple valve.

Key points to remember:

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Continence >95%; stomal stenosis is the commonest complication.

Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types

Asked as:

- Outline management of muscle invasive TCC. give critical evaluation of various types of neobladder. — *Practice Set 12 Paper II Q2 (LAQ)*
- Outline the management of muscle invasive TCC. Discuss the role of adjuvant & neoadjuvant chemotherapy. — *Practice Set 16 Paper II Q1 (LAQ)*

Type: LAQ (25 marks)

(Practice Set 12 Paper II Q2: neobladder also answered in Practice Set 4 Paper 2, Q3e and Practice Set 3 Paper 4, Q2) (Practice Set 16 Paper II Q1: MIBC management also answered in Practice Set 12 Paper 2, Q2; neoadjuvant therapy in 2025-Winter Paper 4, Q1)

Introduction

- MIBC (muscle-invasive bladder cancer; TCC, transitional cell carcinoma, now called urothelial carcinoma) is cT2–T4a disease.
- Nearly half of patients treated by RC (radical cystectomy) alone develop metastases from occult micrometastases, so cure needs local plus systemic therapy, planned by an MDT (multidisciplinary team).

Staging

- Complete TURBT / Re-TURBT (transurethral resection of bladder tumour) with (detrusor) muscle; **MRI (magnetic resonance imaging) with VI-RADS** (Vesical Imaging-Reporting and Data System; ≥3 suggests invasion); CT (computed tomography) urogram and chest CT; bone scan if symptoms (symptomatic).
- Assess fitness and cisplatin eligibility.

Management algorithm

```
cT2–T4a N0 M0
 ├─ Cisplatin-eligible → cisplatin NAC (neoadjuvant chemotherapy)
 │      → RC + bilateral PLND (pelvic lymph node dissection)
 ├─ Cisplatin-ineligible → RC + PLND directly
 │      (pT3–4 or N+ after either → adjuvant nivolumab or chemotherapy)
 ├─ Selected, motivated → trimodality therapy (TMT)
 └─ Unfit → chemoradiation / palliative radiotherapy
cN+ → induction chemotherapy → consolidation RC in responders
M1  → systemic therapy
```

1. Cystectomy pathway: radical cystectomy (standard of care for cT2–T4a N0 M0)

- **Cisplatin-based NAC (neoadjuvant chemotherapy):** ddMVAC (dose-dense methotrexate, vinblastine, doxorubicin, cisplatin) or GC (gemcitabine–cisplatin) ×4 (details below).
- **Cisplatin-ineligible** → direct cystectomy.
- **RC (radical cystectomy) + standard bilateral PLND (pelvic lymph node dissection):** obturator, external and internal iliac nodes. **SWOG S1011:** extended PLND gave no survival gain (benefit).
- Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases.
- Within 12 weeks of diagnosis: do not delay beyond **12 weeks** from diagnosis (worse survival). Diversion: ileal conduit, orthotopic neobladder or continent pouch.

- **Adjuvant:** cisplatin chemotherapy if no NAC; **nivolumab** for pT3–4 or N+ (**CheckMate 274:** disease-free survival 20.8 vs 10.8 months) (details below).

2. Bladder preservation: TMT (trimodality therapy)

- Maximal (visibly complete) TURBT + RT (radiotherapy) (64 Gy, or 55 Gy in 20 fractions) + radiosensitiser. **BC2001:** mitomycin C + 5-FU (fluorouracil) improved 2-year locoregional disease-free survival (67% vs 48%).
- Best candidates (select): cT2, solitary tumour, **no CIS (carcinoma in situ), no hydronephrosis**, good bladder function.
- 5-year cancer-specific survival 50–84%; salvage cystectomy 10–30%.
- **Partial cystectomy:** solitary dome or urachal tumour, no CIS.
- **Also know:** chemotherapy alone or radical TURBT alone is not recommended.

3. Metastatic disease

- First line: **EV (enfortumab vedotin) + pembrolizumab** (EV-302: about doubled overall survival (OS) vs platinum). Alternatives: platinum chemotherapy then **avelumab maintenance** (JAVELIN Bladder 100); erdafitinib for FGFR / FGFR3 (fibroblast growth factor receptor 3) alterations.

Neoadjuvant chemotherapy (NAC)

Rationale

- Treats micrometastases early, while tumour burden is low, and is better tolerated before surgery.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

Eligibility: cT2–T4a N0 M0 and cisplatin-fit. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, grade ≥2 hearing loss or neuropathy, NYHA (New York Heart Association) class III–IV failure → direct cystectomy.

Trial	Regimen	Result
SWOG 8710	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC	pT0 38% vs 15%; median survival 77 vs 46 months; P = 0.06
BA06 30894 (EORTC/MRC)	CMV (cisplatin, methotrexate, vinblastine) ×3	16% relative reduction in mortality; 10-year OS (overall survival) 36% vs 30%
ABC meta-analysis (Advanced Bladder Cancer)	Cisplatin combinations	OS HR (hazard ratio) 0.86: **5% absolute 5-year survival gain**
VESPER	ddMVAC (dose-dense MVAC) ×6 vs GC (gemcitabine–cisplatin) ×4	ddMVAC better PFS than GC: organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity
NIAGARA	Perioperative durvalumab + GC, then adjuvant durvalumab, vs GC	Improved (better) event-free survival (HR 0.68) and OS (HR 0.75)

- **Regimens:** ddMVAC 3–6 cycles or GC 4 cycles (EAU, AUA, NCCN: European Association of Urology, American Urological Association, National Comprehensive Cancer Network).
- **No benefit:** single-agent cisplatin, carboplatin regimens, neoadjuvant radiotherapy. Cisplatin-ineligible patients go straight to RC.
- The chapter calls neoadjuvant immunotherapy unproven; NIAGARA has since made perioperative durvalumab + GC an option.

Adjuvant therapy

	Neoadjuvant	Adjuvant
Selection	Clinical stage (understaging risk)	True pathological stage, but still overtreats
Delivery	Better tolerated	Post-RC morbidity; a quarter never receive it
Evidence	Level 1	Weaker: small, early-closed trials

Adjuvant chemotherapy

- Trials were small or closed early. **ABC 2022 update:** cisplatin-based adjuvant chemotherapy improves 5-year OS by about 6%. Benefit greatest in node-positive and extravesical disease.
- Use: cisplatin-naive patients (if no NAC) with **pT3–4 or N+**: ddMVAC or GC.

Adjuvant immunotherapy

- **CheckMate 274:** nivolumab for 1 year after RC (pT3–4a or N+, with or without prior NAC): median disease-free survival **20.8 vs 10.8 months**.
- **IMvigor010** (atezolizumab): negative overall (included pT2); benefit in ctDNA (circulating tumour DNA)-positive patients.
- Use nivolumab if cisplatin-ineligible or declining chemotherapy, or with ypT3 or worse after NAC.
- **Also know:** adjuvant pelvic radiotherapy (Cairo trials) improves pelvic control in pT3–4; ctDNA-guided therapy is emerging.

B. Urinary diversion options

- Ileal conduit (simplest, commonest), continent cutaneous pouch, or **orthotopic neobladder**. No diversion is proven superior for QoL (quality of life).

C. Neobladder: principles

1. **Reliable outlet:** intact rhabdosphincter, tumour-free urethra (frozen section).
2. **Low pressure:** complete **detubularisation** and cross-folding abolish peristalsis; a **sphere** gives most volume (Laplace).
3. **Capacity:** 44 cm of folded ileum holds <200 mL at first, **500–600 mL by 1 year**.

- **Contraindications:** positive urethral margin, stricture, eGFR (estimated GFR) <35–40 mL/min, liver failure, inability to self-catheterise. Age, obesity and nodes are not bars.

D. Critical evaluation of neobladder types

Type	Bowel; ureteric technique	Strengths	Weaknesses
Studer	40–44 cm ileum + isoperistaltic afferent limb; refluxing	Simple, no staples, takes short ureters; stricture <3%; good long-term function	Night incontinence; reflux on cystogram (harmless)
Hautmann W	About 70 cm ileum; refluxing since 1997	Largest capacity, early continence; stricture fell 9.5% → 1%	More retention, acidosis, B12 loss
T pouch (Stein–Skinner)	44 cm + tunnelled T-limb flap valve	Antireflux; accepts dilated ureters	Complex; more secondary procedures (USC [University of Southern California] RCT)
Abol-Enein serous tunnel	40 cm W, serosal troughs	Antireflux; stricture about 3.8%	Long ureters needed; no dilated ureters

Type	Bowel; ureteric technique	Strengths	Weaknesses
Hemi-Kock	Intussuscepted nipple	Good early results	Late nipple stenosis, stones; abandoned
Camey II; Padua (VIP)	65 cm or 42 cm ileum; Le Duc tunnel	Early designs	Le Duc stricture 7–29%
Mainz III; right colon; sigmoid (Reddy)	Colon ± ileum	When ileum short or diseased	High-pressure contractions, poor compliance, more leakage

Critical points

- **Reflux vs obstruction:** Valsalva voiding pressurises the whole system equally, so reflux is benign; every antireflux valve can obstruct. **USC RCT** (484 patients): Studer = T pouch for renal function and infection; T pouch needed more secondary procedures. Hence **Studer and Hautmann** are the most popular.
- **Segment:** ileum is most compliant; colon and stomach give high pressures.
- **Capacity trade-off:** a larger pouch gives earlier continence but more retention and metabolic load.
- **Robotic intracorporeal neobladder:** comparable outcomes; long-term data awaited.

E. Neobladder outcomes and complications

- **Day continence 80–90%; night 20–50%.** Retention: men 4–10%, **women 20–30%**.
- **Late:** hyperchloraemic metabolic acidosis, B12 deficiency, mucus, stones, ureteroileal stricture, pouch rupture, urethral recurrence (men about 7% at 5 years).
- **Follow-up:** blood gas, B12, upper-tract imaging, urethral cytology if CIS.

Diagram to draw: bladder wall T-staging (T2a, T2b, T3, T4) beside the standard PLND template; Studer pouch with afferent limb and urethral anastomosis.

Key points to remember:

- NAC: cisplatin combinations only; ABC 5% absolute survival gain; SWOG 8710 pT0 38% vs 15%.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab (adds durvalumab) improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Section 12 · Bladder Cancer & Diversion — Short answer questions (SAQ)

Bladder diverticulum.

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3c

Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

Classification

Feature	Congenital (primary)	Acquired (secondary)
Age	Children (<10 years)	Men >60 years
Cause	Weak detrusor at the ureteric hiatus	**BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder
Number, site	Solitary; **paraureteral (Hutch)**	Often multiple, anywhere
Bladder wall	Smooth	Trabeculated
Associations	VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes	Stones, cancer

- **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum.

Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.
- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.

- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.
- **Tumour inside**: partial or radical cystectomy.

Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.
- Complications: ureteric injury, leak.

Diagram to draw: paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

Intravesical gene therapy.

Type: SAQ (10 marks)

Source: 2022-Summer Paper 2, Q3a

Concept

- A gene is instilled into the bladder so that urothelial and tumour cells make an anticancer product, or the vector itself kills them.
- **Why the bladder suits it:** direct contact, little systemic exposure, response checked by cystoscopy and cytology.
- **Barrier:** the glycosaminoglycan layer resists vectors → **transduction enhancers** (Syn3; DDM, n-dodecyl-β-D-maltoside).
- **Indication:** high-risk **BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer)** in patients unfit for or declining radical cystectomy. BCG-unresponsive: persistent or recurrent CIS (carcinoma in situ) or high-grade Ta/T1 after adequate BCG, or relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.

Agents

Agent	Vector, mechanism	Key data
Nadofaragene firadenovec	**Replication-incompetent adenovirus type 5** carrying the **interferon-α2b gene**, with **Syn3** surfactant. Urothelium keeps secreting interferon → apoptosis (TRAIL, TNF-related apoptosis-inducing ligand) and immunity, overcoming the short dwell of interferon protein	75 mL every **3 months**, 1-hour dwell. CIS: **complete response 53% at 3 months, 24% at 12 months**; papillary high-grade RFS (recurrence-free survival) 44% at 12 months; serious events 2%. **FDA (Food and Drug Administration) approved December 2022**, first bladder gene therapy
Cretostimogene grenadenorepvec (CG0070)	**Replication-competent oncolytic adenovirus**: E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity	Two 6-week inductions + maintenance; single-arm phase III BOND-003 in BCG-unresponsive CIS: high CR (complete response) rate; combination with pembrolizumab under study
Inodiftagene vixteplasmid (BC-819)	Plasmid: **diphtheria toxin A** gene under tumour-active **H19** promoter	Investigational

Administration and toxicity

- Catheter instillation, patient turned during dwell; biosafety precautions.
- Mostly grade 1–2: **bladder spasm, urgency, discharge around catheter, fatigue**.

Place in practice

- Alternatives: radical cystectomy (standard), **pembrolizumab** (KEYNOTE-057), gemcitabine–docetaxel, **N-803** (interleukin-15 superagonist) + BCG.
- Limits: responses wane (median CR duration about 10 months); single-arm trials; cost.

Also know: non-viral IL-12 (interleukin-12) plasmid (EG-70); early adenoviral attempts failed through poor transduction.

Diagram to draw: adenovirus transducing a urothelial cell → interferon secretion → tumour apoptosis.

Key points to remember:

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

BCG failure

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 2, Q3a (BCG therapy also answered in 2025-Winter Paper 1, Q3f)

Introduction

- BCG (bacille Calmette–Guérin) is standard for high-risk NMIBC (non-muscle-invasive bladder cancer). "Failure" was poorly defined; consensus categories now guide treatment.

Definitions

Category	Definition	Implication
Adequate BCG	≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction; for T1, ≥5 of 6 induction alone	Threshold for "unresponsive"
BCG-unresponsive	Persistent or recurrent CIS (carcinoma in situ) or HG (high-grade) Ta/T1 despite adequate BCG; or HG relapse within **6 months (Ta/T1) or 12 months (CIS)** of last dose; **HG T1 at first 3-month check**	More BCG futile; cystectomy
BCG-resistant	CIS or HG Ta at 3 months after induction	Reinduction: 30–50% respond
Delayed relapse	HG relapse beyond these windows	Rechallenge may work
BCG-intolerant	Toxicity prevented adequate BCG	Salvage chemotherapy

- **Why BCG fails:** inadequate dosing; PD-L1 (programmed death-ligand 1) upregulation on tumour cells.

Evaluation

- Cystoscopy, cytology, **re-TURBT (transurethral resection of bladder tumour)** and biopsies to exclude invasion.
- **Extravesical disease:** prostatic urethral biopsy (24–30%) and CT urography (upper tract 0.8–25%).

Management

1. **Timely radical cystectomy** (EAU [European Association of Urology], AUA [American Urological Association]): HG T1 at 3 months progresses in 82%; CIS failure in about 50%.
2. **Bladder sparing** (unfit or refusing):

Option	Key result (CIS complete response, CR)
Pembrolizumab IV (KEYNOTE-057)	41% at 3 months, 19% at 12 months; grade 3–4 toxicity 13%
Nadofaragene firadenovec (adenovirus carrying IFN-α2b [interferon alpha-2b] gene) every 3 months	53% at 3 months, 24% at 12 months
Gemcitabine–docetaxel sequential	60% at 1 year; 84% avoid cystectomy (retrospective)
N-803 + BCG (IL-15 [interleukin-15] superagonist)	45% at 12 months

- **Also know:** BCG-refractory (older term); gemcitabine, gemcitabine–mitomycin, valrubicin, chemohyperthermia, EMDA (electromotive drug administration), chemoradiation; TAR-200; cretostimogene.

Diagram to draw: flowchart: HG recurrence after BCG → adequate? → unresponsive → cystectomy or bladder sparing.

Key points to remember:

- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

Intravesical therapy for TCC of bladder

Type: SAQ (10 marks)

Source: Practice Set 8 Paper 2, Q3f (also answered in 2017-Summer Paper 4, Q1)

Introduction

- For NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) after complete TURBT (transurethral resection of bladder tumour). Problems: recurrence and progression.
- **Chemotherapy reduces recurrence; only BCG (bacille Calmette–Guérin) also reduces progression.**

Risk-adapted use

Risk	Therapy
Low (solitary low-grade Ta ≤3 cm)	Single post-op instillation
Intermediate	Chemotherapy or BCG induction + 1-year maintenance
High (HG [high-grade] T1, CIS, multifocal HG)	BCG induction + 3-year maintenance

1. Single immediate instillation

- Kills floating cells before implantation, so give **within 6 hours** (not at 24 hours). MMC (mitomycin C) 40 mg, epirubicin or **gemcitabine 2 g (SWOG S0337)**.
- Avoid after perforation. **Never BCG** immediately (sepsis).

2. Adjuvant chemotherapy

- MMC weekly ×6, optimised (empty bladder, fluid restriction, alkalinised urine). Gemcitabine; **Gem/Doce** (gemcitabine–docetaxel) for BCG shortage or unresponsive disease.

3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin** → cytokines → **Th1 (T-helper 1)** response kills tumour.
- Start 2–4 weeks after TURBT; retain 1–2 hours.
- **SWOG schedule:** 6 weekly induction, then 3 weekly doses at 3, 6 months and every 6 months for 3 years. **NIMBUS:** reduced frequency inferior.
- Contraindications: gross haematuria, traumatic catheterisation, active UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis (symptomatic); persistent fever: isoniazid + rifampicin; sepsis (0.5%): stop BCG, triple therapy (not pyrazinamide) ± steroids.

BCG-unresponsive disease

- Recurrent or persistent HG disease despite **adequate BCG** (≥5 of 6 induction + ≥2 of 3 maintenance).
- **Timely radical cystectomy** is standard.
- Unfit or refusing: **pembrolizumab** (KEYNOTE-057; CIS complete response 41%), **nadofaragene firadenovec**, **N-803 + BCG**, Gem/Doce, TAR-200 device.

Device-assisted

- EMDA (electromotive drug administration) MMC; chemohyperthermia (RITE, HIVEC); UGN-102 gel.

Diagram to draw: BCG mechanism and SWOG schedule timeline.

Key points to remember:

- Single instillation within 6 hours; never BCG.
- Only maintenance BCG reduces progression.
- SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab 41% CR.

What is optical coherence tomography. What is its potential role in management of CA bladder?

Type: SAQ (10 marks)

Source: Practice Set 9 Paper 2, Q3c (also answered in Practice Set 5 Paper 4, Q3c and 2017-Summer Paper 4, Q3c)

What is OCT (optical coherence tomography)?

- An **"optical biopsy"** technique: the **optical analogue of B-mode ultrasound**. It measures the echo delay of back-scattered **near-infrared light**, using low-coherence interferometry.
- Gives real-time cross-sectional images with about 10–20 μm resolution but only **1–2 mm depth**. First described in 1991 (ophthalmology).
- **Technique:** a fibre-optic probe passes through the cystoscope working channel and is placed **in contact** with the lesion found by WLC (white-light cystoscopy) or blue light.

Image interpretation

Tissue	OCT appearance
Normal	Three layers: thin low-scatter urothelium, bright lamina propria, darker muscularis
Inflammation	Layers preserved but thickened
Ta papillary tumour	Thick urothelium; lamina propria boundary intact
T1 tumour	**Loss of the urothelium–lamina propria boundary**

Potential role in carcinoma bladder

1. **Diagnosis:** separates benign from malignant flat lesions in real time; fewer unnecessary biopsies.
2. **Staging aid:** suggests Ta versus T1 during TURBT (transurethral resection of bladder tumour).
3. **Resection margins:** checks completeness of TURBT.
4. **Surveillance** of NMIBC (non-muscle-invasive bladder cancer), including after BCG (bacille Calmette–Guérin).
5. **Upper tract:** through the ureteroscope, OCT correlates well with final grade and stage of UTUC (upper tract urothelial carcinoma).

Evidence and limitations

- Small studies: **high sensitivity, lower specificity**. Inflammation, scar and post-BCG change mimic tumour; adding OCT to blue light improves specificity.
- Small field; too shallow for muscle invasion; cannot grade; cost; learning curve. **Investigational.**

Related techniques

- **Blue-light (hexaminolevulinate) cystoscopy:** detects more CIS (carcinoma in situ).
- **NBI (narrow-band imaging):** 415/540 nm light absorbed by haemoglobin.
- **CLE (confocal laser endomicroscopy):** cellular images after fluorescein.

Diagram to draw: OCT image of three bladder layers beside a tumour with loss of layering.

Key points to remember:

- Near-infrared "ultrasound with light"; depth 1–2 mm.
- Loss of layering suggests tumour; boundary loss suggests T1.
- Sensitive but false positives from inflammation; investigational.

Von Bruns Nest

Type: SAQ (10 marks)

Source: Practice Set 10 Paper 2, Q3e

Read as: von Brunn nests of the urothelium.

Definition

- **Von Brunn nests** are solid nests of benign urothelial cells lying in the **lamina propria**. They form when the basal layer of the surface urothelium buds downward and the buds pinch off.
- A common normal or reactive finding in most adult bladders, especially at the **trigone and bladder neck**; also in the ureter and renal pelvis.

Pathogenesis: the proliferative cystitis spectrum

```
Chronic irritation (infection, stones, catheter, outlet obstruction)
   → urothelial buds → von Brunn nests (solid)
   → central degeneration and fluid → cystitis cystica (cysts)
   → glandular (columnar / goblet cell) metaplasia → cystitis glandularis
```

Lesion	Histology	Malignant potential
Von Brunn nests	Solid urothelial nests, lamina propria	None
Cystitis cystica	Nests with central cyst, urothelial lining	None
Cystitis glandularis (typical)	Cuboidal or columnar lining	Low
Cystitis glandularis, **intestinal type**	Goblet cells, mucin	Possible link to **adenocarcinoma**

Clinical features

- Usually incidental. When florid: irritative symptoms, haematuria, infection; cystoscopy shows submucosal nodules or cysts that can mimic tumour.
- **Associations:** chronic inflammation, outlet obstruction, **pelvic lipomatosis** (florid cystitis glandularis), bladder exstrophy.

Differential diagnosis (pathologist's trap)

- **Nested variant of urothelial carcinoma:** deceptively bland but aggressive (male:female 6:1). Clues: deep infiltrative nests reaching muscle, nuclear atypia.
- **Inverted papilloma:** endophytic nests with trabeculae, usually at the trigone.
- Nephrogenic adenoma; urothelial carcinoma with glandular differentiation.

Management

- None for isolated nests; treat the cause (infection, stone, catheter, obstruction).
- **TURBT (transurethral resection) or biopsy** if a mass is present, to exclude carcinoma.
- **Cystitis glandularis:** remove the irritant and arrange **endoscopic surveillance**, because rare change to adenocarcinoma is reported (especially the intestinal type and in exstrophy).
- Upper tract: benign, but coexisting upper-tract urothelial cancer cannot be excluded, so follow up.

Diagram to draw: urothelium budding into lamina propria, forming a solid nest, then a cyst, then glandular metaplasia.

Key points to remember:

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

Mitomycin

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 2, Q3b

Pharmacology

- MMC (mitomycin C) is an antitumour antibiotic from *Streptomyces caespitosus*.
- **Mechanism:** after reductive activation in the cell it acts as an **alkylating agent**, cross-linking DNA and stopping DNA synthesis; cell-cycle non-specific.
- Molecular weight 334 Da: **little systemic absorption** from the bladder.

Uses in NMIBC (non-muscle-invasive bladder cancer)

1. **Single immediate post-TURBT (transurethral resection of bladder tumour) instillation:** 40 mg. Given **within 6 hours** it reduces recurrence; at 24 hours it does not, because it works by preventing implantation of floating cells. Best for a solitary low-grade papillary tumour. Avoid if perforation or extensive resection is suspected.
2. **Adjuvant induction:** 40 mg weekly for 6 weeks (± maintenance) in low- and intermediate-risk disease. Reduces recurrence but **not progression**; BCG (bacille Calmette–Guérin) with maintenance is superior in high risk.
3. **Optimised delivery (Au, 2001):** empty bladder, overnight fluid restriction, sodium bicarbonate (less drug breakdown), 40 mg in 20 mL; retain 1–2 hours.
4. **Device-assisted:** EMDA (electromotive drug administration) increases tissue uptake; chemohyperthermia at 42 °C (RITE [radiofrequency-induced thermochemotherapeutic effect], HIVEC [hyperthermic intravesical chemotherapy]).
5. **Sequential gemcitabine then MMC** for BCG-exposed disease.

Other urological uses

- **Upper tract:** single bladder dose after nephroureterectomy (ODMIT-C: bladder recurrence 28% to 16% at 1 year). **UGN-101 (mitomycin gel)**, FDA (Food and Drug Administration) approved 2020 for low-grade UTUC (upper tract urothelial carcinoma): complete response 59%, ureteric stricture about 44%. UGN-102 for low-grade intermediate-risk bladder tumours.
- **Bladder-sparing chemoradiation:** 5-FU (5-fluorouracil) + MMC (BC2001: better locoregional control than radiation alone).
- **Antifibrotic:** injected after incision of recurrent bladder neck contracture or urethral stricture.

Side effects

- Chemical cystitis (30–40%); **skin rash, palmar or genital dermatitis** (signature toxicity; wash after voiding); contracted bladder (about 5%); encrustation cystitis; myelosuppression rare.
- Extravasation through a perforation can cause pelvic necrosis.

Diagram to draw: timeline: TURBT, single MMC within 6 hours, then weekly induction ×6.

> **Key points to remember:**
> - Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
> - Reduces recurrence, not progression.
> - Signature toxicity: skin rash; contracted bladder about 5%.

Mullerionosis of UB

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 2, Q3b

"Mullerionosis of UB" is read as müllerianosis of the urinary bladder.

Definition

- Benign lesions of **müllerian-type epithelium** in the bladder wall, in three forms:

Lesion	Tissue
Endometriosis (commonest)	Endometrial glands **and stroma**
Endocervicosis	Endocervical mucinous glands
Endosalpingiosis	Ciliated tubal epithelium

- **Müllerianosis** strictly means two or more of these together.
- Women of reproductive age, often after **caesarean section**. Usually **posterior wall or dome** (vesicouterine pouch).
- Bladder is the commonest urinary tract site of endometriosis, ahead of the ureter.

Pathogenesis

- **Implantation:** retrograde menstruation (Sampson) into the vesicouterine pouch.
- **Iatrogenic implantation** at caesarean or pelvic surgery (primary vs iatrogenic forms).
- **Metaplasia** of coelomic or subperitoneal müllerian remnants (the "secondary müllerian system").
- Lymphatic or vascular spread; direct extension of anterior uterine adenomyosis.

Clinical features

- **Cyclical haematuria** or irritative LUTS (lower urinary tract symptoms) with menses; suprapubic pain, dysmenorrhoea, dyspareunia.
- Endocervicosis: non-cyclical mass or LUTS.
- **Also know:** menouria differential includes vesicouterine fistula (Youssef syndrome).

Evaluation

- **Ultrasound** (transabdominal plus transvaginal): size and depth of a hypoechoic nodule.
- **MRI (magnetic resonance imaging):** extent; haemorrhagic foci **bright on T1**, fibrotic thickening dark on T2.
- **Cystoscopy:** bluish submucosal nodule with neovascularity, best seen during menses. Of **limited value**, because disease is mainly **extravesical** (from serosa inwards).
- **Biopsy:** excludes adenocarcinoma, especially with endocervicosis. Immunohistochemistry: ER/PR (oestrogen/progesterone receptor) positive; CD10 marks endometrial stroma.
- Ultrasound of upper tracts for ureteric involvement.
- **Differential:** adenocarcinoma, cystitis glandularis, nephrogenic adenoma, urachal remnant.

Management

- **Hormonal suppression** (combined pill, progestins, GnRH [gonadotropin-releasing hormone] agonists): symptom relief; recurrence after stopping.
- **TUR (transurethral resection):** incomplete for transmural disease; perforation risk; recurs.

- **Partial cystectomy** (laparoscopic or robotic) with a margin: **primary treatment**. Ureteric stent or reimplantation if the orifice is involved.
- Excise other pelvic endometriosis with the gynaecologist.
- Malignant change (endometrioid or clear cell adenocarcinoma) is rare.

Diagram to draw: sagittal pelvis with a dome nodule growing from the vesicouterine pouch into the detrusor.

> **Key points to remember:**
> - Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
> - Cyclical haematuria; MRI bright on T1.
> - Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

Rapid Revision Sheet – Section 12 · Bladder Cancer & Diversion

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

CA bladder, etiology, newer investigations and treatment. (LAQ)

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman (LAQ)

- Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

Use of intestinal segments in urology, and orthotopic neobladder (LAQ)

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Classification of bladder cancer and bladder-preserving treatment of TCC (LAQ)

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; T2, no hydronephrosis, no CIS.
- TMT: BC2001 MMC/5-FU (mitomycin C/5-FU) + RT, DFS 67% vs 48%; salvage cystectomy 10–30%.

Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction. (LAQ)

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Continence >95%; stomal stenosis is the commonest complication.

Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types (LAQ)

- NAC: cisplatin combinations only; ABC 5% absolute survival gain; SWOG 8710 pT0 38% vs 15%.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab (adds durvalumab) improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Bladder diverticulum. (SAQ)

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

Intravesical gene therapy. (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

BCG failure (SAQ)

- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

Intravesical therapy for TCC of bladder (SAQ)

- Single instillation within 6 hours; never BCG.
- Only maintenance BCG reduces progression.
- SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab 41% CR.

What is optical coherence tomography. What is its potential role in management of CA bladder? (SAQ)

- Near-infrared "ultrasound with light"; depth 1–2 mm.
- Loss of layering suggests tumour; boundary loss suggests T1.
- Sensitive but false positives from inflammation; investigational.

Von Bruns Nest (SAQ)

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

Mitomycin (SAQ)

- Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
- Reduces recurrence, not progression.
- Signature toxicity: skin rash; contracted bladder about 5%.

Mullerionosis of UB (SAQ)

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

Section 13 · Kidney & UTUC — Long answer questions (LAQ)

Metastatic RCC (mRCC): assessment and management, including recent concepts

Asked as:

- Describe assessment and recent concept in management of metastatic RCC. — *Summer 2020 Paper II Q1 (LAQ)*
- Management of metastatic RCC — *Practice Set 11 Paper II Q1 (LAQ)*
- mRCC. — *Practice Set 5 Paper II Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q1; 2024-Winter Paper 3, Q3b; immunotherapy in detail in 2024-Winter Paper 2, Q3e)

Assumed meaning (mRCC.): management of metastatic renal cell carcinoma (mRCC).

Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis; 20–40% of localised tumours metastasise later.
- Five-year survival is about 12%, but median survival now approaches 3–5 years.
- Therapy (treatment) moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

Assessment (before treatment)

- **Clinical:** symptoms of metastases (cough, bone pain, neurological deficit), weight loss; paraneoplastic syndromes (hypercalcaemia, polycythaemia, hypertension, Stauffer syndrome); performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]); comorbidity.
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), ALP (alkaline phosphatase), creatinine, liver tests.
- **Imaging:** contrast CT (computed tomography) of chest, abdomen and pelvis (or MRI, also for IVC [inferior vena cava] thrombus). Bone scan only with bone pain, raised ALP or poor performance status; brain imaging (brain MRI [magnetic resonance imaging]) with neurological symptoms; bone scan or brain MRI only when symptoms or tests suggest spread (if symptomatic); PET not routine.
- **Histology:** nephrectomy specimen or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change, which alters drug choice.
- **Prognostic risk group** (decides first-line therapy):

Adverse factor	MSKCC (Memorial Sloan-Kettering), cytokine era	IMDC (International mRCC Database Consortium [International Metastatic RCC Database Consortium]), TKI era
Karnofsky <80%	Yes	Yes
Haemoglobin < lower limit of normal (below normal)	Yes	Yes

Adverse factor	MSKCC (Memorial Sloan-Kettering), cytokine era	IMDC (International mRCC Database Consortium [International Metastatic RCC Database Consortium]), TKI era
Corrected calcium (raised)	>10 mg/dL	> upper limit of normal (ULN)
LDH >1.5 × ULN (upper limit of normal)	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to systemic therapy
Neutrophils, platelets > ULN (raised)	No	Yes (two factors)

- IMDC 6 factors; MSKCC (Memorial Sloan Kettering) model: 5 factors including LDH (lactate dehydrogenase).
- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

Management: recent concepts

- Multidisciplinary care: systemic therapy is the backbone; surgery and radiotherapy are added selectively.

1. First-line (systemic) therapy, clear cell RCC (NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

IMDC risk	Preferred	Alternatives (if IO not possible)
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib or pazopanib if IO (immuno-oncology) not possible; active surveillance for slow, low-volume (indolent low-volume) disease
Intermediate/poor	The same three, **or nivolumab + ipilimumab**	Cabozantinib; sunitinib; pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (the TKI target). ICIs (checkpoint inhibitors) block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing the brake on T cells (release T-cell brakes) and giving durable responses.
- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab: OS [overall survival] HR [hazard ratio] 0.63; overall survival benefit in intermediate/poor risk); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib: PFS [progression-free survival] 23.9 vs 9.2 months). PD-L1 (programmed death ligand-1) status does not select patients; benefit is independent of PD-L1 status.

2. Later lines: switch (the) mechanism

- After IO (an IO combination): a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor). IO rechallenge is not advised (CONTACT-03 negative).
- After both IO and TKI: **belzutifan** (HIF-2α inhibitor); LITESPARK-005 showed better PFS (progression-free survival) than everolimus.
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with OS (survival) gain.

3. Cytoreductive nephrectomy (CN): (from) routine to selective

- Cytokine era: SWOG 8949 and EORTC (European Organisation for Research and Treatment of Cancer) trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (OS / median survival 18.4 vs 13.9 months).

- **SURTIME:** deferred CN after sunitinib was safe; OS (survival) favoured deferral.
- **Now:** immediate (upfront) CN for fit patients with low metastatic burden (low-burden) not needing (who do not need) immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk, poor performance status or brain metastases.
- Palliative nephrectomy or angioembolisation for haematuria or pain.

4. Metastasectomy and local therapy (radiotherapy)

- Complete resection of solitary or few (limited) metastases gives 5-year survival of 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year, metachronous disease.
- **SBRT (stereotactic body radiotherapy)** (for oligometastases), radiosurgery for brain metastases, or ablation for oligometastatic disease; palliative radiotherapy or surgery for cord compression and weight-bearing bones; palliative radiotherapy for bone pain.

5. Adjuvant immunotherapy (new) / adjuvant therapy after complete resection

- **KEYNOTE-564:** pembrolizumab for 1 year after surgery in high-risk clear cell RCC, including **M1 NED** (metastases completely resected; no evidence of disease after metastasectomy; adjuvant pembrolizumab after complete metastasectomy): disease-free survival HR 0.63 (disease-free survival improved); FDA (Food and Drug Administration) approval 2021; a later report showed OS (overall survival) benefit.

6. Supportive and special groups (non-clear cell RCC and supportive care)

- Zoledronic acid or denosumab for bone metastases. Immune-related adverse events (colitis, hepatitis, hypophysitis, thyroiditis) need steroids and drug interruption.
- Non-clear cell RCC: clinical trial (trials) preferred; cabozantinib or sunitinib; gemcitabine + carboplatin (platinum) for collecting duct carcinoma.
- **Also know:** high-dose IL-2 (interleukin-2) gives durable complete response 7–9% (another source: in a few percent); temsirolimus for poor risk; chemotherapy and hormones are ineffective.

Follow-up

- CT every 8–12 weeks on therapy (response by RECIST [Response Evaluation Criteria in Solid Tumours]); monitor thyroid, liver, blood pressure and cortisol.

Recent advances

- Belzutifan, also approved for VHL-associated RCC (tumours); neoadjuvant ICI trials; biomarkers to choose IO/IO or IO/TKI.

```
Metastatic RCC -> biopsy + IMDC risk
   -> IO/TKI (all risks) or nivo + ipi (intermediate/poor)
   -> deferred CN / metastasectomy / SBRT in responders
   -> progression: switch class (TKI, belzutifan)
```

Diagram to draw: the flowchart above (metastatic RCC → biopsy and IMDC risk → IO combination ± deferred CN in responders → metastasectomy or SBRT → switch drug class on progression).

Key points to remember:

- IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI (IO combinations) for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
- Belzutifan after IO and TKI.

Etiopathogenesis and management of Upper tract TCC

Type: LAQ (25 marks)

Source: 2023-Winter Paper 2, Q1 (management also answered in 2016-Summer Paper 3, Q1)

TCC (transitional cell carcinoma) is taken as UTUC (upper tract urothelial carcinoma).

Introduction

- UTUC is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Bladder recurrence follows in 22–47%.

Etiopathogenesis

Factor	Key facts and mechanism
Tobacco	Most important modifiable factor; risk 2.5–7-fold. Aromatic amines are activated to mutagenic N-hydroxylamines
Aristolochic acid	Balkan endemic nephropathy (UTUC 40% of renal tumours there) and Chinese herb nephropathy. Aristolactam–DNA adducts cause A:T → T:A transversions in TP53. Familial but **not inherited**; often bilateral
Lynch syndrome (HNPCC, hereditary nonpolyposis colorectal cancer)	Germline MMR (mismatch repair) mutation: MLH1, MSH2, MSH6, PMS2. Risk ~14-fold, highest with MSH2; found in 5–9% of screened UTUC
Occupation	"Amino tumours": benzidine, β-naphthylamine (dyes, rubber, petrochemicals); latency up to 20 years
Analgesic abuse	Phenacetin: papillary necrosis; capillarosclerosis is pathognomonic
Arsenic	Well water in south-west Taiwan (blackfoot disease)

- **Also know:** cyclophosphamide (acrolein); chronic infection and stones (squamous carcinoma); alcohol.
- **Lynch screening:** suspect if age <60, or a personal or family history of Lynch-spectrum cancer (colon, endometrium). Test the tumour for MSI (microsatellite instability) or MMR immunohistochemistry, then germline testing.
- **Pathogenesis:** two theories both operate. **Intraluminal seeding** (one clone spreading downstream) explains frequent bladder recurrence. **Field effect** (carcinogen in urine, independent clones) explains multifocality.
- Sequence: hyperplasia → dysplasia → CIS (carcinoma in situ). Variant histology in ~25%. Molecular subtypes include FGFR3-mutant (low grade), TP53/MDM2-mutant and hypermutated (Lynch).

Diagnosis

- Haematuria (56–98%), flank pain (20–32%).
- **CTU (computed tomography urography):** best test (sensitivity ~92%). **Cystoscopy** always (concurrent bladder cancer 17%). Selective cytology.
- **Flexible URS (ureteroscopy) with biopsy:** grade is accurate in ~90%, but **stage cannot be assessed**. Staging is TNM (renal parenchymal invasion by a pelvic tumour = T3).

Risk stratification (EAU, European Association of Urology)

Low risk: all present	High risk: any one
Unifocal	Multifocal
Tumour <2 cm	Tumour >2 cm
Low-grade cytology and biopsy	High-grade cytology or biopsy
No invasion on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

- AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023 splits each group into favourable and unfavourable.

Management

A. Kidney-sparing surgery

- **Elective:** low-risk disease, whatever the other kidney. **Imperative:** solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit patient.
- **Ureteroscopic ablation:** biopsy, then holmium or thulium laser. Closed system; avoid circumferential ablation (stricture).
- **Percutaneous resection:** large pelvic or lower-pole tumours; risk of tract seeding.
- **Distal ureterectomy** with reimplantation (psoas hitch, Boari flap): gives a staging specimen.
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); **UGN-101** mitomycin gel for low grade.
- Lifelong endoscopic surveillance: upper-tract recurrence in about half.

B. RNU (radical nephroureterectomy) with bladder cuff: standard for high risk

- Clip the ureter early; kidney outside Gerota's fascia; never open the urinary tract; remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction.
- Open, laparoscopic and robotic routes are oncologically equivalent. Abandoned: endoscopic "pluck", laparoscopic stapling.
- **Template lymphadenectomy** for high risk: hilar plus paracaval (right) or para-aortic (left); pelvic nodes for lower ureter.
- **Single postoperative intravesical instillation:** ODMIT-C (MMC 40 mg) cut bladder recurrence 28% → 16% at 1 year.

C. Perioperative chemotherapy

- **POUT (2020):** adjuvant platinum after RNU for pT2–T4 or N+: 3-year DFS (disease-free survival) **71% vs 46%**. Gemcitabine–cisplatin; carboplatin if creatinine clearance 30–49.
- **Neoadjuvant** platinum when post-RNU renal function would rule out cisplatin.
- **Also know:** adjuvant nivolumab (CheckMate 274) if platinum-ineligible, but no clear benefit in the UTUC subgroup.

D. Metastatic disease (brief)

- First line: **enfortumab vedotin + pembrolizumab** (EV-302, 2023): overall survival ~31 vs ~16 months versus platinum chemotherapy. Alternative: gemcitabine–platinum, then avelumab maintenance.
- Later: pembrolizumab (KEYNOTE-045), erdafitinib for FGFR3 alterations. Node-positive: chemotherapy first, then consolidative surgery.

Follow-up

- After RNU: cystoscopy and cytology from 3 months; CTU yearly. After kidney-sparing: URS and CTU at closer intervals for life.

Diagram to draw: RNU specimen with bladder cuff, beside lymph node templates.

Key points to remember:

- Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
- EAU low risk needs every favourable feature; one adverse feature means high risk.
- RNU with complete bladder cuff; single postoperative MMC (ODMIT-C 28% → 16%).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches

Asked as:

- Describe role of nephron sparing surgery in management of RCC — *Practice Set 13 Paper II Q1 (LAQ)*
- Partial nephrectomy – indications and approaches. — *Practice Set 6 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2022-Summer Paper 3, Q1 and 2025-Winter Paper 1, Q2; nephron-sparing methods also in 2022-Summer Paper 3, Q1)

Definition and rationale

- NSS (nephron-sparing surgery) / PN (partial nephrectomy) is complete removal of the tumour with **maximal preservation of functioning parenchyma**. PN is the standard form; thermal ablation is the non-excisional form.
- Equal cancer control to RN (radical nephrectomy) with less CKD (chronic kidney disease) and cardiovascular risk.
- **Why it matters:** RN can cause CKD, linked to cardiovascular events and death. PN keeps about **90%** of global function (RN 65–70%) with equal cancer control for small masses.
- **CKD from surgery alone** is stable, with near-normal survival if the new GFR (glomerular filtration rate) stays >45; **CKD from medical disease** (diabetes, hypertension) progresses. So NSS matters most when medical renal disease coexists.

Role by clinical setting

Setting	Role of NSS
cT1a (≤4 cm)	PN is the **standard of care** when intervention is needed; RN is overtreatment
cT1b–T2	PN when feasible and function matters; otherwise debatable
Hereditary RCC (renal cell carcinoma)	Observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision)
Bilateral tumours, solitary kidney	Imperative PN, staged if bilateral; avoids dialysis
Frail, comorbid, cT1a <3 cm	Thermal ablation or active surveillance
Locally advanced, thrombus, cN1, metastatic	Generally RN or cytoreductive surgery

Indications

Category	Situations
Imperative	Bilateral RCC (tumours); solitary (functioning) kidney; RN would mean (cause) dialysis
Relative	Contralateral kidney threatened: stones, renal artery stenosis, diabetes, hypertension, proteinuria; hereditary RCC
Elective	Localised tumour with a normal contralateral kidney; **cT1a with normal contralateral kidney: standard of care**; cT1b–T2 when feasible
Benign	Symptomatic AML (angiomyolipoma); focal atrophy in duplex kidney; infected calyceal diverticulum

- **AUA (American Urological Association) 2021:** RN is preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new GFR likely

>45 after RN. Otherwise PN.
- **Relative contraindications: <20% of nephron mass retained (left)**; pedicle encasement; venous tumour thrombus; cT4; nodes.

Preoperative evaluation (planning)

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging); chest imaging; eGFR (estimated GFR) and urine protein.
- **R.E.N.A.L. nephrometry score** (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines) predicts complexity: **4–6 low, 7–9 moderate, 10–12 high** complexity. PADUA is an alternative. 3D models help.
- **Biopsy** when it changes management (about **20%** of enhancing cT1 masses are benign). Genetic counselling if age ≤46, multifocal or bilateral.

Techniques of PN

Approaches: open (flank extraperitoneal), laparoscopic, or **RAPN (robot-assisted PN)**: transperitoneal for anterior and retroperitoneal for posterior tumours.

Approach	Features
Open, extraperitoneal flank (supracostal 11th/12th rib, 8 cm mini-flank)	Posterior exposure; solitary kidney, complex or central tumours, cold ischaemia
Open transperitoneal (subcostal, chevron, midline)	Bilateral or large tumours, vascular reconstruction
Laparoscopic transperitoneal	Large space, easier suturing; anterior tumours
Laparoscopic retroperitoneal	No bowel mobilisation, direct hilum; posterior tumours
Robot-assisted (RAPN)	Easier suturing, shorter warm ischaemia, fewer conversions

Steps (key technical steps)

1. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
2. Defat the kidney; **intraoperative ultrasound** maps borders and depth.
3. Vascular control: clamp the artery (artery ± vein clamp); off-clamp or selective clamping for small tumours. Use **ice slush** if cold ischaemia needed / if ischaemia is expected to exceed **25–30 minutes** (solitary kidney, CKD).
4. Excise with a thin rim (rim of normal tissue), or **enucleation** (hereditary, multifocal). **A negative margin matters; its width does not** (margin width is unimportant if negative).
5. **Two-layer renorrhaphy:** inner layer closes vessels and collecting system; outer capsular layer (sliding-clip) over a haemostatic bolster.
6. Unclamp; check haemostasis; close Gerota fascia; drain.

	Standard (wedge) resection	Enucleation
Plane	Through normal parenchyma	Along the pseudocapsule
Advantage	Fewer positive margins	Less bleeding, maximal parenchyma
Use	Default	Hereditary, multifocal, severe CKD

- **Ischaemia:** warm ischaemia **<25–30 minutes**. Function depends mainly on the **parenchyma preserved**, less on clamp time; off-clamp, selective clamping and mannitol show no long-term eGFR benefit.

Thermal ablation

- **Cryoablation** (double freeze–thaw) or **RFA (radiofrequency ablation)**; percutaneous preferred.

- AUA: alternative for **cT1a <3 cm**; biopsy before (preferred) or at ablation. Local recurrence is higher than PN but treatable by repeat ablation.
- **Also know:** MWA (microwave ablation), SBRT (stereotactic body radiotherapy) for non-surgical candidates.

Outcomes and evidence

- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.
- Elective PN: local recurrence **1–2%**; imperative 3–5%.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN had better function and similar oncology; 10-year overall survival 81% RN vs 76% PN. The trial is flawed.

Complications

- **Urine leak** (drain → stent); **haemorrhage (bleeding), pseudoaneurysm or AV (arteriovenous) fistula** (delayed haematuria: angioembolisation); AKI (acute kidney injury); temporary dialysis in a solitary kidney.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** when >50% of a solitary kidney is removed: proteinuria, FSGS (focal segmental glomerulosclerosis); give an ACE (angiotensin-converting enzyme) inhibitor.
- **Follow-up:** risk-based imaging; eGFR and urine protein.

Recent advances

- 3D (three-dimensional) models; ICG (indocyanine green) for selective clamping; neoadjuvant TKI (tyrosine kinase inhibitor) downsizing in imperative cases.

Diagram to draw: clamped renal artery, tumour excised with a rim, two-layer renorrhaphy.

Key points to remember:

- Imperative, relative, elective; PN is the standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
- AUA: RN only if complex + no CKD + new GFR likely >45.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- EORTC 30904: better function, similar cancer control.

Section 13 · Kidney & UTUC — Short answer questions (SAQ)

Acquired renal cystic disease

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3b

Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

Epidemiology

Duration of renal failure	Prevalence
Before dialysis	About 10%
3 years of dialysis	44%
5 years	60%
≥10 years	>90%

- Male:female 3:1; independent of the cause of renal failure.

Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate crystals, good prognosis); papillary RCC is also common.

Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

Feature	ARCD	ADPKD (autosomal dominant polycystic kidney disease)
Kidney size	Small (<300 g)	Large (>800 g)
Cysts	Small	Large
Extrarenal	None	Liver cysts, aneurysms

Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.
- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

Diagram to draw: small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

ADPKD (Adult polycystic kidney disease)

Type: SAQ (10 marks)

Source: 2017-Winter Paper 2, Q3d

Genetics

- ADPKD (autosomal dominant polycystic kidney disease): commonest inherited cystic kidney disease (about 1 in 400–1000); 7–15% of patients on haemodialysis.

Gene	Chromosome	Share	ESRD (end-stage renal disease)
PKD1 (polycystin-1)	16	~85%	50s
PKD2 (polycystin-2)	4	~15%	70s

- Polycystins (primary cilium) restrain proliferation.

Clinical features (30–50 years)

- **Pain** (commonest): mass, cyst haemorrhage, infection, stones. **Haematuria** about 50%; **stones** 20–30%.
- **Hypertension:** about 50% at 20–35 years with normal function; stretched vessels release renin.
- **Renal failure:** tracks kidney volume; worse with PKD1, male sex.
- **Extrarenal:** liver cysts (commonest; up to 90% by 50); **intracranial aneurysm 10–30%**; seminal vesicle (40%), pancreatic, arachnoid cysts; mitral valve prolapse; diverticulosis.
- RCC (renal cell carcinoma) is not commoner, but is more often bilateral and sarcomatoid.

Diagnosis (ultrasound in at-risk relatives)

Age	Ravine (PKD1 families)	Pei unified (2009)
15–29	≥2 cysts, one or both kidneys	≥3 cysts in total (15–39)
30–59	≥2 cysts in each kidney	≥2 in each kidney (40–59)
≥60	≥4 cysts in each kidney	≥4 in each kidney

- CT or MRI shows TKV (total kidney volume), which predicts progression. Genetic testing if equivocal or for a living donor.

Management

- **BP <130/80 mmHg** with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker); HALT-PKD: tighter control slowed TKV growth.
- **Tolvaptan** (V2-receptor antagonist; lowers cyclic adenosine monophosphate): **TEMPO 3:4 (2012)** slowed TKV growth and function loss; **REPRISE (2017)** slowed eGFR (estimated glomerular filtration rate) decline in later disease. Causes polyuria and liver toxicity.
- **Pain:** exclude infection, stone, tumour; avoid NSAIDs (non-steroidal anti-inflammatory drugs); aspiration, then laparoscopic unroofing.
- **Infected cyst:** lipid-soluble antibiotics (trimethoprim–sulfamethoxazole, fluoroquinolones); drain if no response.
- **Haemorrhage:** usually settles with rest. **Aneurysm screening** (MR angiography) if family history.
- **ESRD:** dialysis or transplant (no recurrence). **Native nephrectomy** before or at transplant for recurrent infection, bleeding, pain, suspected tumour or no room for the graft.

Diagram to draw: large cystic kidneys, liver cysts and berry aneurysm.

Key points to remember:

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

Biomarkers for patients with advanced RCC

Type: SAQ (10 marks)

Source: 2023-Winter Paper 2, Q3d

Introduction

- A **prognostic** biomarker predicts outcome regardless of treatment; a **predictive** biomarker predicts benefit from a specific drug.
- In advanced RCC (renal cell carcinoma), only clinical risk models are used routinely. No molecular biomarker is yet validated to select treatment (EAU, European Association of Urology; NCCN, National Comprehensive Cancer Network).

Clinical and laboratory: risk models

Factor	MSKCC (Memorial Sloan-Kettering Cancer Center)	IMDC (International Metastatic RCC Database Consortium)
Karnofsky performance status <80%	Yes	Yes
Haemoglobin < lower limit of normal	Yes	Yes
Corrected calcium high	>10 mg/dL	> upper limit of normal
LDH (lactate dehydrogenase) >1.5 × normal	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to treatment
Neutrophils, platelets > normal	No	Yes

- IMDC: favourable 0 factors (median survival 43 months), intermediate 1–2 (22.5 months), poor ≥3 (7.8 months).
- IMDC group guides first-line choice (nivolumab–ipilimumab not for favourable risk).
- **Also know:** neutrophil-to-lymphocyte ratio, CRP (C-reactive protein), liver or bone metastases (poor prognosis).

Tissue biomarkers

- **PD-L1 (programmed death-ligand 1):** benefit of checkpoint combinations is **independent of PD-L1**, so it is not used to select patients.
- **Sarcomatoid features:** respond well to checkpoint inhibitors.
- **Gene expression signatures:** IMmotion150 showed high T-effector signature tumours do better with atezolizumab–bevacizumab; high angiogenic signature tumours do well with sunitinib. Investigational.
- **Genomic:** BAP1 loss and CDKN2A loss indicate poor prognosis; PBRM1 data conflicting. MET alterations in papillary RCC (cabozantinib); FH (fumarate hydratase) deficiency. TMB (tumour mutational burden) is low and not predictive.

Circulating and other biomarkers

- **KIM-1 (kidney injury molecule-1):** high serum levels predict recurrence and benefit from adjuvant immunotherapy (investigational).
- **ctDNA (circulating tumour DNA):** RCC sheds little DNA; methylation assays are promising.
- **Also know:** antibiotics during checkpoint therapy reduced survival (microbiome); sunitinib-induced hypertension is linked to better outcome.

Diagram to draw: IMDC risk groups with survival curves.

Key points to remember:

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

Tivozanib

Type: SAQ (10 marks)

Source: 2024-Winter Paper 2, Q3b

Introduction

- Oral **VEGFR TKI (vascular endothelial growth factor receptor tyrosine kinase inhibitor)** used in advanced RCC (renal cell carcinoma).
- Targets **VEGFR-1, -2 and -3** (also c-KIT and PDGFR-β, platelet-derived growth factor receptor-β); highly potent and selective.

Mechanism and rationale

- In clear cell RCC, **VHL (von Hippel–Lindau) loss** lets HIF (hypoxia-inducible factor) accumulate, driving VEGF and angiogenesis.
- Tivozanib blocks VEGFR signalling, starving the tumour of new vessels.
- High selectivity means fewer off-target effects (less hand-foot skin reaction and diarrhoea than sorafenib).

Dose and pharmacology

- **1.34 mg orally once daily, 21 days of a 28-day cycle.**
- Long half-life (about 4–5 days); dose reduction for toxicity or hepatic impairment.

Key trials

Trial	Setting	Result
TIVO-1 (phase III)	0–1 prior therapy, versus sorafenib	PFS (progression-free survival) 11.9 vs 9.1 months; response 33% vs 23%; OS (overall survival) not better (28.8 vs 29.3 months)
TIVO-3 (phase III)	After 2–3 prior systemic therapies including a VEGFR TKI, versus sorafenib	PFS 5.6 vs 3.9 months (HR, hazard ratio, 0.73); response 18% vs 8%; OS not different (HR 0.97); benefit also after prior checkpoint inhibitor
TiNivo-2 (phase III)	After checkpoint inhibitor; tivozanib ± nivolumab	**Negative:** adding nivolumab did not improve PFS (no immunotherapy re-challenge)

Approval and place

- FDA (Food and Drug Administration) declined in 2013 because of the OS concern in TIVO-1; EMA (European Medicines Agency) approved first-line use in 2017.
- **FDA March 2021:** relapsed or refractory advanced RCC **after ≥2 prior systemic therapies**.
- NCCN (National Comprehensive Cancer Network): later-line option, mainly after immunotherapy combinations.

Adverse effects

- **Hypertension** (commonest; control before starting), dysphonia, fatigue, diarrhoea, proteinuria, hypothyroidism.
- Bleeding, arterial thrombosis, cardiac failure, **impaired wound healing** (stop before surgery), RPLS (reversible posterior leukoencephalopathy syndrome).
- **Monitoring:** blood pressure, thyroid tests, urine protein, liver tests.

Diagram to draw: VHL loss → HIF → VEGF → VEGFR, blocked by tivozanib.

Key points to remember:

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

Immunotherapy in (metastatic) RCC

Asked as:

- Immunotherapy therapy for metastatic RCC — *Winter 2024 Paper II Q3e (SAQ)* (overall management of metastatic RCC: see 2024-Winter Paper 3, Q3b)
- Role of immunotherapy in RCC — *Practice Set 4 Paper II Q3f (SAQ)* (also answered in 2024-Winter Paper 2, Q3e and 2025-Winter Paper 4, Q3a)

Type: SAQ (10 marks)

Rationale

- RCC (renal cell carcinoma) is immunogenic (spontaneous regressions / rare spontaneous regression, T-cell infiltration) and chemoresistant.
- **Checkpoints:** PD-1 (programmed death-1) on CD8 T cells binds tumour PD-L1 and switches the T cell off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) on regulatory T cells damps priming. Blocking them (blockade) **releases the brake** on existing anti-tumour immunity.
- VEGF (vascular endothelial growth factor) is immunosuppressive (suppresses immunity), so TKIs (tyrosine kinase inhibitors) synergise.

Cytokine era

- Interferon-α (IFN-α, interferon-alpha): response 10–15%; obsolete.
- **High-dose IL-2 (interleukin-2):** durable complete response **7–9%**, but capillary leak needs intensive care; selected fit patients only.

Checkpoint inhibitors: trials (all clear cell)

Trial	Regimen	Result
CheckMate 025	Nivolumab vs everolimus, after TKI	OS (overall survival) 25.0 vs 19.6 months
CheckMate 214	Nivolumab + ipilimumab vs sunitinib	Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9%
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	Better PFS (progression-free survival) and OS
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3 months
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2 months; response 71%

- **First line:** an IO/TKI (immuno-oncology plus TKI) combination for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF TKI; IO re-challenge (rechallenge) failed (CONTACT-03, TiNivo-2). Belzutifan after IO and TKI.

Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases (M1 NED, no evidence of disease); DFS (disease-free survival) HR 0.68 (another source: HR 0.63), later OS benefit (HR 0.62). Atezolizumab and nivolumab-based adjuvant trials were negative.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies: hormone replacement.

Biomarkers

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well. Gene signatures investigational; antibiotics may reduce benefit.
- **Also know:** avelumab + axitinib (JAVELIN, no OS benefit); non-clear cell data scant.

Diagram to draw: T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

Key points to remember:

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- CheckMate 214: OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

VHL gene and von Hippel–Lindau disease

Asked as:

- VHL gene. — *Practice Set 2 Paper II Q3f (SAQ)*
- Von Hippel Lindau disease — *Practice Set 8 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

(Practice Set 8 Paper II Q3c: VHL gene also answered in Practice Set 2 Paper 2, Q3f)

Introduction

- **VHL (von Hippel–Lindau)** is a **tumour suppressor gene on chromosome 3p25**, identified in 1993; three exons, 213-amino-acid protein.
- Germline mutation causes **VHL disease**: an **autosomal dominant** familial cancer syndrome (about 1 in 36,000), penetrance >95% by 65 years.
- It is the familial form of **clear cell RCC** (renal cell carcinoma). Somatic VHL loss occurs in most sporadic clear cell RCC; chromosome 3p is altered in over 90%.

Function / genetics and pathogenesis: the oxygen-sensing pathway

- VHL protein binds elongin B, elongin C, cullin-2 and RBX1 to form (VHL protein forms) an **E3 ubiquitin ligase** that degrades **HIF (hypoxia-inducible factor)** in normoxia.
- **Normoxia:** hydroxylated **HIF -1α and -2α** are tagged for proteasomal destruction.
- **VHL loss (or hypoxia):** loss of VHL stabilises HIF, which (chiefly HIF-2α) accumulates, joins ARNT (aryl hydrocarbon receptor nuclear translocator) and switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α (transforming growth factor-α), GLUT1, erythropoietin and CA-IX (carbonic anhydrase IX).
- Hence the **hypervascularity** (**highly vascular** tumours) of clear cell RCC.
- **Knudson two-hit:** one germline plus one somatic hit, so VHL tumours are **early, bilateral and multifocal**.

VHL disease manifestations

Lesion	Frequency
Craniospinal haemangioblastoma (cerebellum commonest)	60–80%
Retinal angioma (earliest)	25–60%
Clear cell RCC (mean age 39; commonest cause of death)	25–75%
Renal cysts (commonest renal lesion)	Up to 76%
Phaeochromocytoma	10–25%
Pancreatic cysts / neuroendocrine tumours	45–75% / 11–17%
Epididymal cystadenoma	25–60% of men

- Also endolymphatic sac tumour.
- **RCC is now the commonest cause of death.**

Genotype–phenotype

Type	Mutation	Phaeochromocytoma	RCC
1	Deletions, truncations	Low	High
2A	Missense	High	Low
2B	Missense	High	High
2C	Missense	Phaeochromocytoma only	Absent

Diagnosis

- Family history plus one lesion; otherwise ≥2 haemangioblastomas, or 1 plus a visceral lesion. Genetic testing of relatives.

Surveillance (VHL Alliance 2020)

- Eye check (examination) yearly from infancy; metanephrines yearly from 5 years; CNS (brain and spine) MRI every 2 years from 11; **abdominal MRI every 2 years from 15**.

Clinical implications / urological management

- **3 cm rule:** observe solid renal tumours <3 cm; intervene when largest reaches 3 cm; then nephron-sparing surgery (enucleation) or ablation, after excluding phaeochromocytoma.
- **Nephron-sparing surgery** (enucleation) is standard; not curative: local recurrence 75–85%, yet 10-year disease-specific survival 81–94%. Ablation for repeated small tumours.
- Exclude phaeochromocytoma before any surgery (normetanephrine raised; α-blockade). **Partial (cortical-sparing) adrenalectomy.**
- **Targeted therapy:** VEGF TKIs (tyrosine kinase inhibitors), bevacizumab, mTOR (mammalian target of rapamycin) inhibitors.
- **Belzutifan** (HIF-2α inhibitor): FDA (US Food and Drug Administration) approved 2021 for VHL disease; renal response 30–50% (another source: responses in about half of renal tumours); anaemia.
- **Also know:** PBRM1, BAP1, SETD2 on 3p.

Diagram to draw: VHL–HIF pathway in normoxia versus (vs) VHL loss, with drug targets.

> **Key points to remember:**
> - AD; 3p25 tumour suppressor; VHL protein degrades HIF.
> - VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
> - Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
> - Belzutifan (HIF-2α inhibitor) approved 2021.

Renal aangiomyolipoma

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 2, Q3f (also answered in 2021-Summer Paper 3, Q3d)

Assumed meaning: renal angiomyolipoma (AML).

Introduction

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin) or TSC1 (hamartin) → unchecked **mTOR (mammalian target of rapamycin)** → growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis); treat as RCC (renal cell carcinoma).

Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock). CKD (chronic kidney disease) in TSC.

Diagnosis

- **CT (computed tomography): macroscopic fat (below about −15 to −20 HU [Hounsfield units]) on non-contrast images is diagnostic.**
- US (ultrasound) hyperechoic (so are some RCCs); MRI (magnetic resonance imaging): signal loss on fat suppression, India-ink artefact.
- Fat-poor lesion → MRI ± biopsy (HMB-45 [human melanoma black-45], Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

Management

Situation	Management
Acute haemorrhage	Resuscitate → **SAE (selective arterial embolisation)** → nephrectomy if SAE fails
Symptomatic	SAE, NSS (nephron-sparing surgery), ablation in selected cases
Asymptomatic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy**	Active treatment
None of these	**Surveillance**: imaging at 6 and 12 months, then yearly
TSC, asymptomatic, **>3 cm**	**Everolimus**; MRI every 1–3 years

- The old **4 cm** cut-off overtreats; it should not be a general trigger.

Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis), abscess, infarction.
- **Partial nephrectomy:** preferred surgery, essential in TSC (multifocal, recurrent).
- Thermal ablation: limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%**; stomatitis, acne, hyperlipidaemia.

Diagram to draw: management algorithm by bleeding, symptoms, size and TSC.

Key points to remember:

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Rapid Revision Sheet – Section 13 · Kidney & UTUC

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Metastatic RCC (mRCC): assessment and management, including recent concepts (LAQ)

- IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI (IO combinations) for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
- Belzutifan after IO and TKI.

Etiopathogenesis and management of Upper tract TCC (LAQ)

- Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
- EAU low risk needs every favourable feature; one adverse feature means high risk.
- RNU with complete bladder cuff; single postoperative MMC (ODMIT-C 28% → 16%).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches (LAQ)

- Imperative, relative, elective; PN is the standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
- AUA: RN only if complex + no CKD + new GFR likely >45.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- EORTC 30904: better function, similar cancer control.

Acquired renal cystic disease (SAQ)

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

ADPKD (Adult polycystic kidney disease) (SAQ)

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

Biomarkers for patients with advanced RCC (SAQ)

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

Tivozanib (SAQ)

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

Immunotherapy in (metastatic) RCC (SAQ)

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- CheckMate 214: OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

VHL gene and von Hippel–Lindau disease (SAQ)

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

Renal aangiomyolipoma (SAQ)

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

Section 14 · Adrenal & Retroperitoneum — Long answer questions (LAQ)

How will you evaluate and manage 38 yr male with uncontrolled hypertension

Type: LAQ (25 marks)

Source: Practice Set 7 Paper 2, Q1 (renovascular hypertension also answered in Practice Set 2 Paper 2, Q1; phaeochromocytoma in Practice Set 6 Paper 1, Q1)

Assumed focus: secondary (urological, renal and adrenal) causes of hypertension in a young adult.

Introduction

- **Resistant hypertension:** BP (blood pressure) above goal despite 3 drugs of different classes, including a diuretic, at optimal doses; or needing 4 or more drugs.
- Poor control below 40 years is a classic trigger to search for a **secondary cause**; many are curable.

Causes

Group	Causes	Clues
Renovascular	FMD (fibromuscular dysplasia), atherosclerotic RAS (renal artery stenosis), Takayasu arteritis, reninoma	Bruit; creatinine rise >30% on ACE (angiotensin-converting enzyme) inhibitor; flash pulmonary oedema; kidneys differing >1.5 cm
Renal parenchymal	CKD (chronic kidney disease), ADPKD (autosomal dominant polycystic kidney disease), reflux nephropathy, glomerulonephritis, Page kidney	Raised creatinine, proteinuria, haematuria
Obstructive uropathy	Bilateral obstruction (about 77% hypertensive), unilateral (about 20%), chronic retention	LUTS (lower urinary tract symptoms), palpable bladder, hydronephrosis
Primary aldosteronism (PA)	Adenoma (Conn), bilateral hyperplasia	Hypokalaemia (present in <40% only)
Phaeochromocytoma, paraganglioma	Sporadic or hereditary	Paroxysmal headache, sweating, palpitations
Cushing syndrome	Exogenous steroid, adrenal adenoma, pituitary	Central obesity, striae, diabetes

- **Also know:** sleep apnoea; drugs (NSAIDs [non-steroidal anti-inflammatory drugs], steroids, decongestants, liquorice); alcohol; thyroid and parathyroid disease; coarctation.

Evaluation

Step 1: Exclude pseudo-resistance

- Correct cuff technique; home or ambulatory monitoring to exclude white-coat effect; check adherence, doses, interfering drugs, salt and alcohol.

Step 2: History and examination

- Paroxysms, weakness or polyuria (hypokalaemia), stones, UTI (urinary tract infection) or childhood reflux, snoring, family history.

- BP both arms and radio-femoral delay; abdominal bruit; palpable kidneys or bladder; cushingoid habitus; café-au-lait spots (NF1 [neurofibromatosis type 1]); fundus.

Step 3: Baseline tests and target-organ damage

- Creatinine, eGFR (estimated glomerular filtration rate), electrolytes, urinalysis with albumin–creatinine ratio, glucose, lipids, ECG (electrocardiogram), echocardiography.
- **Ultrasound KUB (kidneys, ureters, bladder):** renal size and asymmetry, scars, cysts, masses, hydronephrosis, residual urine.

Step 4: Targeted screening (method and interpretation)

Cause	Screen	Confirm or localise
PA	Morning **ARR (aldosterone-to-renin ratio)** >20–30 with aldosterone ≥15–20 ng/dL; stop spironolactone first; correct potassium	**Intravenous saline 2 L over 4 h** (aldosterone stays >5 ng/dL) or oral salt loading → adrenal CT (computed tomography) → **AVS (adrenal vein sampling)**; skip AVS if age <40 with a clear unilateral adenoma >1 cm
Phaeochromocytoma	**Plasma free or 24-h urinary fractionated metanephrines**	CT or MRI (magnetic resonance imaging); ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); genetic testing
Cushing	**1-mg overnight dexamethasone suppression** (cortisol ≤1.8 µg/dL excludes); late-night salivary cortisol; 24-h urinary free cortisol	ACTH (adrenocorticotropic hormone): low = adrenal, high = pituitary or ectopic
Renovascular	**Duplex Doppler**: peak systolic velocity >180–200 cm/s, renal/aortic ratio >3.5	CT or MR angiography; catheter angiography with pressure gradient if intervention planned
Parenchymal, obstructive	Ultrasound, urine tests	Renogram (DTPA [diethylenetriaminepentaacetic acid] or MAG3 [mercaptoacetyltriglycine]) for split function and drainage; CT urography

```
Uncontrolled BP, age 38 → exclude pseudo-resistance
 → baseline tests + ultrasound KUB
 → ARR, metanephrines, dexamethasone test, renal Doppler
    ├─ positive → confirm → localise → treat cause
    └─ all negative → essential: optimise drugs
```

Management

1. General (all patients)

- Lifestyle: salt restriction, weight loss, exercise, limit alcohol.
- Optimise **ACE inhibitor or ARB (angiotensin receptor blocker) + calcium channel blocker + thiazide-like diuretic**. Fourth drug: **spironolactone** (PATHWAY-2, 2015: best add-on in resistant hypertension).

2. Cause-specific

Cause	Treatment
PA, lateralised	**Laparoscopic adrenalectomy**: biochemical cure about 94%, but only 37–52% need no antihypertensives
PA, bilateral or unfit	Spironolactone 25–50 mg/day (up to 400 mg) or eplerenone
Phaeochromocytoma	**α-blockade 7–14 days**, β-blocker only after α, volume expansion → laparoscopic adrenalectomy; metanephrines yearly for life

Cause	Treatment
Adrenal Cushing	Adrenalectomy with steroid cover; pituitary disease → trans-sphenoidal surgery
FMD	**Angioplasty alone** (cure common in the young)
Atherosclerotic RAS	Medical therapy (**ASTRAL 2009, CORAL 2014**: stenting no benefit); stent for flash pulmonary oedema or falling function
Branch disease, aneurysm	Surgical bypass or ex vivo repair with autotransplantation
Reninoma	Partial nephrectomy (cures most)
Small non-functioning kidney (reflux, segmental hypoplasia)	**Nephrectomy** if renal vein renin lateralises (ratio >1.5)
Page kidney	Drain or decapsulate
Obstruction	**Relieve it** (catheter, nephrostomy, pyeloplasty, prostate surgery); watch post-obstructive diuresis
CKD	ACE inhibitor or ARB for proteinuria; scheduled diuretics

3. Follow-up: home BP, creatinine, potassium; duplex after angioplasty.

Recent advances

- **Catheter renal denervation** (radiofrequency or ultrasound), approved in the USA in 2023 for selected resistant hypertension; SPARTACUS trial questioned routine AVS.

Diagram to draw: stepwise algorithm from pseudo-resistance to cause-specific treatment.

> **Key points to remember:**
>
> - Hypertension below 40 years: screen for secondary causes.
> - ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
> - Metanephrines for phaeochromocytoma; α before β.
> - FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Section 14 · Adrenal & Retroperitoneum — Short answer questions (SAQ)

Phaeochromocytoma: overview, pre-operative and intra-operative management

Asked as:

- Pre operative management of pheochromocytoma. — *Summer 2020 Paper II Q3c (SAQ)*
- Intra op mgt of pheo. — *Practice Set 7 Paper II Q3c (SAQ)*
- Pheochromocytoma — *Practice Set 1 Paper II Q3d (SAQ)*

Type: SAQ (10 marks)
(Practice Set 7 Paper II Q3c: preoperative management also answered in 2020-Summer Paper 2, Q3c; perioperative in Practice Set 6 Paper 1, Q1) (Practice Set 1 Paper II Q3d: pre-operative management in detail: 2020-Summer Paper 2, Q3c)

Definition and epidemiology

- Catecholamine-secreting tumour of adrenal medullary chromaffin cells; **paraganglioma** is the extra-adrenal form (organ of Zuckerkandl, bladder).
- 1–2 per 100,000 a year; about 5% of adrenal incidentalomas.
- **"Rule of 10" is outdated:** extra-adrenal up to 25%, **hereditary about one-third**, bilateral commoner in hereditary disease.

Genetics

- **MEN2 (multiple endocrine neoplasia type 2)**, RET: about 50% risk, adrenal, low malignancy. **VHL (von Hippel–Lindau)**: normetanephrine only. **NF1 (neurofibromatosis type 1)**. **SDHB (succinate dehydrogenase subunit B):** extra-adrenal, **30–50% metastatic**. (SDHB/SDHD [succinate dehydrogenase subunits].)
- Offer genetic testing to all: about one-third hereditary (MEN2, VHL, NF1, SDHB/SDHD).

Clinical features

- Triad: **headache, sweating, palpitations**. Sustained hypertension is commoner than paroxysmal; pallor, hyperglycaemia; >20% asymptomatic. Epinephrine-dominant tumours can cause hypotensive episodes.

Why prepare / why it is difficult

- Handling the tumour releases catecholamines: hypertensive crisis, arrhythmia, MI (myocardial infarction), stroke. After vein ligation, sudden withdrawal causes hypotension and hypoglycaemia.
- Phaeochromocytoma surgery is a **two-phase haemodynamic problem**: catecholamine **surge** while the tumour is handled, then catecholamine **withdrawal** once the vein is tied (hypotension, hypoglycaemia).
- Surges occur at intubation, positioning, pneumoperitoneum and tumour manipulation.
- Mortality fell from up to 50% to <3% with blockade, now advised for **all** proven tumours, even normotensive.

Step 1: Confirm, localise, test genes (diagnosis)

- **Biochemistry first:** plasma free or 24-hour urinary fractionated **metanephrines**. Tumours convert catecholamines continuously, so metanephrines are more sensitive than episodic catecholamines. Chromogranin A; clonidine suppression for borderline results.

- **Imaging:** CT (computed tomography): >10 HU (Hounsfield units) unenhanced, no rapid washout; MRI (magnetic resonance imaging). **Functional:** ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG for extra-adrenal or hereditary disease, FDG (fluorodeoxyglucose) PET (best for SDHB), ^{123}I-MIBG (metaiodobenzylguanidine) (best in MEN2).
- **Never biopsy** (hypertensive crisis). Iodinated contrast is safe.

Step 2: Cardiac assessment

- ECG (electrocardiogram) and echocardiography (catecholamine cardiomyopathy); joint plan with the anaesthetist.

Step 3: Drug blockade (pre-operative preparation, 7–14 days)

Drug	Action	Points
Phenoxybenzamine (first line)	Irreversible non-selective α-blocker	10 mg twice daily, increase every 2–3 days (titrated); about 1 mg/kg total
Doxazosin, prazosin	Selective α1-blockers	Fewer side effects
β1-blocker (atenolol, metoprolol)	Controls tachycardia	**Only after adequate α-blockade**
Calcium channel blocker	Smooth muscle relaxation	Alternative or add-on
Metyrosine	Blocks tyrosine hydroxylase	Add-on; sedation, extrapyramidal effects

- **α-blockade for all**, 7–14 days.
- **β-blocker only after α-blockade**, because unopposed α-stimulation causes crisis.
- **Why α before β:** β-blockade removes β2 vasodilatation, leaving α-vasoconstriction unopposed (unopposed α-stimulation): crisis.
- **Target:** seated BP (blood pressure) 120–130/80 mmHg; mild postural drop (systolic >90 mmHg) shows adequate blockade.

Step 4: Volume expansion

- High-salt diet and fluids once blockade starts; 1–2 L IV fluid the night before. This prevents post-excision hypotension.

Step 5: Day of surgery

- Last phenoxybenzamine dose the night before; omit the morning dose.
- Arterial and central lines; nitroprusside, nicardipine or phentolamine for surges; esmolol for tachyarrhythmia; early adrenal vein ligation.

Intra-operative management

Prerequisites

- **α-blockade 7–14 days** (phenoxybenzamine or doxazosin), β-blocker only after α, volume expansion; ECG and echocardiography; joint plan with the anaesthetist.

Monitoring

- **Arterial line before induction**, central venous line, ECG, urine output, frequent glucose; cardiac output or transoesophageal echo if cardiomyopathy.

Anaesthesia

- Deep, smooth induction (propofol, opioid such as fentanyl or remifentanil); attenuate intubation response.

- **Avoid triggers:** histamine-releasing drugs (morphine, atracurium), ketamine, ephedrine, metoclopramide, droperidol, desflurane.
- Sevoflurane or isoflurane; vecuronium or rocuronium; magnesium sulphate as an adjunct (reduces catecholamine release and arrhythmia).

Haemodynamic control

Event	Action
Hypertensive surge	**Pause manipulation**; nitroprusside, nicardipine or clevidipine, nitroglycerin, phentolamine
Tachyarrhythmia	**Esmolol**, lidocaine; amiodarone if needed
Hypotension after vein ligation	**Aggressive volume**, then α-agonist (phenylephrine, noradrenaline); vasopressin if refractory
Hypoglycaemia (insulin rebound)	Dextrose infusion, glucose checks

Surgical principles

- **Laparoscopic adrenalectomy** (transperitoneal or posterior retroperitoneoscopic) is standard; open for large or invasive tumours.
- **Minimal handling**; "no-touch" dissection away from the tumour.
- **Early adrenal vein ligation** where possible, after warning the anaesthetist.
- Remove intact in a bag (avoid capsule rupture and seeding).
- **Cortical-sparing** (partial) adrenalectomy in MEN2 (multiple endocrine neoplasia type 2), VHL (von Hippel–Lindau) and bilateral tumours.
- Steroid cover if bilateral adrenalectomy.

Postoperative

- Fluids ± vasopressors for **hypotension**; glucose monitoring (**rebound hypoglycaemia**); HDU (high dependency unit) overnight: blood pressure, heart rate, glucose; steroid cover after bilateral adrenalectomy.
- Metanephrines after surgery, then yearly for life (**lifelong yearly metanephrines**; recurrence up to 16%).

Metastatic disease and follow-up

- Follow-up: lifelong yearly metanephrines (see Postoperative).
- Malignancy is defined only by **metastases**. Treat with metastasectomy, ^{131}I-MIBG, CVD (cyclophosphamide, vincristine, dacarbazine).

Diagram to draw: flowchart: metanephrines → imaging → α-blockade → β-blockade → salt and fluids → surgery. Intraoperative BP curve: surges during handling, fall after vein ligation, with drugs marked. Catecholamine pathway to metanephrines (COMT (catechol-O-methyltransferase)).

Key points to remember:

- Metanephrines first; never biopsy.
- α-blockade 7–14 days for all.
- Never β before α (α before β), plus volume expansion.
- Volume expansion prevents hypotension; watch glucose.
- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.
- One-third hereditary; SDHB most often metastatic.

Cushing syndrome and the dexamethasone suppression test

Asked as:

- Cushing Syndrome — *Practice Set 2 Paper II Q3c (SAQ)*
- Dexamethasone suppression test — *Practice Set 5 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

Definitions

- **Cushing syndrome:** symptomatic hypercortisolism. **Cushing disease:** due to a pituitary ACTH (adrenocorticotropic hormone)-secreting adenoma.
- **MACS (mild autonomous cortisol secretion):** autonomous cortisol without clinical signs (replaces "subclinical Cushing"); raises cardiovascular, bone and diabetes risk.

Causes

Group	Cause	Share
Exogenous	Glucocorticoids (oral, inhaled, topical)	Commonest overall
ACTH-dependent (80–85% of endogenous)	Pituitary adenoma (Cushing disease)	About 70%
	Ectopic ACTH (bronchial carcinoid, small cell lung cancer, phaeochromocytoma)	10–15%
ACTH-independent	Adrenal adenoma, ACC (adrenocortical carcinoma), AIMAH (macronodular hyperplasia), PPNAD (pigmented nodular disease; Carney complex)	About 10%

Clinical features

- Central obesity, moon face, plethora (90–100%); purple striae, hypertension, hirsutism, ED (70–90%).
- **Discriminating:** proximal weakness, bruising.
- **Urological:** nephrolithiasis in up to 50%; hypogonadotropic hypogonadism (low libido, ED).

Diagnosis

Step 1: exclude exogenous steroids (including creams and sprays) first. Step 2: confirm cortisol excess (two tests)

Test	Method	Abnormal
Overnight 1-mg DST (dexamethasone suppression test)	1 mg at 11 pm; 8 am cortisol	>1.8 µg/dL (50 nmol/L)
Late-night salivary cortisol	11 pm, twice	Raised
24-h UFC (urinary free cortisol)	Two collections	Raised; used in pregnancy

Step 3: locate the source

- **Low ACTH:** adrenal; adrenal CT or MRI.
- **High ACTH:** pituitary MRI; **IPSS (inferior petrosal sinus sampling) with CRH (corticotropin-releasing hormone)**: central gradient = pituitary; none = ectopic (chest CT).
- Pitfalls: pseudo-Cushing (depression, alcohol), oestrogens, enzyme inducers.

Dexamethasone suppression test

Principle

- **Dexamethasone** is a potent synthetic glucocorticoid that is **not detected by the cortisol assay**. In a normal axis it suppresses pituitary ACTH (adrenocorticotropic hormone) by negative feedback, so morning cortisol falls.
- **Autonomous** cortisol (adrenal adenoma) or excess ACTH drive (pituitary, ectopic) fails to suppress.
- "Low-dose" 1 mg is still 3–4 times physiological glucocorticoid.

Indications

- Suspected Cushing syndrome; **every adrenal incidentaloma** (5–8% secrete cortisol); suspected ACC (adrenocortical carcinoma).

Types

Test	Method	Abnormal
Overnight 1-mg LD-DST (low-dose DST; preferred screen)	Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am	See below
48-hour 2-mg LD-DST	0.5 mg 6-hourly for 48 hours; cortisol at 48 hours	>1.8 µg/dL; improves accuracy
High-dose DST	8 mg overnight or 2 mg 6-hourly for 48 hours	Over 50% suppression suggests pituitary source; **no longer routine** (some ectopic tumours also suppress)

Interpreting the 1-mg test

Morning cortisol	Meaning
≤1.8 µg/dL (50 nmol/L)	Normal; >90% sensitive, rules out
1.8–5.0 µg/dL	Possible autonomous cortisol secretion
>5.0 µg/dL (140 nmol/L)	Autonomous secretion; >95% specific

- ESE (European Society of Endocrinology) 2023 incidentaloma guideline: cortisol >1.8 µg/dL without overt signs = **MACS (mild autonomous cortisol secretion)**.
- If abnormal: repeat testing and measure **morning ACTH** (low = adrenal; high = pituitary or ectopic, IPSS [inferior petrosal sinus sampling]).

Pitfalls

- **False positive:** oral contraceptives (raise cortisol-binding globulin; up to 50%); CYP3A4 inducers (phenytoin, carbamazepine, rifampicin); missed dose; pseudo-Cushing states (depression, alcohol, obesity, stress).
- **False negative:** CYP3A4 inhibitors (itraconazole, ritonavir, diltiazem).
- **Choice of test:** pregnancy → UFC (urinary free cortisol); epilepsy → LNSC (late-night salivary cortisol); renal failure → DST (not affected by GFR [glomerular filtration rate]).

Management

Cause	Treatment
Cushing disease	Trans-sphenoidal surgery (cure 60–80%); radiotherapy
Adrenal adenoma	**Laparoscopic adrenalectomy**
ACC	Open radical adrenalectomy, mitotane
Ectopic	Resect source; otherwise bilateral adrenalectomy
AIMAH, PPNAD	Bilateral adrenalectomy

- **Bilateral adrenalectomy:** for failed pituitary surgery or life-threatening hypercortisolism. Cost: lifelong replacement, **Nelson syndrome** (8–29%).
- **Medical (bridge):** metyrapone, ketoconazole, etomidate, mitotane, mifepristone.

Perioperative care (urological relevance)

- **Contralateral adrenal is suppressed:** autonomous cortisol suppresses the opposite adrenal; adrenal insufficiency in about 99.7% after surgery for overt Cushing.
- Give **perioperative steroid cover** and taper after adrenalectomy: **Hydrocortisone 100 mg IV** on the day of surgery, then slow oral taper; recovery takes a median 11 months.

Diagram to draw: HPA (hypothalamic–pituitary–adrenal) axis with dexamethasone feedback and with ACTH-dependent and ACTH-independent sites.

Key points to remember:

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
- High-dose DST no longer routine.
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

Algorithm for management of an incidentaloma

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 2, Q3d (also answered in 2022-Summer Paper 4, Q3f)

Assumed meaning: adrenal incidentaloma.

Definition

- Adrenal mass **≥1 cm** found on imaging done for another reason. Two questions: **Is it malignant (imaging)? Is it functioning (biochemistry)?**
- Causes: non-functioning adenoma (commonest); cortisol-secreting adenoma 5.3%; phaeochromocytoma 5.1%; aldosteronoma 1%; ACC (adrenocortical carcinoma) 4.7%; metastasis 2.5%; myelolipoma, cyst.

Algorithm

```
Adrenal mass ≥1 cm
 ├─ IMAGING: unenhanced CT (computed tomography)
 │    ≤10 HU (Hounsfield units), homogeneous → benign adenoma
 │    >10 HU → washout CT (15 min) or chemical-shift MRI
 │         APW >60% or RPW >40% → adenoma
 │         indeterminate → MDT: repeat CT 6-12 months, FDG-PET or surgery
 │    Macroscopic fat → myelolipoma
 ├─ HORMONES (all patients)
 │    1-mg overnight DST (dexamethasone suppression test)
 │    plasma or urine metanephrines
 │    ARR (aldosterone-renin ratio) if hypertension or low potassium
 │    sex steroids only if ACC suspected
 └─ DECISION
      functioning, suspicious, >4 cm, or growing → adrenalectomy
      benign, non-functioning → discharge
```

- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.

Interpreting the tests

- **DST:** cortisol ≤1.8 μg/dL (50 nmol/L) excludes; 1.8–5.0 = possible autonomous secretion (**MACS**, mild autonomous cortisol secretion); >5.0 μg/dL autonomous.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5%. Benign imaging overrides size.
- **Never biopsy or operate before excluding phaeochromocytoma.** Biopsy only for suspected metastasis in known cancer.

Management

- **Surgery:** cortisol excess, phaeochromocytoma (after alpha-blockade), aldosteronoma (after AVS, adrenal vein sampling), malignant features, >4 cm in fit patients, significant growth. Laparoscopic for benign; open for suspected ACC.
- **MACS:** screen for hypertension, diabetes, osteoporosis; surgery individualised.
- **Follow-up (ESE–ENSAT 2023, European Society of Endocrinology–European Network for the Study of Adrenal Tumors):** homogeneous ≤10 HU, non-functioning → no further imaging. Indeterminate non-resected → re-image at 6–12 months; growth >20% and ≥5 mm is concerning.
- **Bilateral:** metastases, lymphoma, tuberculosis, haemorrhage, CAH (congenital adrenal hyperplasia; 17-hydroxyprogesterone); exclude adrenal insufficiency.

Diagram to draw: the algorithm.

Key points to remember:

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery: functioning, suspicious, >4 cm or growing.

MIBG scan

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 2, Q3c

Principle

- MIBG (meta-iodobenzylguanidine) is a guanethidine analogue that resembles **noradrenaline**.
- It enters chromaffin and sympathetic cells through the **noradrenaline transporter** and is stored in **adrenergic secretory vesicles**. So it maps catecholamine-handling tissue: function, not just anatomy.

Isotopes

	^{123}**I-MIBG**	^{131}**I-MIBG**
Use	**Diagnostic** (preferred)	Older diagnostic; mainly **therapeutic**
Image quality	Better; allows SPECT-CT (single-photon emission computed tomography)	Poorer
Emission	Gamma	Beta (therapy) plus gamma

Technique

- **Thyroid blockade** with potassium iodide before and after injection (free iodine).
- **Stop interfering drugs:** labetalol, tricyclic antidepressants, sympathomimetics, reserpine, some calcium channel blockers.
- Intravenous injection; planar and SPECT-CT images at about 24 hours.
- **Normal uptake:** salivary glands, liver, heart, bladder, thyroid if unblocked; normal adrenals faint.

Uses

1. Phaeochromocytoma and paraganglioma

- **Extra-adrenal, multiple, metastatic or recurrent disease**; biochemistry positive but imaging negative.
- **Isolated normetanephrine excess** may indicate an extra-adrenal tumour; MIBG finds it.
- Tumour >5 cm: MIBG or FDG (fluorodeoxyglucose) PET (positron emission tomography) to exclude metastases before surgery.
- **Best in MEN-2 (multiple endocrine neoplasia type 2).** Raised metanephrines after surgery: MIBG can unmask metastases.

2. Neuroblastoma

- ^{123}I-MIBG is **required by COG (Children's Oncology Group) for staging**, replacing skeletal survey and bone scan. More sensitive for relapse. Non-avid tumours: PET.

3. Therapy (^{131}I-MIBG)

- Confirm uptake first. Metastatic phaeochromocytoma: symptom response in up to two-thirds; complete response <5%. High-specific-activity form FDA (Food and Drug Administration)-approved 2018.
- Relapsed or refractory neuroblastoma: response about 28–39%.

Limitations

- **Cold scans in SDHB (succinate dehydrogenase subunit B) and VHL (von Hippel–Lindau)**; aggressive metastases lose uptake.
- **FDG PET is now gold standard** for staging, and **^{68}Ga-DOTATATE PET/CT** is replacing it at some centres.
- **Also know:** carcinoid and medullary thyroid cancer imaging; cardiac sympathetic imaging.

Diagram to draw: whole-body MIBG showing an adrenal hot spot with normal liver, heart and bladder uptake.

> **Key points to remember:**
>
> - Noradrenaline analogue; block thyroid; stop labetalol.
> - ^{123}I diagnostic, ^{131}I therapeutic.
> - Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Adrenal Sx.

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 2, Q3d (preoperative phaeochromocytoma management also in 2020-Summer Paper 2, Q3c)

Assumed meaning: surgery of the adrenal gland (adrenalectomy).

Surgical anatomy

- Right: pyramidal; **short vein into posterior IVC (inferior vena cava)**, so it is the riskier side. Left: crescentic; long vein into left renal vein. Three arteries (inferior phrenic, aorta, renal), one vein.

Indications (AAES [American Association of Endocrine Surgeons] 2022)

- **Functional, any size:** aldosteronoma (Conn; after AVS [adrenal venous sampling]), cortisol-secreting adenoma, phaeochromocytoma, sex-hormone tumour, failed pituitary surgery for Cushing disease.
- **Malignant potential:** ≥4 cm; growth; >10 HU (Hounsfield units) unenhanced; washout <50%; suspected ACC (adrenocortical carcinoma); solitary metastasis.
- Myelolipoma >7 cm or symptomatic; cysts.
- **Contraindications (correct first):** uncontrolled hypertension, coagulopathy, peritonitis (transperitoneal); pregnancy relative.

Preoperative preparation

- Biochemistry: metanephrines in **every** case before surgery.
- **Phaeochromocytoma:** α-blockade 2 weeks (phenoxybenzamine or doxazosin), β-blocker only after α, salt and fluids.
- Cushing: hydrocortisone 100 mg IV on the day of surgery. Conn: correct potassium.

Approaches

Approach	Use
Laparoscopic transperitoneal (gold standard since 1992)	Familiar view, larger tumours
Retroperitoneoscopic (lateral or prone)	Prior abdominal surgery, small tumours
Robotic	Either route
Open (subcostal, chevron, thoracoabdominal, flank)	**ACC**, large or invasive tumours

Steps (laparoscopic transperitoneal)

- **Right:** elevate liver → Kocher manoeuvre → follow IVC → **control adrenal vein early** (warn anaesthetist) → free gland to psoas → bag intact.
- **Left:** reflect colon and spleen → follow left renal vein to adrenal vein → clip → free from pancreas and splenic vessels.
- Kidney attachment last (handle). ACC: open en bloc, no rupture.
- **Partial adrenalectomy:** bilateral or hereditary phaeochromocytoma, solitary gland.

Complications

- Bleeding (adrenal vein avulsion, IVC); spleen, pancreatic tail, liver, bowel injury; pneumothorax.
- **Phaeochromocytoma:** surges during handling, **hypotension after vein ligation**, rebound hypoglycaemia.
- **Adrenal insufficiency** after Cushing surgery (~99.7%): steroid taper.

- Conn: **hyperkalaemia** for weeks (high-salt diet).

Diagram to draw: right and left adrenal veins and relations.

> **Key points to remember:**
> - Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
> - Control the vein early; right vein short into IVC.
> - Open en bloc surgery for ACC.

Rapid Revision Sheet – Section 14 · Adrenal & Retroperitoneum

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

How will you evaluate and manage 38 yr male with uncontrolled hypertension (LAQ)

- Hypertension below 40 years: screen for secondary causes.
- ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
- Metanephrines for phaeochromocytoma; α before β.
- FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Phaeochromocytoma: overview, pre-operative and intra-operative management (SAQ)

- Metanephrines first; never biopsy.
- α-blockade 7–14 days for all.
- Never β before α (α before β), plus volume expansion.
- Volume expansion prevents hypotension; watch glucose.
- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.
- One-third hereditary; SDHB most often metastatic.

Cushing syndrome and the dexamethasone suppression test (SAQ)

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
- High-dose DST no longer routine.
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

Algorithm for management of an incidentaloma (SAQ)

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery: functioning, suspicious, >4 cm or growing.

MIBG scan (SAQ)

- Noradrenaline analogue; block thyroid; stop labetalol.
- ^{123}I diagnostic, ^{131}I therapeutic.
- Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Adrenal Sx. (SAQ)

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC.

Section 15 · Renal, Nephrology & Transplant — Long answer questions (LAQ)

Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD)

Asked as:

- Discuss principles of hemodialysis and vasular access in details. — *Winter 2017 Paper II Q1 (LAQ)*
- Vascular access for hemodialysis — *Winter 2025 Paper II Q3c (SAQ)*
- Hemodialysis access. — *Practice Set 10 Paper II Q3c (SAQ)*
- Access techniques in dialysis — *Practice Set 12 Paper II Q3f (SAQ)*
- Discuss techniques of dialysis — *Practice Set 5 Paper II Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(Winter 2025 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b) (Practice Set 10 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c) (Practice Set 12 Paper II Q3f: CAPD also answered in Practice Set 1 Paper 2, Q3a) (Practice Set 5 Paper II Q3e: HD principles also answered in 2017-Winter Paper 2, Q1; CAPD in 2018-Summer Paper 2, Q3e)

Assumed meaning (Winter 2017): "vasular" is a typo for "vascular" access for haemodialysis. Assumed meaning (Practice Set 12): access for PD (peritoneal dialysis) and HD (haemodialysis); emphasis on PD catheter insertion, since HD vascular access is answered in full in Practice Set 12 Paper 2, Q3a. Assumed meaning (Practice Set 5): principles and techniques of HD (haemodialysis) and PD (peritoneal dialysis), with continuous therapies.

Introduction

- Haemodialysis (HD) removes solutes and water from blood across a semipermeable membrane in an extracorporeal circuit. It is the commonest KRT (kidney replacement therapy) for ESKD (end-stage kidney disease) and severe AKI (acute kidney injury). It needs an access giving about **250–350 mL/min** of blood flow.
- Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of haemodialysis (physical principles)

Process	Driving force	What it removes
Diffusion	Concentration gradient across the membrane	Small solutes (urea, creatinine, potassium); bicarbonate enters blood
Convection	Transmembrane pressure; solute is dragged with water	Middle molecules (e.g. β2-microglobulin)
Ultrafiltration	Transmembrane (hydrostatic) pressure (HD) or osmotic (PD) gradient	Excess water, down to "dry weight"

- **Countercurrent flow:** dialysate runs opposite to blood (**countercurrent**), keeping (to keep) a gradient along the whole dialyser. Clearance rises with flows, membrane area and time.
- **Dialyser:** blood (250–350 mL/min) passes through hollow fibres (hollow-fibre) of **synthetic membrane**; high-flux membranes clear more middle molecules.
- **Dialysate:** ultrapure (reverse-osmosis) water with sodium about 140 mmol/L, potassium about 2 mmol/L, bicarbonate and calcium, adjusted to the patient.

- **Anticoagulation:** heparin; heparin-free or regional citrate if bleeding risk is high (e.g. after surgery).
- **Schedule:** about 4 h (hours) three times (3 times) a week. **Also know:** haemofiltration, haemodiafiltration (HDF), CRRT (continuous renal replacement therapy).
- **Access:** AVF (arteriovenous fistula), AVG (arteriovenous graft) or CVC (central venous catheter).

Adequacy

- **Kt/V** (dialyser urea clearance × time ÷ urea distribution volume): target 1.4, minimum 1.2 (Kt/V ≥1.2) per thrice-weekly session (KDOQI [Kidney Disease Outcomes Quality Initiative] 2015). **URR (urea reduction ratio) ≥65%.**

Indications

- **Acute ("AEIOU"), refractory to medical treatment:** Acidosis; Electrolytes (hyperkalaemia); Ingestions (lithium, salicylates, alcohols); Overload with respiratory distress; Uraemia (encephalopathy, pericarditis, bleeding).
- **Chronic:** symptomatic ESKD, usually at eGFR (estimated glomerular filtration rate) 5–10; IDEAL (2010): early start gave no benefit.

Complications of haemodialysis

- **During dialysis:** hypotension (commonest), cramps, arrhythmia, air embolism, bleeding. **Disequilibrium syndrome:** rapid urea removal in early sessions shifts water into the brain (cerebral oedema); prevent with short, gentle first sessions.
- **Long term:** cardiovascular disease (main cause of death), mineral–bone disorder, calciphylaxis, β2-microglobulin amyloidosis (carpal tunnel), infection, hepatitis B and C; acquired cystic kidney disease (about 80% in ESKD; RCC [renal cell carcinoma] in 1–2% of these).
- **Access-related:** access infection, thrombosis, steal.

Peritoneal dialysis

- The **peritoneum** is the membrane: diffusion for solutes; **glucose or icodextrin** draws water osmotically.
- **Tenckhoff catheter** (two Dacron cuffs), placed open, laparoscopically (omentopexy) or percutaneously (see PD access below).
- **CAPD (continuous ambulatory PD):** about 4 exchanges of 2 L daily. **APD (automated PD):** night cycler.
- **Adequacy:** weekly Kt/V ≥1.7; PET (peritoneal equilibration test) classifies transport.
- **Complications:** peritonitis, exit-site infection, catheter malfunction, hernia, hyperglycaemia, peritoneal fibrosis.

Continuous and hybrid therapies

- **CRRT (continuous renal replacement therapy):** CVVH (continuous veno-venous haemofiltration; convection), CVVHD (haemodialysis; diffusion), CVVHDF (both). For haemodynamically unstable AKI (acute kidney injury).
- **Also know:** SLED (sustained low-efficiency dialysis); HDF (haemodiafiltration: CONVINCE 2023 lowered mortality versus high-flux HD); home and nocturnal HD.

HD versus PD

	HD	PD
Setting	Centre or home	Home, continuous

	HD	PD
Haemodynamics	Rapid shifts	Gentle; preserves residual function
Favoured in	Abdominal surgery, failed PD	Children, cardiac instability, no vascular access

Types of vascular access

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at the wrist, end-to-side (now usually vein end to artery side)	≥6–8 weeks	**First choice**: best patency, fewest infections; highest failure to mature
Brachiocephalic AVF	Brachial artery to cephalic vein at the elbow	≥6–8 weeks	Higher flow; more steal
Brachiobasilic (transposition) AVF	Deep basilic vein transposed superficially	Longer	When the cephalic vein is unusable
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene), forearm loop or upper-arm straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; highest infection and central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters are for acute dialysis only.

Planning the access

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first** (US **Fistula First**, 2003). **KDOQI 2019** asks for (favours) an individual "ESKD Life-Plan": the right access for the right patient at the right time.
- **Timing:** refer at eGFR <30; create an AVF at least (about) **6 months** before dialysis is expected.
- **Vein preservation:** no venepuncture, cannula or PICC (peripherally inserted central catheter) in the non-dominant arm (that arm); avoid subclavian catheters (lines).
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm), venography if central stenosis is suspected.
- **Maturity, "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, by (at) 6 weeks.

Access complications

Complication	Key point	Treatment
Failure to mature	Juxta-anastomotic stenosis; side veins	Angioplasty; ligate (side) branches
Stenosis	Neointimal hyperplasia; AVG at venous anastomosis	PTA (percutaneous transluminal angioplasty), stent graft, revision
Thrombosis	Usually on an underlying stenosis	Thrombectomy, then treat (fix) the stenosis
Steal syndrome	Hand ischaemia; diabetes, brachial-based access	Banding, DRIL (distal revascularisation–interval ligation), ligation
Aneurysm	Repeated needling at one site	Repair if skin is thin
Infection	CVC > AVG > AVF	Antibiotics; excise graft; remove catheter

- **Also know:** brachiobasilic transposition AVF; venous hypertension (central stenosis); high-output cardiac failure (banding); ischaemic monomelic neuropathy; graft seroma.

Monitoring

- Each session: thrill and bruit, arm-elevation test (a healthy fistula collapses), prolonged bleeding after needling, rising venous pressure, falling Kt/V. Duplex scan when abnormal; treat a stenosis causing clinical change. Rotate needle sites ("rope-ladder").

Peritoneal dialysis access

Catheter: Tenckhoff silicone catheter with **two Dacron cuffs**. The deep cuff sits in the rectus muscle; the superficial cuff about 2–4 cm from the exit. The tip lies in the **pelvis (rectovesical or rectouterine pouch)**; the exit site points **downwards** to reduce infection.

Insertion techniques

Technique	Steps	Comment
Open (mini-laparotomy)	Paramedian incision → split rectus → open posterior sheath and peritoneum → catheter to pelvis on a stylet → purse-string round deep cuff → tunnel → exit site	Simple; general or local anaesthesia
Laparoscopic (advanced)	Rectus sheath tunnel; **omentopexy**, adhesiolysis, hernia repair as needed	Best function; fixes causes of outflow failure
Percutaneous (Seldinger)	Needle, guidewire, dilator and peel-away sheath under local anaesthesia, ± ultrasound or fluoroscopy	Bedside; avoid after major abdominal surgery
Peritoneoscopic	Small-scope guided placement	Variant of percutaneous

- **Break-in:** delay full-volume exchanges about 2 weeks to prevent leak; urgent start uses small supine volumes.
- **Complications:** outflow failure (constipation, omental wrap, migration), leak, hernia, exit-site or tunnel infection, peritonitis, bowel perforation.
- **Contraindications:** peritoneal scarring, uncorrectable hernia, inability to perform exchanges.

Haemodialysis catheter access (brief, insertion)

- **Non-tunnelled CVC (central venous catheter):** dual-lumen 10–12 Fr, internal jugular or femoral vein by **ultrasound-guided Seldinger**; for AKI (acute kidney injury); avoid subclavian (central stenosis).
- **Tunnelled cuffed CVC:** right internal jugular, tip at the cavoatrial junction under fluoroscopy; bridge to AVF.
- **AVF (arteriovenous fistula)** first choice, then **AVG (arteriovenous graft)**; HeRO and endovascular AVF in selected patients (see Practice Set 12 Paper 2, Q3a).

Recent advances

- **Endovascular AVF** (percutaneous creation: WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels; home and nocturnal HD.
- **CONVINCE (2023):** high-dose HDF lowered all-cause mortality compared with high-flux HD.

Diagram to draw: dialyser with countercurrent blood and dialysate (beside a PD catheter in the pelvis), and a radiocephalic end-to-side fistula at the wrist; Tenckhoff catheter with two cuffs, tunnel, downward exit and pelvic tip.

Key points to remember:

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.
- CRRT for unstable AKI; PD preferred in children.

Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval

Asked as:

- Describe selection criteria for cadaver renal donor. Describe in details kidney retrieval surgery. — *Summer 2021 Paper II Q2 (LAQ)*
- Describe procedure of identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt — *Practice Set 3 Paper II Q1 (LAQ)*

Type: LAQ (25 marks)
(Marginal donor in 2016-Summer Paper 3, Q3f; brain death certification in 2018-Summer Paper 4, Q3c.) (also answered in 2021-Summer Paper 2, Q2 and 2022-Summer Paper 4, Q3b)

Assumed meaning: the Indian deceased-donor pathway: identification and brain-stem death certification, donor maintenance, allocation and retrieval.

Introduction

- A cadaver (deceased) donor is **DBD** (donation after brain death: circulation maintained until cold perfusion) or **DCD** (donation after circulatory death: warm ischaemia).
- A 25-year-old brain-dead patient is an ideal **standard-criteria DBD (donation after brain death)** donor: circulation continues until cold perfusion, so warm ischaemia is minimal.
- India: **THOA (Transplantation of Human Organs Act) 1994**, amended 2011, **Rules 2014**, recognises brain-stem death (makes brain-stem death legal death); **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

1. Identification and selection criteria

Potential donor: ventilated patient with catastrophic brain injury (head injury, intracranial bleed, hypoxia). ICU (intensive care unit) informs the **transplant coordinator**; **mandatory request** to the family.

Donor evaluation

- Cause of death, medical and social (high-risk behaviour) history; kidney: terminal creatinine, urine output, hypertension, diabetes, ultrasound.
- Laboratory: ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), EBV (Epstein–Barr virus), syphilis, toxoplasma.

Absolute contraindications

- Active or metastatic cancer, haematological malignancy, history of (past) melanoma, untreatable viral infection.
- Treated bacterial infection (even bacterial meningitis) is acceptable; viral encephalitis needs care.

Standard versus marginal donor

Category	Definition
ECD (expanded criteria donor), UNOS (United Network for Organ Sharing)	Age ≥60, or 50–59 with 2 of: hypertension, creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident)
KDPI (Kidney Donor Profile Index)	Replaced ECD in the US (2014); >85% = marginal; longevity matching with EPTS (Estimated Post-Transplant Survival)
Paediatric <15 kg	En bloc transplant

- **Remuzzi biopsy score** for marginal kidneys: 0–3 single, 4–6 dual, ≥7 discard.

2. Legal death and consent: certification of brain-stem death (THOA)

- **DBD preconditions:** apnoeic coma with a known irreversible cause; exclude drugs (sedatives, relaxants), hypothermia, metabolic causes and shock.
- **Board of four (4-member Board):** doctor in charge of the hospital (hospital in-charge); independent specialist from an approved panel; neurologist or neurosurgeon (else a nominated surgeon, physician, anaesthetist or intensivist); treating doctor. **None from the transplant team.**

All brainstem reflexes absent:

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	Ice-cold water into each ear (drum seen): no eye movement	VIII, III, VI
Motor response in cranial nerve area (cranial motor response)	Supraorbital pressure: no grimace	V, VII
Gag and cough	Touch pharynx; suction catheter in trachea: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**; **Form 10**. Ancillary tests (angiography, perfusion scan) only if examination is impossible.
- **DCD (Maastricht):** I dead on arrival; II failed resuscitation (both uncontrolled); **III arrest after withdrawal of support (controlled, commonest)**; IV arrest in a brain-dead donor; V euthanasia.
- **Consent:** next-of-kin (near relative) consent after a **mandatory request** by ICU staff; in medico-legal cases inform police.

3. Maintenance (donor management)

- **Why:** herniation causes a **catecholamine storm**, then loss of sympathetic tone (hypotension), **diabetes insipidus (DI)**, hypothermia, DIC (disseminated intravascular coagulation), low thyroid hormone and cortisol, and pulmonary oedema.
- **"Rule of 100s":** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Measures:** fluids guided by CVP (central venous pressure), no starch colloids (nephrotoxic); **vasopressin** first (pressor and treats DI), then noradrenaline; methylprednisolone 15 mg/kg, thyroid hormone, insulin; lung-protective ventilation; warming; correct sodium, potassium and clotting.

4. Distribution (allocation)

- **NOTTO (National Organ and Tissue Transplant Organisation)**, New Delhi, with **ROTTOs** (regional) and **SOTTOs** (state organisations), keeps registries and waiting lists.
- **Sequence:** hospital and state list → region → national.
- **Kidney criteria:** ABO compatibility, waiting time on dialysis, HLA match, priority for children and previous living donors, **negative crossmatch** (donor spleen or nodes). Many states let the retrieving hospital keep one kidney.
- 2023 policy removed the domicile requirement and upper age limit for registration.

5. Kidney retrieval surgery (multi-organ, DBD)

1. **Checks:** check (death) certificate, consent, blood group; coordinate thoracic, liver, pancreas and kidney teams.
2. **Incision:** midline xiphoid-to-pubis with median sternotomy.
3. **Exposure:** mobilise right colon and small-bowel mesentery off the retroperitoneum (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava).
4. **Prepare cannulation:** encircle the distal aorta above the bifurcation; ligate the inferior mesenteric artery; prepare the supracoeliac aorta for clamping. **Look for aberrant renal arteries**; keep accessory arteries from the iliacs.
5. **Cold perfusion:** systemic heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC (or right atrium); flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed ice; open Gerota's fascia laterally to cool the kidneys.
6. **Remove heart, lungs, liver and pancreas first.**
7. **Kidneys en bloc:** mobilise (lift) both with perinephric fat off the posterior wall; keep periureteric tissue; divide ureters low. Divide aorta and IVC (upper limit set by the liver team, lower at the bifurcation); lift everything anteriorly and divide prevertebral tissue.
8. **Extras:** iliac vessels for reconstruction; spleen and lymph nodes for crossmatch.

DCD: after death and a no-touch period, **super-rapid** laparotomy, direct aortic cannulation and IVC venting limit warm ischaemia; kidneys usually go on machine perfusion.

6. Back-bench preparation

- Open the aorta posteriorly to see the renal ostia and divide it between them (**Carrel patches**); keep the IVC with the right kidney to lengthen the short right renal vein (its vein).
- Keep an aortic **Carrel patch**; ligate tributaries and lymphatics (prevents lymphocele); trim fat without decapsulation; spare the lower pole–ureter triangle.
- Document anatomy and injuries; biopsy marginal kidneys; triple-bag in cold solution on ice; label.

7. Preservation

- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function) versus static cold storage (Moers, 2009), not long-term survival (graft loss). Keep cold ischaemia short (thrombosis risk >24 hours; cold ischaemia >24 hours raises thrombosis risk).
- **Pitfalls:** polar artery or intimal injury, short right vein, ureteric devascularisation, capsular tears.
- **Also know:** HCV (hepatitis C virus)-viraemic donors with antivirals; normothermic perfusion (experimental).

Diagram to draw: aorta and IVC with distal aortic cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

Key points to remember:

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys en bloc last (kidneys out last); Carrel patch, IVC for right vein; machine perfusion reduces DGF.

Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment

Asked as:

- Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail. — *Summer 2022 Paper II Q2 (LAQ)*
- Discuss evaluation of & management of reno-vascular hypertension — *Practice Set 2 Paper II Q1 (LAQ)*
- Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN. — *Practice Set 4 Paper II Q2 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2022-Summer Paper 2, Q2 and Practice Set 2 Paper 2, Q1)

"Iscemic" is read as ischaemic nephropathy.

Part A: Anatomy of the renal vasculature

Renal arteries

- Arise from the aorta at the **L1–L2 disc**. The longer **right artery passes behind the IVC (inferior vena cava)**. Branches supply the adrenal, renal pelvis and proximal ureter.
- **Hilum, anterior to posterior:** renal vein, renal artery, pelvis; the posterior segmental artery lies behind the pelvis.
- **Anterior division** gives four segmental branches (apical, upper, middle, lower). **Posterior division** gives the posterior segmental artery: the first, most constant branch, passing behind the pelvis. If it crosses in front of the ureter, it can cause UPJO (ureteropelvic junction obstruction).
- **Five segmental arteries are end arteries** with no significant anastomosis. Occlusion causes segmental infarction; each segment can be resected separately.
- **Sequence:** segmental → lobar → interlobar (columns of Bertin) → arcuate (corticomedullary junction) → interlobular → afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries or vasa recta.
- **Brödel line:** avascular plane between anterior and posterior territories, just behind the lateral border. Used for anatrophic nephrolithotomy and posterior calyceal access.
- **Variations (25–40%):** accessory arteries in 25–28% (T11–L4, rarely iliac or SMA [superior mesenteric artery]). They are end arteries, usually to the lower pole. Relevant to donor nephrectomy and endopyelotomy.

Renal veins

- Veins **anastomose freely** (venous collars, subcapsular plexus), so segmental vein ligation is safe. 3–5 segmental veins form the renal vein, anterior to the artery.

	Right renal vein	Left renal vein
Length	2–4 cm	6–10 cm
Tributaries	None	Left adrenal, left gonadal, lumbar
Course	Direct to IVC	Between SMA and aorta (nutcracker syndrome)
Anomalies	Accessory veins commoner	Circumaortic (2–16%); retroaortic

Relevance to RVH (renovascular hypertension): atheroma affects the **ostium and proximal third**; **FMD (fibromuscular dysplasia)** affects the **mid-distal artery and branches**; a stenosed accessory artery can cause segmental RVH.

Definitions

- **RVH (renovascular hypertension) — definition:** hypertension caused by RAS (renal artery stenosis) that improves after (is relieved by) revascularisation or nephrectomy. RAS is common in older adults; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from haemodynamically significant RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS	FMD (fibromuscular dysplasia)
Share	Most (about 90%)	Most of the rest
Patient	Older, smokers (smoker), diabetic, vascular disease	Young women
Site	**Ostium and proximal third** (aortic plaque)	**Mid and distal artery, branches**; "string of beads" (medial fibroplasia, commonest type)
Course	Progresses to occlusion; renal function lost	Rarely occludes
Treatment	Medical first	**Balloon angioplasty without stent**; good BP (blood pressure) response

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), renal artery aneurysm, dissection, embolism, Page kidney, transplant RAS.

Pathophysiology (mechanism) of RVH: Goldblatt models (basis of the tests)

- Goldblatt (1934) clipped the canine renal artery and produced hypertension. Loop: low perfusion pressure releases **juxtaglomerular renin** → angiotensin II (vasoconstriction) → aldosterone (sodium retention).

	Two-kidney, one-clip (unilateral RAS)	One-kidney, one-clip (solitary kidney or bilateral RAS)
Sodium	Normal contralateral kidney excretes the extra sodium (pressure natriuresis)	No normal kidney: sodium and water retained
Renin	**High** from stenotic side; contralateral suppressed	Rises, then returns to normal as volume expands
Type	**Renin (angiotensin)-dependent**	**Volume-dependent**
ACE (angiotensin-converting enzyme) inhibitor	Lowers BP (ACE inhibitors lower BP); captopril test positive	Less BP effect; ACE inhibitors may cause **acute renal failure**

- **Phases:** renin-dependent → sodium retention → **chronic phase** (later): hypertension damages the contralateral kidney (nephrosclerosis), BP no longer depends on the stenosis, and **revascularisation cures less** (cure after revascularisation falls with duration).
- Also: sympathetic activation, oxidative stress, endothelin, less nitric oxide.

Pathophysiology of ischaemic nephropathy

- The kidney receives about **25% of cardiac output**, far more than its metabolic need, and **autoregulation** holds GFR until stenosis is severe (about **70% or more**, or a large translesional gradient).
- Beyond this, the stenotic kidney keeps its GFR by angiotensin II **efferent constriction**; blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR.

- Chronic hypoperfusion → hypoxia, inflammation, microvascular rarefaction → **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive damage add injury. Early disease is reversible ("hibernating" kidney); fibrosis is not.

Part B: Diagnosis (evaluation; investigations)

1. Clinical clues (who to test)

- Onset <30 years (below 30 years) (FMD) or severe hypertension after 55 years; **resistant** (≥3 drugs) or accelerated hypertension; sudden worsening of controlled BP.
- **Creatinine rise >30% after ACE inhibitor/ARB**; **flash pulmonary oedema**; kidney size difference >1.5 cm (kidneys differing by more than 1.5 cm); abdominal bruit; hypokalaemia.

2. Basic tests: creatinine and eGFR (estimated GFR), potassium, urinalysis and proteinuria, ECG; exclude primary aldosteronism and phaeochromocytoma.

3. Imaging and functional tests

Test	Method and finding	Comment
Duplex Doppler US (ultrasound)	High PSV (peak systolic velocity), above about 180–200 cm/s; renal/aortic velocity ratio >3.5; **RI (resistive index) = (PSV − EDV)/PSV**	First line, no contrast; operator dependent. **RI ≥0.8 (>0.8) predicts poor response** (parenchymal disease)
CT angiography	Stenosis, ostial plaque, beads	Best non-invasive anatomy; contrast nephropathy
MR (magnetic resonance) angiography	Gadolinium-enhanced	Caution if eGFR (estimated glomerular filtration rate) <30 mL/min
Captopril renography (DTPA, diethylenetriaminepentaacetic acid)	Day 1 after oral captopril: slow (delayed) uptake, low peak; repeat off ACE inhibitor and calcium channel blocker. Improvement of 10% or more (≥10%) = high probability RVH	Delayed uptake after captopril indicates renin-dependence; captopril removes efferent constriction; DTPA is cleared purely by filtration; use declining
Catheter angiography	Gold standard; **translesional (pressure) gradient** (systolic ≥20 or mean ≥10 mmHg = significant)	Done when intervention is planned
Renal vein renin	Ratio >1.5 lateralises	Before nephrectomy

Algorithm

```
Clinical clue -> Duplex US (or CTA/MRA if eGFR allows)
  -> no RAS: treat as essential hypertension
  -> RAS: FMD -> angioplasty
          Atherosclerotic -> optimal medical therapy
             -> high-risk features + viable kidney -> angiography + gradient -> stent
```

Part C: Treatment (management)

1. Medical (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB** (controls renin-dependent BP; check creatinine and potassium in 1–2 weeks (checked at 1–2 weeks)), calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation (stop smoking), diabetes control**: these patients die mainly of cardiovascular disease (most deaths are cardiovascular).

2. Landmark trials (atherosclerotic RAS)

- **ASTRAL (2009):** stenting plus medical therapy versus medical therapy alone gave **no benefit** in renal function, BP, events or mortality (death); procedural complications.
- **CORAL (2014):** stenting versus optimal medical therapy did **not reduce** (**no reduction**) in cardiovascular or renal events; only a small BP fall.
- ASTRAL and CORAL: stenting added **no benefit** over medical therapy. Hence **no routine revascularisation** of atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

3. Indications for revascularisation

- **FMD with hypertension** (first choice).
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent unexplained heart failure, **rapidly declining function** with bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

4. Endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atherosclerotic lesion (ostial atheroma) → angioplasty plus stent**, because a balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

5. Surgical revascularisation

- Failed endovascular therapy, **branch disease or aneurysm**, children, coexisting aortic surgery.

Technique	Use
Aortorenal bypass (saphenous vein; internal iliac artery in children; PTFE [polytetrafluoroethylene])	Standard
Transaortic endarterectomy	Bilateral ostial plaque
Splenorenal (left) / hepatorenal (right) bypass	Hostile (diseased) aorta
Ex vivo bench repair + autotransplantation	Branch FMD, aneurysms
Nephrectomy (partial or total)	Small non-functioning kidney with lateralising renin

6. Transplant RAS: raised velocity on Doppler, contrast imaging confirms; angioplasty or stent first, surgery if this fails.

Follow-up

- BP, creatinine, duplex at intervals for restenosis; cure is commoner in FMD than atherosclerosis.

Diagram to draw: two-kidney one-clip versus one-kidney one-clip with the renin–angiotensin loop (Goldblatt two-kidney, one-clip loop: low pressure, renin, angiotensin II, aldosterone,

hypertension); renal artery dividing into anterior (four segments) and posterior divisions, with the Brödel line.

Key points to remember:

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- 2K1C = renin-dependent; 1K1C = volume-dependent.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR (function) in a viable kidney (RI <0.8), resistant hypertension.

Acute renal failure (AKI) and acute tubular necrosis

Asked as:

- Etiology, pathogenesis and management of acute renal failure. — *Winter 2024 Paper II Q2 (LAQ)* (acute tubular necrosis also answered in 2022-Summer Paper 2, Q3c)
- Acute tubular necrosis. — *Summer 2022 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Acute renal failure is now called AKI (acute kidney injury).

Definition and staging: KDIGO (Kidney Disease: Improving Global Outcomes)

- AKI (per KDIGO) = rise in serum creatinine **or** fall in urine output; either alone is enough.

Stage	Serum creatinine	Urine output
1	Rise ≥0.3 mg/dL in 48 h, or ×1.5 baseline in 7 days	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or ≥4 mg/dL, or RRT (renal replacement therapy)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- One-fifth to one-third of inpatients; mortality about 25%, about 50% (about half) if RRT is needed.
- **ATN definition:** ATN is the commonest **intrinsic** cause of AKI (acute kidney injury): tubular injury from **ischaemia or toxins**. A misnomer: cell death is mainly **apoptosis**.

Etiology

Type	Causes
Prerenal	Hypovolaemia (bleeding, vomiting, burns), heart failure, sepsis, hepatorenal syndrome; worsened by NSAIDs (non-steroidal anti-inflammatory drugs) and ACE (angiotensin-converting enzyme) inhibitors
Intrinsic	**ATN (acute tubular necrosis):** ischaemia, contrast, aminoglycosides, cisplatin, myoglobin. **AIN (acute interstitial nephritis):** β-lactams, NSAIDs, PPIs (proton pump inhibitors), checkpoint inhibitors. Glomerulonephritis; vascular (renal artery or vein thrombosis, atheroemboli)
Postrenal	BOO (bladder outlet obstruction), bilateral ureteric obstruction or obstructed single kidney: stones, tumours, retroperitoneal fibrosis, ureteric injury at surgery

- Prerenal AKI and ATN cause about three-quarters of hospital AKI.

Causes of ATN

Ischaemic	Toxic
Prolonged prerenal state: shock, sepsis, haemorrhage, burns, cardiac surgery, intraoperative hypotension, renal artery clamping, **transplant (delayed graft function)**	**Exogenous:** aminoglycosides, amphotericin, cisplatin, iodinated contrast. **Endogenous:** myoglobin (rhabdomyolysis, e.g. prolonged flank position), haemoglobin, light chains

Pathogenesis (pathophysiology)

- **Autoregulation** holds GFR (glomerular filtration rate) by afferent dilatation (prostaglandins, myogenic reflex) and efferent constriction (angiotensin II, tubuloglomerular feedback).
- **Prerenal:** low perfusion lowers glomerular pressure. Tubules are intact, so they **conserve sodium and water**. NSAIDs block afferent dilatation; ACE inhibitors block efferent constriction. Prolonged prerenal state becomes ATN.
- **Ischaemic ATN:**

    - **Why tubules:** the medulla works near hypoxia (medullary oxygen tension is only 10–20 mmHg), while the **S3 proximal tubule (segment)** and **medullary thick ascending limb** consume much oxygen and are most vulnerable.
    - **Loss of polarity:** Na^+/K^+-ATPase loses its basolateral position (moves off the basolateral membrane) → sodium reabsorption fails → high sodium at macula densa → **tubuloglomerular feedback** constricts afferent arteriole (afferent constriction) → GFR falls (protective).
    - Cells die (mostly apoptosis) and slough (dead cells slough) → **muddy-brown casts** block (obstruct) tubules; filtrate leaks back; endothelial injury and inflammation worsen hypoxia; autoregulation is lost.
    - Phases (clinical course): initiation → **maintenance** (oliguria, rising creatinine) → **recovery** (polyuria: tubules regenerate before concentrating ability returns).
- **Toxic ATN:** vasoconstriction plus direct tubular injury. **Contrast:** risk with age, diabetes, CKD (chronic kidney disease). **Rhabdomyolysis** (long lateral positioning): myoglobin causes toxicity, cast obstruction and hypoperfusion.
- **Postrenal:** raised tubular pressure, then afferent vasoconstriction; recovery depends on duration (dog studies of complete obstruction: full at 1 week, 70% at 2 weeks, none at 6 weeks).

Evaluation (diagnosis: ATN versus prerenal)

- History (fluid loss, hypotension, drugs, contrast), volume status, bladder scan.
- **Ultrasound early** in unexplained AKI to exclude obstruction.

Index	Prerenal	ATN
Urine sodium	<20 mmol/L	>20 mmol/L
FeNa (fractional excretion of sodium)	<1%	**>1%**
FeUrea (on diuretics)	<35%	>35%
Specific gravity	>1.015	<1.010
Sediment	Bland	**Muddy-brown granular casts**, tubular cells
Fluid challenge (response to fluids)	Quick (rapid) recovery	Lag of days to weeks

- Creatine kinase; white cells (AIN); red cell casts (glomerulonephritis); biopsy only if recovery lags (fails).

Management

General

- Treat (remove) the cause; **avoid hypotension** (MAP, mean arterial pressure, >65 mmHg or near baseline) and nephrotoxins; stop NSAIDs and ACE inhibitors or ARBs (angiotensin receptor blockers) (RAAS (renin–angiotensin–aldosterone system) blockers); adjust drug doses.
- Isotonic crystalloid (fluids) for hypovolaemia; aggressive fluids in rhabdomyolysis; intravenous fluid before contrast (pre-contrast hydration).
- **Postrenal:** urethral or suprapubic catheter; **double-J stent or PCN (percutaneous nephrostomy)**; infected obstructed kidney is an emergency.
- AIN: stop drug ± steroids; glomerulonephritis: nephrology and biopsy.

Complications

- **Hyperkalaemia:** calcium gluconate 10 mL of 10% (protects heart) → insulin 10 U with 50 mL 50% dextrose, salbutamol nebuliser (shift) → binders, diuretics or dialysis (remove).
- **Volume overload:** loop diuretics in escalating doses. **Diuretics (only for volume overload) do not treat ATN.**
- Acidosis: bicarbonate. Uraemic bleeding: desmopressin 0.3 µg/kg.

Dialysis indications (AEIOU), refractory to medical therapy

	Indication
A	Acidosis
E	Electrolytes, mainly hyperkalaemia
I	Intoxications / ingestions (lithium, salicylate, alcohols)
O	Overload with respiratory distress
U	Uraemia: encephalopathy, pericarditis, bleeding

- Intermittent haemodialysis via 10–12 Fr jugular or femoral catheter; **CRRT (continuous RRT)** if haemodynamically unstable. **STARRT-AKI (2020):** early start gave no survival benefit.

Post-obstructive diuresis

- Causes: retained urea (osmotic diuresis), volume expansion with ANP (atrial natriuretic peptide), and down-regulated aquaporin-2 and sodium transport.
- Up to 300 mL/h for about 2 days. Free oral fluids; if unable to drink, replace about 75% of output with 0.45% saline; electrolytes every 4–6 hours. Gradual bladder decompression has no role.

Outcome

- Full or partial recovery, or dialysis dependence; persistence ≥90 days = CKD. ATN recovery takes days to weeks.
- **Also know:** (early) biomarkers NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1); AKD (acute kidney disease, under 3 months); gadolinium (groups I and III) avoided in AKI.

Diagram to draw: tubular cell losing polarity → sloughing → casts and back-leak.

> **Key points to remember:**
> - KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
> - FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
> - S3 segment and thick ascending limb are most vulnerable.
> - Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
> - Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Principles of vascular anastomosis and vascular access surgery for HD

Asked as:

- Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD — *Practice Set 5 Paper II Q2 (LAQ)*
- Describe the principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD — *Practice Set 12 Paper II Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(vascular access also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c)
(Practice Set 12 Paper II Q3a: also answered in Practice Set 5 Paper 2, Q2)

Assumed meaning: HD = haemodialysis.

Introduction

- A vascular anastomosis joins two vessels so that flow is smooth, leak-free and lasting (transplantation, autotransplantation, caval repair, dialysis access).
- **Types:** end-to-end, **end-to-side** (commonest: transplant, fistula), side-to-side, and patch (Carrel aortic patch).

Principles of vascular anastomosis

Principle	How and why
Exposure and control	Proximal and distal control (vessel loops, Rummel tourniquets, bulldog or Satinsky clamps)
Anticoagulation	Systemic heparin before clamping; heparinised saline flush; prevents thrombosis in the static column
Atraumatic handling	Hold the adventitia, never the intima (adventitia only); clamp clear of atheroma, because clamp injury causes intimal dissection and stenosis; avoids intimal dissection
Healthy ends	Trim diseased ends; resect venous valve cusps at the suture line; with no tension, twist or kink, prevents stenosis and leak
No tension, twist or kink	Adequate length; trial the final position before suturing
Intima-to-intima eversion	Smooth endothelium faces the lumen; adventitia and rough surfaces stay outside, as they are thrombogenic
Wide opening	Spatulate or bevel small vessels, or use end-to-side with an elliptical arteriotomy (aortic punch), to avoid stenosis (avoids stricture)
Suture	**Non-absorbable monofilament polypropylene**, double-armed: smooth, low drag, low thrombogenicity. IVC (inferior vena cava) or aorta 3-0/4-0; transplant vessels 5-0/6-0; wrist fistula about 6-0/7-0
Needle direction	Equal, small, close bites; pass **inside-out on the artery** so plaque and intima are tacked down (tacks down plaque and intima), not lifted into a flap
Continuous or interrupted	Continuous is quick and haemostatic; **interrupted** for small vessels and children, allowing growth (allows growth)
Technique	**Carrel triangulation** (three stay sutures keep the back wall clear; clear back wall); **parachute** (sew loosely, then draw up; one knot) or tack both apices and run to the midpoint

Principle	How and why
Flush and release	Before the last knot, briefly release clamps to flush air, clot and debris (removes air and clot); leave slack ("growth factor") to avoid purse-stringing (avoids purse-string); release venous/distal clamp first, then arterial
Confirm	Haemostasis, pulse, thrill, Doppler

- **Failure:** early from technical error (thrombosis, leak, stenosis); late from **intimal hyperplasia**.
- **Also know:** vascular staplers; grafts (autologous vein for small or contaminated fields; PTFE [polytetrafluoroethylene] for large vessels).

Options for vascular access surgery for HD

- Haemodialysis needs a blood flow of about **250–350 mL/min**.

Option	Construction / description	Ready	Comment
1. Non-tunnelled CVC (central venous catheter)	Dual-lumen 10–12 Fr in internal jugular or femoral vein	At once	AKI (acute kidney injury) and emergencies; days only
2. Tunnelled cuffed CVC	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; infection, central stenosis
3. Native AVF (arteriovenous fistula)	Artery joined to a superficial vein, usually vein end to artery side	6–8 weeks (matures)	**First choice:** best patency, fewest infections
a. Snuffbox or **radiocephalic (Brescia–Cimino)**	Radial artery to cephalic vein at wrist		Most distal; preserves proximal sites; higher failure to mature
b. Brachiocephalic	Brachial artery to cephalic vein at elbow		Higher flow; more steal
c. Brachiobasilic transposition	Deep basilic vein mobilised and moved superficially	Longer, often two stages	When the cephalic vein is unusable
4. AVG (arteriovenous graft)	PTFE forearm loop, upper-arm or thigh	2–3 weeks (usable)	Poor veins; more stenosis (venous anastomosis), thrombosis, infection
5. HeRO (Haemodialysis Reliable Outflow) graft	Graft outflow bypasses occluded central veins (central vein occlusion) into the right atrium		Central vein occlusion
6. Endovascular AVF	Percutaneous radiofrequency or thermal anastomosis between adjacent forearm vessels		No incision; selected anatomy

- **Also know:** lower-limb femoral vein transposition when arms are exhausted; peritoneal dialysis needs no vascular access.

Planning

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**; individualised (KDOQI [Kidney Disease Outcomes Quality Initiative] 2019 "Life-Plan").
- **Timing:** refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.
- **Vein preservation:** preserve arm veins; no venepuncture or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).

- **Maturity "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications of access

Complication	Treatment
Failure to mature (non-maturation); stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), ligate side branches, stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then treat the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft, remove catheter

- **Also know:** venous hypertension; high-output failure; monitor thrill and bruit.

Diagram to draw: radiocephalic end-to-side fistula with Carrel triangulation stay sutures.

Key points to remember:

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal → banding or DRIL.

Section 15 · Renal, Nephrology & Transplant — Short answer questions (SAQ)

Role of vascular grafts in urology

Type: SAQ (10 marks)

Source: 2015-Summer Paper 2, Q3e

Introduction

- Grafts and patches replace, bypass, extend or widen vessels when primary repair fails or narrows them.

Types

Type	Examples	Pros	Cons
Autologous vein	Saphenous, gonadal	Resists infection; small vessels	Limited length
Autologous artery	Internal iliac, Carrel aortic patch	Renal arteries	Limited supply
Biological	Bovine pericardium	Pliable patch	Cost
Synthetic	**PTFE (polytetrafluoroethylene)**, Dacron	Off-the-shelf, large calibre	Infection; venous grafts thrombose

Applications

1. **Renal transplantation**: multiple arteries (18–43%) joined by Carrel patch or bench reconstruction, preserving the lower pole artery (supplies the ureter). Short right renal vein extended with donor IVC (inferior vena cava), external iliac or spiral gonadal vein. Transplant artery stenosis: angioplasty first, graft if it fails.
2. **RCC (renal cell carcinoma) with IVC thrombus**: patch cavoplasty if residual lumen would be <50%; **PTFE replacement (16–20 mm)** if the wall is invaded circumferentially.
3. **Post-chemotherapy RPLND (retroperitoneal lymph node dissection)**: IVC: repair if narrowing <25%, patch or graft if more; ligation tolerated with collaterals. Aorta: about 1% need PTFE or Dacron replacement; never ligated.
4. **Renovascular surgery and autotransplantation**: aortorenal bypass (saphenous vein, internal iliac artery, PTFE), splenorenal or hepatorenal bypass, ex vivo bench repair of branch disease.
5. **Dialysis access**: PTFE AV (arteriovenous) graft.

- **Also know:** iatrogenic or traumatic vascular injury (interposition graft); retroperitoneal sarcoma caval resection; superior mesenteric vein injury (vein graft); bland infrarenal thrombus (filter or caval interruption).

Principles

- Autologous graft in contaminated fields and small vessels; synthetic for large high-flow vessels.
- Tension-free, spatulated anastomosis with fine monofilament (polypropylene); heparin before clamping; antiplatelet or anticoagulant afterwards as indicated; omental wrap separates synthetic grafts from bowel and urine.

Complications

- Thrombosis (caval grafts 10–40%), stenosis, infection (often needing removal), enteric or urinary fistula.

Diagram to draw: Bench arterial reconstruction.

> **Key points to remember:**
> - Transplant: Carrel patch, bench reconstruction; preserve the lower pole artery.
> - IVC: patch if lumen <50%; PTFE replacement if wall invaded.
> - Post-chemotherapy RPLND: IVC can be ligated, aorta must be grafted.

Pre-operative evaluation of the living (related and unrelated) renal donor

Asked as:

- Pre operative investigations for renal donor. — *Summer 2017 Paper II Q3b (SAQ)*
- Pre-op assessment of unrelated donor — *Practice Set 8 Paper II Q3a (SAQ)*

Type: SAQ (10 marks)
(related: 2023-Winter Paper 4, Q3e; 2022-Summer Paper 3, Q3a)

Assumed meaning: evaluation of a living kidney donor; for the unrelated question, evaluation of a living unrelated kidney donor (Indian context).

Principles

- **Donor safety first**. **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** is **risk-based**: the donor's lifetime ESKD (end-stage kidney disease) risk is compared with the centre's threshold. Team includes an **independent donor advocate**.
- For an unrelated donor the medical work-up equals that of a related donor; the extra step is **legal and ethical**: proof of genuine, unpaid motive.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011** (Rules 2014): near relatives, or authorisation committee approval for others; swap donation allowed.

Legal assessment of the unrelated donor (THOA)

- Donor "other than near relative" donates for **affection or attachment**: **Authorisation Committee approval is mandatory**.
- Committee: hospital-based if >25 transplants a year, otherwise district or state; members are not from the transplant team.
- Process: joint application; **video-recorded interview** of donor and recipient; verify identity, documents, financial status and long association; exclude middlemen and payment; confirm free, informed consent.
- Foreign recipient with unrelated Indian donor not approved; appeal within 30 days.

Sequence of evaluation (medical assessment)

1. Compatibility (immunological)

- ABO group, HLA (human leukocyte antigen) typing, **crossmatch** (CDC (complement-dependent cytotoxicity) and flow), DSA (donor-specific antibody); incompatible pairs → paired exchange.
- Unrelated donor: more HLA mismatch is expected, yet living unrelated grafts outperform deceased-donor grafts.

2. Kidney function and urine

Test	Accept	Individualise	Decline
GFR (glomerular filtration rate), mL/min/1.73 m²	≥90	60–89	<60
Albuminuria (ACR (albumin–creatinine ratio), mg/g)	<30	30–100	>100

- GFR by two methods, one measured (e.g. DTPA (diethylenetriamine pentaacetic acid) GFR or creatinine clearance).
- Urinalysis, culture; persistent microscopic haematuria → cystoscopy, imaging, ± kidney biopsy.

3. Medical and metabolic

- Blood pressure (ambulatory if borderline), fasting glucose, **HbA1c (glycated haemoglobin)**, OGTT (oral glucose tolerance test) if at risk; lipids, liver tests, coagulation, BMI (body mass index).
- **Type 1 diabetes excludes**; prediabetes, controlled hypertension and obesity: individual risk. (Unrelated-donor source: diabetes, significant obesity and uncontrolled hypertension exclude.)
- **Stones**: metabolic work-up; bilateral or recurrent usually exclude.

4. Infection screen

- HIV (human immunodeficiency virus), HBsAg (hepatitis B surface antigen), anti-HBc, HCV (hepatitis C virus), CMV (cytomegalovirus), syphilis; **tuberculosis** screen.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo; stress test if indicated.

6. Cancer screening by age (age-appropriate): Pap smear, mammogram, PSA (prostate-specific antigen), stool test or colonoscopy.

7. Urological: stones (bilateral exclude), haematuria, malignancy, recurrent UTI (urinary tract infection).

8. Anatomy

- **CT (computed tomography) angiography**: number of vessels, early branching, stones, masses.
- **Split function** (DTPA or DMSA (dimercaptosuccinic acid)) if asymmetric: the **better kidney stays with the donor**; left preferred (longer vein).

9. Psychosocial evaluation: voluntary, no coercion or payment; independent donor advocate.

Counselling figures (Muzaale 2014)

- Function settles at 60–80% of baseline.
- Lifetime ESKD (ESRD, end-stage renal disease) 90 per 10,000 donors vs 14 in matched non-donors; mortality 3 per 10,000.

Also know: APOL1 (apolipoprotein L1) testing in African ancestry; healthy older (marginal) donors acceptable.

Diagram to draw: donor work-up flowchart: legal approval → compatibility → medical → anatomy → surgery.

Key points to remember:

- KDIGO 2017: risk-based; GFR ≥90 accept; decline GFR <60 or ACR >100 (albuminuria >100 mg/g declines).
- ABO and crossmatch first; CT angiography for anatomy.
- Leave the better kidney with the donor (better kidney stays with donor).
- THOA: near relatives or authorisation committee; unrelated donor = Authorisation Committee approval.
- Interview video-recorded; no commerce.

UW solution and organ preservation solutions

Asked as:

- UW solution. — *Summer 2017 Paper II Q3c (SAQ)*
- Organ preservation solution — *Practice Set 7 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)

Assumed context: preservation of the kidney for transplantation.

Introduction

- **University of Wisconsin solution** (Belzer, late 1980s): the **gold standard** cold-storage solution for kidney, liver and pancreas.

Principles of cold preservation

- Cooling / **Hypothermia** to **4 °C** cuts (slows) metabolism to roughly a tenth, but does not stop it, so injury accumulates with time.
- Ischaemia causes ATP (adenosine triphosphate) loss (depletion) → **Na^+/K^+-ATPase pump failure** → **cell swelling**; anaerobic glycolysis → **acidosis**; calcium influx. On reperfusion, hypoxanthine is converted by xanthine oxidase to **oxygen free radicals** → ischaemia–reperfusion injury, seen clinically as (clinically this appears as) **DGF (delayed graft function)**.
- A solution must therefore prevent swelling and oedema, buffer acid, scavenge radicals and supply ATP precursors. It therefore needs:
  - **Impermeants** (lactobionate, raffinose, mannitol, glucose) to prevent cell swelling.
  - **Buffers** (phosphate, histidine) against acidosis.
  - **Colloid** (hydroxyethyl starch, polyethylene glycol) against interstitial oedema.
  - **Antioxidants and energy precursors** (glutathione, allopurinol, adenosine).
  - Electrolytes: "intracellular" (high K^+, low Na^+) or "extracellular" (low K^+, high Na^+) type.

UW composition (per litre)

Component	Amount	Purpose
Lactobionate (K^+ salt)	100 mmol	Impermeant anion; prevents cell swelling; chelates calcium
Raffinose	30 mmol	Impermeant trisaccharide
Hydroxyethyl starch	50 g	Colloid; prevents interstitial oedema
KH_2PO_4	25 mmol	Buffer
$MgSO_4$	5 mmol	Membrane stability
Adenosine	5 mmol	ATP precursor
Glutathione	3 mmol	Antioxidant
Allopurinol	1 mmol	Xanthine oxidase inhibitor; fewer free radicals
Insulin, dexamethasone, antibiotic	Small	Additives

- **Intracellular-type**: K^+ about 125 mmol/L, Na^+ about 30 mmol/L; osmolality ~320 mOsm/kg; pH 7.4.

UW drawbacks

- **Viscous** (starch): slow flush. **High K^+**: flush before reperfusion (hyperkalaemia, vasospasm). May crystallise (filter).

Comparison of solutions

Feature	UW	HTK (Custodiol)	Euro-Collins
Type	Intracellular (high K^+)	Low K^+, low Na^+	Intracellular
Impermeant	Lactobionate, raffinose	Mannitol	Glucose (enters cells; weaker)
Buffer	Phosphate	**Histidine** (strong)	Phosphate
Colloid	HES (hydroxyethyl starch)	None (low viscosity)	None
Others	Antioxidants, adenosine	Tryptophan, ketoglutarate	—

Solution	Type and key components	Points
UW (University of Wisconsin, Belzer)	Intracellular; lactobionate, raffinose, hydroxyethyl starch, glutathione, adenosine, allopurinol	Standard for static storage; viscous; flush out high K^+ before reperfusion
HTK (histidine–tryptophan–ketoglutarate, Custodiol)	Low K^+; histidine buffer, tryptophan (membrane stabiliser), ketoglutarate (substrate), mannitol	Low viscosity, fast cooling; large volume; outcomes similar to UW
Euro-Collins	Intracellular; glucose impermeant, phosphate	First generation; largely abandoned
Celsior	Extracellular; lactobionate, mannitol, glutathione, histidine	Designed for heart; also abdominal organs
IGL-1 (Institut Georges Lopez)	UW-like, but polyethylene glycol and reversed Na^+/K^+	Extracellular; less viscous

- Kidney outcomes with UW and HTK (histidine–tryptophan–ketoglutarate) are similar; Euro-Collins largely abandoned. **Also know:** Celsior, IGL-1, Marshall's (hypertonic) citrate.

Practical use and methods

- Living donor, very short cold ischaemia: cold lactated Ringer's suffices; deceased donor or exchange: UW or HTK.
- **Static cold storage:** in situ aortic flush, IVC (inferior vena cava) venting, ice slush; storage at 4 °C. Keep (aim) cold ischaemia short (ideally <24 h / hours): DGF and thrombosis rise with time (beyond).
- **Hypothermic machine perfusion** (Belzer MPS (machine perfusion solution), gluconate-based): pulsatile cold perfusion; flow and resistance help assess viability. Reduces DGF vs static storage; **Moers (2009)**: DGF 20.8% vs 26.5%, 1-year graft survival 94% vs 90%, without long-term survival gain (the chapter notes no difference in long-term survival). Used especially for DCD (donation after circulatory death) kidneys.
- **Normothermic perfusion:** experimental; allows assessment and repair; limited experience.

Diagram to draw: ischaemia(–reperfusion) cascade with the UW / solution component acting at (that blocks) each step.

Key points to remember:

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

CAPD (Continuous Ambulatory Peritoneal Dialysis)

Asked as:

- CAPD (Continuous Ambulatory Peritoneal Dialysis). — *Summer 2018 Paper II Q3e (SAQ)*
- CAPD — *Practice Set 1 Paper II Q3a (SAQ)*

Type: SAQ (10 marks)
(also answered in 2018-Summer Paper 2, Q3e)

Definition and principle

- CAPD (continuous ambulatory peritoneal dialysis): home dialysis using the **peritoneum** (about 1–2 m^2) as the membrane; dialysate dwells in the abdomen.
- **Diffusion:** urea, creatinine and potassium move into dialysate down a concentration gradient.
- **Ultrafiltration:** an osmotic agent draws water (removes water by osmosis): **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** (for the long dwell).
- **CAPD:** about **4 exchanges of 2 L daily**, dwell 4–6 hours (4–6-hour dwells), overnight long dwell. **APD** (automated peritoneal dialysis) uses a night cycler.

Catheter

- **Tenckhoff catheter:** silicone with **two Dacron cuffs** (deep cuff in rectus muscle; superficial cuff 2 cm from (near) the exit), exit site pointing downward, tip in the pelvis.
- **Insertion:** open; **laparoscopic** (rectus sheath tunnelling, omentopexy, adhesiolysis); percutaneous Seldinger. Wait (delay full volumes) about 2 weeks before full volumes to prevent leaks.

Adequacy

- Weekly **Kt/V urea ≥1.7** (peritoneal plus residual renal urea clearance × time ÷ urea distribution volume), with clinical well-being.
- **PET** (peritoneal equilibration test) identifies high transporters, who suit (do better on) short dwells (APD).

Advantages versus (over) haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Preferred (favoured) in children, poor vascular access and unstable heart disease.

Contraindications

- **Absolute:** peritoneal adhesions or fibrosis (scarring), loss of membrane function (membrane failure), uncorrectable hernia, no one (nobody) able to do exchanges.
- **Relative:** recent abdominal surgery, stoma, abdominal wall infection, massive polycystic kidneys, morbid obesity, severe malnutrition.

Complications

- **Peritonitis:** diagnosis needs 2 of (3): (1) abdominal pain or **cloudy effluent**; (2) effluent **WCC (white cell count) >100/µL with >50% neutrophils**; (3) positive culture. Commonest: coagulase-negative *Staphylococcus* (touch contamination). Give **intraperitoneal** antibiotics (Gram-positive and Gram-negative cover) covering Gram-positives (vancomycin or cefazolin) and Gram-negatives (ceftazidime or aminoglycoside). Remove the catheter if (for) fungal or refractory peritonitis (no response in 5 days).
- **Exit-site and tunnel infection:** prevent with topical mupirocin or gentamicin.

- **Mechanical:** outflow failure (constipation, omental wrap, migration), pericatheter **leak**, **hernias** (raised abdominal pressure), hydrothorax, scrotal oedema (patent processus vaginalis).
- **Metabolic:** hyperglycaemia, weight gain, hypokalaemia, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**: rare, presents as bowel obstruction.
- **Also know:** PD (peritoneal dialysis) catheter is removed after a successful transplant; retroperitoneal laparoscopy protects the peritoneum for PD patients.

Diagram to draw: Tenckhoff catheter: deep and superficial (two) cuffs, tunnel, downward exit site, tip in the pelvis (pelvic tip).

Key points to remember:

- Peritoneum is the membrane; glucose or icodextrin gives (drives) ultrafiltration.
- Tenckhoff catheter with two cuffs (two-cuff); Kt/V ≥1.7 per week (weekly).
- Peritonitis: cloudy fluid, WCC >100/µL, >50% neutrophils; give intraperitoneal antibiotics.

Pre transplant nephrectomy

Type: SAQ (10 marks)

Source: 2022-Summer Paper 2, Q3e

Assumed meaning: removal of the native kidney(s) before or at kidney transplantation.

Principle

- **Default: leave native kidneys.** They give residual urine (less fluid restriction), erythropoietin and vitamin D activation; nephrectomy adds morbidity.
- Remove them only when they will threaten the patient or graft under **immunosuppression**.

Indications

Group	Indication	Reason
Infection	Chronic pyelonephritis, infected stones, recurrent upper UTI (urinary tract infection) with **high-grade VUR (vesicoureteral reflux)** or obstruction, XGP (xanthogranulomatous pyelonephritis)	Sepsis under immunosuppression
ADPKD (autosomal dominant polycystic kidney disease)	No room for graft (kidney below iliac crest), recurrent cyst infection or bleeding, pain, early satiety	Space and symptoms
Malignancy	Suspicious mass or complex cyst (acquired cystic disease), high tumour risk (e.g. Denys–Drash)	Most RCC (renal cell carcinoma) in recipients arises in native kidneys
Hypertension	Uncontrolled despite multiple drugs	Renin-driven
Heavy proteinuria	Nephrotic syndrome with malnutrition or thrombosis (e.g. congenital nephrotic syndrome)	Hides graft proteinuria, hypercoagulability
Others	Massive polyuria, large stone burden	Graft fluid balance, infection

Timing

- **Before transplantation:** infection or tumour; reconstruction 6–12 weeks earlier.
- **Simultaneous:** ADPKD needing space, ipsilateral through the transplant incision; avoided in small children.
- **After transplantation:** symptomatic ADPKD once the graft works.

Technique

- **Laparoscopic** (transperitoneal or retroperitoneal), open, or **embolisation** in high-risk patients.
- **PD (peritoneal dialysis) patients:** retroperitoneal route keeps PD going; transperitoneal surgery needs temporary haemodialysis.
- Bilateral: prone approach for small kidneys; morcellation or open for large ADPKD kidneys.
- **Ureter:** transect high and leave a normal ureter (for later ureteroureterostomy); **remove a dilated refluxing ureter completely**; keep it as a possible continent stoma if catheterisation may be needed.

Complications

- Bleeding, transfusion (sensitisation), infection, bowel injury, hypotension after bilateral removal, loss of residual function.

Also know: pre-transplant urodynamics; bladder augmentation if capacity <75% expected or pressure >40 cmH_2O.

Diagram to draw: enlarged ADPKD kidney reaching the iliac fossa beside the planned graft site.

> **Key points to remember:**
> - Default: leave native kidneys.
> - Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
> - Retroperitoneal route for PD; leave a normal ureter.

GFR estimation: clinical assessment and in the prospective transplant donor

Asked as:

- Methods for GFR estimation in prospective tx donor. — *Practice Set 9 Paper II Q3a (SAQ)*
- GFR clinical assessment — *Winter 2023 Paper II Q3a (SAQ)*

Type: SAQ (10 marks)
(also answered in 2019-Summer Paper 1, Q3a; donor setting answered in 2019-Summer Paper 1, Q3a)

Assumed meaning: tx = transplant (prospective living kidney donor).

Definition

- GFR (glomerular filtration rate) is the plasma volume filtered by both kidneys per minute; normal about 120 mL/min/1.73 m². It is the standard index of kidney function. The ideal marker is freely filtered and neither secreted nor reabsorbed.

Why it matters (donor)

- The donor loses half the nephron mass; function settles at 60–80% of the pre-donation level. GFR (glomerular filtration rate) is assessed by at least two methods, **including a measured one**.

Methods

1. Serum creatinine and eGFR (estimated GFR)

- Creatinine comes from muscle; freely filtered but also secreted, so it **overestimates GFR** as function falls.
- Creatinine is freely filtered but depends on **muscle mass, diet and age**, and is also secreted, so it overestimates GFR. Varies with **muscle mass, diet and age**; trimethoprim blocks secretion.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021** equation: age and sex, **no race term**; currently recommended. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate above 60 mL/min, where donors lie: a screening test only.

2. Cystatin C

- 13-kDa protein from all nucleated cells; freely filtered, then (fully) reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- The **combined creatinine–cystatin C equation** is more accurate than either alone; it is the most accurate estimate.

3. 24-hour creatinine clearance

- (Urine creatinine × urine volume) ÷ serum creatinine; collection errors common; secretion causes overestimation.

4. Measured GFR (mGFR): reference

Marker	Method
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe and accurate (another source: plasma or urinary clearance)
Iothalamate	Urinary or plasma clearance

Marker	Method
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method); purely filtered; also gives **split function**

5. Split function

- DTPA renogram (or DMSA, dimercaptosuccinic acid) gives each kidney's share. If the difference exceeds 10%, **leave the better kidney with the donor**.

Clinical use

- **CKD (chronic kidney disease) staging (KDIGO, Kidney Disease: Improving Global Outcomes):** G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m², always with albuminuria (ACR, albumin-to-creatinine ratio).
- **Living donor (KDIGO 2017):** see table below; confirm with mGFR.
- **Urology:** drug and contrast dosing, cisplatin eligibility, continent diversion (>35 mL/min), partial versus radical nephrectomy.
- **Acute change:** eGFR is invalid when function is unstable; AKI (acute kidney injury) uses creatinine change and urine output. NGAL (neutrophil gelatinase-associated lipocalin) is emerging.

KDIGO (Kidney Disease: Improving Global Outcomes) 2017 donor decision

GFR (mL/min/1.73 m²)	Decision
≥90	Accept
60–89	Individualise (age, lifetime risk)
<60	Decline

- Initial eGFR is confirmed by mGFR, measured creatinine clearance or creatinine–cystatin C eGFR.
- Indian practice: DTPA GFR with split function, plus CT (computed tomography) angiography.

Diagram to draw: plasma tracer clearance curve after a single injection.

Key points to remember:

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.

Post transplant malignancy

Type: SAQ (10 marks)

Source: 2023-Winter Paper 2, Q3c

Introduction

- Cancer risk after kidney transplantation is about 2–3 times the general population; a leading cause of late death with a functioning graft.

Causes

- **Lost immune surveillance** from immunosuppression (dose and duration matter; depleting antibodies add risk).
- **Oncogenic viruses:** EBV (Epstein–Barr virus) → PTLD (post-transplant lymphoproliferative disorder); HHV-8 (human herpesvirus-8) → Kaposi sarcoma; HPV (human papillomavirus) → skin and anogenital cancer; hepatitis B/C → liver cancer.
- **Direct drug effects:** azathioprine sensitises skin to UV (ultraviolet) light; calcineurin inhibitors promote tumour growth. mTOR (mammalian target of rapamycin) inhibitors are **antitumour**.
- **Pre-existing risk:** ESRD (end-stage renal disease), acquired cystic disease, analgesic nephropathy; rarely donor-transmitted.

Common cancers

Cancer	Key facts
Non-melanoma skin	Commonest; squamous exceeds basal cell (reverse of general population); multiple, aggressive
PTLD	Mostly B-cell, EBV-driven; highest in first year, children and **EBV-negative recipient of EBV-positive donor**; may present as graft mass
Kaposi sarcoma	HHV-8; regresses with less immunosuppression and switch to sirolimus
RCC (renal cell carcinoma)	4.6% of post-transplant cancers; **90% in native kidneys**; graft tumours mostly papillary, low grade
Bladder cancer	About **3-fold** risk; more aggressive
Prostate	Same incidence as general population

Management

- **Reduce immunosuppression** and switch to an mTOR inhibitor; sirolimus conversion reduced new skin squamous cancers (Euvrard, 2012).
- **PTLD:** reduce immunosuppression, rituximab, then chemotherapy (R-CHOP).
- RCC, prostate and bladder: treat as in the general population; protect graft vessels and ureter at surgery. BCG (bacillus Calmette–Guérin) is usable in selected bladder cancer.

Prevention and screening

- Pre-transplant cancer screening and waiting periods (none after curative treatment of small RCC <3 cm, Gleason 6 prostate cancer, superficial bladder cancer).
- Sun protection, yearly skin examination, age-appropriate screening.
- KDIGO (Kidney Disease: Improving Global Outcomes) does **not** endorse routine native-kidney screening; screen only high-risk recipients.

Diagram to draw: pie chart of post-transplant cancers (skin largest).

Key points to remember:

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Pre transplant Immunosupression treatment options

Type: SAQ (10 marks)

Source: 2023-Winter Paper 2, Q3e (maintenance drugs answered in 2020-Summer Paper 4, Q1)

Taken as induction immunosuppression and desensitisation given before or at transplantation.

Principle

- Rejection risk is highest in the first weeks. **Induction** gives intense immunosuppression around transplant; **desensitisation** removes preformed antibody in HLA (human leucocyte antigen)- or ABO-incompatible recipients.
- Immunological risk is assessed first: cPRA (calculated panel-reactive antibody), DSA (donor-specific antibody) on Luminex beads, crossmatch (CDC, complement-dependent cytotoxicity, and flow).

Induction agents

Agent	Mechanism	Use and drawbacks
Basiliximab	Anti-CD25 (IL-2, interleukin-2, receptor); non-depleting	Low-risk recipients; 20 mg on days 0 and 4; few side effects
rATG (rabbit antithymocyte globulin)	Polyclonal T-cell depletion	High-risk (sensitised, retransplant, delayed graft function); less rejection than basiliximab; cytokine release, leucopenia, infection, PTLD (post-transplant lymphoproliferative disorder)
Alemtuzumab	Anti-CD52; profound depletion	Allows steroid avoidance; off-label
IV methylprednisolone	Cytokine suppression	Given intraoperatively with all regimens

- Over 90% receive induction, then **tacrolimus + mycophenolate** ± steroids. In living-donor transplants, tacrolimus and mycophenolate often start days before surgery.

Desensitisation: HLA-incompatible

- **Plasmapheresis** (removes antibody) plus **low-dose IVIG (intravenous immunoglobulin)** after each session; or **high-dose IVIG** (immunomodulation).
- **Rituximab** (anti-CD20) reduces B cells that renew antibody. Bortezomib (plasma cells) and eculizumab (anti-C5) in selected cases.
- **Imlifidase:** bacterial enzyme that **cleaves all IgG** within hours, turning a positive crossmatch negative; EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients.
- Desensitised live-donor transplant gives better survival than staying on dialysis (Montgomery, 2011).

Desensitisation: ABO-incompatible

- **Rituximab** 2–4 weeks before; **plasmapheresis or immunoadsorption** to lower anti-A/B titres to a centre target (commonly ≤1:8); IVIG; tacrolimus–mycophenolate started before surgery. Splenectomy is now obsolete.
- Post-transplant titre monitoring; A2 donors carry lower risk.

Alternatives

- **Kidney paired exchange** avoids desensitisation; acceptable-mismatch programmes.

Diagram to draw: antibody titre falling with plasmapheresis sessions to target before transplant.

Key points to remember:

- Basiliximab for low risk; rATG for high immunological risk.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.

Brainstem death: biology and clinical implications

Asked as:

- Brain stem Death: Biology and clinical implications. — *Winter 2024 Paper II Q3d (SAQ)*
- Brainstem death — *Practice Set 13 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)
(Winter 2024 Paper II Q3d: also answered in 2022-Summer Paper 4, Q3b and 2015-Summer Paper 1, Q3d) (Practice Set 13 Paper II Q3c: also answered in 2022-Summer Paper 4, Q3b and 2024-Winter Paper 2, Q3d)

Definition and law

- **Irreversible loss of the capacity for consciousness and breathing**; legal death (legally death) in India: THOA (Transplantation of Human Organs Act) 1994, amended 2011, Rules 2014.

Biology (pathophysiology)

```
Brain injury -> oedema -> raised ICP (intracranial pressure)
 -> falling cerebral perfusion -> herniation ("coning")
 -> Cushing reflex (hypertension, bradycardia)
 -> catecholamine storm (hypertension, tachyarrhythmia)
 -> loss of sympathetic tone -> vasodilatation, hypotension
```

System	Consequence
Heart and vessels	Storm causes myocardial injury and arrhythmia, then **hypotension**
Lungs	Neurogenic pulmonary oedema
Pituitary	**Diabetes insipidus (DI)** (no ADH, antidiuretic hormone): polyuria, hypernatraemia; low thyroid hormone and cortisol
Hypothalamus	**Hypothermia** (poikilothermia)
Blood	DIC (disseminated intravascular coagulation) from brain thromboplastin
Whole body	Cytokine injury to organs; hyperglycaemia

- Consequences (summary): **diabetes insipidus** (polyuria, hypernatraemia), hypothermia, DIC, neurogenic pulmonary oedema.

Diagnosis

Preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude drugs (sedatives, relaxants), hypothermia, metabolic or endocrine causes, shock.

Tests: all brainstem reflexes absent

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic	Turn head: eyes move with head	VIII, III, VI
Oculovestibular	50 mL ice-cold water in each ear: no eye movement	VIII, III, VI
Motor in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Pharynx touch, tracheal suction: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect ventilator with oxygen insufflation; **no breathing at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Ancillary tests** if examination is impossible: angiography, perfusion scan, EEG (electroencephalogram).

Indian Board of Medical Experts

- **India:** 4-member Board (four members): hospital in-charge; independent specialist nominated from an approved panel; neurologist or neurosurgeon; treating doctor. **No transplant-team member.**
- **Two sets of tests at least 6 hours apart (two tests ≥6 hours apart)**; recorded on **Form 10**.

Clinical implications: donor management

- **Mandatory request** for donation; next-of-kin consent; NOTTO (National Organ and Tissue Transplant Organisation) allocates.
- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** (pressor, treats DI); thyroid hormone, methylprednisolone (steroids), insulin; warming; protective ventilation.
- **DBD (donation after brain death)** keeps circulation, so warm ischaemia is minimal.

Diagram to draw: herniation sequence leading to storm, DI, hypothermia, DIC. Flowchart: coma → preconditions → test 1 → 6 h → test 2 → Form 10 → consent → NOTTO.

Key points to remember:

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Six (brainstem) reflexes absent; apnoea at $PaCO_2$ ≥60 mmHg.
- Four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

Anomalies of renal artery.

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 2, Q3c

Normal pattern

- One renal artery from the aorta at L1–L2; anterior and posterior divisions; **five segmental end arteries** (apical, upper, middle, lower, posterior). The right artery passes behind the IVC (inferior vena cava).

Terminology

Term	Definition
Multiple renal arteries (preferred)	More than one vessel from the aorta to one kidney
Aberrant	Arises from a vessel other than aorta or main renal artery (iliac, SMA (superior mesenteric artery), gonadal)
Accessory	Two or more branches supplying the same segment

Incidence

- Single artery in **71–85%**; variants in 25–40%. Accessory arteries 25–28% (aorta T11–L4), usually to the **lower pole**. True aberrant vessels are rare except in **ectopic and horseshoe kidneys**.
- **Also know:** early prehilar branching; right artery anterior to IVC in about 5%.

Embryology

- The ascending kidney takes successive aortic branches; the lower ones normally degenerate. **Failure of degeneration** gives polar or multiple arteries.

Clinical significance

- **PUJ (pelviureteric junction) obstruction** from a lower-pole crossing vessel; but crossing vessels occur in a fifth to a quarter of normal PUJs.
- Compression of an infundibulum: hydrocalyx, pain, stones, UTI (urinary tract infection).
- **Surgery:** bleeding at endopyelotomy (incise posterolaterally); polar arteries are end arteries, so ligation infarcts that segment; **donor nephrectomy** needs CT (computed tomography) angiography; aortic surgery.
- Not linked to hypertension.

Renal artery aneurysm

- About **0.1%**. **Abeshouse types:** saccular (about 93%), fusiform, dissecting, arteriovenous.
- Mostly asymptomatic; hypertension, haematuria, pain; wreath-like calcification.
- **Treat** (endovascular or open) if rupture, **>2 cm**, expanding, **pregnancy or planned pregnancy**, symptoms or emboli.

Renal arteriovenous fistula

	Congenital (cirsoid AVM (arteriovenous malformation))	Acquired
Share	<25%	Majority (biopsy, PCNL (percutaneous nephrolithotomy), trauma)
Features	Bruit, haematuria, high-output failure	Often resolves

	Congenital (cirsoid AVM (arteriovenous malformation))	Acquired
Treatment	Selective embolisation	Observe; embolise if bleeding

Diagram to draw: kidney with main artery, lower-pole accessory artery crossing the PUJ.

Key points to remember:

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

Mycophenolate Mofetil.

Type: SAQ (10 marks)

Source: Practice Set 2 Paper 2, Q3b (immunosuppressants also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

Introduction

- **MMF (mycophenolate mofetil):** an antimetabolite immunosuppressant; morpholinoethyl ester **prodrug** of MPA (mycophenolic acid). With tacrolimus it forms maintenance therapy in **over 90%** of kidney recipients.

Mechanism

- MPA is a reversible, non-competitive inhibitor of **IMPDH (inosine monophosphate dehydrogenase)**, the key enzyme of **de novo guanosine nucleotide synthesis**.
- **Why lymphocyte-selective:** T and B lymphocytes depend on the de novo pathway; other cells use the salvage pathway. MPA also inhibits the type II isoform found in activated lymphocytes.
- Result: arrest of T- and B-cell proliferation and reduced antibody formation.

Pharmacology

Feature	Detail
Dose (kidney)	**1 g twice daily** oral or IV; EC-MPS (enteric-coated mycophenolate sodium) 720 mg twice daily is equivalent
Metabolism	Hydrolysed to MPA; liver glucuronidation to inactive MPAG (MPA glucuronide); **enterohepatic recirculation**
Excretion	MPAG in urine; accumulates in renal failure
Interactions	**Ciclosporin lowers MPA** (blocks enterohepatic recirculation), tacrolimus does not; antacids, cholestyramine reduce absorption; never with azathioprine

Uses

- Kidney transplant maintenance (tacrolimus + MMF ± steroids); other solid organs.
- Lupus nephritis, vasculitis; steroid-refractory immune-related adverse events of checkpoint inhibitors.

Adverse effects

- **GI (gastrointestinal):** diarrhoea, nausea, colitis (commonest; dose split or EC-MPS).
- **Marrow:** leucopenia, anaemia, thrombocytopenia: blood count weekly in month 1.
- **Infection:** CMV, BK virus (ureteric stricture), rarely PML (progressive multifocal leucoencephalopathy).
- **Malignancy:** skin cancer, PTLD (post-transplant lymphoproliferative disorder).
- **Teratogenic:** miscarriage, ear and facial malformations; contraception mandatory; switch to azathioprine before conception.
- Advantage: **not nephrotoxic**; no hypertension, diabetes or dyslipidaemia.

Evidence

- Three mid-1990s randomised trials (US, European, Tricontinental) showed fewer early acute rejections than azathioprine or placebo, so MMF replaced azathioprine.
- **ELITE-Symphony (2007):** low-dose tacrolimus + MMF + daclizumab + steroids gave the lowest rejection and best GFR.

Diagram to draw: purine synthesis pathway showing IMPDH block (de novo) and the salvage pathway spared.

Key points to remember:

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Metabolic abnormalities in CKD

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 2, Q3e

Definition

- **CKD (chronic kidney disease), KDIGO (Kidney Disease: Improving Global Outcomes):** abnormal kidney structure or function for **≥3 months**. Staged by GFR (glomerular filtration rate) G1–G5 (G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m²) and albuminuria A1–A3.
- Each abnormality is a lost kidney function: excretion, acid–base balance, hormone production.

Abnormalities

Abnormality	Mechanism	Management
Sodium and water	Fixed urine concentration; sodium retention → overload, hypertension	Salt restriction; diuretics (chlorthalidone works even in advanced CKD)
Hyperkalaemia	Less distal sodium delivery; RAAS (renin–angiotensin–aldosterone system) blockers lower aldosterone	Potassium binders and diuretics, so RAAS blockers can continue
Metabolic acidosis	Less ammonium excretion; retained sulphate and phosphate (anion gap rises late); bicarbonate loss	Sodium bicarbonate or citrate to keep bicarbonate **≥22 mEq/L**; slows progression, protects bone and muscle
CKD-MBD (mineral and bone disorder)	Phosphate retention → FGF-23 (fibroblast growth factor 23) rises → less **calcitriol** → hypocalcaemia → **secondary hyperparathyroidism**	Phosphate restriction, binders with meals, active vitamin D; calcimimetic; parathyroidectomy if tertiary
Anaemia	Erythropoietin falls, mainly below GFR 30	Iron first, then erythropoietin; Hb (haemoglobin) target 10–11.5 g/dL (harm above 13)
Hyperuricaemia	Reduced excretion	Treat gout; asymptomatic lowering unproven (CKD-FIX)
Lipids	High triglycerides, low HDL (high-density lipoprotein)	Statin ± ezetimibe if ≥50 years, not on dialysis (SHARP)
Glucose	Insulin resistance, but slower insulin clearance → hypoglycaemia	Lower insulin doses

Bone disease (renal osteodystrophy)

- **High turnover:** osteitis fibrosa (PTH [parathyroid hormone] excess).
- **Low turnover:** adynamic bone (oversuppressed PTH), osteomalacia.
- **Tertiary hyperparathyroidism:** autonomous gland, hypercalcaemia; persists after transplant.
- **Vascular calcification** adds cardiovascular death.

Other endocrine and uraemic effects

- **Also know:** hypogonadism, hyperprolactinaemia, erectile dysfunction (better after transplant); growth failure in children; protein-energy wasting; dialysis amyloid; uraemic platelet dysfunction (desmopressin).

- **Urological relevance:** dose drugs by GFR; avoid gadolinium if eGFR (estimated GFR) <30; each AKI (acute kidney injury) speeds progression.

Diagram to draw: CKD-MBD cascade: phosphate ↑ → FGF-23 ↑ → calcitriol ↓ → calcium ↓ → PTH ↑ → bone disease.

Key points to remember:

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

Renal transplant in neurogenic bladder

Type: SAQ (10 marks)

Source: Practice Set 4 Paper 2, Q3a

Why the bladder matters

- A neurogenic bladder with **poor compliance, low capacity or detrusor overactivity** transmits high pressure to the graft even without reflux. Poor emptying adds infection. Recurrent pyelonephritis, not augmentation, causes graft loss.
- **Five goals:** unimpeded drainage, low-pressure capacious storage, effective emptying, no infection, fewest procedures.

Pretransplant evaluation

- Renal ultrasound, VCUG (voiding cystourethrogram) and **urodynamics** (capacity, compliance, overactivity, emptying, sphincter). Repeat them; the bladder changes over time.
- **Target:** storage to expected capacity at detrusor pressure **<40 cm H_2O**.
- **Defunctionalised bladder** (anuria or diversion): **bladder cycling** (saline fills of increasing volume via catheter), then urodynamics.

Pretransplant optimisation (stepwise)

```
Antimuscarinic (oxybutynin), mirabegron, botulinum toxin
  → CIC (clean intermittent catheterisation) ± overnight drainage
  → bladder cycling if defunctionalised
  → still >40 cm H₂O, <75% expected capacity or CIC ≤3-hourly → augmentation
```

- **Augmentation:** ileum or sigmoid most used; dilated ureter (ureterocystoplasty) preferred when available (no mucus or metabolic effects). **Stomach avoided** (haematuria–dysuria syndrome in anuric patients). Augmentation does not clearly raise transplant complications.
- **Mitrofanoff** channel if urethral CIC is difficult; site it **medial** to the planned transplant incision.
- Do reconstruction **before** transplant; allow **6–12 weeks** to heal. Neuropathic bladders may deteriorate under high graft output, so earlier reconstruction is justified.
- **Native nephrectomy** only for named reasons: recurrent upper-tract infection, massive reflux, stones, uncontrolled hypertension.

At transplant

- Extravesical non-refluxing ureteroneocystostomy into bladder or augment (know the augment pedicle). Ureteroureterostomy to a native ureter is an option.
- Ileal conduit drainage if the bladder is unusable.

Post-transplant care

- Continue CIC; colonisation is expected, so treat only symptomatic UTI (urinary tract infection).
- **Creatinine rise with hydronephrosis:** catheter drainage for 1–2 weeks first. If it improves, the bladder is the cause; otherwise stent or nephrostomy and consider rejection.
- Febrile UTI with graft reflux: reimplant after optimising the bladder.
- Lifelong surveillance of augments (malignancy, stones).

Diagram to draw: stepwise bladder optimisation ladder.

Key points to remember:

- Target <40 cm H_2O storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

Pre-emptive renal transplant

Type: SAQ (10 marks)

Source: Practice Set 6 Paper 2, Q3a

Definition

- Kidney transplantation **before starting maintenance dialysis** in a patient with advanced CKD (chronic kidney disease).

Rationale

- Every year on dialysis adds cardiovascular disease, vascular calcification and bone–mineral damage that the graft later inherits. Pre-emptive transplantation **avoids this exposure**.
- Transplantation beats dialysis: Wolfe (1999) showed early risk is higher, equal at 106 days, then 68% lower long term.

Advantages

- Better patient and graft survival than transplantation after dialysis; longer dialysis time worsens outcome (registry data).
- No dialysis access surgery, access infection or catheter complications.
- Fewer transfusions, so less HLA (human leucocyte antigen) sensitisation.
- Better quality of life, employment; lower cost.
- Children: better growth and development.

Frequency

- 28.7% of US transplants in 2021 (living donor 51%, deceased donor 21%, children 56%). Commonest with a **living donor**.

Timing and evaluation

- Evaluate early, ideally before dialysis. KDIGO (Kidney Disease: Improving Global Outcomes) 2020: refer when eGFR (estimated glomerular filtration rate) is <30 mL/min/1.73 m², 6–12 months before expected dialysis. Living-donor transplant is usually done at eGFR <20 with progressive decline.
- Deceased-donor listing (US) at eGFR ≤20.
- Full recipient work-up: cardiac, vascular (iliac calcification), infection (TB [tuberculosis], hepatitis B and C, HIV [human immunodeficiency virus]), malignancy, psychosocial; recurrence risk (primary FSGS [focal segmental glomerulosclerosis]).
- **Urological:** bladder function and lower tract (valves, neurogenic bladder); native nephrectomy only for specific indications.
- India: mainly living related donors under THOA (Transplantation of Human Organs Act).

Limitations

- Needs early referral and a ready donor.
- Transplant may come earlier than needed, using up dialysis-free native function; adherence untested.
- In advanced CKD, bowel augmentation may speed decline.

Also know: ABO-incompatible or paired exchange programmes widen pre-emptive options.

Diagram to draw: Wolfe curve: relative risk of death after transplant vs dialysis over time.

Key points to remember:

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules

Asked as:

- Amendments to THOA: 2011. — *Practice Set 9 Paper II Q3f (SAQ)*
- HOTA and its modification — *Practice Set 11 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

Indian law; not covered in the chapter notes. Assumed meaning (Practice Set 11): India's Transplantation of Human Organs Act (THOA) 1994, its 2011 amendment and 2014 Rules.

Background: THOA 1994 aims

- **THOA (Transplantation of Human Organs Act) 1994** regulated (regulates) removal, storage and transplantation of organs, banned commercial dealing (**prevents commercial dealing**) and legalised **brain-stem death** (**recognised brain-stem death as legal death**, allowing deceased donation).
- Organ trade continued and deceased donation stayed low. The **Transplantation of Human Organs (Amendment) Act 2011** responded; **Rules followed in 2014**.

Main provisions (1994)

- **Living donors: near relatives** (spouse, parents, children, siblings) after verification. Others, donating "by reason of affection or attachment", need **Authorisation Committee** approval.
- **Appropriate Authority** registers and inspects hospitals; only registered centres may transplant.
- **Brain-stem death certification by a Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon; treating doctor. **None from the transplant team.** Two sets of tests ≥6 hours apart.
- **Organ sale is a punishable offence.**

Main amendments: 2011 amendment (THOTA, Transplantation of Human Organs and Tissues Act)

Area	Change	Purpose
Scope	**Tissues** added (included); renamed the **Transplantation of Human Organs and Tissues Act (THOTA)**; **tissue banks** registered	Regulates tissue banks
Near relatives	**Grandparents and grandchildren** added to spouse, children, parents and siblings (become near relatives)	Wider related donor pool
Swap donation	**Paired exchange** (swap (paired) donation) between near-relative pairs allowed, with Authorisation Committee approval	Overcomes ABO or crossmatch incompatibility
Retrieval centres	Hospitals with ICU (intensive care unit) facilities can register as **non-transplant organ retrieval centres**	
Transplant coordinator	Mandatory in every registered transplant hospital	Organises donation
Mandatory ("required") request	ICU doctor (staff) must ask relatives (families) of a potential brain-dead donor (brain-dead patients) about donation and inform the retrieval centre	More deceased donors
Brain-death board	If a neurologist or neurosurgeon is unavailable (if no neurologist), a surgeon, physician, anaesthetist or intensivist nominated by the hospital can serve	Smaller hospitals can certify

Area	Change	Purpose
Eye donation	Trained technicians may remove corneas	
Registry and network	National human organ and tissue removal and storage network and **national registry** of donors and recipients (now run by NOTTO, the National Organ and Tissue Transplant Organisation)	Fair allocation
Penalties	Raised: **higher penalties** for organ trade (up to 10 years' imprisonment; fine up to ₹1 crore) (see below)	Deterrence

Penalties under the amendment

- **Removal of organs without authority:** imprisonment up to **10 years** and fine up to ₹20 lakh.
- **Commercial dealing in organs:** imprisonment **5–10 years** and fine **₹20 lakh to ₹1 crore**.
- Registered medical practitioner: name removed from the register for **3 years** for a first offence and **permanently** for a later offence.

Safeguards retained

- Unrelated donors and foreigners need **Authorisation Committee** approval.
- Minors and mentally ill persons cannot be living donors except in exceptional, approved cases.

Rules 2014 (THOT Rules)

- Standard forms for consent and certification (**Form 10** for brain-stem death); brainstem reflexes and apnoea test set out.
- Authorisation Committee interviews donor and recipient to exclude coercion or payment.
- **Also know:** regional and state networks (ROTTO, SOTTO); 2023 guidelines removed the domicile rule and age limit for deceased-donor listing.

Diagram to draw: flowchart: potential donor (brain-dead patient) in ICU → required request → transplant coordinator → brain-stem death board (Board of four) → family consent → retrieval → national registry and allocation (NOTTO allocation).

Key points to remember:

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors.
- 2011: tissues added; grandparents and grandchildren became near relatives.
- Swap donation, retrieval centres, transplant coordinators (coordinator), required request, higher penalties.
- Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.

Marginal kidney donor

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 2, Q3d (also answered in 2024-Winter Paper 4, Q3e and 2016-Summer Paper 3, Q3f)

Definition

- A **marginal (extended) donor** is one whose kidney carries **more DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- They still give better survival than staying on dialysis (Ojo, 2001).

Deceased marginal donors

- **ECD (expanded criteria donor), UNOS (United Network for Organ Sharing):** age **60 or over**, or age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**. Graft-loss risk about 1.7 times a standard donor.
- **KDPI (Kidney Donor Profile Index) >85%** replaced ECD in the US (2014); it uses 10 donor factors. Longevity matching pairs low-KDPI kidneys with recipients with the best EPTS (Estimated Post-Transplant Survival).
- **DCD (donation after circulatory death):** warm ischaemia raises DGF; long-term survival similar to DBD (donation after brain death).
- **Others:** donor AKI (acute kidney injury); HCV (hepatitis C virus)-positive donors with antiviral treatment; paediatric donors under 15 kg (en bloc).

Living marginal donors

- Older donors, controlled hypertension, obesity, small stones, multiple renal arteries. Acceptable after careful evaluation and consent.

Evaluation: Remuzzi biopsy score

- Pre-implantation biopsy; four compartments, each 0–3 (total 0–12):

Component	0	1	2	3
Glomerular sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial narrowing	None	Wall < lumen	Wall = or slightly > lumen	Severe narrowing

- **0–3: single** transplant; **4–6: dual** kidney transplant; **7–12: discard**.

Optimising use

- Short cold ischaemia; **HMP (hypothermic machine perfusion)** reduces DGF without changing long-term survival.
- **Old-for-old** allocation (Eurotransplant Senior Program, 65 or over).
- Informed consent; induction; delayed calcineurin inhibitor if DGF.
- **Urological point:** older kidneys have more urine leak and ureteric stricture.

Diagram to draw: Remuzzi grid with four compartments scored 0–3.

Key points to remember:

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.

Rapid Revision Sheet – Section 15 · Renal, Nephrology & Transplant

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD) (LAQ)

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.
- CRRT for unstable AKI; PD preferred in children.

Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval (LAQ)

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys en bloc last (kidneys out last); Carrel patch, IVC for right vein; machine perfusion reduces DGF.

Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment (LAQ)

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- 2K1C = renin-dependent; 1K1C = volume-dependent.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR (function) in a viable kidney (RI <0.8), resistant hypertension.

Acute renal failure (AKI) and acute tubular necrosis (LAQ)

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Principles of vascular anastomosis and vascular access surgery for HD (LAQ)

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal → banding or DRIL.

Role of vascular grafts in urology (SAQ)

- Transplant: Carrel patch, bench reconstruction; preserve the lower pole artery.
- IVC: patch if lumen <50%; PTFE replacement if wall invaded.
- Post-chemotherapy RPLND: IVC can be ligated, aorta must be grafted.

Pre-operative evaluation of the living (related and unrelated) renal donor (SAQ)

- KDIGO 2017: risk-based; GFR ≥90 accept; decline GFR <60 or ACR >100 (albuminuria >100 mg/g declines).
- ABO and crossmatch first; CT angiography for anatomy.
- Leave the better kidney with the donor (better kidney stays with donor).
- THOA: near relatives or authorisation committee; unrelated donor = Authorisation Committee approval.
- Interview video-recorded; no commerce.

UW solution and organ preservation solutions (SAQ)

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

CAPD (Continuous Ambulatory Peritoneal Dialysis) (SAQ)

- Peritoneum is the membrane; glucose or icodextrin gives (drives) ultrafiltration.
- Tenckhoff catheter with two cuffs (two-cuff); Kt/V ≥1.7 per week (weekly).
- Peritonitis: cloudy fluid, WCC >100/µL, >50% neutrophils; give intraperitoneal antibiotics.

Pre transplant nephrectomy (SAQ)

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

GFR estimation: clinical assessment and in the prospective transplant donor (SAQ)

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.

Post transplant malignancy (SAQ)

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Pre transplant Immunosupression treatment options (SAQ)

- Basiliximab for low risk; rATG for high immunological risk.

- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.

Brainstem death: biology and clinical implications (SAQ)

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Six (brainstem) reflexes absent; apnoea at $PaCO_2$ ≥60 mmHg.
- Four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

Anomalies of renal artery. (SAQ)

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

Mycophenolate Mofetil. (SAQ)

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Metabolic abnormalities in CKD (SAQ)

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

Renal transplant in neurogenic bladder (SAQ)

- Target <40 cm H_2O storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

Pre-emptive renal transplant (SAQ)

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules (SAQ)

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors.
- 2011: tissues added; grandparents and grandchildren became near relatives.
- Swap donation, retrieval centres, transplant coordinators (coordinator), required request, higher penalties.
- Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.

Marginal kidney donor (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.

Section 16 · BPH — Long answer questions (LAQ)

BPH: medical management and surgical modalities

Asked as:

- Describe medical management of BPH and role of various surgical modalities in its management. — *Summer 2015 Paper II Q1 (LAQ)*
- Medical treatment of BPH — *Practice Set 6 Paper II Q3f (SAQ)* (also answered in 2015-Summer Paper 2, Q1)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction, pathophysiology and rationale

- **BPH (benign prostatic hyperplasia)** is stromal and epithelial hyperplasia of the transition zone; it may cause BPO (benign prostatic obstruction) and **LUTS (lower urinary tract symptoms)**.
- Obstruction has a **static** component (gland bulk, DHT (dihydrotestosterone)-dependent / DHT-driven gland bulk) and a **dynamic** component (dynamic part: α1A-mediated smooth-muscle tone; static part: DHT-driven gland bulk). Drugs target each. The bladder responds with detrusor overactivity or later underactivity.

Evaluation (before treatment)

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, frequency-volume chart, Qmax (maximum flow), PVR (post-void residual); PVR >300 mL is managed as chronic retention (PVR >300 mL = chronic retention).
- Measure prostate volume before surgery (AUA (American Urological Association)); pressure-flow study if unclear.

Medical management

Option / class	Doses	Key facts
Watchful waiting	—	Mild or non-bothersome LUTS; lifestyle advice
α1-blockers (first line)	Tamsulosin 0.4 mg; alfuzosin 10 mg; silodosin 8 mg; doxazosin 4–8 mg (titrate)	First line; act within days; no effect on gland size, AUR (acute urinary retention) or surgery (do not prevent AUR or surgery). Hypotension, ejaculatory dysfunction; silodosin causes most anejaculation. **IFIS (intraoperative floppy iris syndrome)**
5-ARIs (5α-reductase inhibitors)	Finasteride 5 mg; dutasteride 0.5 mg	For glands (prostate) >30–40 mL or PSA >1.5 ng/mL. Shrink the gland about 18–25% (another source: ~20%) (maximal at 6 months; act after 6 months), halve PSA. PLESS (Proscar Long-Term Efficacy and Safety Study): AUR or surgery down (↓) 51%. Low libido, ED (erectile dysfunction). Also BPH haematuria
Combination	—	For high-risk men. The α-blocker gives early relief; the 5-ARI prevents hard events
Antimuscarinic or β3-agonist (mirabegron, vibegron)	Add-on	Add for persisting storage symptoms; trials included only men with low PVR (PVR <150 mL)

Option / class	Doses	Key facts
Tadalafil 5 mg daily	PDE5 (phosphodiesterase-5) inhibitor	LUTS with or without (±) ED; not with nitrates
Desmopressin	—	Nocturnal polyuria (night urine >33% of 24-h output); check (monitor) sodium, especially ≥65 years

Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell (↓) 39% (doxazosin), 34% (finasteride) and **66% (combination)**; only finasteride or combination (finasteride-containing arms) reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.
- Combine when **PV (prostate volume) >30–40 mL and PSA >1.5 ng/mL**; α-blocker may stop (withdrawable) after 6 months of combination (SMART (Symptoms Management After Reducing Therapy) trial).
- Tadalafil + finasteride fixed dose approved (FDA [Food and Drug Administration] 2021).

Side effects and important cautions

- **Side effects**: α-blockers cause dizziness, postural hypotension, ejaculatory dysfunction and **IFIS (intraoperative floppy iris syndrome)**, which persists after stopping, so avoid starting them before planned cataract surgery. 5-ARIs: low libido, ED, gynaecomastia.
- **IFIS:** billowing iris, prolapse, miosis; stopping the drug does not prevent it.
- 5-ARIs: double the PSA; not for cancer prevention (more high-grade tumours in PCPT [Prostate Cancer Prevention Trial] and REDUCE).
- **AUR**: catheterise (catheter), then TWOC (trial without catheter) on an α-blocker (ALFAUR (Alfuzosin in Acute Urinary Retention): success 62% vs 48%).
- **Also know:** phytotherapy (saw palmetto) not recommended by guidelines.

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, failed medical therapy (surgery if refractory).

Modality	Size (AUA 2023)	Key feature
TURP (transurethral resection), monopolar or bipolar	30–80 mL	Reference standard; retrograde ejaculation ~two-thirds
TUIP (transurethral incision)	≤30 mL	Less retrograde ejaculation; more reoperation
PVP (photoselective vaporization)	≤80 mL	Safe when anticoagulation cannot stop; no histology
HoLEP/ThuLEP (laser enucleation)	Any size	Lowest retreatment, least bleeding
Simple prostatectomy (open or robotic)	>80 mL	Complete removal; diverticula or stones
Aquablation	30–80 mL (consider 80–150)	Heat-free; preserves ejaculation
PUL (prostatic urethral lift), Rezūm (water vapour)	30–80 mL (PUL: no median lobe)	Preserve sexual function; higher retreatment
iTind (temporary implant)	25–75 mL, no median lobe	Removed after a few days; modest efficacy

Modality	Size (AUA 2023)	Key feature
PAE (prostate artery embolisation)	Larger glands (AUA 2023: may be offered)	Local anaesthesia; less effective than TURP; radiation, non-target embolisation

- **TUR syndrome** (monopolar TURP): hypo-osmolar glycine or water is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with gland >45 mL and resection >90 minutes. Treat with diuretics and, if severe, 3% saline, correcting slowly. **Bipolar TURP** uses saline and eliminates it.
- **Complications**: bleeding, stricture, bladder neck contracture, retreatment.
- **GOLIATH**: 180 W PVP non-inferior to TURP. **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior, with less retrograde ejaculation (10% vs 36%).
- **Also know:** TUVP (transurethral vaporization); TUMT (transurethral microwave therapy) dropped as legacy; Optilume paclitaxel-coated balloon (2023).

Follow-up

- Reassess at 4–12 weeks (α-blocker), 3–6 months (5-ARI) (another source: 6 months for 5-ARIs), then yearly; surgery if refractory.

Diagram to draw: Size-based surgical algorithm (≤30, 30–80, >80 mL); algorithm: LUTS → size, PSA, storage symptoms → drug choice.

Key points to remember:

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Section 16 · BPH — Short answer questions (SAQ)

Millin's prostatectomy

Type: SAQ (10 marks)

Source: 2015-Summer Paper 2, Q3c

Definition

- **Retropubic extravesical simple prostatectomy** (Millin, 1945): enucleation through a transverse anterior capsulotomy without opening the bladder. The plane is the surgical capsule (compressed peripheral zone).

Indications and work-up

- LUTS (lower urinary tract symptoms) or obstructive complications with gland **>80–100 g**, or when lithotomy is impossible. Prefer suprapubic (Freyer) for a large median lobe, diverticulum or stones.
- Work-up: exclude cancer (PSA (prostate-specific antigen), MRI (magnetic resonance imaging)), TRUS (transrectal ultrasound) volume, cystoscopy, ± urodynamics.

Steps

1. Extraperitoneal approach to the **space of Retzius**; control the dorsal venous complex.
2. **Transverse capsulotomy ~2 cm below the bladder neck** (a longitudinal cut may extend into the sphincter).
3. **Finger enucleation** in the adenoma–capsule plane.
4. **Divide the apical urethra under vision**, sparing the sphincter.
5. Trim the bladder neck lip; advance mucosa into the fossa.
6. **Figure-of-eight sutures at 5 and 7 o'clock**, where prostatic arteries enter.
7. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Fossa haemostasis	Directly visible	Less visible
Stones, diverticulum, median lobe	Difficult	Ideal
Specific risk	Dorsal venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%), urine leak, infection, ureteric injury, bladder neck contracture (2–6%), stricture, thromboembolism.
- Retrograde ejaculation 80–90%; ED (erectile dysfunction) 3–5%; incontinence rare.
- **Also know:** bleeding control by catheter traction, Malament purse-string or O'Conor capsular plication; vesicocapsular variant; versus TURP (transurethral resection of prostate): lower retreatment, no TUR syndrome, longer stay.

Modern alternatives

- HoLEP (holmium laser enucleation), robotic (multiport, single-port) prostatectomy: less bleeding.

Diagram to draw: Sagittal capsulotomy view.

Key points to remember:

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

Catheter free trial

Type: SAQ (10 marks)

Source: 2016-Summer Paper 2, Q3e

Assumed meaning: trial without catheter (TWOC) after acute urinary retention.

Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

Types of AUR

Spontaneous	Precipitated
BPH (benign prostatic hyperplasia) progression	Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs
TWOC fails more often; about 75% later need surgery	Better TWOC outcome; about a quarter need surgery

Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.

- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

Diagram to draw: algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

> **Key points to remember:**
> - TWOC is first choice after AUR; success about 60%.
> - ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
> - Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

Irrigation fluids for TURP.

Type: SAQ (10 marks)

Source: 2017-Summer Paper 2, Q3a

Ideal irrigant

- Transparent, **non-conducting** (monopolar), isotonic, non-toxic, non-haemolytic, cheap.
- **Why non-conducting?** In monopolar TURP (transurethral resection of the prostate), current flows loop → patient → pad; ionic saline would disperse it. **Bipolar** TURP completes the circuit in the instrument, so it uses **normal saline**.

Fluids

Fluid	Osmolality (approx.)	Points
Sterile water	0	Best vision, cheap; haemolysis, hyponatraemia; now avoided for TURP
Glycine 1.5%	~200 mOsm/kg	Commonest (UK, North America); glycine → ammonia, serine; transient blindness (retinal inhibitory neurotransmitter), encephalopathy
Sorbitol 3%	~165	Metabolised to fructose and glucose → hyperglycaemia; lactic acidosis
Mannitol 5%	~275	Near iso-osmolar; not metabolised; osmotic diuresis; volume overload in renal failure
Sorbitol 2.7% + mannitol 0.54%	~180	Continental Europe
Normal saline 0.9%	~308	Bipolar TURP, lasers (HoLEP, PVP); no TUR syndrome

- Plasma is about 290 mOsm/kg, so monopolar fluids are **hypo-osmolar**.

TUR (transurethral resection) syndrome

- **Mechanism**: prostatic venous pressure is ~10 mmHg; irrigant at higher pressure enters open sinuses → volume overload, **dilutional hyponatraemia** and hypo-osmolality → cerebral oedema; plus glycine toxicity.
- **Features**: confusion, nausea, vomiting, visual disturbance, hypertension then hypotension, bradycardia, seizures, coma.
- **Risk factors**: gland >45 mL, resection >90 minutes (VA (Veterans Affairs) study), high irrigation pressure, open sinuses or capsular perforation.
- **Prevention**: bag height ~60 cm (each extra 10 cm more than doubles absorption), continuous-flow resectoscope, time limit, spinal anaesthesia (awake patient), bipolar TURP. Ethanol 1% tagging can monitor absorption.
- **Treatment**: stop surgery, check sodium; Na <120 mEq/L serious. Furosemide for overload; **3% saline** 100 mL boluses (max 300 mL) for neurological signs; raise sodium by 4–6 mEq/L initially, not over 12 mEq/L in 24 h (risk of central pontine myelinolysis).

Bipolar advantage

- Meta-analyses: no TUR syndrome, less transfusion and clot retention; NNT (number needed to treat) about 50 to prevent one TUR syndrome.

Diagram to draw: monopolar vs bipolar current pathways.

Key points to remember:

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

LASERS in BPH.

Type: SAQ (10 marks)

Source: 2017-Winter Paper 2, Q3a

Principle

- LASER (light amplification by stimulated emission of radiation): the tissue effect depends on wavelength and **chromophore** (water or haemoglobin), which set **penetration depth**.
- About 60 °C coagulates tissue (delayed sloughing); ≥100 °C vaporises it (immediate removal).

Lasers used in BPH (benign prostatic hyperplasia)

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet)	~2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium (Tm:YAG, TFL [thulium fibre laser])	2013 / 1940 nm	Water (TFL at peak)	0.2 mm	ThuLEP (thulium laser enucleation), vaporesection, vaporisation
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	940–1470 nm	Water and haemoglobin	0.5–5 mm	Vaporisation, enucleation
Nd:YAG	1064 nm	Weak	About 1 cm	Coagulation; abandoned (sloughing)

Enucleation versus vaporisation

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Tissue	Whole transition zone to the capsule, then morcellated	Removed layer by layer
Histology	Yes	No
Gland size	**Any (size-independent)**	Up to 80 cc
Retreatment	Rare	Higher, especially >80 cc

Evidence

- **HoLEP** (about 2 J × 40–50 Hz): the only endoscopic procedure **superior to TURP** (transurethral resection of prostate) for IPSS (International Prostate Symptom Score) and Qmax. Less bleeding, shorter catheter time and stay.
- **GOLIATH RCT** (randomised controlled trial; 180 W PVP vs TURP, about 270 men, ≤100 cc): PVP non-inferior for IPSS, Qmax and complications at 2 years; shorter catheter time and stay.
- **ThuLEP:** outcomes equal to HoLEP. TFL energy is absorbed about four times more by water than holmium.
- **AUA (American Urological Association):** HoLEP or ThuLEP for any size; PVP for <80 cc; HoLEP, ThuLEP or PVP for men at **higher bleeding risk**.

Complications

- Storage symptoms and dysuria; transient stress incontinence; retrograde ejaculation (about three-quarters after HoLEP); bladder neck contracture (0–3%); stricture; capsular perforation; morcellator bladder injury.
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Diagram to draw: HoLEP planes: enucleated adenoma pushed into the bladder, then morcellated.

> **Key points to remember:**
> - Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - HoLEP is size-independent and superior to TURP.
> - GOLIATH: PVP non-inferior to TURP; best for anticoagulated men.

Silodosin.

Type: SAQ (10 marks)

Source: 2017-Winter Paper 2, Q3c

Pharmacology

- The **most α1A-selective** blocker (well over 100-fold vs α1B; about 50-fold vs α1D), used for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing α1B gives **minimal hypotension**. Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by CYP3A4.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**
CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver impairment; strong CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers, PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Efficacy

- Better than placebo for IPSS (International Prostate Symptom Score) and flow, including severe LUTS. **Non-inferior to tamsulosin 0.4 mg**, not better (European trial, 2011).
- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers.
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.

Adverse effects

- **Anejaculation:** over a quarter of men in trials (about 14% vs 2% with tamsulosin head-to-head), yet few stop. Men with anejaculation had better responses.
- No excess hypotension in the elderly or with antihypertensives; no QT prolongation.
- **IFIS (intraoperative floppy iris syndrome):** not specifically studied but reported; warn before cataract surgery.

Other uses

- **MET (medical expulsive therapy)** for distal ureteric stones 5–10 mm. α-blockers reduce ureteric spasm. Silodosin gave faster expulsion than tamsulosin in meta-analyses; **SUSPEND (2015)** found no overall tamsulosin benefit.
- **Also know:** before ureteroscopy (easier access); recent-onset CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome).

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction
Silodosin	α1A (highest)	8 mg	Lowest	**Highest**
Tamsulosin	α1A, α1D > α1B	0.4 mg	Low	Moderate
Alfuzosin	Clinically uroselective	10 mg	Low	Lowest

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low

Diagram to draw: α1A in prostate and vas, α1B in vessels, α1D in bladder.

Key points to remember:

- 8 mg with a meal; 4 mg if CrCl 30–50; avoid if <30.
- Equal to tamsulosin 0.4 mg; greatest BOOI fall.
- Anejaculation is the signature effect; least hypotension.

Dutasteride.

Type: SAQ (10 marks)

Source: 2018-Summer Paper 2, Q3c

Pharmacology

- A **dual 5ARI (5α-reductase inhibitor)**: it blocks **type 1** (liver, skin) and **type 2** (prostate stroma, genital tissue) isoenzymes. Finasteride blocks type 2 only.
- **Mechanism:** blocks conversion of testosterone to DHT (dihydrotestosterone), the main prostatic androgen; serum DHT falls about **90%**. Epithelial atrophy shrinks the gland and relieves the **static** component of obstruction.
- **Dose: 0.5 mg once daily**; hepatic metabolism; **half-life about 5 weeks**.

Effects

- **Prostate volume falls about 25%** (maximum by 6 months).
- Symptoms and Qmax (maximum flow rate) improve modestly and slowly (**at least 6 months**).
- **Changes natural history:** pivotal trials: AUR (acute urinary retention) down 57%, BPH (benign prostatic hyperplasia) surgery down 48%.
- **PSA (prostate-specific antigen) halves** by 6–12 months: **double the PSA** to interpret it. A rise from the nadir on treatment needs evaluation.

Indications

- Moderate-to-severe LUTS (lower urinary tract symptoms) with an enlarged prostate: **>40 mL (EAU [European Association of Urology])** or PSA about >1.5 ng/mL; AUA (American Urological Association): >30 mL.
- **Combination with tamsulosin** for men at risk of progression (fixed-dose capsule available).
- Off-label: BPH haematuria (lowers prostatic microvessel density); before TURP (transurethral resection of the prostate) to reduce bleeding (inconsistent evidence).

Landmark trials

Trial	Result
CombAT (Combination of Avodart and Tamsulosin, 4 years; PV [prostate volume] ≥30 mL, PSA ≥1.5)	Combination best for symptoms and progression; dutasteride (and combination) beat tamsulosin for AUR and surgery
REDUCE (Reduction by Dutasteride of Prostate Cancer Events, 4 years)	Prostate cancer down about 23%, but slightly more Gleason 8–10; **not licensed for prevention**

Side effects

- **Sexual:** ED (erectile dysfunction), low libido, ejaculatory disorders; lessen with time.
- Gynaecomastia and breast tenderness; depression (FDA [Food and Drug Administration] warning).
- **No IFIS** (intraoperative floppy iris syndrome) and no hypotension, unlike α-blockers.

Precautions

- **Teratogenic:** pregnant women must not handle leaking capsules. **No blood donation for 6 months** after the last dose.

Diagram to draw: testosterone to DHT pathway with 5α-reductase types 1 and 2 blocked by dutasteride.

Key points to remember:

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Measures to decrease complications of TURP

Type: SAQ (10 marks)

Source: Practice Set 9 Paper 2, Q3b (TUR syndrome also answered in 2017-Winter Paper 3, Q3f)

Introduction

- TURP (transurethral resection of the prostate) is the benchmark; overall morbidity approaches one-fifth, mostly preventable.

Preoperative measures

- **Selection:** AUA (American Urological Association) 2023: TURP for glands up to 80 mL; larger glands need HoLEP (holmium laser enucleation) or simple prostatectomy.
- **Sterile urine:** treat UTI (urinary tract infection) first; single-dose prophylaxis covering gram-negatives and enterococci.
- **Anticoagulants:** stop after advice from the prescriber; if they cannot be stopped, choose HoLEP, ThuLEP or PVP (photoselective vaporisation).
- **Finasteride** for recurrent BPH (benign prostatic hyperplasia) bleeding.
- Spinal anaesthesia shows early TUR syndrome; pad the peroneal nerve.

Intraoperative measures

Complication	Prevention
TUR syndrome (hypo-osmolar fluid absorbed, dilutional hyponatraemia)	**Bipolar TURP with saline**; else irrigant ≤60 cm high, resection <90 min, avoid capsular perforation, check sodium in long resections
Bleeding	Systematic resection to capsule; control arterial bleeders as you go; bipolar "cut and seal"; catheter traction for venous ooze
Ureteric orifice injury	Identify orifices first; take a large median lobe down first
Perforation	Respect the capsule; visible fat means perforation
Incontinence	**Stop at the verumontanum**; resect apex last in a bloodless field
Stricture	Calibrate or dilate meatus, lubricate generously, correct sheath size
Bladder neck contracture	Avoid over-resection and excess fulguration; TUIP (transurethral incision of the prostate) for small glands

Postoperative measures

- Large-bore three-way catheter; careful **continuous bladder irrigation** (blocked outflow distends the bladder and worsens bleeding).
- Even routine cases absorb 800–1000 mL of irrigant; check sodium if confused.
- TUR syndrome treatment: **3% saline 100 mL** boluses; raise sodium by no more than 12 mEq/L in 24 h (pontine myelinolysis).
- Stool softeners; no cycling for 4–6 weeks; counsel about **retrograde ejaculation** (about two-thirds).

Diagram to draw: resection field from bladder neck to verumontanum with the external sphincter marked.

Key points to remember:

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

Tamsulosin

Type: SAQ (10 marks)

Source: Practice Set 12 Paper 2, Q3d

Introduction

- A **uroselective α1-adrenergic antagonist**, first-line drug for bothersome male LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia) (AUA [American Urological Association], EAU [European Association of Urology]).

Receptor basis

Subtype	Site	Effect of blockade
α1a (about 70% in prostate)	Prostate, bladder neck, prostatic urethra, iris dilator	Relieves dynamic obstruction; ejaculatory dysfunction; IFIS (intraoperative floppy iris syndrome)
α1b	Blood vessels	Hypotension (tamsulosin has 10-fold less affinity)
α1d	Detrusor, spinal afferents	Storage symptom relief

Mechanism

- Blocks noradrenaline at α1a receptors → relaxes prostatic and bladder-neck smooth muscle → lowers the **dynamic component** of BOO (bladder outlet obstruction). It does not shrink the prostate.

Pharmacology and dose

- **0.4 mg once daily** after the same meal (up to 0.8 mg); no titration.
- Modified release; hepatic metabolism (CYP3A4, CYP2D6); avoid strong CYP3A4 inhibitors (ketoconazole).
- Flow rises within 4–8 hours, symptoms by 1 week; 0.2 mg is not better than placebo.

Indications

1. **BPH/LUTS:** IPSS (International Prostate Symptom Score) falls about 30–40%; Qmax rises; storage and nocturia improve.
2. **Combination with 5ARI** (5α-reductase inhibitor) for prostate >30–40 mL: **CombAT** (dutasteride + tamsulosin) and MTOPS (combination cut progression by 66%).
3. **MET (medical expulsive therapy):** distal ureteric stones 5–10 mm; SUSPEND (2015) negative overall but included all sizes.
4. **AUR (acute urinary retention):** before trial without catheter.
5. **Also know:** ureteric stent symptoms; female functional outlet obstruction.

Adverse effects

- **Ejaculatory dysfunction** (dose-dependent; less than silodosin): retrograde ejaculation from bladder-neck relaxation, or anejaculation (emission failure).
- **IFIS:** floppy, billowing iris, iris prolapse, progressive miosis during cataract surgery; **highest risk with tamsulosin**. Stopping the drug does not remove the risk, so warn the ophthalmologist.
- Dizziness, rhinitis, headache; less orthostatic hypotension than non-selective agents. Caution: sulfonamide allergy.

Comparison

	Tamsulosin	Silodosin	Alfuzosin
Selectivity	α1a/α1d	Most α1a	Clinical only
Ejaculatory effect	Moderate	Highest	Lowest

Diagram to draw: prostate cross-section showing α1a receptors in stroma and bladder neck.

Key points to remember:

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

Prostatic stents

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 2, Q3f (urethral stents also answered in Practice Set 4 Paper 4, Q3f and 2015-Summer Paper 4, Q3d)

Definition

- A device placed in the **prostatic urethra**, between bladder neck and verumontanum, to relieve BOO (bladder outlet obstruction) without resection. Mainly for men **unfit for surgery**.

Classification

Type	Examples	Features
Temporary, first generation	Urospiral, Prostakath (spirals)	Steel or gold-plated coil; not epithelialised
Temporary, second generation	**Memokath** (thermo-expandable nitinol); Allium (covered nitinol)	Expands with warm water, softens with cold water for removal
Temporary polyurethane	Intraurethral catheter, Spanner	Bulbar anchor; sphincter left free
Biodegradable	Polylactic acid stents	Dissolve; trial use
Permanent, epithelialising	**UroLume** (woven mesh)	Urothelium covers the mesh; withdrawn

Indications

- Retention or severe BOO from BPH (benign prostatic hyperplasia) in men unfit for surgery: avoids a long-term catheter.
- Drainage after thermal treatments; trial of de-obstruction.
- **Also know:** DSD (detrusor–sphincter dyssynergia), now abandoned; bulbar stricture.

Contraindications

- Poor detrusor contractility (do urodynamics if in doubt); untreated UTI (urinary tract infection); large median lobe; bladder stones; stricture.

Technique

- Measure prostatic urethral length cystoscopically. Place under local anaesthesia with endoscopic or fluoroscopic guidance. The stent must **end above the external sphincter**, or incontinence follows.

Complications

- **Migration** (commonest with temporary stents); **encrustation**; UTI; irritative symptoms, perineal pain, haematuria.
- **Hyperplastic ingrowth** and restenosis with permanent stents (difficult removal); incontinence if the sphincter is crossed.

Current status

- Most unfit men void after placement, but many stents are later removed.
- **EAU (European Association of Urology):** alternative to catheterisation in men unfit for surgery (weak). Not in the **AUA (American Urological Association) 2023** BPH list.
- **Recent: iTind**: temporary nitinol device left 5–7 days; it remodels the urethra by ischaemic pressure necrosis. AUA 2023 option for 25–75 mL without median lobe. **Optilume BPH**

(paclitaxel balloon).

Diagram to draw: sagittal prostate with a Memokath from bladder neck to verumontanum, above the external sphincter.

> **Key points to remember:**
>
> - Temporary (Memokath) vs permanent (UroLume, withdrawn).
> - Need detrusor contractility; never cross the sphincter.
> - Migration, encrustation, ingrowth; iTind is the modern temporary device.

Role of UDS before prostate surgery

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 2, Q3a

Why consider UDS (urodynamic study)

- Symptoms, flow rate, PVR (post-void residual), prostate volume and cystoscopy **cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**: low Qmax (maximum flow rate) occurs in both. DUA is found in 9–48% of men with LUTS (lower urinary tract symptoms) studied.
- Surgery helps obstruction, not a weak bladder. **PFS (pressure–flow study) is the gold standard** for BOO.

Components

- **Filling cystometry:** sensation, compliance, **DO (detrusor overactivity)**.
- **PFS:** detrusor pressure at Qmax plotted against flow (ICS (International Continence Society) nomogram).

Index	Formula	Interpretation
BOOI (bladder outlet obstruction index)	PdetQmax − 2Qmax	>40 obstructed; 20–40 equivocal; <20 unobstructed
BCI (bladder contractility index)	PdetQmax + 5Qmax	>150 strong; 100–150 normal; <100 weak

- High pressure, low flow = BOO; low pressure, low flow = DUA.
- **Videourodynamics** adds anatomy (diverticula, reflux).

Indications before BPH (benign prostatic hyperplasia) surgery (EAU (European Association of Urology))

- Previous failed invasive treatment.
- Cannot void >150 mL (unreliable flow).
- PVR >300 mL.
- Age >80 years with mainly voiding LUTS.
- Age <50 years with mainly voiding LUTS.
- **Also:** suspected neurogenic bladder, prior pelvic surgery or radiation, Qmax >10 mL/s (AUA (American Urological Association): diagnostic uncertainty).

How results change management

- **Unobstructed or DUA:** poorer surgical results; counsel, consider CIC (clean intermittent catheterisation).
- **DO with BOO:** storage symptoms persist in about one-third after TURP (transurethral resection of the prostate); warn the patient.
- **Low compliance:** the only "absolute" urodynamic indication to relieve BOO (protects upper tracts).

Evidence and limitations

- Not routine. **UPSTREAM RCT (randomised controlled trial):** routine UDS did not change IPSS (International Prostate Symptom Score) or surgery rates.
- Invasive: UTI (urinary tract infection), discomfort, cost. Before **radical prostatectomy**, not routine.

Diagram to draw: ICS nomogram with obstructed, equivocal and unobstructed zones.

Key points to remember:

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

Monopolar v/s Bipolar TURP.

Type: SAQ (10 marks)

Source: Practice Set 15 Paper 2, Q3a

Introduction

- TURP (transurethral resection of the prostate) removes transition-zone tissue from bladder neck to verumontanum, down to the surgical capsule; the benchmark BPH (benign prostatic hyperplasia) operation.
- **AUA (American Urological Association):** no preference; choose by expertise.

Principles

- **Monopolar (m-TURP):** current runs from the loop through the patient to a grounding pad. Saline would disperse the current, so a **non-conducting, hypo-osmolar irrigant** (glycine 1.5%, sorbitol–mannitol, water) is needed.
- **Bipolar (b-TURP):** active and return poles are on the same instrument, so current stays local and **normal saline** can be used. The Plasma system turns saline into a plasma layer (orange glow) that cuts at lower temperature and seals vessels.

Comparison

Feature	Monopolar	Bipolar
Current path	Loop → patient → pad	Within the loop
Irrigant	Glycine, sorbitol, water	Normal saline
TUR syndrome	Yes	Practically eliminated
Resection time	Limited (speed over haemostasis)	No time pressure; large glands
Loop	Tungsten	Platinum–iridium (Plasma)
Bleeding, transfusion, clot retention	Higher	Lower
Catheter time, hospital stay	Longer	Shorter
Obturator reflex	Risk	Reduced
IPSS (International Prostate Symptom Score), QoL, PVR	Equivalent (Qmax marginally better with bipolar)	Equivalent
Stricture, BNC (bladder neck contracture), sexual function, retreatment	Similar	Similar

TUR syndrome (monopolar)

- Irrigant absorbed through open sinuses (venous pressure about 10 mmHg) → **dilutional hyponatraemia**: confusion, visual disturbance, nausea, bradycardia, hypertension, seizures; glycine also yields ammonia.
- **Risk factors:** gland >45 mL, resection >90 minutes, high irrigation bag.
- **Treatment:** stop surgery, check sodium; 3% saline 100 mL boluses for neurological symptoms; avoid over-rapid correction (central pontine myelinolysis).

Evidence

- Meta-analyses: equal efficacy up to 5 years; b-TURP safer (NNT about 50 to prevent one TUR syndrome, about 20 to prevent one clot retention).

- **Also know:** TURis (sheath as return) caused strictures; bipolar enucleation and vaporisation (PlasmaButton).

Diagram to draw: monopolar circuit (loop → patient → pad) versus bipolar (loop to return electrode on the same resectoscope).

> **Key points to remember:**
> - Bipolar allows saline; TUR syndrome practically eliminated.
> - Equal efficacy; bipolar has less bleeding and shorter catheter time.
> - TUR syndrome risk: >45 mL gland, >90 minutes.

Rapid Revision Sheet – Section 16 · BPH

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

BPH: medical management and surgical modalities (LAQ)

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Millin's prostatectomy (SAQ)

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

Catheter free trial (SAQ)

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

Irrigation fluids for TURP. (SAQ)

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

LASERS in BPH. (SAQ)

- Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- HoLEP is size-independent and superior to TURP.
- GOLIATH: PVP non-inferior to TURP; best for anticoagulated men.

Silodosin. (SAQ)

- 8 mg with a meal; 4 mg if CrCl 30–50; avoid if <30.
- Equal to tamsulosin 0.4 mg; greatest BOOI fall.
- Anejaculation is the signature effect; least hypotension.

Dutasteride. (SAQ)

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Measures to decrease complications of TURP (SAQ)

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

Tamsulosin (SAQ)

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.

- MET benefit in distal stones 5–10 mm.

Prostatic stents (SAQ)

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.

Role of UDS before prostate surgery (SAQ)

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

Monopolar v/s Bipolar TURP. (SAQ)

- Bipolar allows saline; TUR syndrome practically eliminated.
- Equal efficacy; bipolar has less bleeding and shorter catheter time.
- TUR syndrome risk: >45 mL gland, >90 minutes.

Section 17 · Prostate Cancer — Long answer questions (LAQ)

Castration-resistant prostate cancer (CRPC): definition and treatment

Asked as:

- CRPC in Ca prostate. — *Summer 2020 Paper II Q3a (SAQ)*
- Discuss treatment of CRPC — *Practice Set 6 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q3a)

Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
  - **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, with PSA >2 ng/mL; or
  - **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

Why it happens: the AR still drives growth

- Castration-resistant is **not** androgen-independent, so treatment first targets the AR (androgen receptor) more deeply.
- Mechanisms: AR overexpression and amplification; AR mutation (activated by other steroids); **splice variants lacking the ligand-binding domain**; adrenal and intratumoural androgen synthesis (CYP17 overexpressed); bypass pathways (PI3K/AKT, PTEN loss in ~40%).
- **Lineage plasticity:** neuroendocrine change with low PSA, visceral and lytic disease (visceral disease).

Evaluation

- Testosterone (exclude non-compliance), PSA, **PSADT (PSA doubling time)**; CT and bone scan; PSMA (prostate-specific membrane antigen) PET increasingly.
- Prognosis: performance status, pain, haemoglobin, ALP (alkaline phosphatase), LDH (lactate dehydrogenase), visceral disease.
- **Germline testing** in all metastatic disease; somatic sequencing of tumour tissue (germline and tumour testing) for HRR (homologous recombination repair) genes such as BRCA2, and MSI (microsatellite instability).
- Biopsy a metastasis if disease progresses fast with a low PSA (neuroendocrine change).
- PCWG3 (Prostate Cancer Clinical Trials Working Group 3): early bone-scan lesions may be **flare**; re-baseline at 12–16 weeks, then progression = ≥2 new lesions.

Clinical states model (PCWG3):

```
Localised → rising PSA → mCSPC (castration-sensitive)
                  ↓ (on ADT)
              nmCRPC → mCRPC 1st line → 2nd line → 3rd line
```

General principles of management

1. **Continue ADT (androgen deprivation therapy) lifelong**, including during chemotherapy; try antiandrogen withdrawal.
2. **Antiandrogen withdrawal** (e.g. bicalutamide): watch PSA 4–8 weeks; brief responses in a minority.
3. Choose by prior therapy, symptoms, visceral disease, genomics and comorbidity.
4. **Sequencing: switch mechanism**; a second ARPI rarely works (cross-resistance). Cross-resistance means a second ARPI (AR-pathway inhibitor) helps only 15–30%.

nmCRPC (non-metastatic)

- **PSADT ≤10 months:** add an ARPI. Each improved MFS (metastasis-free survival) and overall survival (improve metastasis-free and overall survival):
  - Apalutamide 240 mg (**SPARTAN**: MFS HR 0.28); enzalutamide 160 mg (**PROSPER**); darolutamide 300 mg twice daily (**ARAMIS**: MFS 40.4 vs 18.4 months).
- Slower PSADT: continue ADT and monitor.

mCRPC: life-prolonging options

Class	Agent and dose	Key trial (overall survival)
ARPI (AR-pathway inhibitor)	**Abiraterone** 1000 mg fasting + prednisone	COU-AA-301: 14.8 vs 10.9 months; COU-AA-302 (pre-chemotherapy)
	Enzalutamide 160 mg	AFFIRM: 18.4 vs 13.6 months; PREVAIL
Taxane	**Docetaxel** 75 mg/m² 3-weekly + prednisone	TAX 327: 18.9 vs 16.4 months (mitoxantrone)
	Cabazitaxel 20–25 mg/m² after docetaxel	TROPIC: 15.1 vs 12.7; CARD: beats a second ARPI
Radioligand (mCRPC)	**Radium-223** 50 kBq/kg 4-weekly ×6 (symptomatic bone-only, no visceral)	ALSYMPCA: 14.9 vs 11.3 months
	^{177}Lu-PSMA-617 (PSMA-positive, after ARPI + taxane)	VISION: 15.3 vs 11.3 months
PARP (poly-ADP-ribose polymerase) inhibitor	Olaparib (HRR genes); rucaparib (BRCA)	PROfound; TRITON3
	Olaparib + abiraterone; niraparib + abiraterone; talazoparib + enzalutamide (first line)	PROpel; MAGNITUDE; TALAPRO-2
Immunotherapy	**Sipuleucel-T** (asymptomatic, no visceral disease)	IMPACT: 25.8 vs 21.7 months
	Pembrolizumab for MSI-high/dMMR (mismatch repair-deficient), 2–5%	Tumour-agnostic approval

- **Abiraterone** blocks CYP17, which also lowers cortisol. ACTH (adrenocorticotropic hormone) then rises and causes mineralocorticoid excess (hypokalaemia, hypertension, oedema), so prednisone is co-prescribed. Never combine with radium-223 (more deaths and fractures).
- **Enzalutamide:** seizures (~1%), fatigue, falls. Apalutamide: rash, hypothyroidism.
- **PARP inhibitors** cause synthetic lethality in HRR-deficient cells; benefit is greatest with BRCA2 and weak with ATM.

Sequencing by prior mCSPC therapy

Prior therapy	Next option
Docetaxel only	ARPI
ARPI only	Docetaxel, radium-223, sipuleucel-T
Both	Cabazitaxel, ^{177}Lu-PSMA, PARP inhibitor if HRR mutation

Neuroendocrine/aggressive variant

- Neuroendocrine variant: low PSA, liver or brain metastases, lytic bone lesions with hypercalcaemia. Treat with **platinum + etoposide** (or carboplatin + taxane) ± radiotherapy; survival <12 months.

Supportive and palliative care

- **Bone health:** denosumab 120 mg SC or zoledronic acid 4 mg IV (renal dose adjustment) every 4 weeks against skeletal events, with calcium and vitamin D. Denosumab delayed the first SRE (skeletal-related event) longer (20.7 vs 17.1 months); neither prolongs life. Risk: osteonecrosis of the jaw.
- Focal bone pain: external-beam radiotherapy (radiotherapy for bone pain).
- **Cord compression** needs high-dose dexamethasone → urgent MRI → radiotherapy ± decompression (Patchell trial: surgery + radiotherapy better).
- Ureteric obstruction: stent or nephrostomy. Anaemia; hospice care.

Recent advances

- PSMA-ligand therapy earlier (PSMAfore); AR degraders (ARV-110); AKT inhibition in PTEN loss; bipolar androgen therapy; antibody–drug conjugates.

Diagram to draw: clinical states: rising PSA → nmCRPC → mCRPC first, second, third line.

Diagram to draw: clinical states model with drugs placed at each state.

Key points to remember:

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- AR still drives CRPC.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid)

Asked as:

- Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain — *Practice Set 1 Paper II Q1 (LAQ)*
- Management of bone mets in prostatic CA — *Practice Set 3 Paper II Q2 (LAQ)*
- Zolendronic acid — *Practice Set 14 Paper II Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(mHSPC systemic therapy also answered in 2025-Winter Paper 3, Q3b) (mHSPC with retention and bone pain also answered in Practice Set 1 Paper 2, Q1; related: bone metastases in Practice Set 3 Paper 2, Q2)

Zolendronic acid: read as zoledronic acid.

Introduction

- Bone is the commonest metastatic site; visceral metastasis occurs in <20%.
- Aims: control the cancer, relieve pain, **prevent SREs (skeletal-related events)** and treat emergencies (cord compression, fracture).
- **Retention + bone pain case:** this is newly diagnosed (synchronous) **mHSPC (metastatic hormone-sensitive prostate cancer)** with two urgent problems: an obstructed outlet and painful bone metastases. Treat the emergencies first, start castration safely (no flare), then intensify systemic therapy.

Pathophysiology

- Tumour cells lodge in axial marrow, adhering through integrins and collagen receptors.
- Metastases are mainly **osteoblastic (sclerotic)**, but resorption also rises. Tumour cytokines induce **RANKL (receptor activator of nuclear factor-κB ligand)**, which activates osteoclasts; released growth factors feed the tumour (**vicious cycle**: tumour cytokines activate osteoclasts, and resorbed bone releases growth factors that feed the tumour). ADT (androgen deprivation therapy) adds osteoporosis through loss of oestrogen.
- **SREs:** pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone. Marrow replacement causes anaemia. Hypercalcaemia is rare and suggests neuroendocrine disease.

Evaluation (immediate assessment)

- **Clinical:** pain, neurological signs, performance status. **Ask and examine for cord compression:** back pain, leg weakness, sensory level, bowel symptoms. DRE (digital rectal examination).
- **Bloods:** PSA (prostate-specific antigen), ALP (alkaline phosphatase), calcium, creatinine, haemoglobin, testosterone.
- **Imaging:** ultrasound for hydronephrosis and residual; **bone scan and CT (computed tomography)** define volume; PSMA (prostate-specific membrane antigen) PET is more sensitive, but trials used conventional imaging. **Plain X-ray or CT of painful weight-bearing sites** (lytic lesion, impending fracture). **MRI (magnetic resonance imaging) of the whole spine (whole-spine MRI)** if any back pain or neurological sign (for back pain with neurological signs).
- **Histology:** prostate biopsy (or TURP (transurethral resection of the prostate) chips). **Germline genetic testing** for all men with metastatic disease.
- **Volume (CHAARTED):** high volume = ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Step 1: Relieve the retention

- **Urethral catheter** (suprapubic if urethral fails). Watch for post-obstructive diuresis.
- Persisting renal impairment with hydronephrosis means ureteric obstruction by tumour at the trigone: **nephrostomy or stent** (stent or nephrostomy for obstruction).
- ADT (androgen deprivation therapy) shrinks the prostate, so try **TWOC (trial without catheter)** after a few months of ADT.
- If TWOC fails or symptoms persist: **channel (palliative) TURP**. ADT alone does not prevent local progression (older series: about a third needed TURP within 60 days of orchidectomy).

Step 2 / 1. Systemic treatment of the cancer

Hormone-sensitive (mHSPC): start ADT at once, without flare

- **ADT** (orchidectomy, LHRH [luteinising hormone-releasing hormone] agonist or antagonist):

Option	Point
Bilateral orchidectomy	Castrate within hours; best for impending cord compression
GnRH (gonadotropin-releasing hormone) antagonist: degarelix 240 mg subcutaneously, then 80 mg monthly; oral relugolix	**No flare**; castrate by day 3 in >95% (degarelix)
LHRH (luteinising hormone-releasing hormone) agonist + antiandrogen	Agonist raises LH (luteinising hormone) up to 10-fold (raises testosterone) for 10–20 days (**flare**). Give bicalutamide 50 mg daily with or before the agonist, for 21–28 days

- **Why avoid flare:** the testosterone surge can worsen bone pain, retention and cord compression in spinal metastases. Cover it with an antiandrogen for 21–28 days, or use an antagonist (degarelix) or orchidectomy.
- Target castrate testosterone **<50 ng/dL**. Antiandrogen alone is not recommended.

Intensify systemic therapy (mHSPC)

- ADT alone is no longer standard. Classify volume (CHAARTED, above).

Add to ADT	Trial (overall survival HR (hazard ratio))
Abiraterone 1000 mg + prednisone 5 mg	LATITUDE (0.62); STAMPEDE arm G
Enzalutamide or apalutamide (ARPIs (androgen receptor pathway inhibitors))	ENZAMET (0.67), ARCHES (enzalutamide); TITAN (0.67)
Docetaxel 75 mg/m² × 6 + abiraterone or darolutamide (**triplet**)	PEACE-1; ARASENS (0.68)

- Hormonal agents work regardless of volume; docetaxel helps only high volume, so it is now given within a triplet.
- **Low-volume synchronous disease:** add **RT to the prostate** (**prostate radiotherapy**; STAMPEDE arm H, 3-year survival 81% vs 73%); it also helps local control.

Castration-resistant (mCRPC):

- ARPIs (androgen receptor pathway inhibitors), docetaxel then cabazitaxel, PARP (poly-ADP-ribose polymerase) inhibitors for HRR (homologous recombination repair) mutations.
- **Radium-223** (alpha emitter, path <100 µm, spares marrow): for **symptomatic bone-only** disease (licensed only in castration-resistant disease); 50 kBq/kg every 4 weeks × 6. **ALSYMPCA:** overall survival 14.9 vs 11.3 months; delayed SREs. **Never with abiraterone** (more fractures and deaths).

- **^{177}Lu-PSMA-617** (beta emitter, PSMA-avid disease at any site): after ARPI and taxane; **VISION:** survival 15.3 vs 11.3 months.

Step 3 / 3. Bone pain

- **Analgesia** by the WHO (World Health Organization) ladder (analgesic ladder); neuropathic pain needs gabapentinoids or tricyclics; steroids for diffuse pain.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals multiple fractions (30 Gy in 10 fractions, RTOG 9714); retreatment is commoner.
- X-ray or CT of painful weight-bearing sites to exclude lytic lesions and impending fracture; **orthopaedic fixation** then RT (radiotherapy).
- **Diffuse pain:** steroids, wide-field RT; radiopharmaceuticals (strontium-89, samarium-153 relieve pain only; radium-223 adds survival). Radium-223 is licensed only in castration-resistant disease.
- **SBRT (stereotactic body RT)** for oligometastases (STOMP, ORIOLE) and spinal re-irradiation.

2. Bone-targeted agents

Feature	Zoledronic acid	Denosumab
Mechanism / target	Bisphosphonate: inhibits osteoclasts (osteoclast enzyme)	Antibody against RANKL (receptor activator of nuclear factor-κB ligand)
SRE dose (mCRPC) / route	4 mg IV every 4 weeks (IV 4-weekly), adjust for creatinine clearance	120 mg SC (subcutaneous) every 4 weeks (4-weekly), no renal adjustment
Renal adjustment	Yes	No
Evidence	Fewer SREs than placebo	Time to first SRE 20.7 vs 17.1 months vs zoledronate
Toxicity	Fever, myalgia, renal impairment, hypocalcaemia, ONJ	Hypocalcaemia, hypophosphataemia, ONJ 2–4%

- **No survival benefit** (**neither improves survival**); they protect the skeleton only.
- **In mHSPC, SRE doses are not indicated** (no benefit, e.g. CALGB 90202, STAMPEDE). SRE doses (zoledronic acid 4 mg or denosumab 120 mg 4-weekly) are for CRPC (castration-resistant prostate cancer). Use osteoporosis doses for ADT bone loss: denosumab 60 mg every 6 months (6-monthly) or zoledronic acid yearly, by DEXA (dual-energy X-ray absorptiometry) and FRAX; these protect against ADT osteoporosis.
- **ONJ (osteonecrosis of the jaw) prevention:** dental check before starting; calcium 1000 mg and vitamin D 800 IU daily.
- Give a bone agent with radium-223 or any ARPI.

Zoledronic acid (in detail)

- **Introduction:** a potent IV (intravenous) **nitrogen-containing bisphosphonate**.
- **Mechanism:**
  - Binds hydroxyapatite at sites of active bone turnover and is taken up by **osteoclasts**.
  - Inhibits **farnesyl pyrophosphate synthase** (mevalonate pathway), causing osteoclast dysfunction and apoptosis, so resorption falls.
  - Breaks the "vicious cycle".
  - Excreted unchanged by the kidneys.

Indication (indications and doses)	Dose
SRE (skeletal-related event) prevention, mCRPC (metastatic castration-resistant prostate cancer) and other bone metastases (RCC (renal cell carcinoma), bladder)	**4 mg IV over ≥15 min every 4 weeks**, adjusted for creatinine clearance

Indication (indications and doses)	Dose
ADT (androgen deprivation therapy) bone loss (by DEXA (dual-energy X-ray absorptiometry) and FRAX)	4–5 mg IV yearly
Hypercalcaemia of malignancy	4 mg IV with saline hydration

- 12-weekly dosing after the first doses is non-inferior to 4-weekly (CALGB 70604).
- **Not indicated at SRE doses in mHSPC (metastatic hormone-sensitive prostate cancer):** no benefit (CALGB 90202, STAMPEDE).
- **Evidence: Zometa 039 trial (mCRPC):** fewer SREs than placebo (about 33% vs 44%). **Denosumab vs zoledronate (1904 men, mCRPC):** time to first SRE 20.7 vs 17.1 months. **Neither improves survival.**
- **Precautions and monitoring:** creatinine before every dose; avoid if creatinine clearance <30 mL/min. **Dental review before starting**; avoid invasive dental work during therapy. Calcium and vitamin D supplements.
- **Adverse effects: acute-phase reaction** (fever, myalgia) after the first infusion; nephrotoxicity, hypocalcaemia, hypophosphataemia; **ONJ (osteonecrosis of the jaw)**, atypical femoral fracture; ocular inflammation (rare).

4. Emergencies

Spinal cord compression: watch constantly (every focal back pain is cord compression until proven otherwise):

```
Back pain ± weakness, sensory level, sphincter change
 → dexamethasone 10 mg IV load, then 4–10 mg 6–12-hourly
 → urgent whole-spine MRI
 → decompressive surgery + RT (unstable spine, progression on RT, prior RT)
   or RT alone
 → orchidectomy or antagonist if hormone-naive (never agonist alone)
```

- **Dexamethasone** at once (load 10 mg intravenously), urgent whole-spine MRI, then **RT ± decompressive surgery**. Surgery plus RT beats (beat) RT alone (for walking ability) (**Patchell, 2005**).
- Castrate with orchidectomy or antagonist, never an agonist alone.

Pathological or impending fracture: orthopaedic fixation then RT. **Mirels score** ≥9 (site, pain, lytic type, size) favours prophylactic fixation.

Follow-up

- PSA, ALP, liver tests, calcium, renal function and blood count every 3–6 months; testosterone if PSA rises; new scan if symptoms or PSA rise. Rising PSA with castrate testosterone = CRPC.
- PSA nadir ≤0.2 ng/mL at 7 months predicts long survival (SWOG 9346).
- PCWG3 (Prostate Cancer Working Group 3): bone scan progression needs ≥2 new lesions, confirmed later (early "flare" misleads).
- Manage ADT effects: bone loss (DEXA (dual-energy X-ray absorptiometry)), diabetes, cardiovascular risk, hot flushes.
- **Also know:** stent or nephrostomy for obstruction; marrow-failure anaemia; hospice care.

```
Retention + bone pain + metastases
 ├─ Catheter ± nephrostomy;   cord signs? → dexamethasone → MRI → RT ± surgery
 ├─ Orchidectomy / antagonist / agonist + antiandrogen
 ├─ Analgesia + 8 Gy single-fraction RT; bone protection
 ├─ Add ARPI or triplet; prostate RT if low volume
 └─ TWOC after ADT → channel TURP if it fails
```

Diagram to draw: the flowchart above; vicious cycle: tumour → RANKL → osteoclast → growth factors → tumour, with the sites blocked by denosumab and zoledronate (tumour–osteoclast vicious cycle with zoledronate's site of action).

Key points to remember:

- Never start an LHRH agonist alone with spinal metastases: cover the flare or use an antagonist (avoid flare: antiandrogen cover, antagonist or orchidectomy).
- Single 8 Gy fraction gives the same pain relief as longer courses (for focal pain); dexamethasone and MRI for cord compression.
- Focal back pain = cord compression until MRI proves otherwise.
- ADT plus abiraterone or an ARPI, or a triplet, for every fit man.
- Denosumab 120 mg or zoledronate 4 mg (IV) 4-weekly prevent SREs in CRPC (mCRPC), not in HSPC; adjust zoledronate for creatinine clearance.
- No survival benefit; not for SREs in mHSPC.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

PSA screening for prostate cancer: evidence, problems and solutions

Asked as:

- What are the various problems in PSA screening — *Practice Set 8 Paper II Q1 (LAQ)*
- Screening of CA prostate. — *Practice Set 13 Paper II Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction (concept)

- Screening tests healthy, symptom-free (asymptomatic) men; its aim is lower prostate cancer mortality through earlier treatment (cure). The price is **overdiagnosis** (cancers that would never cause harm) and **overtreatment**.
- PSA (prostate-specific antigen) is a kallikrein (hK3) made by prostatic epithelium. It reaches blood when glandular architecture is disrupted. It is **organ-specific, not cancer-specific**.
- Since PSA testing began, (US) mortality has fallen by >40% (over 40%; SEER data), but at the price of overdiagnosis.

Tools

- **PSA:** biopsy threshold 4 ng/mL (many use 2.5–3; ERSPC used 3). Repeat a raised PSA before biopsy.
- **DRE (digital rectal examination):** little value alone; abnormal DRE raises risk.
- **Refinements:** PSA density ≥0.15; free PSA ≤25%; PHI (Prostate Health Index), 4Kscore, Stockholm3; urine tests (PCA3 [prostate cancer antigen 3], ExoDx, SelectMDx).
- **MRI (magnetic resonance imaging) before biopsy:** fewer biopsies and less low-grade overdiagnosis.

Evidence from screening trials

Trial	Design	Result
ERSPC (European Randomised Study of Screening for Prostate Cancer)	>182,000 men; PSA every 2–4 years vs none	**20% relative fall** in (prostate) cancer death at 16 years; 570 invited and 18 extra diagnoses per death prevented
PLCO (Prostate, Lung, Colorectal and Ovarian), USA	Annual PSA vs usual care	**No difference** at 17 years: control arm heavily PSA-tested (contamination), lower power
Göteborg-1 (Sweden)	Biennial PSA, men 50–64	44% fall in cancer death at 14 years (larger mortality reduction with biennial screening)
CAP (Cluster randomised trial of PSA testing for prostate cancer), UK	Single PSA invitation	No mortality difference at 10 years; only a small fall at 15 years
Göteborg-2 (2022)	MRI-targeted biopsy only	Halved overdiagnosis

- Corrected for non-attendance and contamination, ERSPC benefit is 27–31%; it appears only after about 10 years.

Problems in PSA screening

1. Poor specificity (false positives)

- BPH (benign prostatic hyperplasia), prostatitis, retention, catheterisation, biopsy and TURP (transurethral resection of prostate) all raise PSA.
- At PSA 4–10 ng/mL with a normal DRE (digital rectal examination), only 17–32% of biopsies find cancer. Most men biopsied therefore have no cancer.

2. Poor sensitivity: no safe cut-off

- PCPT (Prostate Cancer Prevention Trial): risk is continuous; about 1 in 11 men biopsied at PSA 0–1 ng/mL has cancer.
- A high threshold misses curable cancer; a low threshold adds biopsies and overdiagnosis. Thresholds used: 4, 3 or 2.5 ng/mL.

3. Biological variability of PSA

- Rises with age, prostate volume (about 4% per mL) and Black race; physiological fluctuation; assay differences.
- Lowered by 5α-reductase inhibitors (about 50% at 12 months, so double the value), obesity, hypogonadism and statins.

4. Overdiagnosis (the central problem)

- Detection of cancers that would never have caused symptoms or death. Autopsy prevalence is high: 29% at 60–69 years and 59% after 79.
- Mostly Grade Group 1 cancers.

5. Overtreatment

- Overdiagnosed men treated radically suffer ED (erectile dysfunction), incontinence and bowel toxicity for no gain, especially above 65 years.

6. Harms of biopsy

- Pain, haematuria, haematospermia, retention and sepsis (transrectal route); anxiety from repeat biopsies.

7. Modest and delayed benefit

- Screening prevents only 20–30% of cancer deaths; no overall survival benefit has been shown. Men with life expectancy <10 years cannot benefit.

8. Biases in interpreting screening data

- **Lead-time bias**: earlier diagnosis lengthens apparent survival without delaying death.
- **Length-time bias**: screening preferentially finds slow-growing tumours.
- **Healthy-user bias**: health-conscious men seek testing, confounding risk-factor studies.

9. Unsettled policy

- Start age, stop age, interval and threshold are all debated; further large randomised trials are unlikely.
- Opportunistic screening is common and inequitable.

Guideline positions (guidelines)

Body	Recommendation
USPSTF (US Preventive Services Task Force) 2012	Grade D: against screening
USPSTF 2018 (current)	**Grade C, 55–69 years: individual shared decision; Grade D ≥70** (not ≥70; was Grade D in 2012)
AUA (American Urological Association)	Shared decision 55–69; every 2 years (2-yearly interval)
NCCN (National Comprehensive Cancer Network)	Baseline PSA and discussion at 45
EAU (European Association of Urology)	Risk-adapted early detection: informed men from 50; earlier start (45) with family history or African descent, 40 for BRCA2 carriers

- **No body recommends (advises) unselected population screening.**
- **High risk:** Black men, father or brother diagnosed <65, BRCA2 and other DNA-repair mutations: start earlier.

- Stop when life expectancy is under 10–15 years.

Solutions: risk-adapted, MRI-first pathway (reducing harm)

```
Informed man, life expectancy >10–15 y
   → PSA (repeat if raised)
   → risk stratify: PSA density, free/total PSA, risk calculator,
     PHI / 4Kscore / Stockholm3
   → MRI (magnetic resonance imaging)
        PI-RADS 1–2, low density → no biopsy, keep in follow-up
        PI-RADS ≥3 → targeted ± systematic biopsy (transperineal)
   → Grade Group 1 → active surveillance
```

- **Baseline PSA** predicts future risk (Malmö): low PSA at 40–60 allows longer intervals (intervals by baseline PSA).
- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI-targeted pathway found more significant cancer (38% vs 26%) and less insignificant cancer; 28% avoided biopsy.
- **Göteborg-2 (2022):** MRI with targeted biopsy in screening roughly halved overdiagnosis.
- **Active surveillance** of low-risk disease (Grade Group 1) uncouples overdiagnosis from overtreatment; MRI and biomarkers before biopsy.
- **Also know:** germline testing (BRCA2, ATM, HOXB13); polygenic risk scores; transperineal biopsy lowers sepsis.

Diagram to draw: iceberg of latent vs clinically significant cancer, plus the MRI-first flowchart (pathway: PSA → repeat → risk calculator or MRI → targeted biopsy → active surveillance or treatment).

Key points to remember:

- ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
- Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
- USPSTF 2018: Grade C at 55–69, Grade D ≥70.
- Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy

Asked as:

- Describe staging of CA prostate and management of non-organ confirmed CA prostate — *Practice Set 10 Paper II Q1 (LAQ)*
- Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy. — *Practice Set 16 Paper II Q2 (LAQ)*
- Describe management of LAPC — *Practice Set 17 Paper II Q2 (LAQ)*

Type: LAQ (25 marks)
(related: Practice Set 10 Paper 2, Q1)

Read as: staging of prostate cancer, and management of non-organ-confined (locally advanced and node-positive, non-metastatic) disease. Assumed meaning: describe the management options for locally advanced carcinoma prostate, and discuss the complications of bilateral orchidectomy. LAPC is read as locally advanced prostate cancer.

Introduction and definition

- **Locally advanced CaP (carcinoma prostate) / LAPC: cT3–4, N0 or N1 (N+), M0:** extracapsular extension (spread) (cT3a), seminal vesicle invasion (cT3b), fixed or invading adjacent organs (cT4), ± pelvic nodes.
- At least 10% of new prostate cancers (new diagnoses, new cases) are locally advanced (T3–4, N0 or N+, M0); these men contribute a disproportionate share of cancer deaths. Any single treatment carries a high recurrence risk, so **multimodal therapy** (**combined therapy**) is the rule; no single modality suffices.
- In the PSA (prostate-specific antigen) era, high risk is defined more by grade and PSA than by palpation (today most "high-risk" disease is defined by grade and PSA rather than palpation).
- Untreated, 15-year disease-specific survival is 57% for T3–T4 vs 81% for T1–T2 (Swedish cohort). Untreated, Gleason 8–10 disease carries high cancer mortality, so fit men need treatment; no single modality suffices, hence **combined therapy**.

Staging: TNM (tumour, node, metastasis) 2017 (AJCC 8th edition)

Stage	Definition
T1a / T1b	Incidental at TURP (transurethral resection of prostate): ≤5% / >5% of tissue
T1c	Not palpable; found on needle biopsy (for raised PSA)
T2a / T2b / T2c	Palpable, confined: ≤ half of one lobe / > half of one lobe / both lobes
T3a	Extraprostatic extension (unilateral or bilateral)
T3b	**Seminal vesicle invasion** (not T4)
T4	Fixed or invading sphincter, rectum, bladder, levators, pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites ± bone

- **Prognostic stage groups** add PSA and grade group: stage III = T3–4, PSA ≥20 ng/mL or grade group 5; IVA = N1; IVB = M1. pT2 is not subdivided.
- **Grade group (ISUP, International Society of Urological Pathology):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.

Risk groups

System	High risk	Very high / locally advanced
D'Amico	T2c, PSA >20 or Gleason 8–10 (any one)	–
NCCN (National Comprehensive Cancer Network)	cT3a, GG4–5 or PSA >20	cT3b–T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG4–5 (grade group 4–5)
EAU (European Association of Urology)	PSA >20, ISUP 4–5 or cT2c	**Locally advanced: cT3–4 or cN+, any PSA, any grade**

- **Also know:** CAPRA (Cancer of the Prostate Risk Assessment) score; Partin tables; Kattan nomograms; Decipher.

Staging work-up (evaluation)

- **DRE (digital rectal examination)** sets clinical T stage; PSA, biopsy ISUP (International Society of Urological Pathology) grade group. PSA (prostate-specific antigen) >20 ng/mL: nodes in about 20%.
- **mpMRI (multiparametric MRI, magnetic resonance imaging):** for local extent; separates T2 from T3 (PI-RADS v2 reporting); best in higher-risk men.
- **Bone scan + CT/MRI** (CT, computed tomography) for high risk, PSA >20 or GG4–5.
- **PSMA (prostate-specific membrane antigen) PET-CT (positron emission tomography),** which is more accurate; proPSMA trial: accuracy 92% vs 65% for conventional imaging.
- Assess life expectancy: a fit man in his 70s with high-grade disease will live long enough to die of it.

Management of locally advanced N0 (cT3–4 N0 M0)

A (1). Radiotherapy (RT) + long-term ADT (androgen deprivation therapy): standard (the commonest standard)

- Dose-escalated IMRT (intensity-modulated RT, ≥78–81 Gy), or EBRT (external-beam RT, ≥78 Gy) with or without pelvic node RT, with brachytherapy boost (**ASCENDE-RT:** LDR (low-dose-rate) boost halved biochemical failure (**brachytherapy boost** improves biochemical control); overall survival (OS) equal).
- **Why add ADT:** it shrinks the target and acts synergistically (is synergistic) with radiation; ADT works synergistically with RT: neoadjuvant, concurrent and adjuvant, **18–36 months** for high risk.

Trial	Finding
EORTC 22863	3 years of goserelin with RT improved 5-year OS (overall survival), 78% vs 62% (other sources: 79% vs 62%; 78–79% vs 62%)
RTOG 85-31 / 92-02	Long adjuvant ADT better; RTOG 92-02: 28 vs 4 months ADT better on all endpoints except overall OS; OS gain in Gleason 8–10
TROG 96.01	6 months better than 3 months
SPCG-7; MRC PR07	**RT is still needed with ADT:** ADT + RT beats ADT alone (overall survival gain): SPCG-7 (cancer death RR (relative risk) 0.44) and MRC PR07 (OS HR (hazard ratio) 0.77) showed ADT alone is inferior
STAMPEDE (high-risk M0)	Adding 2 years of abiraterone: metastasis-free survival HR (hazard ratio) 0.53

Trial	Finding
RTOG 0521; GETUG 12	Docetaxel added modest benefit

- **Duration:** NCCN: ADT 18–36 months with EBRT (12–36 with brachytherapy boost); 18 vs 36 months gave similar OS.
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT (Tata Memorial) improved biochemical failure-free survival in (favoured it in) high-risk N0.

B. Systemic intensification

- **STAMPEDE (abiraterone, M0):** for N1, or two of T3–4, Gleason ≥8 or PSA ≥40, **abiraterone + prednisolone for 2 years** with ADT + RT gave metastasis-free survival HR 0.53 and OS HR 0.60. NCCN now lists it for very high risk.
- Docetaxel (RTOG 0521 docetaxel + RT, OS HR 0.69): not routine.

C. Radical prostatectomy (RP) + extended PLND (pelvic lymph node dissection, ePLND), within multimodal therapy

- For fit men with a resectable (**completely resectable**) tumour (mainly cT3a; cT3a, selected cT3b, limited nodes). As part of multimodal therapy. cT3 is overstaged in 7–27% (clinical overstaging occurs in 7–27%), so cT3 alone is no bar.
- **Why surgery / advantages:** exact (accurate) pathological staging, removal of the primary (removes the source of local morbidity: an untreated prostate causes retention, haematuria and ureteric obstruction that ADT does not prevent), and RT kept for salvage.
- RP alone gives disease-free survival in at least half at 8–10 years, and 10-year disease-specific survival ~80–90%. Disease-specific survival 85–92% at 5 years and 79–82% at 10 years, but biochemical recurrence is common, so multimodal care is planned.
- **Neoadjuvant ADT before RP:** clinically downstages (clinical downstaging) and fewer positive margins, but **no survival gain** (no survival or PSA benefit).
- **Adjuvant vs early salvage RT (post-RP RT):** EORTC 22911, SWOG 8794 (metastasis-free survival and OS improved) and ARO 96-02 showed benefit / favour adjuvant RT over observation. RADICALS-RT, RAVES and GETUG-AFU 17 favour **early salvage RT** (PSA rise) for most men. Adjuvant RT is still considered for very high risk (pT3b, GG4–5 / Gleason 8–10, positive margins / positive margin). Dose 64–66 Gy; adding short-term ADT to salvage RT improved progression-free survival (GETUG-AFU 16).

D. ADT alone

- Only for men unfit for or refusing (declining) local treatment (therapy). Consider immediate ADT if PSA doubling time is short (<12 months), PSA is very high (>50 ng/mL) or symptoms; otherwise deferred ADT is acceptable. Early ADT delays progression; survival gain mainly in aggressive disease. In fit men it is inferior to ADT + RT.
- Options: bilateral orchidectomy, LHRH (luteinising hormone-releasing hormone) agonist or antagonist; bicalutamide 150 mg monotherapy preserves libido (spares sexual interest; harmful in localised disease); intermittent ADT.

E. Others

- Watchful waiting with deferred ADT in elderly comorbid men.
- **Also know:** cryotherapy and HIFU (high-intensity focused ultrasound) not standard (remain investigational); TURP (transurethral resection of the prostate) or stents for local obstruction; trial enrolment.

Node-positive disease

Setting	Management
cN1 M0	RT to prostate and pelvis + ADT 2–3 years + abiraterone 2 years (STAMPEDE). RP + extended PLND (ePLND) in selected men
pN1 (pN+) after RP	**Immediate ADT** (ECOG 7887: OS benefit; better survival) **or ADT + adjuvant RT** (better survival than ADT alone); observation with undetectable PSA and ≤2 nodes. pN+: adjuvant ADT (ECOG 7887: better survival) ± RT

Algorithm

```
cT3–4 / cN1, M0 (PSMA PET or CT + bone scan)
   ├─ Fit, resectable  → RP + ePLND → early salvage RT ± ADT
   ├─ Most             → RT + ADT 18–36 months (+ abiraterone if N1 / very high risk)
   └─ Unfit / refuses  → ADT alone (immediate or deferred)
```

Complications and local sequelae

- ADT: hot flushes, osteoporosis, metabolic syndrome, cardiovascular risk, loss of libido; give bone protection.
- RT: proctitis, cystitis. RP: incontinence, erectile dysfunction.
- A prostate left in place can cause outflow obstruction, haematuria and ureteric obstruction.

Bilateral orchidectomy

- Removes about 95% of testosterone: **castrate level (<20 ng/dL) within 12 hours**. Cheap, no compliance issues, no flare, so useful in impending cord compression. Simple or subcapsular (all Leydig tissue must be removed).
- Versus LHRH agonists: fewer cardiac events, fractures and peripheral arterial disease reported.

A. Surgical complications

- Scrotal haematoma (commonest), wound infection, chronic scrotal or groin pain.
- Psychological: empty scrotum and altered self-image. Prevent with subcapsular or epididymis-sparing technique, or a testicular prosthesis.
- Irreversible: intermittent ADT is impossible.

B. Systemic effects of castration

Effect	Details and management
Hot flushes	Commonest (50–80%); loss of hypothalamic oestrogen signalling; venlafaxine, gabapentin, megestrol or cyproterone if bothersome
Sexual dysfunction	Loss of libido, ED (erectile dysfunction), penile shortening; ED aids help selected men
Osteoporosis, fractures	Fracture risk up 20–45%; baseline DEXA (dual-energy X-ray absorptiometry), FRAX; calcium, vitamin D, exercise; **denosumab 60 mg 6-monthly** or zoledronic acid
Metabolic	Weight gain, fat gain, **sarcopenia**, insulin resistance, diabetes (HR about 1.34), metabolic syndrome
Cardiovascular	Coronary events, sudden cardiac death, thromboembolism
Anaemia	Normocytic; reversible after stopping
Others	Gynaecomastia (≤5%), fatigue, depression, cognitive change

- **Also know:** risk of castration resistance over time; check testosterone with rising PSA.

Follow-up and recent advances

- PSA every 3–6 months; bone and metabolic health.
- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

Diagram to draw: TNM T-stage diagram (T2 confined, T3a extracapsular, T3b seminal vesicle, T4 adjacent organs) (extension of a prostate tumour), and subcapsular orchidectomy.

Key points to remember:

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
- RT + long-term ADT (18–36 months) is the standard; ADT alone is inferior (SPCG-7).
- RT + 18–36 months ADT: EORTC 22863 overall survival 79% vs 62%; STAMPEDE abiraterone metastasis-free survival HR 0.53.
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Neoadjuvant ADT before RP: no survival benefit (gain); early salvage RT favoured.
- Early salvage RT preferred over adjuvant RT for most men after RP.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis

Type: LAQ (25 marks)

Source: Practice Set 12 Paper 2, Q1 (biochemical recurrence also discussed in 2019-Summer Paper 4, Q2)

Assumed meaning: 6 months after RP (radical prostatectomy) the PSA (prostate-specific antigen) never became undetectable and is now ≥2 ng/mL, i.e. PSA persistence with early rise, suggesting residual or systemic disease.

Introduction

- After RP, PSA (half-life about 3 days) should become undetectable (<0.1 ng/mL) within about 6 weeks.
- **PSA persistence:** failure to become undetectable after RP. **BCR (biochemical recurrence):** a rise after an undetectable nadir.
- **Gleason 4+3 = grade group 3** (unfavourable intermediate risk).

Definitions of BCR after RP

Body	Definition
AUA (American Urological Association)	PSA ≥0.2 ng/mL
NCCN (National Comprehensive Cancer Network)	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1 ng/mL
EAU (European Association of Urology) risk groups	**Low risk:** PSADT (PSA doubling time) >1 year **and** grade group <4. **High risk:** PSADT ≤1 year **or** grade group 4–5

Why is the PSA high? (sources)

1. **Residual local disease:** positive margin (the only modifiable factor), residual benign tissue (low, stable PSA).
2. **Occult nodal metastases** (limited or no lymph node dissection).
3. **Distant micrometastases:** likeliest with persistent PSA and a fast rise.

Evaluation

- **Repeat PSA** to confirm; calculate PSADT.
- **Review pathology:** pT stage, margin (length >1 mm, Gleason 4 at margin), seminal vesicles, nodes. **Also know:** CAPRA-S (Cancer of the Prostate Risk Assessment–Surgery) nomogram; Decipher genomic classifier.
- **PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography):** best test; detection about half below 0.5 ng/mL and nearly 100% above 2 ng/mL. Choline or fluciclovine if PSMA is unavailable.
- **mpMRI (multiparametric MRI) pelvis:** best for local recurrence in the prostatic fossa.
- CT and bone scan are rarely positive below PSA 10 ng/mL.

Treatment

```
Confirm PSA, PSADT, pathology → PSMA PET ± mpMRI
 ├ Negative or fossa only → salvage RT to bed + ADT
 ├ Pelvic nodes only      → bed + pelvic node RT + ADT
 ├ Oligometastatic (≤3)   → ADT ± metastasis-directed therapy
 └ Polymetastatic         → treat as mHSPC: ADT + ARPI ± docetaxel
```

1. Salvage RT (radiotherapy): the only curative option

- Dose **≥64–66 Gy** to the bed (SAKK 09/10: 70 Gy gave no gain over 64 Gy); 40–45 Gy to pelvic nodes.
- Best at the lowest PSA: each 0.1 ng/mL costs about 2.6% recurrence-free survival.
- **RADICALS-RT, RAVES, GETUG-AFU 17 (ARTISTIC meta-analysis):** early salvage at PSA 0.1–0.2 equals adjuvant RT and spares ≥60% from RT. These apply to men who become undetectable, not to this patient.

2. Add ADT (androgen deprivation therapy) to salvage RT: essential here

Trial	Arms	Result
RTOG 9601 (Radiation Therapy Oncology Group)	RT ± 2 years bicalutamide 150 mg; PSA 0.2–4.0	12-year OS (overall survival) 76.3% vs 71.3%
GETUG-AFU 16	RT ± 6 months goserelin; PSA 0.2–2.0	10-year BCR-free 64% vs 49%; MFS (metastasis-free survival) better; OS unchanged
SPPORT (RTOG 0534)	Bed / bed + 4–6 months ADT / pelvis + ADT	5-year freedom from progression 70.9% / 81.3% / 87.4%
RADICALS-HD	None vs 6 vs 24 months ADT	24 months better MFS than 6 months

- At PSA ≥2 ng/mL with grade group 3, give whole-pelvis RT with **long-course (about 2 years) ADT**.

3. Systemic therapy

- **PSMA-positive metastases:** ADT + ARPI (androgen receptor pathway inhibitor: abiraterone, enzalutamide, apalutamide) ± docetaxel (LATITUDE, ARCHES, PEACE-1, ARASENS).
- **Oligometastases (≤3):** MDT (metastasis-directed therapy) by SBRT (stereotactic body RT) delays ADT (**STOMP:** ADT-free survival 21 vs 13 months; **ORIOLE:** progression 19% vs 61%).
- **High-risk BCR not fit for salvage: EMBARK** (PSADT ≤9 months, PSA ≥1 ng/mL): enzalutamide + leuprolide, MFS HR (hazard ratio) 0.42.
- **Also know:** castrate testosterone <50 ng/dL; intermittent ADT acceptable (PR.7); observation only for EAU low-risk BCR or short life expectancy; salvage lymph node dissection is investigational.

Prognosis

- **Untreated BCR (Pound):** metastasis in about one-third, median 8 years; death a median 5 years later. Predictors: **BCR <2 years, PSADT <10 months, Gleason 8–10.**
- **This patient is poor-risk:** persistent PSA, high level within 6 months, likely short PSADT. Persistent PSA carries a higher risk of metastasis than late BCR.
- **Salvage RT:** one-third to one-half BCR-free at 4–5 years; poorer above PSA 0.5–1 ng/mL. Cancer-specific survival gain mainly with PSADT <6 months treated within 2 years.
- Lifelong PSA follow-up.

Diagram to draw: the treatment flowchart above, with PSA kinetics curve after RP.

Key points to remember:

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.

Type: LAQ (25 marks)

Source: Practice Set 14 Paper 2, Q1 (related: prognosis in 2019-Summer Paper 4, Q3e; staging of non-organ-confined disease in Practice Set 10 Paper 2, Q1)

"Mixed voiding symptoms" is read as mixed storage and voiding LUTS (lower urinary tract symptoms).

Introduction

- In a 48-year-old, a PSA (prostate-specific antigen) of 18 ng/mL is far above what BPH (benign prostatic hyperplasia) usually produces. **Prostate cancer must be excluded first**; the LUTS are then treated on their own merits.
- PSA is organ-specific, not cancer-specific: BPH, prostatitis, UTI (urinary tract infection), retention and instrumentation also raise it. At PSA >10 ng/mL, biopsy yield is 43–65% with a normal DRE (digital rectal examination) and 70–90% with a suspicious one.

History

- **LUTS:** IPSS (International Prostate Symptom Score) with bother score; storage versus voiding; retention, haematuria, recurrent UTI.
- **Red flags:** bone pain, weight loss, leg oedema, cord compression.
- Family history (prostate cancer <65 years; BRCA (breast cancer gene)-related cancers), recent catheter, fever or perineal pain (prostatitis), 5ARI (5α-reductase inhibitor) use, neurological disease, erectile function.

Examination

- **DRE:** size, nodule, induration, extension, fixity (clinical T stage). Palpable bladder, neurological and spinal examination.

Investigations

Test	Purpose
Urinalysis, culture	Exclude UTI or prostatitis; treat and repeat PSA later (no empirical antibiotics without symptoms)
Repeat PSA, free PSA	Confirm the rise; low free PSA favours cancer
Creatinine, ultrasound KUB (kidney, ureter, bladder)	Upper tracts, prostate volume (PSA density)
Uroflowmetry, PVR (post-void residual)	Baseline LUTS assessment
mpMRI (multiparametric MRI) before biopsy	PI-RADS (Prostate Imaging Reporting and Data System) score; guides targeted cores; local staging
Biopsy	Targeted (PI-RADS ≥3) **plus systematic** 12 cores; **transperineal** preferred (lower sepsis)

- At PSA 18, biopsy is needed **even if MRI is negative**. Blood or urine biomarkers (PHI (Prostate Health Index), 4Kscore, PCA3) matter mainly in the 2–10 ng/mL grey zone.

Staging if cancer is found

- Grade group, cores involved, PSA, T stage → risk group. PSA 10–20 makes him at least **intermediate risk** (D'Amico, NCCN (National Comprehensive Cancer Network)).

- **High risk:** bone scan with CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**. Organ-confined disease is <50% at PSA >10; nodes about 20% at PSA >20.
- **Germline testing** (BRCA2, mismatch repair genes) for high-risk or metastatic disease or strong family history.

Management of the cancer (life expectancy >25 years)

Situation	Management
Low or favourable intermediate risk	AS (active surveillance) is possible, but uncommon at PSA 18
Localised intermediate or high risk	**RP (radical prostatectomy)**, usually robotic, with extended PLND (pelvic lymph node dissection) when node risk exceeds the nomogram threshold; or **EBRT (external beam radiotherapy) + ADT (androgen deprivation therapy)**: 4–6 months intermediate, 18–36 months high risk
Locally advanced (T3–T4, N1)	Multimodal: RP ± adjuvant or early salvage RT, or EBRT + long-term ADT; abiraterone added for high-risk node-positive or high-risk N0 disease (STAMPEDE)
Metastatic	ADT + ARPI (androgen receptor pathway inhibitor) ± docetaxel; RT to the prostate for low volume

- **Why surgery suits him:** young, gives pathological staging, and **removes the obstructing gland**, curing the LUTS. SPCG-4: surgical benefit is concentrated in men <65.
- Brachytherapy is relatively contraindicated with severe LUTS, a large gland or prior TURP (transurethral resection of the prostate) (retention, incontinence).

If biopsy is negative

- PSA 18 still needs explaining (prostatitis, large gland, missed anterior tumour): repeat MRI-targeted transperineal biopsy; %free PSA, PCA3, ConfirmMDx; close PSA follow-up.

Management of LUTS

- **Behavioural measures** (fluid timing, less caffeine) and an **α-blocker** (e.g. tamsulosin) at once; it does not alter PSA.
- **Avoid starting a 5ARI until cancer is excluded:** it halves PSA and masks cancer; if used later, record a baseline PSA and double readings after 12 months.
- Persistent storage symptoms with low PVR: add an antimuscarinic or β3-agonist (mirabegron).
- **Surgery** (TURP, HoLEP (holmium laser enucleation of the prostate)) for retention, refractory symptoms or complications, only once cancer is excluded. In locally advanced cancer with retention, a channel TURP relieves obstruction.
- **Also know:** pressure-flow study before surgery at age <50 with mainly voiding LUTS (EAU (European Association of Urology)).

```
PSA 18 + LUTS → history, DRE, urine culture → repeat PSA
      → mpMRI → targeted + systematic biopsy (transperineal)
   Cancer → risk group + staging → RP / RT + ADT / systemic
   Benign → explain PSA, re-biopsy if needed → BPH pathway
LUTS throughout: α-blocker; no 5ARI until cancer excluded
```

Diagram to draw: the flowchart above.

Key points to remember:

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

Discuss management of organ confined CA prostate

Type: LAQ (25 marks)

Source: Practice Set 15 Paper 2, Q2

Introduction

- Organ-confined prostate cancer is **cT1–T2 N0 M0**. Treatment must balance cancer mortality against **competing mortality**: life expectancy, comorbidity, baseline urinary, sexual and bowel function, and patient preference (shared decision-making).

Risk stratification (NCCN, National Comprehensive Cancer Network)

Risk group	Criteria
Very low	All of: T1c, GG (grade group) 1, PSA (prostate-specific antigen) <10 ng/mL, <3 positive cores with ≤50% cancer in each, PSA density <0.15
Low	All of: T1–T2a, GG1, PSA <10
Favourable intermediate	1 intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	2–3 intermediate factors, GG3, or ≥50% cores positive
High	Any of: T3a, GG4–5, PSA >20

- **Grade groups:** GG1 = 3+3; GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10.

Evaluation

- PSA, DRE (digital rectal examination), mpMRI (multiparametric MRI) with targeted and systematic biopsy.
- **Staging imaging** from unfavourable intermediate risk: bone scan + CT/MRI, or PSMA (prostate-specific membrane antigen) PET (positron emission tomography). **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Life expectancy (life tables, Charlson index); IPSS (International Prostate Symptom Score) and erectile function.
- **Also know:** Partin tables, Kattan nomogram, CAPRA (Cancer of the Prostate Risk Assessment) score; genomic tests (Decipher, Prolaris, Oncotype DX) for borderline surveillance decisions.

Options by risk

Risk	Preferred options
Very low / low	**Active surveillance (AS)**; watchful waiting if life expectancy short
Favourable intermediate	AS in selected men; RP (radical prostatectomy); EBRT (external-beam radiotherapy) or brachytherapy alone
Unfavourable intermediate	RP + PLND (pelvic lymph node dissection); EBRT + **4–6 months ADT** (androgen deprivation therapy)
High (organ-confined, e.g. GG4–5 or PSA >20)	RP + extended PLND (multimodal); EBRT + **18–36 months ADT**

1. Active surveillance vs watchful waiting

- **AS:** deferred treatment with **curative intent**. Monitor by PSA every 3–6 months, DRE, **confirmatory biopsy or MRI within 6–12 months**, then periodic MRI and biopsy. Trigger for treatment: grade progression (GG ≥2–3), increasing volume or patient choice.

- Entry criteria: Johns Hopkins (strict, Epstein), PRIAS (Prostate Cancer Research International Active Surveillance), Toronto (inclusive). About half are treated by 10–15 years.
- **Watchful waiting:** no curative intent; ADT for symptomatic progression. For limited life expectancy.

2. Radical prostatectomy

- Open retropubic or **robot-assisted** (similar outcomes in the randomised trial). Nerve sparing when there is no same-side extracapsular risk.
- **PLND** when nodal risk is significant on nomograms (extended template).
- **Complications:** incontinence, ED (erectile dysfunction), anastomotic stricture, rectal injury.
- **Advantages:** pathological staging; relieves obstruction; RT kept for salvage.

3. Radiotherapy

- **EBRT:** IMRT (intensity-modulated RT) with image guidance. Fractionation: conventional 78 Gy in 39; **moderate hypofractionation 60 Gy in 20** (CHHiP: non-inferior); **SBRT (stereotactic body RT) 36.25 Gy in 5** (PACE-B: similar toxicity).
- **Why add ADT:** synergistic cell kill and target shrinkage. **RTOG 94-08:** 4 months ADT improved OS in intermediate risk; **DFCI 95-096:** 6 months ADT improved 8-year OS (61% → 74%).
- **Brachytherapy:** LDR (low-dose-rate) seeds for low and favourable intermediate risk. Relative contraindications: gland >60 mL, high IPSS, recent TURP. HDR (high-dose-rate) or LDR **boost** with EBRT for higher risk (ASCENDE-RT: half the biochemical failure).
- **Toxicity:** proctitis, irritative urinary symptoms, late ED, second cancers (controversial).

4. Focal and whole-gland ablation

- HIFU (high-intensity focused ultrasound), cryotherapy, focal laser, IRE (irreversible electroporation), VTP (vascular-targeted photodynamic therapy).
- Based on the **index lesion hypothesis**. Best for MRI-visible, biopsy-confirmed, unifocal **intermediate-risk (GG2)** disease. Investigational; ideally in trials. VTP is the only modality with a randomised trial (versus AS in low risk).

Landmark trials

Trial	Comparison	Result
SPCG-4 (pre-PSA era)	RP vs watchful waiting	Cancer death RR (relative risk) 0.56 at 23 years; NNT 8; benefit <65 years
PIVOT	RP vs observation (PSA era)	No mortality difference; benefit only in intermediate risk
ProtecT	Monitoring vs RP vs RT + ADT	15-year cancer mortality 2.7%, 2.2%, 2.9% (no difference); more metastases with monitoring

- **ProtecT function:** RP worst for continence and sexual function; RT transient bowel effects.

Algorithm

```
Organ-confined CaP → risk group + life expectancy
 ├─ short life expectancy → watchful waiting
 ├─ very low / low → active surveillance
 ├─ favourable intermediate → AS (selected), RP, EBRT, brachytherapy
 └─ unfavourable intermediate / high → RP + PLND
                                    or EBRT + ADT (4–6 or 18–36 months)
```

Follow-up

- PSA every 6–12 months. **Biochemical recurrence:** after RP, PSA ≥0.2 ng/mL confirmed; after RT, **nadir + 2 ng/mL (Phoenix)**.

Recent advances

- PSMA PET staging; MRI-led surveillance; SBRT; genomic classifiers.

Diagram to draw: stepwise decision tree (treat or not → partial or whole gland → surgery or radiation).

> **Key points to remember:**
> - Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
> - AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
> - EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
> - Focal therapy rests on the index lesion hypothesis; still investigational.

Section 17 · Prostate Cancer — Short answer questions (SAQ)

Transrectal ultrasonography (TRUS)

Asked as:

- Transrectal ultrasonography — *Summer 2015 Paper II Q3f (SAQ)*
- TRUS — *Practice Set 10 Paper II Q3d (SAQ)*

Type: SAQ (10 marks)

(also answered in 2017-Winter Paper 1, Q3e and 2015-Summer Paper 2, Q3f)

Read as: transrectal ultrasonography (TRUS) of the prostate.

Physics, principle and technique

- Piezoelectric crystals; higher frequency: better resolution, less penetration (higher frequency gives better resolution but less depth).
- The prostate lies directly in front of the rectum. Probes: **6–10 MHz** end-fire, side-fire or biplane probes; left lateral position after DRE (digital rectal examination); transverse and sagittal planes, base to apex.

Sonographic anatomy (normal)

Zone (McNeal)	Glandular volume (share)	TRUS appearance	Cancers (share)
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Transition	5–10% (BPH (benign prostatic hyperplasia))	Heterogeneous, anterior	~20–25%
Central	~25%	Homogeneous, echogenic, at base	~5–10%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone).
- **Corpora amylacea** mark the surgical capsule; seminal vesicles about 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** for brachytherapy.
- **PSAD (PSA (prostate-specific antigen) density)** = PSA/volume; **>0.15** is suspicious (suggests cancer).

Findings

- **Cancer**: classically a hypoechoic peripheral-zone (PZ) lesion, but **39% are isoechoic**; false positives are BPH nodules, prostatitis and infarcts. Capsular bulge suggests extracapsular extension.
- **Cysts**: midline Müllerian; off-midline ejaculatory duct cyst (near midline); seminal vesicle cyst (lateral; renal agenesis).

- Abscess, calculi.

Indications

- **Guided biopsy:** 12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion.
- Volume before BPH surgery, brachytherapy or cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicles >1.5 cm wide, dilated ducts, midline cyst).
- Abscess drainage (drainage of abscess or cysts); haematospermia.

TRUS (transrectal ultrasound)-guided biopsy (biopsy notes)

- **12-core** systematic biopsy: medial and lateral cores at base, mid-gland and apex (sextant alone is inadequate). Periprostatic lidocaine block at the junction of prostate base and seminal vesicle.
- Contraindications: coagulopathy, acute prostatitis, severe immunosuppression. Continue low-dose aspirin; stop warfarin or clopidogrel 5–7 days.
- Complications: haematuria, haematospermia, rectal bleeding, **sepsis** (0.3–3%) from FQ (fluoroquinolone)-resistant *E. coli*; povidone-iodine rectal cleansing plus targeted or augmented prophylaxis.

Current shift and advances

- **Transperineal biopsy** preferred by EAU (European Association of Urology) for lower infection risk (the transperineal route lowers infection).
- **MRI-first pathway**: PROMIS (2017): mpMRI (multiparametric MRI) sensitivity 93% versus 48% for TRUS biopsy. PRECISION (2018): MRI-targeted biopsy found more significant cancer (38% vs 26%). Targeting: cognitive, fusion or in-bore.
- **Also know:** colour Doppler, contrast-enhanced TRUS, **elastography**; **micro-ultrasound (29 MHz)** with PRI-MUS scoring, detection comparable to mpMRI; saturation biopsy after negative biopsies; treated glands become diffusely hypoechoic.

Diagram to draw: 12-core biopsy template.

Diagram to draw: transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

Key points to remember:

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic; 12-core biopsy; sepsis 0.3–3%.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
- PROMIS and PRECISION established the MRI-first pathway.

Radiotherapy and brachytherapy for CA prostate

Asked as:

- Radiotherapy for CA prostate. — *Summer 2017 Paper II Q3f (SAQ)*
- Brachytherapy for CA prostate — *Practice Set 8 Paper II Q3e (SAQ)*

Type: SAQ (10 marks)

Radiobiology

- Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum, so larger fractions favour tumour kill over late toxicity: the basis of **hypofractionation**. Prostate cancer's low α/β ratio also favours high dose per fraction (HDR).

EBRT (external beam radiotherapy)

- **IMRT (intensity-modulated radiotherapy)**/VMAT (volumetric modulated arc therapy) with **IGRT (image-guided radiotherapy)** (fiducials, cone-beam CT (computed tomography)) is standard, allowing dose escalation with less toxicity.

Schedule	Dose per fraction	Example	Trial
Conventional	1.8–2 Gy	78 Gy in 39	—
Moderate hypofractionation (standard)	2.25–4 Gy	**60 Gy in 20**	**CHHiP**: non-inferior to 74 Gy in 37
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**PACE-B**, HYPO-RT-PC

- **Also know:** focal MRI boost (FLAME); protons; **hydrogel rectal spacer** reduces rectal toxicity.

Brachytherapy

Definition and principle

- Brachytherapy (BT; Greek *brakhus*, short): radioactive sources placed **within the prostate**.
- **Steep dose fall-off** gives an intraprostatic dose far above what EBRT (external beam radiotherapy) can safely give.

Types

Feature	LDR (low-dose-rate), permanent seeds	HDR (high-dose-rate), temporary
Source	^{125}I (half-life 59.6 days), ^{103}Pd (17 days), ^{131}Cs (9.7 days)	**^{192}Ir** in remote afterloader (temporary afterloading)
Monotherapy dose	^{125}I 140–160 Gy; ^{103}Pd 110–125 Gy	27 Gy/2 (e.g. 27 Gy in 2), 31.5 Gy/3, 34–38 Gy/4 fractions
Dosimetry	Post-implant CT at 14–30 days	Real time; none afterwards

- **LDR (low dose rate)**: permanent ^{125}I (140–160 Gy) or ^{103}Pd seeds via TRUS (transrectal ultrasound)-guided perineal template; monotherapy for low and favourable intermediate risk.

Technique

- **TRUS (transrectal ultrasound)-guided transperineal** implant under anaesthesia: high lithotomy, stepper, template grid, needles, seeds deployed on withdrawal.

- **Peripheral loading** spares the urethra. Intraoperative planning gives best coverage.
- Quality: **D90 >90%** of prescription and **V100 >85%** of gland; urethral and rectal (V100 <1 cm^3) limits. Low D90 predicts failure.

Indications

- **Monotherapy:** low and favourable intermediate risk (**RTOG 0232**: adding EBRT gave no benefit, more toxicity).
- **Boost with EBRT 45 Gy (+ implant) ± ADT (androgen deprivation therapy):** unfavourable intermediate and high risk. **ASCENDE-RT:** LDR boost halved biochemical failure vs EBRT boost, but (with more urinary toxicity) grade 3 urinary toxicity 18.4% vs 5.2%.
- Also salvage and focal BT.

Contraindications (relative)

- **IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s**, large median lobe; recent or large TURP (transurethral resection of the prostate) defect; gland >60 cm^3 (cytoreduce with ADT); pubic arch interference; prior pelvic radiation (RT [radiotherapy]); inflammatory bowel disease; cannot lie in lithotomy.

Complications of brachytherapy

- Acute obstructive and irritative symptoms; retention (29% if IPSS >20, 2% if <10); α-blockers help.
- Chronic grade ≥2 urinary toxicity about 20%; urethral stricture; rectal complications 1–2%; rare rectourethral fistula; ED (erectile dysfunction).

Follow-up after brachytherapy

- Nadir is deeper and later (about 2.5 years). **PSA bounce** in about one-third: observe for 30 months before biopsy. Failure: Phoenix (nadir + 2 ng/mL).

ADT (androgen deprivation therapy) with RT, by risk

Risk	ADT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High / locally advanced	**24–36 months** (EORTC 22863); abiraterone if very high risk (STAMPEDE)

- RT added to ADT improves survival in locally advanced disease (**SPCG-7, NCIC PR.3/MRC PR07**).

Post-prostatectomy RT

- **Adjuvant RT** (pT3, positive margin) improves control (EORTC 22911, ARO 96-02) and survival (SWOG 8794).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares most men RT.
- Salvage 64–66 Gy + ADT (**RTOG 9601**: 2 years bicalutamide improved survival) ± pelvic nodes (SPPORT).

Outcomes

- **ProtecT** (2016; 15-year 2023): equally low prostate-cancer mortality with RT, surgery or active monitoring; RT more bowel effects, surgery more incontinence and ED (erectile dysfunction).
- Failure after RT: **Phoenix definition** = nadir + 2 ng/mL.

Toxicity

- Acute: frequency, dysuria, diarrhoea. Late: **proctitis** with bleeding (6–18 months; sucralfate, argon plasma coagulation), cystitis, stricture, ED, second cancers.

Diagram to draw: TRUS-guided perineal template brachytherapy (TRUS-template transperineal implant set-up).

Key points to remember:

- α/β ~1.5 Gy: basis of hypofractionation (CHHiP 60 Gy in 20).
- ADT: 4–6 months unfavourable intermediate, 24–36 months high risk.
- Early salvage (PSA 0.1–0.2) = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

TRUS / prostate biopsy

Asked as:

- TRUS biopsy. — *Summer 2020 Paper II Q3d (SAQ)*
- Prostate biopsy — *Practice Set 4 Paper II Q3d (SAQ)* (TRUS biopsy also answered in 2020-Summer Paper 2, Q3d)

Type: SAQ (10 marks)

Introduction

- TRUS (transrectal ultrasound)-guided systematic 12-core biopsy has been the standard for diagnosing prostate cancer; about 30% are positive at first biopsy.

Indications and contraindications

- Raised PSA (prostate-specific antigen) with risk factors; suspicious DRE (digital rectal examination); MRI (magnetic resonance imaging) PIRADS (Prostate Imaging Reporting and Data System) ≥3; suspected metastasis.
- **Repeat:** rising PSA; ASAP (atypical small acinar proliferation; 50% cancer risk); positive biomarkers (PHI [Prostate Health Index], 4Kscore, PCA3); active surveillance.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression; no anus (use transperineal route).

MRI first

- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy. **PRECISION (2018):** targeted biopsy found more significant cancer (38% vs 26%) and avoided biopsy in 28% (avoided 28% of biopsies).
- Targeting: cognitive, MRI/US fusion or in-bore; 2–3 cores per target ± systematic cores.

Preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Transrectal: antibiotics for all** (contaminated route), ≤24 hours: fluoroquinolone, or cephalosporin ± aminoglycoside. Augmented (augment) or swab-directed if risk factors (recent antibiotics / antibiotics in last 6 months, diabetes, previous infection).
- Enema optional. **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the seminal vesicle–prostate junction.

Technique

	Transrectal	Transperineal
Position	Left lateral	Lithotomy
Needle route	Rectal wall (contaminated)	Sterile perineal skin
Antibiotics	Mandatory	Skin-flora only or none
Anterior/apex	Poor	Good
Anaesthesia	Periprostatic block	Perineal and periapical triangle block

- Left lateral position; DRE; 6–10 MHz probe; measure volume (PSA density >0.15 ng/mL/cc suggests cancer).
- 18-gauge spring-loaded needle; touch (indent) the capsule, withdraw slightly, then fire, so the core samples peripheral zone.

- **12 cores**: base, mid and apex on each side, medial and lateral (sextant missed lateral cancers). Add MRI-targeted cores (cognitive or fusion). Saturation only after negative biopsies.
- **EAU (European Association of Urology) 2023 prefers transperineal biopsy.**

Complications (transrectal)

Complication	Rate
Haematuria	23–63%
Haematospermia	10–50%
Rectal bleeding	2–22%
Infection	0.1–7%; sepsis 0.3–3%
Retention	0.2–0.4%
Vasovagal reaction	1–5%

- **Sepsis** from fluoroquinolone-resistant *E. coli*: carbapenem until cultures return. Rectal bleeding: probe pressure, balloon tamponade, endoscopy, embolisation.

Recent advances

- **MRI first:** PROMIS, PRECISION (above).
- **Transperineal biopsy** under local anaesthesia: fewer infections, better apex and anterior sampling; preferred by EAU (European Association of Urology) 2023.

Also know: PSA density; TZ (transition zone) cores if gland >50 mL; PSMA (prostate-specific membrane antigen) PET.

Diagram to draw: 12-core template on a transverse prostate outline.

> **Key points to remember:**
> - MRI before biopsy (PROMIS, PRECISION).
> - 12 cores; antibiotics ≤24 hours for all (transrectal).
> - Bleeding commonest; sepsis most serious.
> - Transperineal now preferred.

PIN

Type: SAQ (10 marks)

Source: Practice Set 1 Paper 2, Q3f

Assumed meaning: prostatic intraepithelial neoplasia.

Definition

- PIN (prostatic intraepithelial neoplasia): **architecturally benign** acini or ducts lined by **cytologically atypical** cells, with an intact or patchy **basal cell layer**.
- Graded **LGPIN (low-grade PIN)** and **HGPIN (high-grade PIN)**. LGPIN is not reported: it is irreproducible and carries no excess risk.

Histology

- HGPIN: nuclear enlargement, hyperchromasia, **prominent nucleoli**.
- Patterns: tufting, micropapillary, flat. **Cribriform lesions are now classed as IDC-P (intraductal carcinoma of the prostate) or AIP (atypical intraductal proliferation).**
- Immunohistochemistry: basal cells stain with **HMWCK (high-molecular-weight cytokeratin) and p63**; AMACR (α-methylacyl-CoA racemase) positive. Mainly in the peripheral zone.

Evidence that HGPIN is a precursor

- More and larger HGPIN foci in cancerous prostates; carcinoma seen budding from HGPIN.
- About 20% carry **TMPRSS2:ERG fusion** (about 50% of cancers). Extra genomic changes are needed for invasion.

Incidence

- About **4–5%** of biopsies (range 0–25%, due to observer variation).

Related lesions

Lesion	Meaning	Action on isolated finding
HGPIN	Weak risk marker	**No repeat biopsy**
PINATYP (HGPIN with adjacent atypical glands)	As "atypical"	Manage as ASAP (atypical small acinar proliferation)
ASAP	Cancer on re-biopsy about 40%	Repeat biopsy ± MRI (magnetic resonance imaging)
AIP	May herald IDC-P	Consider re-biopsy
IDC-P	Usually high-grade cancer present	Definitive treatment; excludes active surveillance

Management

- Isolated HGPIN, unifocal or multifocal: **no immediate repeat biopsy**. Decide further testing on PSA (prostate-specific antigen), DRE (digital rectal examination) and **mpMRI (multiparametric MRI)**.
- HGPIN on TURP (transurethral resection of the prostate) chips: no further sampling on current data.
- **Chemoprevention:** none proven; green tea showed reduced progression in small trials.
- *Guideline note:* EAU (European Association of Urology) guidelines still list extensive HGPIN (≥3 biopsy sites) as a reason for repeat biopsy; the chapter does not.

Diagram to draw: benign gland, HGPIN gland (atypical cells, intact basal layer) and cancer (no basal layer).

Key points to remember:

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

Maximum (combined) androgen blockade and bicalutamide

Asked as:

- Maximum androgen blockade — *Practice Set 2 Paper II Q3d (SAQ)*
- Bicalutamide — *Practice Set 13 Paper II Q3a (SAQ)*

Type: SAQ (10 marks)
(Practice Set 2 Paper II Q3d: also answered as Combined Androgen Blockade in 2018-Summer Paper 4, Q3f) (Practice Set 13 Paper II Q3a: combined androgen blockade is answered in 2018-Summer Paper 4, Q3f)

Definition and rationale of MAB

- **MAB (maximum androgen blockade)**, also CAB (combined) or total androgen blockade: castration (orchidectomy or LHRH [luteinising hormone-releasing hormone] agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes about 90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) in the prostate. The antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg, then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; no CAB benefit

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Bicalutamide: introduction and mechanism

- Non-steroidal **AR (androgen receptor) antagonist**, called first- or second-generation. It competes with testosterone and DHT (dihydrotestosterone) at the ligand-binding domain.
- Given alone, it blocks central feedback, so **LH (luteinising hormone) and testosterone rise** (about 1.5 times); aromatisation to oestradiol causes gynaecomastia; libido is partly preserved.
- **Half-life about 6 days** (once daily); unaffected by age or renal impairment. Most potent and best tolerated of its class.

Evidence for MAB

- **Crawford (1989):** leuprolide + flutamide improved survival (partly flare control); **Eisenberger (1998):** orchidectomy + flutamide, **no** benefit.
- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, about 8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **about 2–3%**; cyproterone slightly worse.
- Small gain, more cost and toxicity: **AUA (American Urological Association) advises against long-term CAB**.

Uses of bicalutamide

Use	Dose	Evidence
CAB (combined androgen blockade) with LHRH (LH-releasing hormone) agonist, metastatic	50 mg daily	Small survival gain (2–3%, PCTCG [Prostate Cancer Trialists' Collaborative Group])
Flare prevention	50 mg, with or before the agonist, 2–4 weeks	Main use today
Monotherapy, locally advanced	150 mg daily	**EPC (Early Prostate Cancer) programme:** better progression-free survival; **harm in localised disease** (death HR [hazard ratio] 1.23)
With salvage radiotherapy	150 mg for 2 years	**RTOG 9601:** 12-year overall survival 76.3% vs 71.3%
CRPC (castration-resistant prostate cancer)	50 mg	Second-line step; largely replaced by newer agents

- 150 mg monotherapy roughly equals castration in locally advanced disease, with better sexual interest; not FDA (US Food and Drug Administration) approved alone, but used in some countries.
- **Also know:** stuttering priapism; feminising hormone therapy; studied in BPH (benign prostatic hyperplasia).

Current uses of MAB

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) up to 10-fold for 10–20 days, risking cord compression, pain or obstruction with bone metastases. Start antiandrogen **with or before** the agonist for **21–28 days**. Antagonists (degarelix, relugolix) or orchidectomy cause no flare.
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in **15–30%** for 3.5–5 months (mutant AR turns the drug into an agonist).

Adverse effects of bicalutamide

- **Gynaecomastia and breast pain** (two-thirds at 150 mg): prevent with breast radiotherapy (10–15 Gy single fraction) or tamoxifen 10 mg daily.
- **Hepatotoxicity** (monitor liver tests); rare **interstitial pneumonitis**; erectile dysfunction; fewer hot flushes than castration.

Modern replacement in mHSPC (metastatic hormone-sensitive prostate cancer)

- ADT (androgen deprivation therapy) plus abiraterone, enzalutamide, apalutamide, docetaxel, or triplets (PEACE-1, ARASENS).
- **ENZAMET:** enzalutamide beat a CAB control arm for overall survival.
- **Comparison:** enzalutamide beat bicalutamide in CRPC (TERRAIN, STRIVE) and in **ENZAMET**.

Diagram to draw: testicular and adrenal androgen sources with LHRH agonist and antiandrogen sites of action. Hypothalamus–pituitary–testis axis with bicalutamide blocking the AR.

Key points to remember:

- MAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- Newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Gleason's score and its significance

Type: SAQ (10 marks)

Source: Practice Set 3 Paper 2, Q3f

Introduction

- **Gleason grading** (Donald Gleason, late 1960s) grades prostate adenocarcinoma by **glandular architecture at low magnification**, not cytology. Patterns run from 1 (most differentiated) to 5. Modified by ISUP (International Society of Urological Pathology) in **2005** and **2014**.

Patterns (modern)

Pattern	Architecture
1–2	Not used on biopsy now
3	Discrete, well-formed glands of any size
4	**Cribriform** (including glomeruloid), poorly formed or fused glands
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

Scoring rules

- **Radical prostatectomy:** primary (commonest) + secondary pattern.
- **Biopsy (2005):** commonest + **highest** pattern, however small. Single pattern is doubled (3+3).
- A tertiary pattern 5 under 5% at prostatectomy is reported as a comment; it adds risk but does not change the grade group.
- Report **% pattern 4** in score 7; report cribriform and intraductal carcinoma (both adverse).

ISUP Grade Groups (2014; WHO [World Health Organization] 2016)

Grade group	Gleason score	5-year BCR (biochemical recurrence)-free after prostatectomy
1	3+3=6	96%
2	3+4=7	88%
3	4+3=7	63%
4	8 (4+4, 3+5, 5+3)	48%
5	9–10	26%

- Grade groups separate 3+4 from 4+3 and 8 from 9–10. Starting at "1" avoids patients misreading the lowest score, 6, as moderate cancer.

Significance

- **Most powerful biopsy predictor** of pathological stage, EPE (extraprostatic extension), nodal spread (0% in GG1, 4.3% in GG3), recurrence and death.
- **Risk groups** (D'Amico, NCCN [National Comprehensive Cancer Network]) combine grade, PSA (prostate-specific antigen) and stage.
- **Treatment choice:** GG1 → active surveillance; selected GG2 with little pattern 4 → surveillance; GG ≥4 → high risk, longer ADT (androgen deprivation therapy) with radiotherapy; also selects for brachytherapy and focal therapy.
- **Limitations:** biopsy undergrades about one-third of GG1 (sampling error); inter-observer variation; after ADT or radiotherapy, grading is unreliable.
- **Also know:** pure GG1 does not metastasise, so "is it cancer?" is debated.

Diagram to draw: Gleason's pattern diagram 3 → 4 → 5.

Key points to remember:

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.

HIFU

Type: SAQ (10 marks)

Source: Practice Set 5 Paper 2, Q3d

Definition and principle

- HIFU (high-intensity focused ultrasound) focuses ultrasound from a bowl-shaped transducer or lens onto a point. Energy is absorbed only where the beams converge, so tissue there is heated while the path is spared.
- **Intensity 1500–10,000 W/cm²** (diagnostic ultrasound causes negligible heating).
- **Mechanisms: thermal coagulative necrosis** (≥55 °C; prostate exceeds 100 °C within seconds) plus **cavitation** (bubbles collapse and damage cells).
- Each lesion is a **rice grain**; the target is "painted" with overlapping lesions. Ablated and viable tissue look alike on ultrasound, so gaps occur.

Devices and routes

- **Transrectal** probe that both monitors and treats (posterior zone).
- **TULSA (MRI [magnetic resonance imaging]-guided transurethral ultrasound ablation): MR thermometry** confirms lethal temperature; reaches the anterior gland.

Indications

Setting	Use
Focal therapy, prostate cancer	MRI-visible, unilateral, grade group 2–3 index lesion; hemiablation; based on the index-lesion hypothesis
Whole-gland ablation	Localised cancer; retreatment 29–38%
Salvage after EBRT (external beam radiotherapy)	Biopsy-proven local recurrence, **≥2 years after EBRT**
BPH (benign prostatic hyperplasia)	Approved
Kidney	Investigational (incomplete ablation)

Limitations

- **Anterior lesions** (energy lost, oedema displaces target), large gland, gland height and **calcification** (some advise TURP [transurethral resection of the prostate] first).

Outcomes

- **Focal HIFU:** continence 90–95%, potency 50–78%; infield positive biopsy 16.7–54% (csPCa [clinically significant prostate cancer] 0–21.8%).
- **HEAT registry** (1379 men): 7-year failure-free survival 69%.
- **Salvage HIFU:** biochemical progression-free survival about 48% at 3 years; continence 64–87.5%.

Complications

- Retention (about 8%), UTI (urinary tract infection) 6–18%, stricture (4%), ED (erectile dysfunction); rectourethral fistula rare (about 2% after salvage); osteitis pubis.

Follow-up

- PSA (prostate-specific antigen) falls only 50–80%; **mpMRI (multiparametric MRI) + targeted biopsy at 3–6 months**, then at 1–2 and 5 years. EAU (European Association of Urology): focal therapy within trials or registries.

Also know: histotripsy (>10,000 W/cm², mechanical liquefaction, visible bubble cloud).

Diagram to draw: transrectal transducer focusing beams on a rice-grain lesion in the posterior prostate.

Key points to remember:

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Saturation biopsy in CA prostate

Type: SAQ (10 marks)

Source: Practice Set 7 Paper 2, Q3d (also answered in 2016-Summer Paper 4, Q3a)

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often 20–30), sampling the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone; the 12-core scheme is today's standard.
- Saturation reduces sampling error, mainly for **anterior and apical cancers** missed transrectally.

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation; **focal therapy** mapping (shows multifocality; TRUS [transrectal ultrasound] biopsy misses almost half of disease relevant to hemiablation).

Templates

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine).

Results

- **Initial biopsy**: 12 to 18–21 cores did not raise detection.
- **Repeat setting**: early series gained; a later large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention**; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** detected more significant cancer (38% vs 26%) and 28% avoided biopsy.
- Still used for MRI-negative high suspicion and focal therapy planning.

Diagram to draw: transperineal grid over the prostate with sampling sectors.

Key points to remember:

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Biological effects of cryosurgery

Type: SAQ (10 marks)

Source: Practice Set 11 Paper 2, Q3a (related: cryotherapy for carcinoma prostate, 2015-Summer Paper 4, Q3f)

Principle

- Tissue destruction by extreme cold. **Argon** expanding in the probe tip cools by the **Joule–Thomson effect**; **helium** expanding warms (active thaw).
- Used for small renal masses and prostate cancer.

Mechanisms of cell death

Tissue dies during **both freezing and thawing**, by four mechanisms:

Mechanism	What happens
1. Direct cell injury	*Freeze:* extracellular water freezes first and becomes hyperosmolar; cells dehydrate and membranes distort. Rapid cooling forms **intracellular ice**, which ruptures organelles. *Thaw:* water rushes back, cells swell and burst; ice recrystallises into larger crystals
2. Vascular injury	Endothelial damage, platelet activation and **thrombosis** cause ischaemic necrosis; the main delayed killing effect
3. Apoptosis	Sublethal cold at the periphery triggers programmed cell death
4. Immunomodulation	Necrotic cells release antigens and DAMPs (damage-associated molecular patterns); dendritic cells activate T cells against tumour

- **End result:** coagulative necrosis, then fibrous scar.

Factors deciding lethality

- **Temperature:** normal renal tissue dies at about −19.4 °C; tumour may need −40 to −50 °C. **Target ≤−40 °C.**
- **Fast freeze, slow thaw** maximise injury.
- **Double freeze–thaw cycle** enlarges the necrotic zone (standard of care); each freeze about 8–10 minutes.
- **Heat sink:** nearby large vessels warm tissue; more probes or longer freezing needed.

Zones of the ice ball

- Centre: complete necrosis. Edge: 0 °C; below −20 °C only about 3 mm inside the edge. The lethal −40 °C isotherm lies well inside the visible ice ball.
- So the ice ball must extend **5–10 mm beyond the tumour**.

Clinical relevance

- Real-time imaging shows the ice ball; thermocouples and a **urethral warmer** protect sphincter and urethra in prostate cryotherapy.
- **Also know:** passive versus active thaw (both used); insulated probe shafts reduce skin injury.

Diagram to draw: ice ball with probe, −40 °C isotherm inside the 0 °C edge and tumour margin.

Key points to remember:

- Argon freezes, helium thaws (Joule–Thomson).
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.

Imaging measures in CA prostate

Type: SAQ (10 marks)

Source: Practice Set 13 Paper 2, Q3e (mpMRI detail in 2024-Winter Paper 1, Q3d; PSMA PET in 2019-Summer Paper 4, Q3a)

1. Ultrasound

- **TRUS (transrectal ultrasound):** guides biopsy, measures volume (PSA [prostate-specific antigen] density). Cancer is a hypoechoic PZ (peripheral zone) lesion, but **39% are isoechoic**.
- **Also know:** Doppler, contrast and elastography (adjuncts); **micro-ultrasound** (29 MHz; PRI-MUS [Prostate Risk Identification using Micro-Ultrasound] score; close to MRI accuracy).

2. MRI: detection and local staging

- **mpMRI (multiparametric magnetic resonance imaging):**

Sequence	Shows	Cancer
T2-weighted	Anatomy; EPE (extraprostatic extension), SVI (seminal vesicle invasion)	Low signal
DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient)	Cellularity	Bright on DWI, dark on ADC
DCE (dynamic contrast-enhanced)	Vascularity	Early enhancement, washout

- **PI-RADS v2.1 (Prostate Imaging Reporting and Data System)** 1–5: DWI dominant in PZ, T2 in TZ (transition zone); DCE only upgrades PZ 3 to 4.
- **PROMIS** (sensitivity 93% vs 48% for TRUS biopsy) and **PRECISION** (more significant, less insignificant cancer): **MRI before biopsy**. Cochrane: sensitivity 0.91, specificity 0.37.
- **Also know:** biparametric MRI; **PI-QUAL** (Prostate Imaging Quality); **PRECISE** (scores change on surveillance); fusion biopsy.

3. Staging by risk group

- **Low or intermediate risk:** no routine CT (computed tomography) or bone scan.
- **High risk:** CT plus **^{99m}Tc-MDP (methylene diphosphonate) bone scan**, or **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**.
- **proPSMA:** accuracy 92% vs 65% for conventional imaging. Nodal sensitivity only 30–40%, specificity >90%: a negative scan does not exclude micrometastases.

4. Biochemical recurrence

- **PSMA PET** works at low PSA. **Calais:** 56% vs 26% for fluciclovine. **CONDOR:** management changed in 63.9%.
- Choline and fluciclovine PET are older; FDG (fluorodeoxyglucose) PET is poor except in neuroendocrine change.

5. Advanced disease

- PSMA PET selects men for **^{177}Lu-PSMA-617** (VISION); whole-body MRI. No proof yet that PSMA-guided treatment improves survival.

Diagram to draw: prostate zones with a PZ lesion dark on ADC; PSMA PET nodal uptake.

Key points to remember:

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

LHRH analogue

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 2, Q3c (also answered in 2019-Summer Paper 4, Q3b and 2017-Winter Paper 4, Q3c)

Physiology

- **LHRH (luteinising hormone-releasing hormone), also GnRH (gonadotrophin-releasing hormone)**, is a hypothalamic decapeptide released in **pulses**. It drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone); LH stimulates Leydig cells (about 95% of testosterone).
- **Continuous** exposure desensitises and down-regulates the pituitary receptor: the basis of agonist therapy.

Agonists

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year

- Substitution at **position 6** raises potency and half-life.

```
Agonist → LH surge (flare) → receptor down-regulation
        → LH falls → castrate testosterone in 2–4 weeks
```

- Survival **equals orchidectomy**. Castrate level **<50 ng/dL**; aim **<20 ng/dL** (orchidectomy median about 15).

Flare phenomenon

- LH rises up to **10-fold for 10–20 days**: bone pain, **spinal cord compression**, retention.
- **Prevent:** antiandrogen (bicalutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → antagonist or orchidectomy.

Antagonists

- Immediate receptor block, **no flare**, FSH falls.
- **Degarelix** (SC monthly): castrate by day 3 in >95%.
- **Relugolix** (oral). **HERO** vs leuprolide: sustained castration 97% vs 89%, faster recovery, fewer MACE (major adverse cardiovascular events).

Indications

- Metastatic hormone-sensitive prostate cancer (with ARPI (androgen receptor pathway inhibitor) or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** intermittent ADT (androgen deprivation therapy) acceptable; precocious puberty, endometriosis.

Side effects

- Hot flushes, loss of libido, ED (erectile dysfunction).
- **Osteoporosis:** DEXA (dual-energy X-ray absorptiometry), calcium, vitamin D, denosumab or zoledronic acid.
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%); slow testosterone recovery.

Diagram to draw: hypothalamic–pituitary–testis axis showing agonist and antagonist sites.

Key points to remember:

- Agonist: surge, then down-regulation; castrate in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; antiandrogen 21–28 days.
- Castrate <50 ng/dL; aim <20 ng/dL.
- Antagonist: no flare; preferred in cord compression.

Expectant treatment of CA prostate

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 2, Q3d

Rationale

- Overdiagnosis: autopsy prevalence of microfocal cancer roughly equals age as a percentage; lifetime risk of death is about 3%.
- **Grade group 1 (Gleason 6) has virtually no metastatic potential.** Deaths on observation come from higher grade missed at biopsy (about 30% upgraded at prostatectomy).

Two forms of expectant treatment

	AS (active surveillance)	WW (watchful waiting)
Patient	Fit for radical therapy, life expectancy ≥10 years	Limited life expectancy (age, comorbidity)
Intent	**Curative, deferred**	**Palliative**
Monitoring	PSA (prostate-specific antigen), DRE (digital rectal examination), MRI, biopsy	Infrequent PSA and review; no biopsy
Action	Radical treatment on reclassification	ADT (androgen deprivation therapy) or palliation for symptoms

Evidence

Trial	Result
SPCG-4 (pre-PSA)	Prostatectomy cut cancer death (19.6% vs 31.3%), mainly <65 years and higher grade
PIVOT	No significant mortality benefit; low-risk cancer death 2.7%
ProtecT (15 years)	Cancer death <3% in all arms; metastasis doubled on active monitoring (9.4% vs about 5%)

Selection for AS

- **Very low risk (Epstein):** grade group 1, ≤2 positive cores, no core >50%, PSA density <0.15.
- **Low risk:** grade group 1, PSA <10 ng/mL; all guidelines prefer AS.
- **Favourable intermediate risk** (small-volume grade group 2): selected cases.
- Caution: **BRCA2** carriers, high-volume disease.

AS protocol

- MRI and **confirmatory biopsy** within 1–3 years; PSA 6-monthly; DRE every 1–2 years; MRI every 18–24 months; biopsy every 3–5 years.
- **Triggers:** grade group ≥3 → treat; volume or PSA rise → MRI or biopsy first.

Outcomes

- Cancer mortality **0.5–5% at 15 years** by entry criteria (Johns Hopkins vs Toronto); about **25% treated by 5 years**.

Watchful waiting

- Switch AS to WW when life expectancy falls below about 5–10 years.

- **Also know:** genomic tests refine selection; 5ARI (5α-reductase inhibitor) has no proven role.

Diagram to draw: AS flowchart: diagnosis → MRI + confirmatory biopsy → surveillance → trigger → treatment.

Key points to remember:

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Molecular staging of CA prostate

Type: SAQ (10 marks)

Source: Practice Set 14 Paper 2, Q3e (related: prognosis in 2019-Summer Paper 4, Q3e)

Definition and rationale

- Clinical staging (DRE (digital rectal examination), PSA (prostate-specific antigen), grade, imaging) misses occult spread and cannot fully predict behaviour.
- **Molecular staging** uses DNA, RNA or protein markers in tissue, blood or nodes to find hidden disease and predict outcome.

1. Detection of occult spread

- **RT-PCR (reverse transcription polymerase chain reaction)** for PSA or PSMA (prostate-specific membrane antigen) mRNA in blood, marrow or "negative" nodes: detects micrometastases, but inconsistent, so not used clinically.
- **CTCs (circulating tumour cells):** counts are prognostic in metastatic disease; **AR-V7** (splice variant lacking the ligand-binding domain) predicts resistance to ARPIs (androgen receptor pathway inhibitors).
- **ctDNA (circulating tumour DNA)** liquid biopsy: DNA-repair mutations; clonal haematopoiesis gives false calls.

2. Tissue genomic classifiers

Test	Basis	Predicts
Decipher	22-RNA classifier	Metastasis, cancer death; post-prostatectomy radiotherapy decision
Oncotype DX GPS (Genomic Prostate Score)	17 genes (12 cancer, 5 reference) by RT-PCR	Adverse pathology; AS (active surveillance) choice
Prolaris	31 cell-cycle progression genes	Progression, cancer death

- **Also know:** PORTOS (24-gene radiotherapy response score); ConfirmMDx (methylation, after negative biopsy).

3. Genetic markers

- **Germline:** BRCA2 (aggressive disease), BRCA1, ATM, CHEK2, Lynch genes, HOXB13 (risk, not aggressiveness); test in metastatic or high-risk disease.
- **Somatic:** PTEN loss, TMPRSS2:ERG fusion, homologous recombination repair defects (guide PARP (poly ADP-ribose polymerase) inhibitors).

4. Molecular imaging

- **PSMA PET (positron emission tomography):** more sensitive and specific than CT and bone scan for nodes and metastases (proPSMA), but cannot exclude micrometastases.

Clinical use

- Refines **AS vs treatment** and post-operative radiotherapy decisions; guides targeted therapy.

Limitations

- Mostly retrospective; **multifocality and heterogeneity** (the sampled focus may not be the worst); cost; added value still being tested (G-MAJOR trial).

Diagram to draw: clinical stage vs molecular stage (tissue, blood, nodes, imaging).

Key points to remember:

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data.

Rapid Revision Sheet – Section 17 · Prostate Cancer

All the key points from this section in one place. Read this in the last 30 minutes before the exam.

Castration-resistant prostate cancer (CRPC): definition and treatment (LAQ)

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- AR still drives CRPC.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid) (LAQ)

- Never start an LHRH agonist alone with spinal metastases: cover the flare or use an antagonist (avoid flare: antiandrogen cover, antagonist or orchidectomy).
- Single 8 Gy fraction gives the same pain relief as longer courses (for focal pain); dexamethasone and MRI for cord compression.
- Focal back pain = cord compression until MRI proves otherwise.
- ADT plus abiraterone or an ARPI, or a triplet, for every fit man.
- Denosumab 120 mg or zoledronate 4 mg (IV) 4-weekly prevent SREs in CRPC (mCRPC), not in HSPC; adjust zoledronate for creatinine clearance.
- No survival benefit; not for SREs in mHSPC.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

PSA screening for prostate cancer: evidence, problems and solutions (LAQ)

- ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
- Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
- USPSTF 2018: Grade C at 55–69, Grade D ≥70.
- Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy (LAQ)

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
- RT + long-term ADT (18–36 months) is the standard; ADT alone is inferior (SPCG-7).
- RT + 18–36 months ADT: EORTC 22863 overall survival 79% vs 62%; STAMPEDE abiraterone metastasis-free survival HR 0.53.
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Neoadjuvant ADT before RP: no survival benefit (gain); early salvage RT favoured.
- Early salvage RT preferred over adjuvant RT for most men after RP.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis (LAQ)

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief. (LAQ)

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

Discuss management of organ confined CA prostate (LAQ)

- Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
- AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
- EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
- Focal therapy rests on the index lesion hypothesis; still investigational.

Transrectal ultrasonography (TRUS) (SAQ)

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic; 12-core biopsy; sepsis 0.3–3%.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
- PROMIS and PRECISION established the MRI-first pathway.

Radiotherapy and brachytherapy for CA prostate (SAQ)

- α/β ~1.5 Gy: basis of hypofractionation (CHHiP 60 Gy in 20).
- ADT: 4–6 months unfavourable intermediate, 24–36 months high risk.
- Early salvage (PSA 0.1–0.2) = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

TRUS / prostate biopsy (SAQ)

- MRI before biopsy (PROMIS, PRECISION).
- 12 cores; antibiotics ≤24 hours for all (transrectal).
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

PIN (SAQ)

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

Maximum (combined) androgen blockade and bicalutamide (SAQ)

- MAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.

- Newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Gleason's score and its significance (SAQ)

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.

HIFU (SAQ)

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Saturation biopsy in CA prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Biological effects of cryosurgery (SAQ)

- Argon freezes, helium thaws (Joule–Thomson).
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.

Imaging measures in CA prostate (SAQ)

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

LHRH analogue (SAQ)

- Agonist: surge, then down-regulation; castrate in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; antiandrogen 21–28 days.
- Castrate <50 ng/dL; aim <20 ng/dL.
- Antagonist: no flare; preferred in cord compression.

Expectant treatment of CA prostate (SAQ)

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Molecular staging of CA prostate (SAQ)

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data.

Appendix · Original papers: headings, notes and contents

Each source file's own title, note and contents list, exactly as printed, so every word of the originals is kept.

2015-Summer Paper 2 – Urology

Contents

1. Q1. BPH: medical and surgical management (LAQ)
2. Q2. Metabolic evaluation of stones (LAQ)
3. Q3a. Management of priapism (SAQ)
4. Q3b. Renal tubular acidosis (SAQ)
5. Q3c. Millin's prostatectomy (SAQ)
6. Q3d. Laparoscopic energy sources (SAQ)
7. Q3e. Vascular grafts in urology (SAQ)
8. Q3f. Transrectal ultrasonography (SAQ)

2016-Summer Paper 2 – Urology

Contents

1. Q1. Carcinoma bladder: aetiology, newer investigations and treatment (LAQ)
2. Q2. Urethral stricture: aetiology, presentation and management (LAQ)
3. Q3a. Transureteroureterostomy
4. Q3b. Acquired renal cystic disease
5. Q3c. Bladder diverticulum
6. Q3d. Botulinum neurotoxin
7. Q3e. Catheter-free trial (trial without catheter)
8. Q3f. Urethral caruncle

2017-Summer Paper 2 – Urology

Contents

1. Q1. Inguinal lymph nodes in carcinoma penis: importance and management (LAQ)
2. Q2. Traumatic renal parenchymal and vascular injuries: diagnosis and management (LAQ)
3. Q3a. Irrigation fluids for TURP (SAQ)
4. Q3b. Preoperative investigations for a living renal donor (SAQ)
5. Q3c. University of Wisconsin (UW) solution (SAQ)
6. Q3d. Urodynamic studies (SAQ)
7. Q3e. Azoospermia (SAQ)
8. Q3f. Radiotherapy for carcinoma prostate (SAQ)

2017-Winter Paper 2 – Urology

Contents

1. Q1. Principles of haemodialysis and vascular access (LAQ)
2. Q2. Evaluation for urolithiasis; management of bilateral renal staghorn calculi (LAQ)
3. Q3a. LASERS in BPH (SAQ)
4. Q3b. Interstitial cystitis (bladder pain syndrome) (SAQ)
5. Q3c. Silodosin (SAQ)
6. Q3d. ADPKD (adult polycystic kidney disease) (SAQ)
7. Q3e. RIRS (retrograde intrarenal surgery) (SAQ)
8. Q3f. Renal isotope scans (SAQ)

2018-Summer Paper 2 – Urology

Contents

1. Q1. Pathophysiology of genitourinary tuberculosis; management of thimble bladder in a 30-year-old man (LAQ)
2. Q2. Lymphatic drainage of testis and scrotum; management of stage I seminoma (LAQ)
3. Q3a. Oligospermia (SAQ)
4. Q3b. DMSA renal scan (SAQ)
5. Q3c. Dutasteride (SAQ)
6. Q3d. Penile prosthesis (SAQ)
7. Q3e. CAPD (continuous ambulatory peritoneal dialysis) (SAQ)
8. Q3f. Uroflowmetry (SAQ)

2019-Summer Paper 2 – Urology

Contents

1. Q1. Management of T1G3 TCC of the urinary bladder (LAQ)
2. Q2. ESWL: indications, mechanism of stone fragmentation and current status (LAQ)
3. Q3a. Management of urethral meatal stenosis (SAQ)
4. Q3b. Flexible ureteroscopy (SAQ)
5. Q3c. Testicular aspiration of sperm and assisted reproduction (SAQ)
6. Q3d. Management of detrusor underactivity (SAQ)
7. Q3e. Retroperitoneal fibrosis: medical management (SAQ)
8. Q3f. Female circumcision (female genital mutilation) (SAQ)

2020-Summer Paper 2 – Urology

Contents

1. Q1. Assessment and recent concepts in management of metastatic RCC (LAQ)
2. Q2. Surgical management of erectile dysfunction (LAQ)
3. Q3a. CRPC in Ca prostate (SAQ)
4. Q3b. Retractile testis (SAQ)
5. Q3c. Preoperative management of pheochromocytoma (SAQ)
6. Q3d. TRUS biopsy (SAQ)
7. Q3e. Management of valve bladder (SAQ)
8. Q3f. TOT (transobturator tape) (SAQ)

2021-Summer Paper 2 – Urology

Contents

1. Q1. Factors affecting outcome of ESWL; complications and strategies to avoid them (LAQ)
2. Q2. Selection criteria for cadaver renal donor; kidney retrieval surgery (LAQ)
3. Q3a. Post prostatectomy incontinence (SAQ)
4. Q3b. Early versus delayed VVF repair (SAQ)
5. Q3c. Perineal urethrostomy (SAQ)
6. Q3d. Medical expulsive therapy in urinary calculi (SAQ)
7. Q3e. TOT versus TVT in SUI (SAQ)
8. Q3f. Hereditary urological malignancies (SAQ)

2022-Summer Paper 2 – Urology

Contents

1. Q1. Role of the urologist in reproductive endocrinology and male infertility practice (LAQ)
2. Q2. Pathophysiology of renovascular hypertension and ischaemic nephropathy; treatment of renovascular hypertension (LAQ)

3. Q3a. Intravesical gene therapy (SAQ)
4. Q3b. Penile inversion vaginoplasty technique (SAQ)
5. Q3c. Acute tubular necrosis (SAQ)
6. Q3d. Management of stones in pregnancy (SAQ)
7. Q3e. Pre-transplant nephrectomy (SAQ)
8. Q3f. Uretero-calycostomy (SAQ)

2023-Winter Paper 2 – Urology

Contents

1. Q1. Etiopathogenesis and management of Upper tract TCC (LAQ)
2. Q2. Use of intestinal segments in urology (LAQ)
3. Q3a. GFR clinical assessment (SAQ)
4. Q3b. Neuromodulation (SAQ)
5. Q3c. Post transplant malignancy (SAQ)
6. Q3d. Biomarkers for patients with advanced RCC (SAQ)
7. Q3e. Pre transplant Immunosupression treatment options (SAQ)
8. Q3f. Physiology of UPJ in relation to urine transport (SAQ)

2024-Winter Paper 2 – Urology

Contents

1. Q1. Pathophysiology, classification and management of LUTS (LAQ)
2. Q2. Etiology, pathogenesis and management of acute renal failure (LAQ)
3. Q3a. ALPP (SAQ)
4. Q3b. Tivozanib (SAQ)
5. Q3c. 46 XX Male (SAQ)
6. Q3d. Brain stem Death: Biology and clinical implications (SAQ)
7. Q3e. Immunotherapy therapy for metastatic RCC (SAQ)
8. Q3f. Sperm extraction in Obstructive azoospermia (SAQ)

2025-Winter Paper 2 – Urology

Contents

1. Q1. Urethral stricture disease: etiology, presentation, recent advances in management (LAQ)
2. Q2. Immunology and immunotherapy in urological oncology (LAQ)
3. Q3a. Non obstructive azoospermia (SAQ)
4. Q3b. Retroileal (retroiliac) ureter (SAQ)
5. Q3c. Vascular access for hemodialysis (SAQ)
6. Q3d. Botulinum toxin and its role in urology (SAQ)
7. Q3e. Tubeless PCNL (SAQ)
8. Q3f. Surgery for recurrent vesicovaginal fistula (SAQ)

Practice Set 1 Paper 2 – Urology

Contents

1. Q1. Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain (LAQ)
2. Q2. Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)
3. Q3a. CAPD (SAQ)
4. Q3b. Flexible URS (SAQ)
5. Q3c. Anomalies of renal artery. (SAQ)

6. Q3d. Pheochromocytoma (SAQ)
7. Q3e. Microscopic Hematuria (SAQ)
8. Q3f. PIN (SAQ)

Practice paper compiled from the question bank.

Practice Set 2 Paper 2 – Urology

Contents

1. Q1. Evaluation and management of renovascular hypertension (LAQ)
2. Q2. Management of upper ureteric stones (LAQ)
3. Q3a. Bladder dysfunction in non-genitourinary systemic disease (SAQ)
4. Q3b. Mycophenolate mofetil (SAQ)
5. Q3c. Cushing syndrome (SAQ)
6. Q3d. Maximum androgen blockade (SAQ)
7. Q3e. Complications of PCNL (SAQ)
8. Q3f. VHL gene (SAQ)

Practice paper compiled from the question bank.

Practice Set 3 Paper 2 – Urology

Contents

1. Q1. Identification, maintenance, distribution and retrieval of kidney in a 25-year-old brain-dead patient (LAQ)
2. Q2. Management of bone metastases in prostate cancer (LAQ)
3. Q3a. BCG failure (SAQ)
4. Q3b. Patient positions in PCNL (SAQ)
5. Q3c. Grade 4 renal trauma (SAQ)
6. Q3d. Algorithm for management of an adrenal incidentaloma (SAQ)
7. Q3e. Metabolic abnormalities in CKD (SAQ)
8. Q3f. Gleason score and its significance (SAQ)

Practice paper compiled from the question bank.

Practice Set 4 Paper 2 – Urology

Contents

1. Q1. Treatment of a 25-year-old woman with T1N0M0 bladder TCC and CIS (LAQ)
2. Q2. Anatomy of the renal vasculature; diagnosis and treatment of renovascular hypertension (LAQ)
3. Q3a. Renal transplant in neurogenic bladder (SAQ)
4. Q3b. Urolithiasis in pregnancy (SAQ)
5. Q3c. MIBG scan (SAQ)
6. Q3d. Prostate biopsy (SAQ)
7. Q3e. Neobladder (SAQ)
8. Q3f. Role of immunotherapy in RCC (SAQ)

Practice paper compiled from the question bank.

Practice Set 5 Paper 2 – Urology

1. Q1. Describe metabolic evaluation in stone disease
2. Q2. Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD
3. Q3a. Bladder preserving approach in invasive TCC of bladder
4. Q3b. mRCC.
5. Q3c. Dexamethasone suppression test

6. Q3d. HIFU
7. Q3e. Discuss techniques of dialysis
8. Q3f. RIRS

Practice paper compiled from the question bank.

Practice Set 6 Paper 2 – Urology

Contents

1. Q1. Discuss treatment of CRPC (LAQ)
2. Q2. Classification and treatment of lower track trauma in # pelvis (LAQ)
3. Q3a. Pre-emptive renal transplant (SAQ)
4. Q3b. Causes of bleeding in PCNL; management of persistent hematuria 6 days after PCNL (SAQ)
5. Q3c. Partial nephrectomy – indications and approaches (SAQ)
6. Q3d. Adrenal Sx (SAQ)
7. Q3e. Malakoplakia (SAQ)
8. Q3f. Medical treatment of BPH (SAQ)

Practice paper compiled from the question bank.

Practice Set 7 Paper 2 – Urology

Contents

1. Q1. Evaluation and management of a 38-year-old man with uncontrolled hypertension (LAQ)
2. Q2. Etiopathogenesis of stone formation and medical management of urolithiasis (LAQ)
3. Q3a. Management of T1G3 bladder tumour (SAQ)
4. Q3b. Organ preservation solution (SAQ)
5. Q3c. Intraoperative management of phaeochromocytoma (SAQ)
6. Q3d. Saturation biopsy in carcinoma prostate (SAQ)
7. Q3e. Steinstrasse (SAQ)
8. Q3f. Renal angiomyolipoma (SAQ)

Practice paper compiled from the question bank.

Practice Set 8 Paper 2 – Urology

Contents

1. Q1. What are the various problems in PSA screening (LAQ)
2. Q2. Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction. (LAQ)
3. Q3a. Pre-op assessment of unrelated donor (SAQ)
4. Q3b. Treatment of inferior calyceal calculus (SAQ)
5. Q3c. Von Hippel Lindau disease (SAQ)
6. Q3d. Pelvic # assessment and management (SAQ)
7. Q3e. Brachytherapy for CA prostate (SAQ)
8. Q3f. Intravesical therapy for TCC of bladder (SAQ)

Practice paper compiled from the question bank.

Practice Set 9 Paper 2 – Urology

Contents

1. Q1. Diagnosis and Treatment of renal trauma (LAQ, 25)
2. Q2. Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, (LAQ, 25)
3. Q3a. Methods for GFR estimation in prospective tx donor. (SAQ, 10)

4. Q3b. Measures to decrease complications of TURP (SAQ, 10)
5. Q3c. What is optical coherence tomography. What is its potential role in management of CA bladder? (SAQ, 10)
6. Q3d. Tubeless PCNL (SAQ, 10)
7. Q3e. Medullary sponge kidney (SAQ, 10)
8. Q3f. Amendments to THOA: 2011. (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 10 Paper 2 – Urology

Contents

1. Q1. Describe staging of CA prostate and management of non-organ confirmed CA prostate (LAQ)
2. Q2. Classify CA bladder. Discuss in details organ preserving surgeries for TCC of bladder. (LAQ)
3. Q3a. Allopurinol (SAQ)
4. Q3b. Wilm's tumor (SAQ)
5. Q3c. Hemodialysis access. (SAQ)
6. Q3d. TRUS (SAQ)
7. Q3e. Von Bruns Nest (SAQ)
8. Q3f. Primary Hyperoxaluria (SAQ)

Practice paper compiled from the question bank.

Practice Set 11 Paper 2 – Urology

Contents

1. Q1. Management of metastatic RCC (LAQ, 25)
2. Q2. Urinary crystals (LAQ, 25)
3. Q3a. Biological effects of cryosurgery (SAQ, 10)
4. Q3b. Mitomycin (SAQ, 10)
5. Q3c. HOTA and its modification (SAQ, 10)
6. Q3d. Marginal kidney donor (SAQ, 10)
7. Q3e. Medical expulsive Rx for urinary calculi. (SAQ, 10)
8. Q3f. Prostatectomy incontinence (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 12 Paper 2 – Urology

1. Q1. 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis
2. Q2. Outline management of muscle invasive TCC. give critical evaluation of various types of neobladder.
3. Q3a. Describe the principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD
4. Q3b. Describe medical management of renal calculi
5. Q3c. Embryogenesis of urinary bladder
6. Q3d. Tamsulosin
7. Q3e. Valve bladder.
8. Q3f. Access techniques in dialysis

Practice paper compiled from the question bank.

Practice Set 13 Paper 2 – Urology

Contents

1. Q1. Describe role of nephron sparing surgery in management of RCC (LAQ)
2. Q2. Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. (LAQ)
3. Q3a. Bicalutamide (SAQ)
4. Q3b. Mullerionosis of UB (SAQ)
5. Q3c. Brainstem death (SAQ)
6. Q3d. Screening of CA prostate. (SAQ)
7. Q3e. Imaging measures in CA prostate (SAQ)
8. Q3f. Prostatic stents (SAQ)

Practice paper compiled from the question bank.

Practice Set 14 Paper 2 – Urology

1. Q1. 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.
2. Q2. Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder.
3. Q3a. Role of UDS before prostate surgery
4. Q3b. Zolendronic acid
5. Q3c. LHRH analogue
6. Q3d. Expectant treatment of CA prostate
7. Q3e. Molecular staging of CA prostate
8. Q3f. CRE & BEP

Practice paper compiled from the question bank.

Practice Set 15 Paper 2 – Urology

Contents

1. Q1. Describe classification and management of Wilm's bladder (LAQ)
2. Q2. Discuss management of organ confined CA prostate (LAQ)
3. Q3a. Monopolar v/s Bipolar TURP. (SAQ)
4. Q3b. Overactive bladder (SAQ)
5. Q3c. Fracture penis (SAQ)
6. Q3d. Describe various penile prosthesis in ED (SAQ)
7. Q3e. Ovarian vein syndromes (SAQ)
8. Q3f. Megameatus intact prepuce (SAQ)

Practice paper compiled from the question bank.

Practice Set 16 Paper 2 – Urology

Contents

1. Q1. Outline the management of muscle invasive TCC. Discuss the role of adjuvant & neoadjuvant chemotherapy. (LAQ, 25)
2. Q2. Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy. (LAQ, 25)
3. Q3a. Vesicourodynamics (SAQ, 10)
4. Q3b. Early surgical intervention in urethral injuries (SAQ, 10)
5. Q3c. TVT (SAQ, 10)
6. Q3d. Undescended testis (SAQ, 10)
7. Q3e. Sperm Retrival techniques (SAQ, 10)
8. Q3f. Salvage procedures for Complicated PUJO. (SAQ, 10)

Practice paper compiled from the question bank.

Practice Set 17 Paper 2 – Urology

Contents

1. Q1. Various uses of intestines in Uro (LAQ, 25)
2. Q2. Describe management of LAPC (LAQ, 25)
3. Q3a. Drug therapy in OAB (SAQ, 10)
4. Q3b. Techniques of chordee correction (SAQ, 10)
5. Q3c. Anterior urethral valve (SAQ, 10)
6. Q3d. Vaginal prolapse repair. (SAQ, 10)
7. Q3e. Perineo-scrotal hypospadias (SAQ, 10)
8. Q3f. Surgical management of bilateral varicocele (SAQ, 10)

Practice paper compiled from the question bank.